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2025-06-10

NGA STANDARDIZATION DOCUMENT

SENSOR INDEPENDENT COMPLEX DATA (SICD)

Volume 3

Image Projections Description Document

Exploitation processing description for SICD data products.
Projections from Image-to-Scene & Scene-to-Image.

(2025-06-10)

Version 1.5

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NATIONAL CENTER FOR GEOSPATIAL INTELLIGENCE STANDARDS

FOREWORD

The suite of NGA.STND.0024 standardization documents defines the Sensor Independent Complex Data (SICD) format for complex image data products generated by Synthetic Aperture Radar (SAR) systems and their data processing elements.

A SAR complex image is an intermediate data product. The real utility is in the products and measurements that may be derived from it. The quality of the pixel array (resolution, signal-to-noise ratio, etc.), along with the set of metadata provided, are critical in generating the derived products. The “sensor independence” of the SICD product refers to the ability of the allowed pixel array and metadata options to accurately describe the image products from many sensors and data processing systems. Sensor independence does NOT mean that all products have the same format for the pixel array or the same set of metadata parameters.

The SICD documentation has been organized into three volumes and one XML schema. The three volumes are summarized below. The SICD XML schema document defines the XML metadata structure for the XML metadata instance included in a given product.

Volume 1 Design & Implementation Description Document

Contains the description needed by producers of SAR complex image products to design a SICD product and the set of metadata that describe it.

Volume 2 File Format Description Document

Defines the placement of SICD data products in the allowed image file formats. Also provides the description needed by users of SICD products to read and properly extract the SICD data components from a SICD product file.

Volume 3 Image Projections Description Document

Describes the SICD sensor model and the correct projections from image location to ground point and from ground point to image location for all SICD products.

The companion suite of NGA.STND.0025 standardization documents defines the Sensor Independent Derived Data (SIDDD) format for image data products that may be derived from a SICD data product. The NGA.STND.0024 and NGA.STND.0025 documents and associated XML artifacts are available in the National System for Geospatial-Intelligence (NSG) Standards Registry (<https://nsgreg.nga.mil>).

REVISION HISTORY

Date	Version	Description
2011-10-07	1.0	Initial publication.
2016-03-31	1.1	Section 1: Table 1-2 added to identify input parameters used for error propagation. Section 3.2: Updated to include both RIC ECF and RIC ECI coordinate transformations. Section 8: Updated Figure 8-2 to identify the initial and adjusted COA Projection Sets. Added Sections 9, 10, 11 & 12.
2018-12-13	1.2.1	Version number updated to match the version number of Volume 1 and Volume 2. No technical changes. Contacts updated.
2021-11-30	1.3.0	Version number updated to match the version number of Volume 1 and Volume 2. No technical changes. 10/7/2022: Document was re-released with minor administrative corrections – all equations & symbols now display correctly.
2023-10-26	1.4.0	Projection methods for bistatic images added. Changes include: Table 1-1: Updated to include XML metadata parameter used for bistatic image projections. Table 1-2: Updated to include bistatic error statistics. Table 1-3: Updated to include APOs for monostatic & bistatic images. Sect 1: Summary of projection methods expanded to include both monostatic and bistatic images. Added new Figure 1-2. Sect 2: Modify to specify how COA positions and velocities are computed monostatic & bistatic images. Revised section 2.5 and Figure 2-7. Sect 3: Modified to include SCP COA parameters for both monostatic and bistatic images. Sect 4: Modified to include bistatic images. Sect 4.1 modified to support monostatic & bistatic images formed with the PFA. Sect 5: Revised to include projection to a ground plane for both monostatic & bistatic images. Section 5.2 revised. Section 5.3 added. Sect 6: Revised to support image-to-scene projections for monostatic and bistatic images. Sect 8: Updated to specify the use of APOs for both monostatic and bistatic images. Sect 9: Revised projection method to an HAE surface to support both monostatic & bistatic images. Added List of Abbreviations Note: The methods for projection error propagation were not updated with these changes. The methods will be extended to support bistatic images in a subsequent revision.
2024-05-01	1.4.0	• No changes to this document. • V1.4.0 XML schema corrected to show APOs are optional.

SICD Volume 3 Image Projections Description
NGA.STND.0024-3_1.5 (2025-06-10)

Date	Version	Description
2025-06-10	1.5	Projection methods for bistatic images updated. Changes include: • Version number reduced to two-digit format per NITFS Technical Board direction. Sect 1: Revised Table 1-2 and added Table 1-4 to include all input parameters used for projection error propagation computations. • Sect 3.4: Revised description of RIC coordinates frames to support both monostatic and bistatic images. • Sect 10: Revised projection method to a DEM surface to support both monostatic and bistatic images. • Sect 11: Revised to specify projection parameters and sensitivity matrices for both monostatic and bistatic images and for projection computations that have APOs applied to the COA projection set. Tables 11-1 through 11-5 updated. • Sect 11.5.1: Formula for $cRdotSF = 0$ is a correction that replaces the incorrect expressions in V1.4 and earlier for $\Delta RDOT_CLK$ and for $cAZSF$. For V1.4 and earlier, $cAZSF$ should be set equal to zero. • Sect 12: Revised single image projection error methods to support both monostatic and bistatic images with or without APOs applied.
2026-02-28	1.5	• Administrative changes made following the GEOINT Standards Working Group review. • Typographical errors corrected. No change to SICD V1.5 format design and definition. • No changes made to the SICD V1.5 XML schema.

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LIST OF ABBREVIATIONS

Abbreviation & Acronyms	
Refer to NGA.STND.0024-1 SICD Volume 1 and NGA.STND.0024-3 SICD Volume 3	
Abbreviation	Phrase in Full – Reference for SICD specific description.
ACF	Antenna Coordinate Frame – See Volume 1: Section 4.11
APC	Antenna Phase Center – See Volume 1: Section 4.6
APO	Adjustable Parameter Offset – See Volume 3: Section 1.3
ARP	Aperture Reference Point – See Volume 1: Section 4.6
CCW	Counterclockwise
CA	Closest Approach – See Volume 1: Section 4.16.3
COA	Center of Aperture – See Volume 1: Section 4.6
CST	Collection Start Time
DCA	Doppler Cone Angle
DEM	Digital Elevation Model – See Volume 3: Section 10
DRSF	Doppler Rate Scale Factor – See Volume 1: Section 4.16.3
EB	Electrical Boresight – See Volume 1: Section 4.11
ECF or ECEF	Earth Centered Fixed (ECF) or Earth Centered Earth Fixed (ECEF) – Used interchangeably to refer to the ECEF coordinate frame w/ ECF primarily used in XML parameter names.
ENU	Enumeration Type – See Volume 1: Table 2-1
ETP	Earth Tangent Plane – See Volume 1: Section 4.9.1
FP	Focus Plane – See Volume 1: Section 4.15
GCP	Ground Center Point – See Volume 3: Section 7
GPC	Ground Plane Coordinates
GPN	Ground Plane Normal – See Volume 3: Section 5
GREF	Ground Plane Reference Point – See Volume 3: Section 5
GRP	Ground Reference Point – See Volume 1: Section 4.6
HAE	Height Above the Ellipsoid – Measured relative to the WGS 84 reference ellipsoid.
HMSL	Height Above Mean Sea Level
Hz	Hertz
I2S	Image-to-Scene – See Volume 3: Section 1
IP	Image Plane – See Volume 1: Section 4.15
IFA	Image Formation Algorithm
INCA	Imaging Near Closest Approach – See Volume 1: Section 4.16
IPP	Inter-Pulse Period – See Volume 1: Section 4.5
IRW	Impulse Response Width
Kap	Polar Aperture Spatial Frequency – See Volume 1: Section 4.15.2
KSF	Polar Aperture Spatial Frequency Scale Factor – See Volume 1: Section 4.15.2
LLH	Latitude, Longitude, Height Above the Ellipsoid

Abbreviation & Acronyms	
Refer to NGA.STND.0024-1 SICD Volume 1 and NGA.STND.0024-3 SICD Volume 3	
Abbreviation	Phrase in Full – Reference for SICD specific description.
ML	Main Lobe or Mainlobe
NITF	National Imagery Transmission Format
NSG	National System for Geospatial-Intelligence
PFA	Polar Format Algorithm
PHD	Phase History Data
POI	Plane of Incidence – See Volume 1: Section 4.9
RCS	Radar Cross Section
RIC	Radial, In-Track, Cross-Track – See Volume 3: Section 3.4
RMA	Range Migration Algorithm
RMAT	Range Migration – Along Track
S2I	Scene-to-Image – See Volume 3: Section 1
SAR	Synthetic Aperture Radar
SCP	Scene Center Point – See Volume 1: Section 4.2
SICD	Sensor Independent Complex Data
SIDD	Sensor Independent Derived Data
UTC	Coordinated Universal Time
VARP	Aperture Reference Point Velocity
WGS 84	World Geodetic System 1984
XML	Extensible Markup Language

1 Introduction

The mapping of SAR image pixel locations to geolocated ground points is a fundamental step in the exploitation of the image. The SAR image pixel grid locations represent a two-dimensional projection of the three-dimensional scene that was imaged. The projection is a function of the collection geometry and the image formation processing. The projections from three-dimensional scene points to two-dimensional image grid locations are along contours of constant range and constant range rate. All points in scene that lie along a given contour will have their image signal response centered at the same image location.

The mapping of image pixel grid locations to geolocated points in the imaged scene involves two fundamental projection computations.

- (1) Image-to-Scene: The projection from image pixel grid location to a geolocated point in the imaged scene. A fixed point scatterer located at the point in the scene has its two-dimensional image impulse response centered on the image grid location.
- (2) Scene-to-Image: The projection from geolocated point in the scene to an image pixel grid location. The image pixel grid location is the center of the two-dimensional image signal response of a fixed point scatterer located at the scene point.

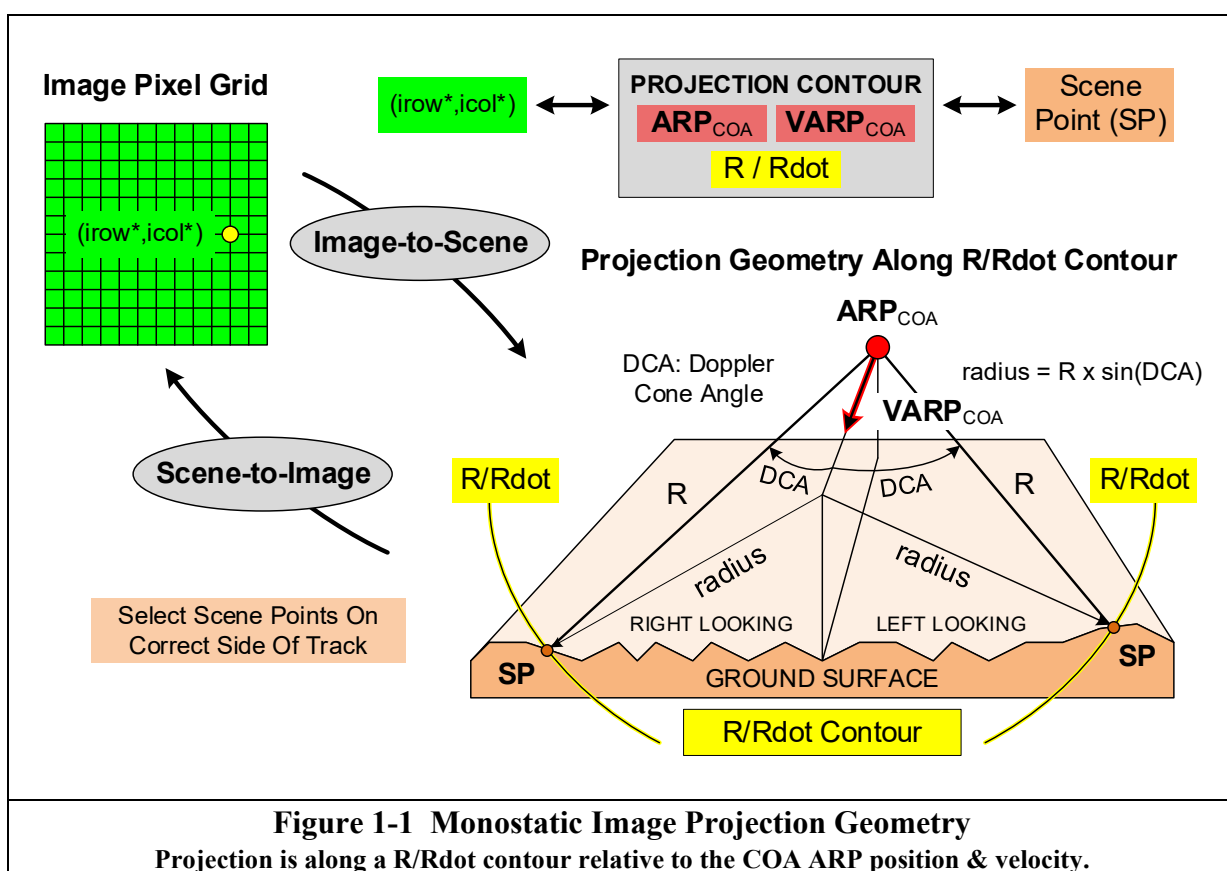
The following sections provide a description of these computations for SICD image products using the metadata supplied with the product. For all computations, the image pixel grid is considered to be a continuous two-dimensional space. Image pixel grid locations expressed in image pixel indices (irow, icol) may include fractional values. See Section 2.2.

The image contained in a SICD product may be formed from a monostatic collection or from a bistatic collection. Projection methods are provided for images formed from both types of collection. For a monostatic image, the initial Phase History Data (PHD) is collected using a single radar sensor on a single radar platform. The projection methods use the position and velocity of a single platform. For a bistatic image, the initial PHD is collected using two radar sensors located on separate transmit and receive platforms. The projection methods use the position and velocity of both the transmit platform and the receive-only platform. The methods provided for monostatic images provide precise results with some simplifications allowed by the single platform collection geometry. The methods provided for bistatic images provide precise results for both monostatic and bistatic images.

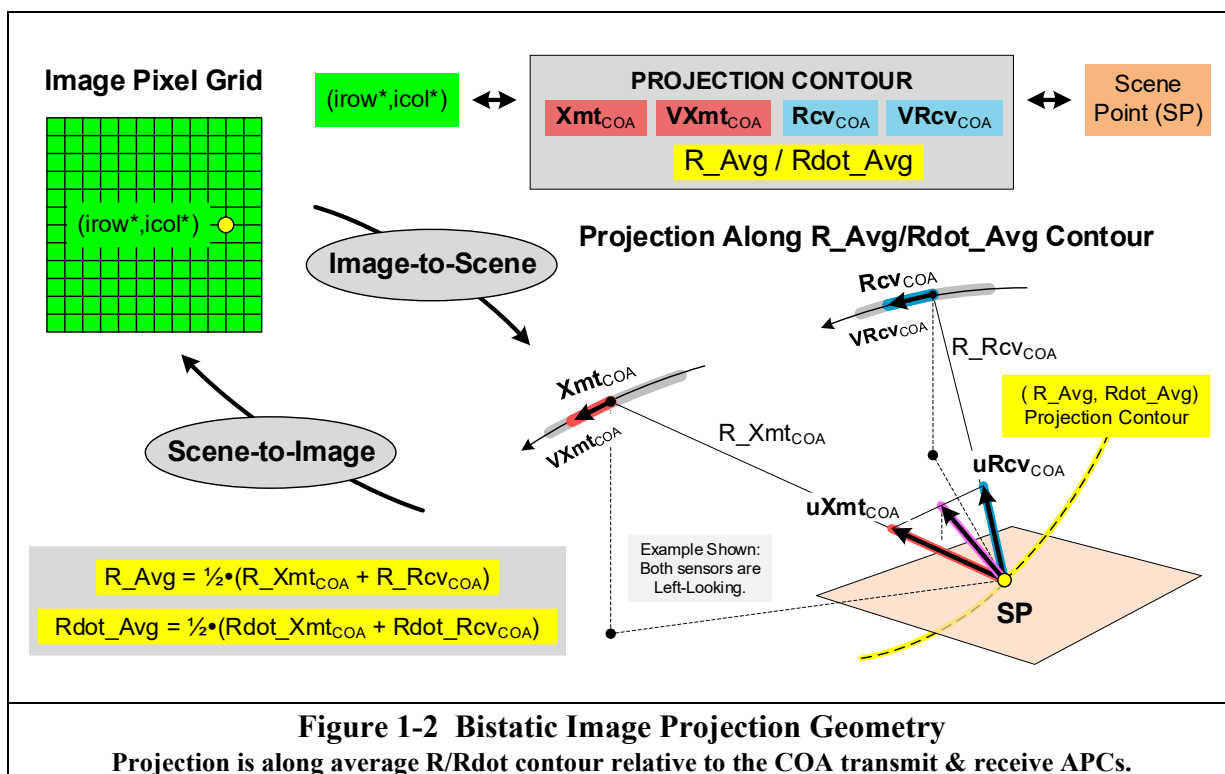
The Image-To-Scene projection maps an image grid location to a scene point on a geolocated surface. The geolocated surface is most commonly a ground plane but may be any of a number of more complex surfaces. For all image products, the image grid location uniquely determines a contour of constant range and constant range rate computed from the Center of Aperture (COA) collection geometry. For a monostatic image, the contour of constant range and constant range rate (referred to as an “R/Rdot” contour) is relative to a single platform position and velocity. For a bistatic image, the R/Rdot contour is a contour of constant “average range” and constant “average range rate” relative to two platform positions and velocities. The intersection

of the R/Rdot contour with the geolocated surface establishes the three-dimensional scene point position.

The projection geometry for a pixel location in a monostatic image is shown in Figure 1-1. The selected pixel location is $(irow^*, icol^*)$ and the projected scene point is SP. The R/Rdot contour is relative to the COA Aperture Reference Point (ARP) position and velocity, denoted ARP_{COA} and $VARP_{COA}$. As shown in the figure, there are two possible scene points – one on each side of the ground track. Only intersection points on the correct Side of Track (either left side or right side) are considered. For simple surfaces such as a ground plane, the R/Rdot contour will have a single intersection and yield a single scene point. For complex surfaces such as a terrain height model of a mountainous scene, the R/Rdot contour may yield multiple intersection points. The multiple scene points may yield an ambiguity in the precise scene location.



The projection geometry for a pixel location in a bistatic image is shown in Figure 1-2. The selected pixel location is $(irow^*, icol^*)$ and the projected scene point is SP. The R/Rdot contour is relative to the COA Antenna Phase Center (APC) positions and velocities for the two radar platforms. The transmit APC position and velocity at COA are denoted Xmt_{COA} and $VXmt_{COA}$. The receive APC position and velocity at COA are denoted Rcv_{COA} and $VRcv_{COA}$. The projection contour is along constant average range and average range rate relative to the two APC positions and velocities. For the example shown, both platforms are left-looking relative to the imaged scene.



The Scene-To-Image projection maps a geolocated point in the three-dimensional scene to a single image grid location. Only scene points on the correct Side of Track yield meaningful image grid locations. The projection computation finds the image grid location for which the COA R/Rdot projection contour passes through the scene point. In general, a precise computation requires an iterative approach. The iterative approach selects (or computes) an initial image grid location estimate. For the initial grid location, the precise R/Rdot contour is computed and projected to a ground plane that contains the scene point. The displacement between the scene point and the R/Rdot contour/ground plane point of intersection is compared to a user selected accuracy threshold. For acceptably small displacements (e.g., less than a fraction of the image resolution), the estimated image location is accepted. For larger than acceptable displacements, the displacement is used to compute a refined estimate of the image grid location.

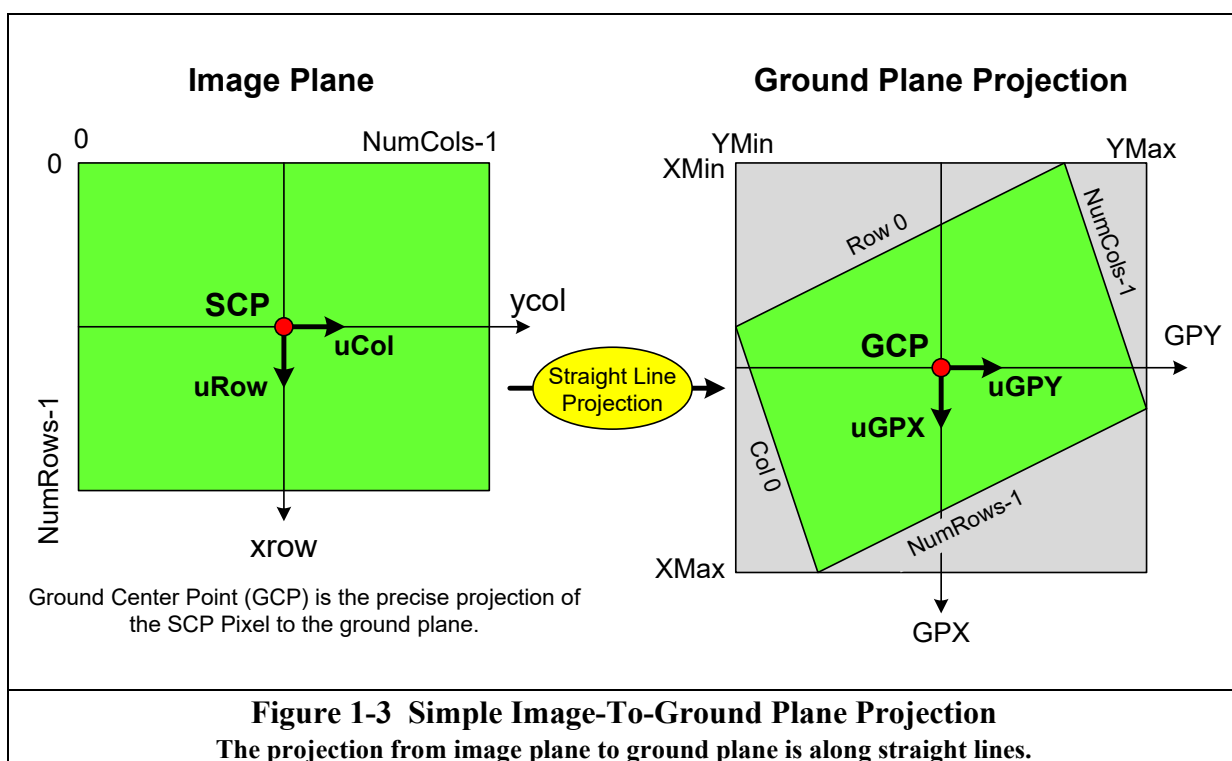
The Image-To-Scene and Scene-To-Image projections do not place any restrictions on the geolocated surface used. Only the portion of the surface on the correct Side of Track and at ranges and angles consistent with the imaging collection and processing are of interest. Also, the surface considered may or may not have been used explicitly in the image formation processing.

1.1 Ground Plane Projections: Precise Versus Simple Projection

A common application of the Image-to-Scene and Scene-To-Image projections is the resampling of the complex image grid to an evenly spaced grid in a ground plane. For each ground plane grid location, the image pixel grid location is computed. The computed image grid location will, in general, not lie precisely on a pixel center. The image signal at the grid location is then computed (via a user selected image interpolation method). The interpolated image signal value is then assigned to the ground plane grid location.

The precise resampling from image grid to ground plane grid can be computationally intensive. For each ground plane location, the image grid location is (for many images) an iterative computation where multiple image grid estimates are computed. For each estimated image grid location, the precise R/Rdot contour is computed and then projected back to the ground plane. The image grid to R/Rdot contour computation is specific to the image grid type and the image formation algorithm.

For many applications, such as generating an overview image for viewing, a less precise but greatly simplified projection algorithm is adequate. A simple projection method is presented that is computationally fast and can be applied to any SICD image product. See Figure 1-3. The projection is precise for the SCP pixel, located near the center of the image, to the ground plane. Absolute accuracy degrades slowly for pixels displaced from the SCP pixel. The simple projection method approximates the image pixel grid as a uniformly spaced grid in the image plane. Ground plane points are projected along straight lines to the image plane to compute the image grid locations. The simple projection is described in Section 7.



1.2 SICD Metadata Parameter List

The image projection computations described in the following sections use the metadata parameters listed in Table 1-1. For each parameter, the parameter name used in the processing description and the corresponding XML metadata parameter path and tag are listed. Also see the SICD Volume 1 Design & Implementation Description Document, Tables 3-1 through 3-16. In the processing description, scalar quantities such as time, t , are presented in normal face type. Vector quantities, such as the Scene Center Point (SCP) position in ECEF coordinates, **SCP**, are listed in bold face type.

Table 1-1 Metadata Parameter List	
Parameter	Description
Collect_Type	Parameter that specifies type of collection for the image contained in the product. Allowed values: MONOSTATIC and BISTATIC. XML: SICD.CollectionInfo.CollectType Note: XML element CollectType is only REQUIRED for a bistatic image. If it is not provided in the XML instance, set Collect Type = MONOSTATIC.
Scene Center Point Parameters	
Parameter	Description
SCP	SCP position in ECEF coordinates (m). XML: SICD.Geodata.SCP.ECF
SCP.Lat, SCP.Lon, SCP.HAE	SCP position in geodetic coordinates (deg, deg, m). May also be computed from SCP position in ECEF coordinates. XML: SICD.Geodata.SCP.LLH
SCP_Row, SCP_Col	SCP image pixel grid global row and column indices. XML: SICD.ImageData.SCPPixel
SCP COA Parameters – Provided for all image products.	
Parameter	Description
t_{COA}^{SCP}	SCP pixel Center Of Aperture time (sec). Time $t = 0$ at Collection Start. XML: SICD.SCPCOA.SCPTime
ARP _{COA} ^{SCP}	ARP position at the SCP COA time in ECEF coordinates (m). XML: SICD.SCPCOA.ARPPos
VARP _{COA} ^{SCP}	ARP velocity at the SCP COA time in ECEF coordinates (m/sec). XML: SICD.SCPCOA.ARPVel
SideOfTrack	Side of Track for imaging collection (L or R). XML: SICD.SCPCOA.SideOfTrack
GRAZ _{COA} ^{SCP}	Grazing angle for the ARP Line-of-Sight to SCP at SCP COA time (deg). XML: SICD.SCPCOA.GrazeAng

Table 1-1 Metadata Parameter List	
SCP COA Parameters – Only provided for Collect_Type = BISTATIC	
Parameter	Description
tx_{COA}^{SCP}	Transmit time for computing bistatic SCP COA geometry (sec). XML: SICD.SCPCOA.Bistatic.TxPlatform.Time
Xmt_{COA}^{SCP}	Transmit APC position in ECEF coordinates (m) at time tx_{COA}^{SCP} . XML: SICD.SCPCOA.Bistatic.TxPlatform.Pos
$VXmt_{COA}^{SCP}$	Transmit APC velocity in ECEF coordinates (m/sec) at time tx_{COA}^{SCP} . XML: SICD.SCPCOA.Bistatic.TxPlatform.Vel
tr_{COA}^{SCP}	Receive time for computing bistatic SCP COA geometry (sec). XML: SICD.SCPCOA.Bistatic.RcvPlatform.Time
Rcv_{COA}^{SCP}	Receive APC position in ECEF coordinates (m) at time tr_{COA}^{SCP} . XML: SICD.SCPCOA.Bistatic.RcvPlatform.Pos
$VRcv_{COA}^{SCP}$	Receive APC velocity in ECEF coordinates (m/sec) at time tr_{COA}^{SCP} . XML: SICD.SCPCOA.Bistatic.RcvPlatform.Vel
Image Data Parameters	
Parameter	Description
NumRows, NumCols	Dimensions of the Image pixel array, $S_{II}(\text{row}, \text{col})$. The discrete elements of the pixel array are indexed (row,col) XML: SICD.ImageData.NumRows & SICD.ImageData.NumCols
FirstRow, FirstCol	Row index of the first row and column index of the first column of the pixel array. For the full image array, FirstRow = 0 and FirstCol = 0. XML: SICD.ImageData.FirstRow & SICD.ImageData.FirstCol
Image Grid Parameters	
Parameter	Description
Grid_Type	Image grid type of the image. Allowed values: RGAZIM, RGZERO, XRGYCR, XATYCT, PLANE XML: SICD.Grid.Type
$uRow$	Unit vector in the increasing row direction in ECEF coordinates. Image plane is spanned by vectors $uRow$ & $uCol$. XML: SICD.Grid.Row.UVectECF
$uCol$	Unit vector in the increasing column direction in ECEF coordinates. Image plane is spanned by vectors $uRow$ & $uCol$. XML: SICD.Grid.Col.UVectECF
Row_SS	Sample spacing in the row direction (m). XML: SICD.Grid.Row.SS

Table 1-1 Metadata Parameter List	
Col_SS	Sample spacing in the column direction (m). XML: SICD.Grid.Col.SS
cT_COA(m,n), M_TCOA, N_TCOA	Center Of Aperture time polynomial coefficients (sec, sec/m, etc). The polynomial yields COA time (t_{COA}) as a function of image coordinates (xrow, ycol). Time $t = 0$ at Collection Start. M_TCOA, N_TCOA = Orders of the polynomial in xrow and ycol. XML: SICD.Grid.TimeCOAPoly
ARP Position Parameters	
Parameter	Description
ARP_Poly(t)	Aperture Reference Point position vs. time polynomial (ARP(t)). The vector polynomial yields ARP position in ECEF coordinates as a function of time. Time $t = 0$ at Collection Start Time. XML: SICD.Position.ARPPoly
APC & GRP Position Parameters – Required for Collect_Type = BISTATIC	
Parameter	Description
Xmt_Poly(t)	Transmit APC position vs. time polynomial (Xmt(t)). The vector polynomial yields transmit APC position in ECEF coordinates as a function of time. Time $t = 0$ at Collection Start Time. XML: SICD.Position.TxAPCPoly
Rcv_Poly(t)	Receive APC position vs. time polynomial (Rcv(t)). The vector polynomial yields receive APC position in ECEF coordinates as a function of time. Time $t = 0$ at Collection Start Time. Note: If multiple receive APCs are used to collect the PHD, the position is for the Receive APC centroid. The Receive APC centroid position is computed from the average of the positions for the APCs used to collect the PHD that was used to form the image. See guidance in SICD Volume 1, Section 4.6.3.
GRP_Poly(t)	Ground Reference Point (GRP) vs. time polynomial (GRP(t)). The vector polynomial yields GRP position in ECEF coordinates as a function of time. Time $t = 0$ at Collection Start Time. XML: SICD.Position.GRPPoly
Image Formation Parameters	
Parameter	Description
IFA	Image Formation Algorithm (IFA) used to form the image. Allowed values: RGAZCOMP, PFA, RMA, OTHER XML: SICD.ImageFormation.ImageFormAlgo
Range & Azimuth Compression Parameters Parameters included if and only if IFA = RGAZCOMP	
Parameter	Description

Table 1-1 Metadata Parameter List	
AzSF	Scale factor that scales image coordinate $az = ycol$ (m) to a delta cosine of the Doppler Cone Angle at COA (1/m). $\Delta\cos DCA_{COA} = AzSF \times az$ XML: SICD.RgAZComp.AzSF
Polar Format Algorithm Parameters Parameters included if and only if IFA = PFA	
Parameter	Description
cPA(n), N_PA	Polar Angle polynomial coefficients (rad, rad/sec, etc). The polynomial yields polar angle θ (rad) from collection time. Collection time in seconds ($t = 0$ at Collection Start). N_PA = Order of polynomial. XML: SICD.PFA.PolarAngPoly
cKSF(n), N_KSF	Spatial Frequency Scale Factor polynomial coefficients (dimensionless, rad^{-1} , rad^{-2} , etc.). The polynomial yields the scale factor KSF from polar angle θ (rad). The scale factor is used to map radio frequency (fx , in Hz) to aperture spatial frequency (Kap, in cycles/m). $Kap = fx \times 2/c \times KSF$ N_KSF = Order of polynomial. XML: SICD.PFA.SpatialFreqSFPoly
Range Migration Algorithm INCA Parameters Parameters included if and only if Grid_Type = RGZERO & IFA = RMA.	
Parameter	Description
cT_CA(n), N_TCA	Time of Closest Approach polynomial coefficients (sec, sec/m, etc) for Grid_Type = RGZERO. The polynomial yields time of closest approach (t_{CA}) as a function of image column coordinate (ycol). Time $t = 0$ at Collection Start. N_TCA = Order of polynomial. XML: SICD.RMA.INCA.TimeCAPoly
R_{CA}^{SCP}	Range to SCP at closest approach (m). XML: SICD.RMA.INCA.R_CA_SCP
cDRSF(m,n), M_DRSF, N_DRSF	Doppler Rate Scale Factor (DRSF) polynomial coefficients (dimensionless, m^{-1} , m^{-2} , etc). The polynomial yields DRSF as a function of image coordinates (xrow,ycol). M_DRSF, N_DRSF = Orders of the polynomial in xrow and ycol. XML: SICD.RMA.INCA.DRateSFPoly

For any image product, an optional set of parameters may be included that may be used to compute the expected accuracy of the image-to-scene and scene-to-image projections. The error propagation methods are described in Section 12. The optional parameters are provided in the XML ErrorStatistics branch. The Error Statistics parameters are described in Table 3-12 of the SICD Volume 1 Design & Implementation Description Document.

The Error Statistics parameters included in a product are dependent on the type of collection (either monostatic or bistatic). For each Collect Type, either of two optional sets of parameters may be provided. Option 1 is a set of statistics for the combined or “composite” effect of all

sources of projection error. The composite statistics are specified for the SCP pixel location and the SCP position. Option 2 is a larger set of statistics for the primary components of image projection error for a SAR system. The component statistics may be used to compute the expected error statistics using the precise geometry for a given pixel location and scene point position of interest. A given product may include both composite and component parameter sets.

For products that include component error statistics, an optional set of statistics for “unmodelled” error may be included. The unmodelled error statistics, if included, are combined with the component error statistics when computing expected projection error statistics. The set of unmodelled error statistics may be included for both monostatic images and bistatic images.

Listed in Table 1-2 are the error parameters that are the inputs to the error propagation methods specified in this document. Each parameter is either the error variance for a single parameter or an error covariance matrix for a vector parameter composed of multiple components. All parameters for which error statistics are provided or computed are assumed to be zero mean. The variance parameters and covariance matrices are computed based on the common definitions for variance, covariance, standard deviation, and correlation coefficient. For each parameter in Table 1-2, the parameters in XML ErrorStatistics branch used to compute it are specified. Guidance is also provided as needed when optional parameters within the block have been omitted.

Table 1-2 Error Statistics Parameters	
Error statistics that apply to projection computations using the Initial COA Projection Set. Adjustable Parameters Offsets are not applied.	
Monostatic Image - Option 1: Composite Error Statistics	
Parameter	Description
$\mathbf{C}_{\text{RGZ}}^{\text{SCP}}$	Covariance matrix for the composite image error vector for the SCP pixel location and the SCP position, $\Delta\text{RGZ}^{\text{SCP}}$. Composite RGAZ Error: $\Delta\text{RGZ}^{\text{SCP}} = \begin{bmatrix} \Delta\text{RG} \\ \Delta\text{AZ} \end{bmatrix}$ Size 2 x 1 $\mathbf{C}_{\text{RGZ}}^{\text{SCP}} = E \{ \Delta\text{RGZ}^{\text{SCP}} \cdot (\Delta\text{RGZ}^{\text{SCP}})^T \}$ Matrix Size: 2 x 2 Entry Units: m² Matrix entries computed from the statistics provided. XML: SICD.ErrorStatistics.CompositeSCP
Monostatic Image - Option 2: Component Error Statistics	
Parameter	Description
$\mathbf{C}_{\text{APV}, \text{AIF}}^{\text{AIF}}$	Covariance matrix for the ARP position & velocity error vector at $t = t_{\text{COA}}^{\text{SCP}}$. Parameter AIF specifies the coordinate frame used for ARP error statistics. Allowed AIF values: “ECF”, “RIC_ECF” and “RIC_ECF” Position Error: $\Delta\mathbf{AP}_{\text{AIF}} = \begin{bmatrix} \Delta\text{AP1} \\ \Delta\text{AP2} \\ \Delta\text{AP3} \end{bmatrix}$ Units: meters Velocity Error: $\Delta\mathbf{AV}_{\text{AIF}} = \begin{bmatrix} \Delta\text{AV1} \\ \Delta\text{AV2} \\ \Delta\text{AV3} \end{bmatrix}$ Units: meters/sec ARP_PV Error: $\Delta\mathbf{APV}_{\text{AIF}} = \begin{bmatrix} \Delta\mathbf{AP}_{\text{AIF}} \\ \Delta\mathbf{AV}_{\text{AIF}} \end{bmatrix}$ Size 6 x 1 $\mathbf{C}_{\text{APV}}^{\text{AIF}} = E \{ \Delta\mathbf{APV}_{\text{AIF}} \cdot (\Delta\mathbf{APV}_{\text{AIF}})^T \}$ Matrix Size: 6 x 6 Entry Units: m², m²/sec and m²/sec² Matrix entries computed from the product metadata. If optional correlation coefficients are omitted, then set all cross correlations to zero. XML: SICD.ErrorStatistics.Components.PosVelErr
VAR_{RB}	Variance of the radar sensor range bias. Range bias error is constant across the imaged scene. Range Bias Error: $\Delta\text{R_BIAS}$ Units: meters Variance Units: m² Variance computed from the product metadata standard deviation. XML: SICD.ErrorStatistics.Components.RadarSensor.RangeBias

Table 1-2 Error Statistics Parameters	
VAR_{CLK_SF}	Variance of the clock frequency error expressed as scale factor for the radar timing clock that controls both “fast time” and “slow time”. Clock Frequency Error: $\Delta f_Clk = f_Clk - f_Clk_0$ Units: Hz Clock Freq. Error Scale Factor: $CLK_SF = \Delta f_Clk / f_Clk_0$ Variance Units: dimensionless – scale factor is Hz/Hz. Variance computed from the product metadata standard deviation. If omitted, then set variance equal to zero. XML: SICD.ErrorStatistics.Components.RadarSensor.ClockFreqSF
VAR_{TROP}	Variance of the error in the troposphere delay expressed as a range error to SCP at COA time, ΔR_TROP. Assumed constant across the scene. Tropo Range Error: ΔR_TROP Units: meters $VAR_{TROP} = \begin{cases} TropoRangeSlant^2 & \text{If TropoRangeSlant present} \\ \frac{TropoRangeVertical^2}{\sin^2(GRAZ_{COA}^{SCP})} & \text{If only TropoRangeVertical present} \\ 0 & \text{Otherwise} \end{cases}$ Variance Units: m^2 Variance computed from the product metadata standard deviation. If omitted, then set variance equal to zero. XML: SICD.ErrorStatistics.Components.RadarSensor.TropoError
VAR_{IONO}	Variance of the error in ionosphere delay. Error expressed as a range error for the SCP incidence, ΔR_IONO. Iono Range Error: ΔR_IONO Units: meters $VAR_{IONO} = \begin{cases} IonoRangeVertical^2 & \text{If IonoRangeVertical present} \\ \frac{IonoRangeVertical^2}{\sin^2(GRAZ_{COA}^{SCP})} & \text{If IonoRangeVertical present} \\ 0 & \text{Otherwise} \end{cases}$ Variance Units: m^2 Variance computed from the product metadata standard deviation. If omitted, then set variance equal to zero. XML: SICD.ErrorStatistics.Components.RadarSensor.IonoError
Bistatic Image - Option 1: Composite Error Statistics	
Parameter	Description
C_{RRdot}^{SCP}	Covariance matrix for the composite image error vector for the SCP pixel and the SCP position, $\Delta RRdot^{SCP}$. Composite RRdot Error: $\Delta RRdot^{SCP} = \begin{bmatrix} \Delta R_Avg \\ \Delta Rdot_Avg \end{bmatrix}$ Size 2 x 1 $C_{RRdot}^{SCP} = E \{ \Delta RRdot^{SCP} \bullet (\Delta RRdot^{SCP})^T \}$ Matrix Size: 2 x 2 Entry Units: $m^2, m^2/sec, m^2/sec^2$ Matrix entries computed from the statistics provided. XML: SICD.ErrorStatistics.BistaticCompositeSCP
Bistatic Image - Option 2: Component Error Statistics	

Table 1-2 Error Statistics Parameters

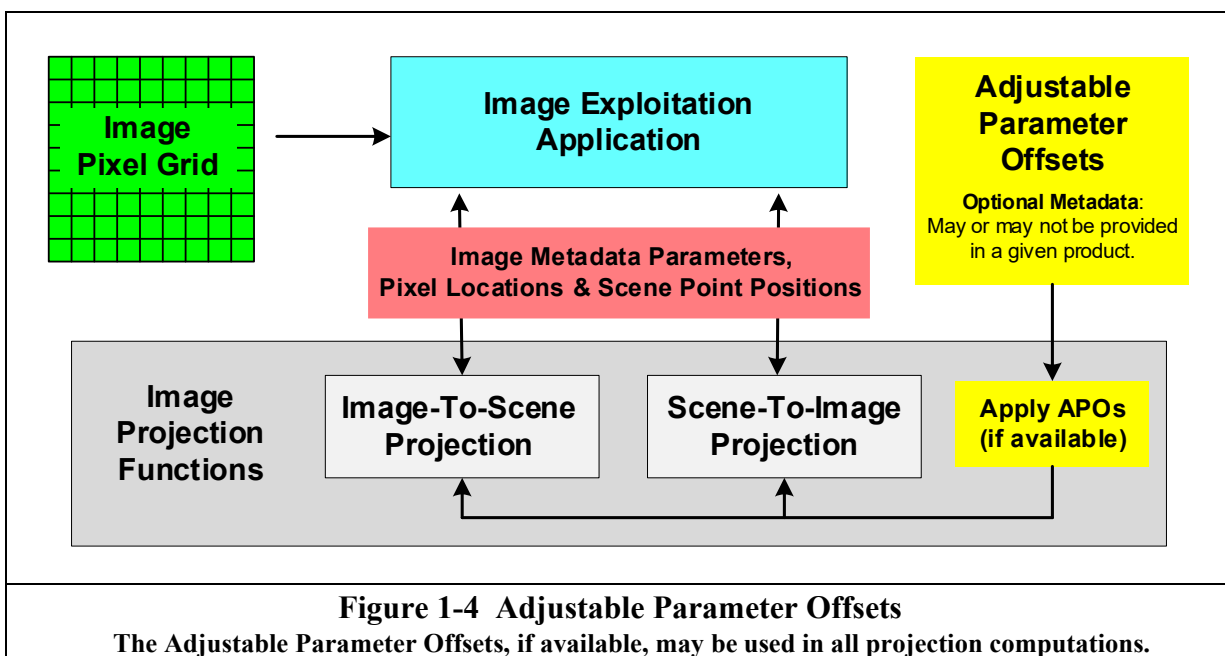
Parameter	Description
$\mathbf{C}_{XPV}^{XIF}$ XIF	Covariance matrix for the transmit APC position & velocity error vector at $t = t_{COA}^{SCP}$. Parameter XIF specifies the coordinate frame used for APC error statistics. Allowed XIF values: “ECF”, “RIC_ECF” and “RIC_ECI” Position Error: $\Delta \mathbf{XP}_{XIF} = \begin{bmatrix} \Delta XP1 \\ \Delta XP2 \\ \Delta XP3 \end{bmatrix}$ Units: meters Velocity Error: $\Delta \mathbf{XV}_{XIF} = \begin{bmatrix} \Delta XV1 \\ \Delta XV2 \\ \Delta XV3 \end{bmatrix}$ Units: meters/sec Xmt_PV Error: $\Delta \mathbf{XPV}_{XIF} = \begin{bmatrix} \Delta \mathbf{XP}_{AIF} \\ \Delta \mathbf{XV}_{AIF} \end{bmatrix}$ Size 6 x 1 $\mathbf{C}_{XPV}^{XIF} = E \{ \Delta \mathbf{XPV}_{XIF} \bullet (\Delta \mathbf{XPV}_{XIF})^T \}$ Matrix Size: 6 x 6 Entry Units: m^2, m^2/sec and m^2/sec^2 XML: SICD.ErrorStatistics.BistaticComponents.PosVelErr.TxPVCov
$\mathbf{C}_{RPV}^{RIF}$ RIF	Covariance matrix for the receive APC position & velocity error vector at $t = t_{COA}^{SCP}$. Parameter RIF specifies the coordinate frame used for APC error statistics. Allowed RIF values: “ECF”, “RIC_ECF” and “RIC_ECI” Position Error: $\Delta \mathbf{RP}_{RIF} = \begin{bmatrix} \Delta RP1 \\ \Delta RP2 \\ \Delta RP3 \end{bmatrix}$ Units: meters Velocity Error: $\Delta \mathbf{RV}_{RIF} = \begin{bmatrix} \Delta RV1 \\ \Delta RV2 \\ \Delta RV3 \end{bmatrix}$ Units: meters/sec Rcv_PV Error: $\Delta \mathbf{RPV}_{RIF} = \begin{bmatrix} \Delta \mathbf{RP}_{AIF} \\ \Delta \mathbf{RV}_{AIF} \end{bmatrix}$ Size 6 x 1 $\mathbf{C}_{RPV}^{RIF} = E \{ \Delta \mathbf{RPV}_{RIF} \bullet (\Delta \mathbf{RPV}_{RIF})^T \}$ Matrix Size: 6 x 6 Entry Units: m^2, m^2/sec and m^2/sec^2 XML: SICD.ErrorStatistics.BistaticComponents.PosVelErr.RcvPVCov
$\mathbf{CC}_{XPV_RPV}^{XIF_RIF}$	Cross-covariance matrix for the transmit APC PV error vector at time $t = t_{COA}^{SCP}$ and the receive APC PV error vector at time $t = t_{COA}^{SCP}$. $\mathbf{CC}_{XPV_RPV}^{XIF_RIF} = E \{ \Delta \mathbf{XPV}_{XIF} \bullet (\Delta \mathbf{RPV}_{RIF})^T \}$ Matrix Size: 6 x 6 Entry Units: m^2, m^2/sec and m^2/sec^2 Error vectors and reference frames as defined above. XML: SICD.ErrorStatistics.BistaticComponents.PosVelErr.TxRcvPVXCov

Table 1-2 Error Statistics Parameters	
C_{XRTF}	Covariance matrix for the Transmit & Receive time and clock frequency error vector. Transmit time error (Δt_x) and receive time error (Δt_r) are at time $t = t_{COA}^{SCP}$. Transmit and receive control clock frequency errors (Δf_Clk_X and Δf_Clk_R) are expressed as scale factors. Time/Freq Error Vector: $\Delta XRTF = \begin{bmatrix} \Delta t_x_{COA}^{SCP} \\ f_Clk_X_SF \\ \Delta t_r_{COA}^{SCP} \\ f_Clk_R_SF \end{bmatrix}$ Size 4 x 1 $C_{XRTF} = E\{ \Delta XRTF \bullet (\Delta XRTF)^T \}$ Transmit Time Error: $\Delta t_x_{COA}^{SCP}$ Units: sec Xmt Clock Error SF: $f_Clk_X_SF = \Delta f_Clk_X / f_Clk_X_0$ Receive Time Error: $\Delta t_r_{COA}^{SCP}$ Units: sec Rev Clock Error SF: $f_Clk_R_SF = \Delta f_Clk_R / f_Clk_R_0$ Matrix Size: 4 x 4 Entry Units: sec^2, sec, 1 (unitless) XML: SICD.ErrorStatistics.BistaticComponents.RadarSensor
C_{ATM}	Covariance matrix for the atmospheric delay compensation error vector. Tropo Error: $\Delta TD_ATM = \begin{bmatrix} \Delta TD_Atm_X \\ \Delta TD_Atm_R \end{bmatrix}$ Size 2 x 1 $C_{ATM} = E\{ \Delta TD_ATM \bullet (\Delta TD_ATM)^T \}$ Matrix Size: 2 x 2 Entry Units: sec^2 Matrix entries computed from the product metadata. XML: SICD.ErrorStatistics.BistaticComponents.AtmosphericError
Unmodeled Error Statistics	
Parameter	Description
C_{UI}	Covariance matrix for the unmodeled image error vector. Errors expressed in image coordinates (Δx_{row}, Δy_{col}). Unmodeled Error: $\Delta IL_{UI} = \begin{bmatrix} \Delta x_{row} \\ \Delta y_{col} \end{bmatrix}$ Size 2 x 1 $C_{UI} = E\{ \Delta IL_{UI} \bullet (\Delta IL_{UI})^T \}$ Matrix Size: 2 x 2 Entry Units: m^2 If omitted from the product metadata, set all enties equal to 0. XML: SICD.ErrorStatistics.Unmodeled

1.3 Adjustable Parameter Offsets

The error in the geolocation estimates derived from a SAR image product are due to errors in the measured or estimated parameters associated with the collection and/or processing of the image. The sources of geolocation error include two primary types. The first type are errors in the measured platform position versus time during the imaging collection. The second type are errors in estimated or measured signal timing parameters within the radar sensor. The errors in estimated signal timing result in errors in the measured range and range rate to scatterers in the imaged scene. For a given image product, the accuracy of the geolocation estimates derived from the product may be improved if estimates of the errors in position versus time and/or errors in the signal timing are available and incorporated into the projection computations.

The Image-To-Scene and Scene-To-Image projection methods described in the following sections are specified using only metadata that is required to be provided with the product. The metadata provided may also include an optional set of parameters that may be used to correct errors in the computed projection parameters. The optional parameters are referred to as Adjustable Parameter Offsets (APOs). The APOs include corrections to the platform positions and velocities and parameters used to correct the computed ranges and range rates. The use of Adjustable Parameter Offsets is conceptually shown in Figure 1-4. In general, the APOs for a given image are determined/computed after the image formation is complete. The APOs may then be included in metadata of the SICD product.



The set of Adjustable Parameter Offsets that may be included in a product is dependent upon the Collect Type. The parameter sets that may be included are listed in Table 1-3. The methods for applying the offsets for the Image-to-Scene and Scene-to-Image projections are specified in the Section 8.

Table 1-3 Adjustable Parameter Offsets	
Parameters that may be included for Collect Type = Monostatic	
XML: SICD.ErrorStatistics.AdjustableParameterOffsets	
Parameter	Description
$\Delta \text{ARP}_{\text{COA}}^{\text{SCP}}$	Aperture Reference Point (ARP) position offset in ECEF coordinates at $t = t_{\text{COA}}^{\text{SCP}}$. Units: meters XML: ARPPosSCPCOA
ΔVARP	Aperture Reference Point (ARP) velocity offset in ECEF coordinates. Units: meters/sec XML: ARPVel
$\Delta t_{\text{COA}}^{\text{SCP}}$	Transmit time offset at $t = t_{\text{COA}}^{\text{SCP}}$. Units: sec XML: TxTimeSCPCOA
$\Delta t_{\text{COA}}^{\text{SCP}}$	Receive time offset at $t = t_{\text{COA}}^{\text{SCP}}$. Units: sec XML: RcvTimeSCPCOA
Parameters that may be included for Collect Type = Bistatic	
XML: SICD.ErrorStatistics.BistaticAdjustableParameterOffsets	
Parameter	Description
$\Delta \text{Xmt}_{\text{COA}}^{\text{SCP}}$	Transmit APC position offset in ECEF coordinates at $t = t_{\text{COA}}^{\text{SCP}}$. Units: meters XML: TxPlatform.APCPosSCPCOA
ΔVXmt	Transmit APC velocity offset in ECEF coordinates. Units: meters/sec XML: TxPlatform.APCVel
$\Delta t_{\text{COA}}^{\text{SCP}}$	Transmit time offset at $t = t_{\text{COA}}^{\text{SCP}}$. Units: sec XML: TxPlatform.TimeSCPCOA
$f_{\text{Clk_X_SF}}$	Transmit payload control clock frequency offset scale factor. Units: Hz/Hz (i.e., dimensionless). XML: TxPlatform.ClockFreqSF
$\Delta \text{Rcv}_{\text{COA}}^{\text{SCP}}$	Receive APC position offset in ECEF coordinates at $t = t_{\text{COA}}^{\text{SCP}}$. Units: meters XML: RcvPlatform.APCPosSCPCOA
ΔVRcv	Receive APC velocity offset in ECEF coordinates. Units: meters/sec XML: RcvPlatform.APCVel
$\Delta t_{\text{COA}}^{\text{SCP}}$	Receive time offset at $t = t_{\text{COA}}^{\text{SCP}}$. Units: sec XML: RcvPlatform.TimeSCPCOA
$f_{\text{Clk_R_SF}}$	Receive payload control clock frequency offset scale factor. Units: Hz/Hz (i.e., dimensionless). XML: RcvPlatform.ClockFreqSF
Note: The method for determining the Adjustable Parameter Offsets is not addressed in this document. For some SAR systems, an improved estimate of the platform position versus time may become available after the image formation processing is complete. The improved position estimate may be used to compute the ARP / APC offset parameters.	

For a product that includes APOs, an optional set of error statistics may also be included in XML ErrorStatistics branch. The error statistics may be used to compute the expected error statistics for projections computed using the APOs. The APO error statistics that may be provided, like the APO input set, are dependent upon the Collect Type. The APO error statistics that may be included are listed in Table 1-4.

Table 1-4 APO Error Parameters	
Error statistics that apply to projection computations when APOs are applied.	
Monostatic Image APO Statistics	
Parameter	Description
C_{RGZ}^{SCPAPO}	Covariance matrix for the composite image error vector for the SCP pixel location and its projected position on the SCP ETP when monostatic APOs are applied. Composite RGAZ Error: $\Delta RGZ_{APO}^{SCP} = \begin{bmatrix} \Delta RG \\ \Delta AZ \end{bmatrix}$ Size 2 x 1 $C_{RGZ}^{APO_M} = E \{ \Delta RGZ_{APO}^{SCP} \bullet (\Delta RGZ_{APO}^{SCP})^T \}$ Matrix size: 2 x 2 Entry Units: m² Matrix entries computed from the statistics provided. XML: SICD.ErrorStatistics.AdjustableParameterOffsets.CompositeSCP
C_{APOM}	Covariance matrix for the monostatic APO input parameter error vector. APO Input Error: $\Delta APO_M = \begin{bmatrix} \delta ARP_{COA}^{SCP} \\ \delta VARP \\ \delta tx_{COA}^{SCP} \\ \delta tr_{COA}^{SCP} \end{bmatrix}$ Size 8 x 1 $C_{APOM} = E \{ \Delta APO_M \bullet (\Delta APO_M)^T \}$ Matrix Size: 8 x 8 Entry Units: m², m²/sec², sec², m²/sec, m•sec, and m. XML: SICD.ErrorStatistics.AdjustableParameterOffsets.APOError
Bistatic Image APO Statistics	
Parameter	Description
C_{RRdot}^{SCPAPO}	Covariance matrix for the composite image error vector for the SCP pixel location and its projected position on the SCP ETP when bistatic APOs are applied. Composite RRdot Error: $\Delta RRdot_{APO}^{SCP} = \begin{bmatrix} \Delta R_Avg \\ \Delta Rdot_Avg \end{bmatrix}$ Size 2 x 1 $C_{RRdot}^{SCPAPO} = E \{ \Delta RRdot_{APO}^{SCP} \bullet (\Delta RRdot_{APO}^{SCP})^T \}$ Matrix size: 2 x 2 Entry Units: m², m²/sec², m²/sec Matrix entries computed from statistics provided. XML: SICD.ErrorStatistics.BistaticAdjustableParameterOffsets.BistaticCompositeSCP

Table 1-4 APO Error Parameters	
C_{APOXR}	Covariance matrix for the bistatic APO input parameter error vector. Xmt APO Error: ΔAPO_X $\begin{bmatrix} \delta X_{mt}^{SCP} \\ \delta V_{Xmt} \\ \delta t_{COA}^{SCP} \\ \delta f_{Clk_X_SF} \end{bmatrix}$ Size 8 x 1 Rcv APO Error: ΔAPO_R $\begin{bmatrix} \delta R_{cv}^{SCP} \\ \delta V_{Rcv} \\ \delta t_{COA}^{SCP} \\ \delta f_{Clk_R_SF} \end{bmatrix}$ Size 8 x 1 Xmt/Rcv APO Error: $\Delta APO_{XR} = \begin{bmatrix} \Delta APO_X \\ \Delta APO_R \end{bmatrix}$ Size 16 x 1 $C_{APOXR} = E\{\Delta APO_{XR} \bullet (\Delta APO_{XR})^T\}$ Matrix Size: 16 x 16 Entry Units: m², m²/sec², sec², m²/sec, m•sec, m, m/sec, sec, and 1 (unitless). XML: SICD.ErrorStatistics.BistaticAdjustableParameterOffsets.APOError

1.4 Image Projection Computations

The image projections computations described in the following sections are summarized below.

- | | |
|------------|---|
| Section 2 | Image Pixel Array & Image Plane Parameters
Describes image pixel array and the basis vectors that define the image plane. Also specifies the method for computing COA times, positions, and velocities. |
| Section 3 | SCP Pixel Projection
Describes the COA projection parameters for the SCP pixel and the SCP position. |
| Section 4 | Image Grid To R/Rdot Contour
Describes the precise image grid location (xrow, ycol) to COA Projection Set computation. The computation is a function of the Collect Type, Grid Type, and the Image Formation Algorithm used.
4.1 Image Grid To R/Rdot: Grid_Type = RGAZIM & IFA = PFA
4.2 Image Grid To R/Rdot: Grid_Type = RGAZIM & IFA = RGAZCOMP
4.3 Image Grid To R/Rdot: Grid_Type = RGZERO
4.4 Image Grid To R/Rdot: Grid_Type = XRGYCR
4.5 Image Grid To R/Rdot: Grid_Type = XCTYAT
4.6 Image Grid To R/Rdot: Grid_Type = PLANE |
| Section 5 | Precise R/Rdot to Ground Plane Projection
Describes the precise projection to a ground plane along an R/Rdot contour. The method is dependent upon the Collect Type. |
| Section 6 | Scene Point to Image Grid Projection
Describes the precise scene point to image grid projection using an iterative algorithm. |
| Section 7 | Simple Ground Plane Projection
Describes the simple ground plane projection along straight lines from the image plane. Allows for fast ground plane resampling for any image product. |
| Section 8 | Adjustable Parameters
Describes the use of Adjustable Parameters in the Image-to-Scene and Scene-to-Image projections. |
| Section 9 | Precise R/Rdot to Constant HAE Surface Projection
Describes the precise projection to a surface of constant height above the WGS-84 reference ellipsoid along an R/Rdot contour. |
| Section 10 | Precise R/Rdot to DEM Surface Projection
Describes the precise projection to a DEM surface model along an R/Rdot contour. |

Section 11 Projection Sensitivity Parameters

Describes the Image-to-Scene and Scene-to-Image projection sensitivities for a given image location/scene point projection pair.

Section 12 Projection Error Propagation

Describes the error propagation for both Image-to-Scene and Scene-to-Image projections. Projection error statistics are computed based upon the error statistics provided in the image metadata.

2 Image Pixel Array & Image Plane Parameters

The image pixel array is a single two-dimensional array of image signal values. The size of the pixel array is NumRows x NumCols. The elements of the signal array are denoted $\{S_{II}(\text{row}, \text{col})\}$. All image products include a Scene Center Point (SCP) position and a SCP pixel location. The spatial sampling represented by the image pixel grid is referred to as the grid type. The image grid type is a function of the imaging collection geometry and the image formation processing that formed the image. Precise image-to-scene and scene-to-image projections dependent upon the image grid type.

Image Pixel Array Parameters:

NumRows	Number of rows in the image pixel array.
NumCols	Number of columns in the image pixel array.
FirstRow	Index of the first row of the image pixel array.
FirstCol	Index of the first column of the image pixel array.
Row_SS	Sample spacing in the row direction.
Col_SS	Sample spacing in the column direction.

The image array may be the original image produced by the image formation processor (referred as the full image) or a sub-image extracted from the original image. For the full image array, index FirstRow = 0 and index FirstCol = 0.

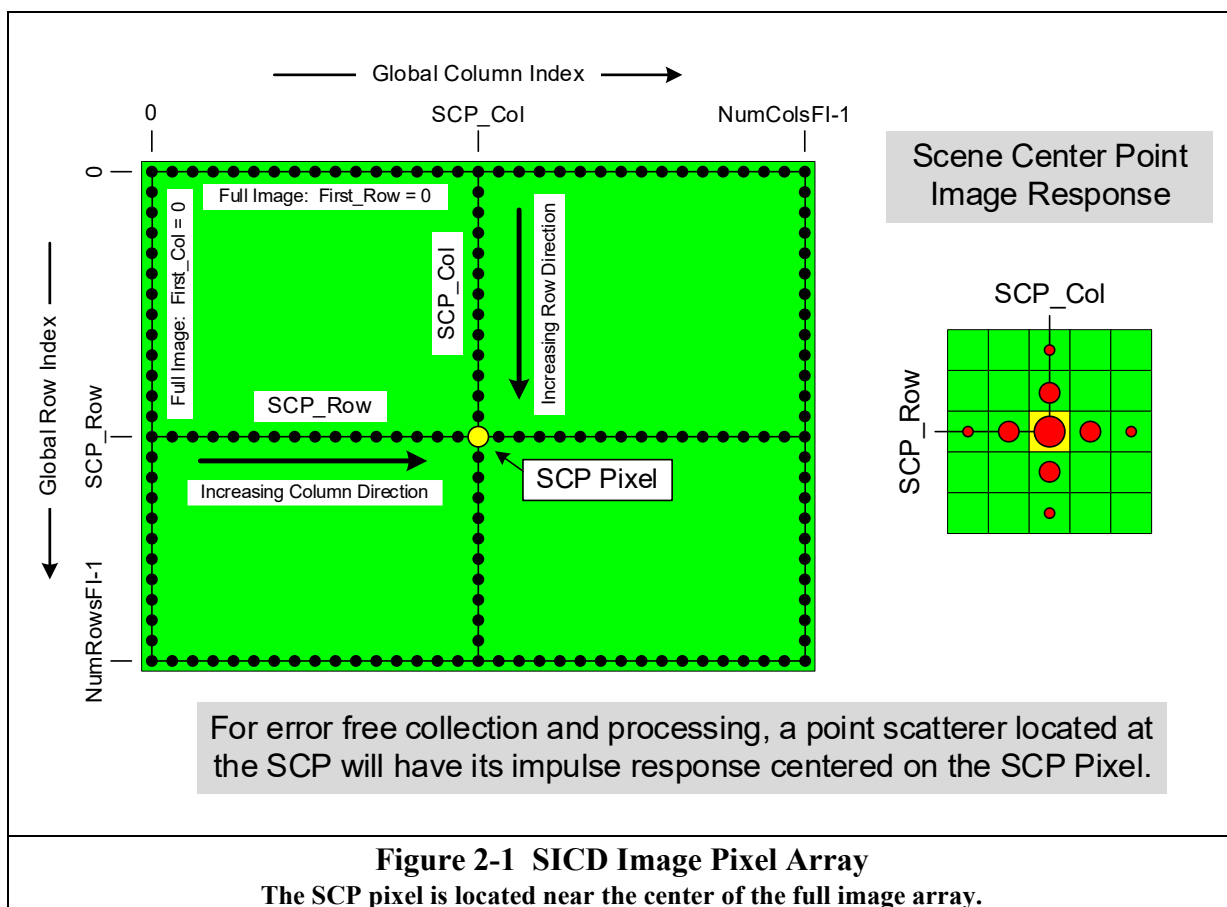
2.1 SCP Position & Pixel

The Scene Center Point (SCP) position and image pixel grid location are specified in every image product. The SCP position and pixel location are the base connection point between the image pixel grid and the geolocated three-dimensional scene that was imaged.

SCP Parameters:

SCP	SCP position in ECF coordinates.
SCP.Lat	SCP geodetic latitude.
SCP.Lon	SCP geodetic longitude.
SCP_Row	SCP image pixel grid global row index.
SCP_Col	SCP image pixel grid global column index.

The image formation processing places the image of the Scene Center Point in the SCP pixel. The SCP pixel is located near the center of the Full Image. For error-free collection and processing, the image impulse response of a point scatterer located at the SCP is centered on the SCP pixel. See Figure 2-1.



2.2 Image Indices & Image Coordinates

The image grid locations are specified using two sets of pixel-based indices: (row, col) and (irow, icol). The image array locations are also specified using distance-based image coordinates (xrow, ycol). Indices (row, col) take on only integer values. Indices (irow, icol) and image coordinates (xrow, ycol) take on continuous values. The image sample spacing in the row dimension is Row_SS. The image sample spacing in the column dimension is Col_SS. The row and column spacings are the precise spacings measured at the SCP pixel.

(1) Global Row, Column Indices: (row, col)

The global row, column indices are used to index the pixels of the image array. The origin (0, 0) is located at the first row (row = 0) and the first column of the first row (col = 0) of the Full Image array. In all descriptions, the (row, col) indices are only used to index individual elements of the pixel array $\{S_{II}(\text{row}, \text{col})\}$. The global row, column indices take on INTEGER values only. The elements of the full image pixel array are indexed as follows.

Global Row Index: $\text{row} = 0, 1, 2, \dots, \text{NumRowsFI} - 1$

Global Column Index: $\text{col} = 0, 1, 2, \dots, \text{NumColsFI} - 1$

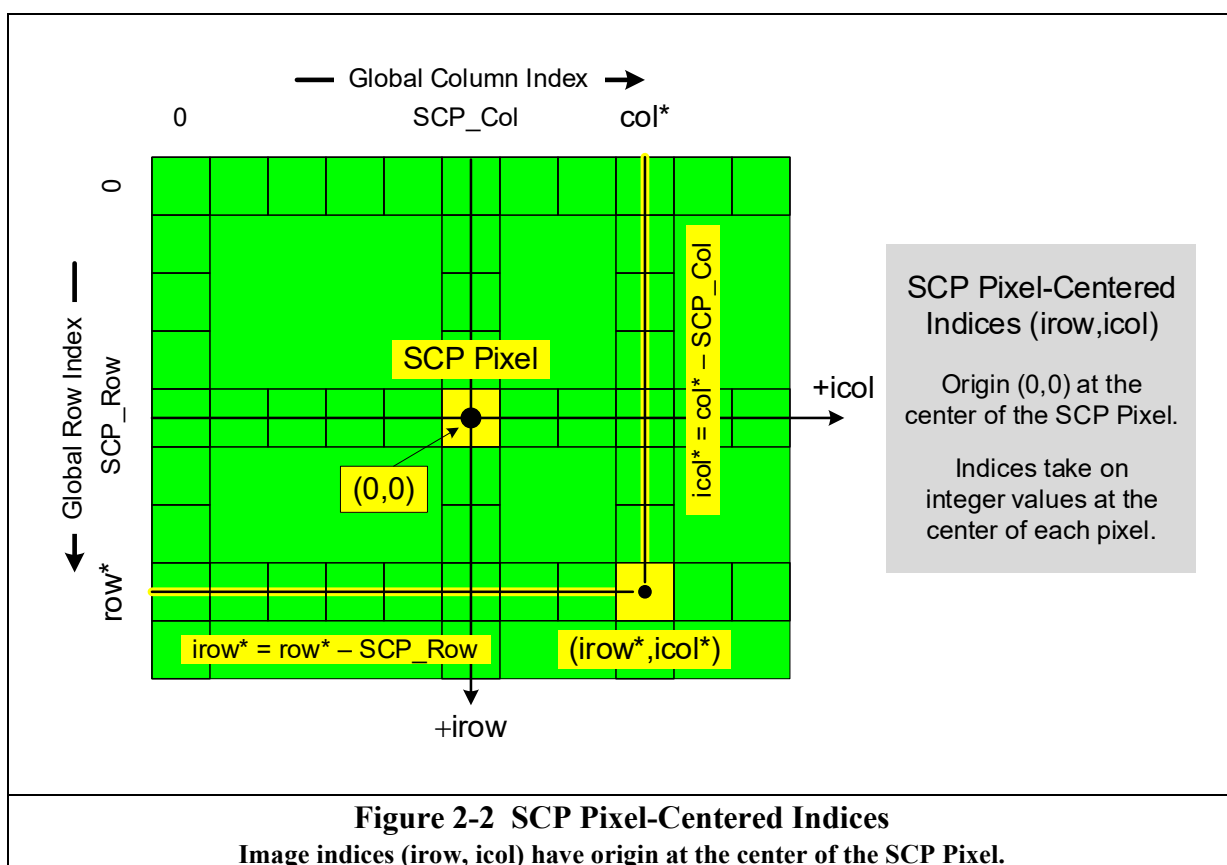
(2) SCP Pixel-Centered Image Indices: (irow, icol)

The SCP pixel-centered image indices are continuous valued indices that are also used to index locations in the image pixel array. The origin (0, 0) is located at the CENTER of the SCP pixel. The image indices (irow, icol) are related to global row, column indices (row, col) as follows. See Figure 2-2.

$$\text{SCP Pixel-Centered Row Index: } irow = row - \text{SCP_Row}$$

$$\text{SCP Pixel-Centered Column Index: } icol = col - \text{SCP_Col}$$

In the descriptions that follow, SCP pixel-centered indices (irow, icol) are used to address the image array as a continuous two-dimensional space. The image pixel values $\{S_{II}(\text{row}, \text{col})\}$ are considered to be the discrete samples of a continuous, two-dimensional image signal at the integer values of indices (irow, icol).

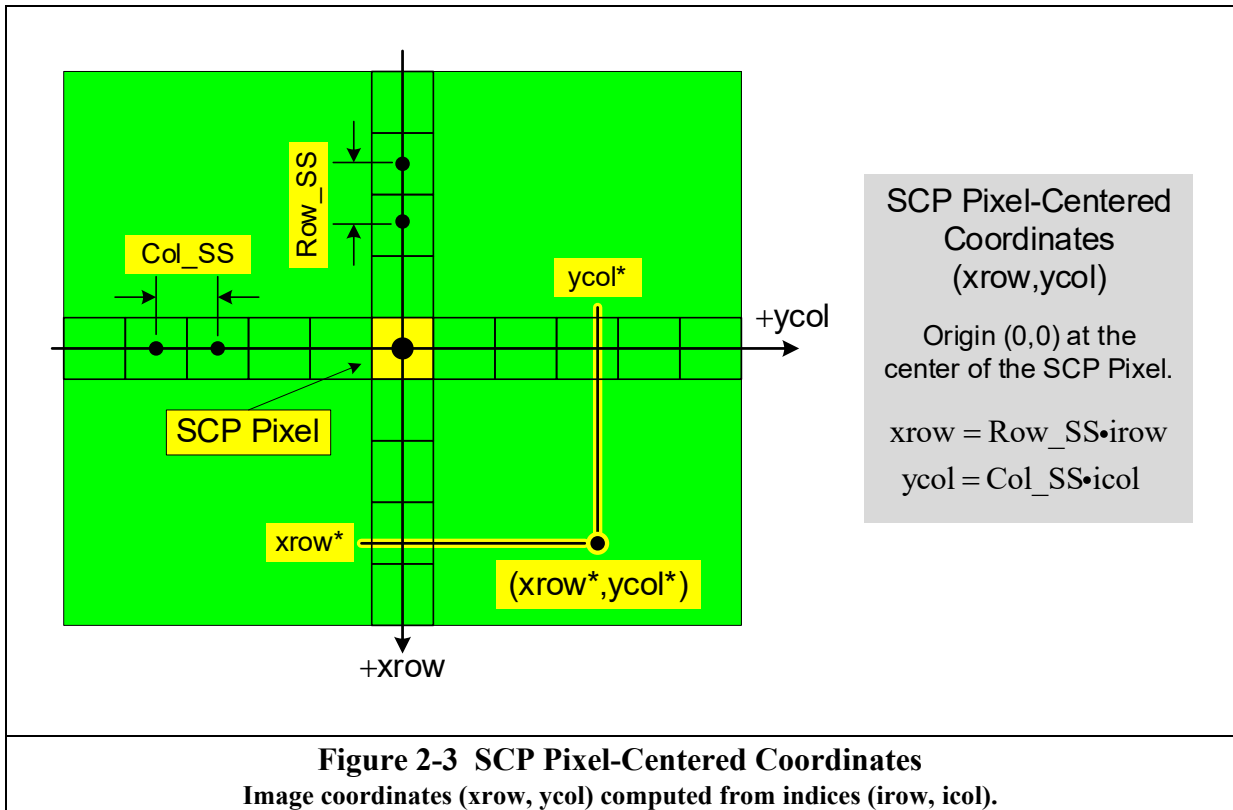


(3) SCP Centered Image Coordinates: (xrow, ycol)

The SCP centered image coordinates are continuous valued distances that treat the image array as equally spaced samples on an orthogonal grid. The origin (0, 0) is located at the CENTER of the SCP pixel. Image coordinate xrow is equal to index irow scaled by the row spacing (Row_SS). Image coordinate ycol is equal to index icol scaled by the column spacing (Col_SS).

$$\text{xrow} = \text{Row_SS} \cdot \text{irow} \qquad \text{ycol} = \text{Col_SS} \cdot \text{icol}$$

- (1) In all SICD descriptions, indices (row, col) are only used to index the discrete elements of the pixel array. Indices (row, col) take on INTEGER values only.
- (2) The SCP Pixel-Centered indices (irow, icol) are used to reference the pixel array as a continuous 2-dimensional space. An index value with zero fractional part references the center of the pixel.
- (3) Image indices (irow, icol) and image coordinates (xrow, ycol) apply to all SICD images and are used with any grid type.



2.3 Image Grid Types

The type of image sample grid is a fundamental attribute of the complex image. The spatial sampling of the complex image is usually oriented along the natural radar coordinates range and cross range. The exact grid is a function of both the collection mode and the image formation algorithm. The parameters that identify the grid type and image formation algorithm are as follows.

Image Grid Parameters:

Grid_Type	Identifies the type of spatial sampling of the image grid. Allowed values: RGAZIM, RGZERO, XRGYC, XCTYAT, PLANE
IFA	Image Formation Algorithm used to form the image. Allowed values: PFA, RMA, RGAZCOMP, OTHER

For a given image grid, the conversion from image grid location to R/Rdot contour is a function of the grid type. For the RGAZIM grid type, it is also dependent upon the image formation algorithm that produced the grid.

(1) Grid_Type = RGAZIM $\Leftrightarrow$ Range & Azimuth

Samples are oriented in range and azimuth with respect to the ARP position and velocity at an image formation reference time (denoted $\mathbf{ARP}_{REF}$ and $\mathbf{VARP}_{REF}$). Image rows are samples along nominally constant range contours relative to the $\mathbf{ARP}_{REF}$. Image columns are nominally oriented along constant azimuth. Along a given row, the image grid samples are evenly spaced increments in an azimuth angle related quantity (usually the cosine of the Doppler cone angle). The RGAZIM grid is the natural output for image formation algorithms RGAZCOMP and PFA. The RGAZCOMP algorithm (simple range and azimuth compression) is only used for coarse resolution or small scene sizes. The PFA algorithm was developed to overcome the limitations of RGAZCOMP to yield fine resolution and large scene sizes.

(2) Grid_Type = RGZERO $\Leftrightarrow$ Range & Zero Doppler

The Range and Zero Doppler grid is for imaging collections near closest approach (i.e. near zero Doppler frequency). The grid is produced using a variation of the RMA for near broadside imaging. Image rows are samples along contours of constant range from the ARP trajectory and are “parallel” to the ARP trajectory. Image columns are samples along a contour of constant time of closest approach with range increasing in the increasing row direction.

(3) Grid_Type = XRGYCR $\Leftrightarrow$ X,Y Grid Oriented Range, Cross Range

The XRGYCR grid is a slant plane grid that is oriented range and cross range. The slant plane is defined by the ARP position and velocity at a reference time ($\mathbf{POS}_{REF}$ and $\mathbf{VEL}_{REF}$) and the SCP. The +XRG direction is oriented from the ARP to the SCP (in the increasing range direction). The +YCR direction is orthogonal to the +XRG direction (in the cross range direction). The rows of the image are along lines of constant XRG and evenly spaced. The columns of the image are along lines of constant YCR and evenly spaced.

- (4) Grid_Type = XCTYAT $\Leftrightarrow$ X,Y Grid Oriented Cross Track, Along Track

The XCTYAT grid is a slant plane grid that is oriented cross track and along track. The slant plane is defined by the ARP position and velocity at a reference time (**POS_{REF}** and **VEL_{REF}**) and the SCP. The +XCT direction is orthogonal to the **VEL_{REF}** and in the increasing distance from the trajectory. The +YAT direction is orthogonal to the +XCT direction and parallel to the **VEL_{REF}**. For right looking collections, the +YAT direction is in the same direction as **VEL_{REF}**. For left looking collections, the +YAT is in the opposite direction as **VEL_{REF}**. The rows of the image are along lines of constant XCT and evenly spaced. The columns of the image are along lines of constant YAT and evenly spaced. The grid is the natural output for image formation using the RMA.

- (5) Grid_Type = PLANE $\Leftrightarrow$ Arbitrary Plane w/ Axes U,V

The PLANE grid is provided for images that do not meet the precise definition of the other grid types. The samples are uniformly spaced in a geolocated image plane that contains the SCP. The +U coordinate is equal to the row coordinate (xrow). The +V coordinate is equal to the column coordinate (ycol). The rows of the image are along lines of constant U coordinate and are evenly spaced. The columns of the image are along lines of constant V coordinate and are evenly spaced. The only restriction is the image be a view from above the earth.

2.4 Image Plane Parameters

The samples of the image pixel grid may be described as a grid of uniformly spaced samples in a geolocated Image Plane. For image grid types XRGYCR and XCTYAT, this is precisely true. For grid types RGAZIM and RGZERO, this is only approximately true but is still a useful representation of the image sampling. The mapping from image grid location (xrow, ycol) to image plane position is described below.

Image Plane Parameters:

uRow	Unit vector in the increasing row direction in ECEF coordinates.
uCol	Unit vector in the increasing column direction in ECEF coordinates.

The Image Plane is defined by the SCP position and unit vectors **uRow** and **uCol** that lie in the plane. The unit vector in the row direction (**uRow**) represents the change in image plane position moving from the SCP pixel in the increasing row direction. The unit vector in the column direction (**uCol**) represents the change in image plane position moving from the SCP pixel in the increasing column direction.

The Image Plane unit normal vector (**uIPN**) in ECEF coordinates is computed as follows. For all image products, the **uIPN** will point away from the center of the earth. Refer to Figure 2.4. For most image products, vectors **uRow** and **uCol** are an orthogonal pair (**uRow**•**uCol** = 0). However, **uRow** and **uCol** are not constrained be an orthogonal pair. Define the angle θ_{COL} as the angle between the vectors **uRow** and **uCol**. See Figure 2-5.

$$\mathbf{IPN} = \mathbf{uRow} \times \mathbf{uCol} \quad \mathbf{uIPN} = \frac{1}{|\mathbf{IPN}|} \bullet \mathbf{IPN} \quad \cos(\theta_{COL}) = \mathbf{uRow} \bullet \mathbf{uCol}$$

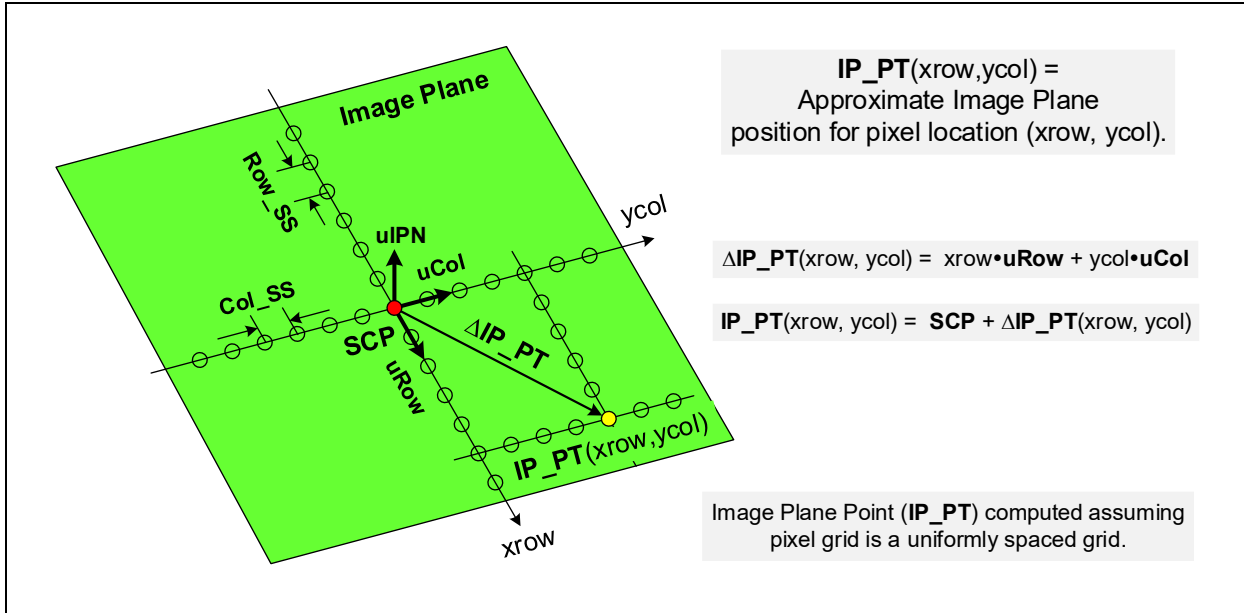


Figure 2-4 Image Plane Parameters
 The Image Plane is defined by the SCP and unit vectors uRow & uCol.

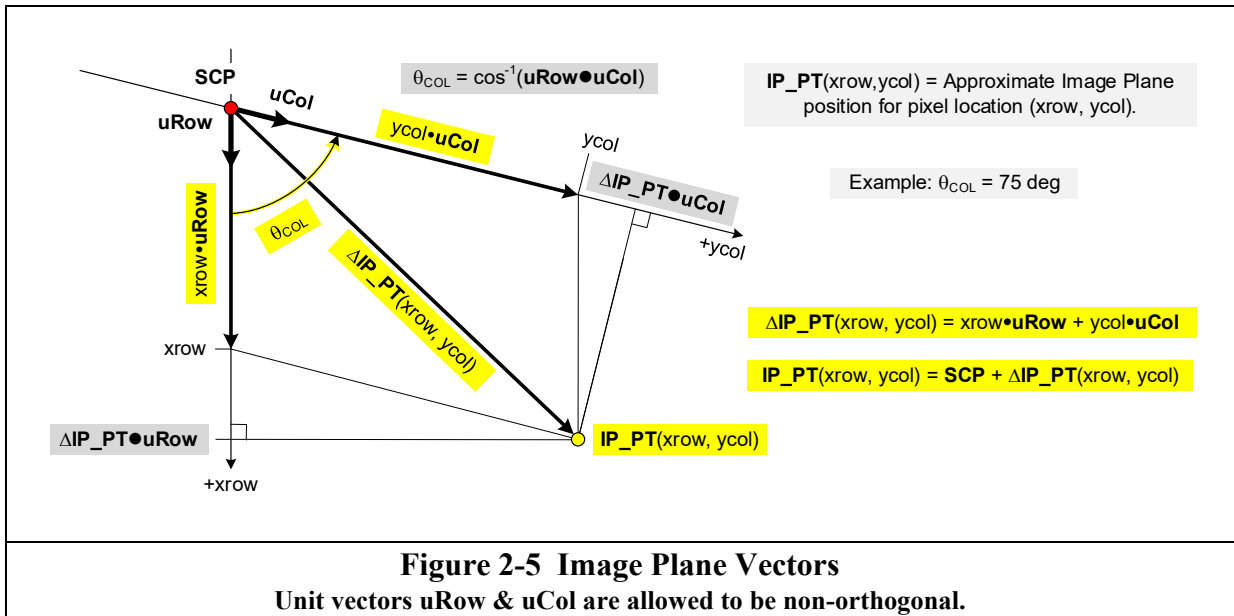


Figure 2-5 Image Plane Vectors
 Unit vectors uRow & uCol are allowed to be non-orthogonal.

The methods that relate an image pixel grid location (xrow, ycol) and the corresponding Image Plane position IP_PT are specified in the following sub-sections. These functions will be used in computing image-to-scene and scene-to-image projections.

2.4.1 Image Grid to Image Plane Point

The method to convert an image pixel grid location (xrow, ycol) to the corresponding image plane position **IP_PT** may be implemented as the function shown below. The inputs are the image grid location, the SCP position, and unit vectors **uRow** and **uCol**. The computed image plane point position is **IP_PT**(xrow, ycol). The input image location is in image coordinates and has units of meters. Refer to Figure 2-4 and Figure 2-5.

$$\left\{ \begin{bmatrix} \text{xrow} \\ \text{ycol} \end{bmatrix} \right\} + \left\{ \text{SCP} \begin{bmatrix} \mathbf{uRow} \\ \mathbf{uCol} \end{bmatrix} \right\} \rightarrow \boxed{\text{Convert (xrow, ycol) to IP_PT}} \rightarrow \left\{ \text{IP_PT}(\text{xrow}, \text{ycol}) \right\}$$

- (1) Compute displacement from the **SCP** to the **IP_PT**, $\Delta \text{IP_PT}(\text{xrow}, \text{ycol})$.

$$\Delta \text{IP_PT}(\text{xrow}, \text{ycol}) = \text{xrow} \bullet \mathbf{uRow} + \text{ycol} \bullet \mathbf{uCol}$$

- (2) Compute the image plane point position, **IP_PT**(xrow, ycol), in ECEF coordinates.

$$\text{IP_PT}(\text{xrow}, \text{ycol}) = \begin{bmatrix} \text{IP_PT.X} \\ \text{IP_PT.Y} \\ \text{IP_PT.Z} \end{bmatrix} = \text{SCP} + \Delta \text{IP_PT}(\text{xrow}, \text{ycol})$$

2.4.2 Image Plane Point to Image Grid Location

The method to convert an image plane position **IP_PT** to the corresponding image pixel grid location (xrow, ycol) may be implemented as the function shown below. The inputs are the image plane point position, the SCP position, and unit vectors **uRow** and **uCol**. The computed image grid location is **IL** = (xrow, ycol). The input image plane point is in ECEF coordinates. Refer to Figure 2-4.

$$\left\{ \text{IP_PT} \right\} + \left\{ \text{SCP} \begin{bmatrix} \mathbf{uRow} \\ \mathbf{uCol} \end{bmatrix} \right\} \rightarrow \boxed{\text{Convert IP_PT to (xrow, ycol)}} \rightarrow \left\{ \text{IL} = \begin{bmatrix} \text{xrow} \\ \text{ycol} \end{bmatrix} \right\}$$

- (1) Compute cosine and sine of the angle θ_{COL} that is the angle between unit vectors **uRow** and **uCol**. Also compute the 2 x 2 matrix $\mathbf{M}_{\text{IP_PT}}^{\text{IL}}$ that is a function of the angle θ_{COL} .

$$\cos(\theta_{\text{COL}}) = \mathbf{uRow} \bullet \mathbf{uCol} \quad \sin(\theta_{\text{COL}}) = +(1 - \cos^2(\theta_{\text{COL}}))^{1/2}$$

$$\mathbf{M}_{\text{IP_PT}}^{\text{IL}} = \frac{1}{(\sin(\theta_{\text{COL}}))^2} \begin{bmatrix} 1 & -\cos(\theta_{\text{COL}}) \\ -\cos(\theta_{\text{COL}}) & 1 \end{bmatrix}$$

- (2) Compute the displacement vector from the SCP to image plane point, $\Delta \text{IP_PT}$. Compute the image grid location **IL** = (xrow, ycol). Image coordinates xrow and ycol are in meters.

$$\Delta \text{IP_PT} = \text{IP_PT} - \text{SCP} \quad \text{IL} = \begin{bmatrix} \text{xrow} \\ \text{ycol} \end{bmatrix} = \mathbf{M}_{\text{IP_PT}}^{\text{IL}} \bullet \begin{bmatrix} \Delta \text{IP_PT} \bullet \mathbf{uRow} \\ \Delta \text{IP_PT} \bullet \mathbf{uCol} \end{bmatrix}$$

2.5 Image Grid to COA Positions & Velocities

For a given image grid location, the Center of Aperture (COA) R/Rdot projection contours are relative to the COA sensor positions and velocities. The parameters that specify the COA positions and velocities are dependent on the Collect Type of the image collection. The following section specifies the method for computing the COA position and velocity parameters as a function of the image grid location. The input image grid location is expressed in image coordinates (xrow, ycol).

Image Parameters:

Collect Type	Collect Type used to collect the PHD used to form the image – either MONOSTATIC or BISTATIC
--------------	---

Center of Aperture Time Parameters:

cT_COA(m, n)	Center of Aperture time polynomial coefficients. The polynomial yields COA time, t_{COA} , as a function of image coordinates (xrow, ycol) in meters. COA time in seconds relative to CST.
M_TCOA	Order of the COA time polynomial in the Row Coordinate.
N_TCOA	Order of the COA time polynomial in the Column Coordinate.

Aperture Reference Point (ARP) Position vs Time:

ARP_Poly(t)	Aperture Reference Point (ARP) position vs. time. Vector polynomial yields ARP position (ARP(t)) in ECEF coordinates as function time t in seconds relative CST.
--------------------	---

Transmit & Receive APC Positions vs. Time – For Collect Type = BISTATIC

Xmt_Poly(t)	Transmit APC position vs. time. Vector polynomial yields transmit APC position (Xmt(t)) in ECEF coordinates as function time t in seconds relative CST.
Rcv_Poly(t)	Receive APC position vs. time. Vector polynomial yields receive APC position (Rcv(t)) in ECEF coordinates as function time t in seconds relative CST. If multiple receive APCs are used to collect the PHD, the position is for the Receive APC centroid.

Ground Reference Point (GRP) Position vs Time – For Collect Type = BISTATIC

GRP_Poly(t)	Ground Reference Point (GRP) position vs. time. Vector polynomial yields GRP position (GRP(t)) in ECEF coordinates as function time t in seconds relative CST.
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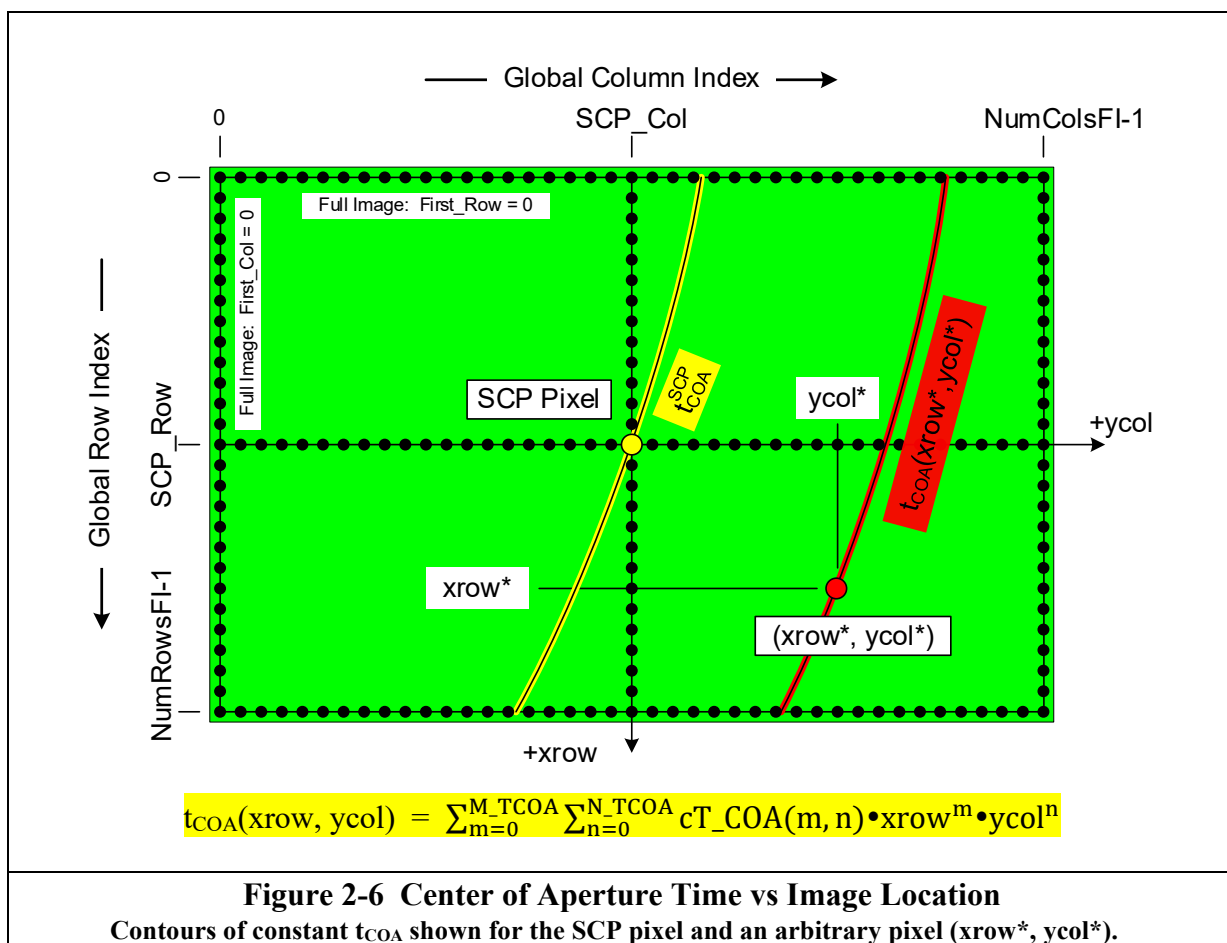
2.5.1 Center of Aperture Time

The t_{COA} is specified as a two-dimensional polynomial of image coordinates (xrow, ycol). Polynomial coefficients are denoted $cT_COA(m,n)$. The polynomial is of order M_TCOA in the row coordinate and N_TCOA in the column coordinate. The t_{COA} for image grid location (xrow, ycol) is computed as shown below.

$$t_{COA}(xrow, ycol) = \sum_{m=0}^{M_TCOA} \sum_{n=0}^{N_TCOA} cT_COA(m,n) \cdot xrow^m \cdot ycol^n$$

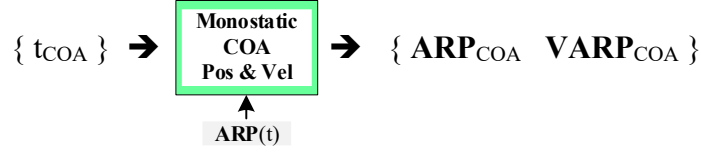
The $t_{COA}(xrow, ycol)$ computed above is the precise value of t_{COA} for the pixel location. Even though image coordinates (xrow, ycol) may be approximate displacements, the computed t_{COA} is the time to be used for accurate exploitation. For all images, coefficient $cT_COA(0,0)$ is the COA time for the SCP pixel (t_{COA}^{SCP}).

Images with constant t_{COA} (typically from spotlight collections) may be stored with only the $cT_COA(0,0)$ coefficient and order parameters $M_TCOA = N_TCOA = 0$. An example image for which the COA time varies across the image is shown in Figure 2-6. Shown in the figure are two contours of constant t_{COA} . For most images, the contours will be approximately linear and equally spaced for even increments in t_{COA} .



2.5.2 Monostatic Image: COA ARP Position & Velocity

For a monostatic image, the COA ARP position and velocity are computed at the COA time. The COA ARP position and velocity may be implemented as a simple function as shown. Time t_{COA} is the COA time for a selected image grid location $\mathbf{IL} = (xrow, ycol)$.



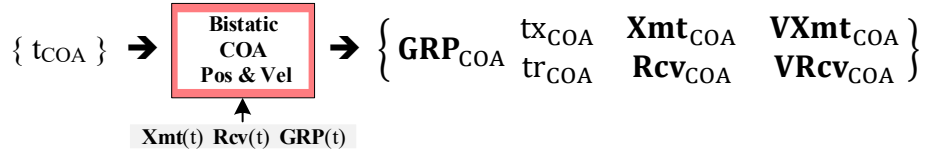
- (1) Compute ARP position and velocity at time $t = t_{COA}$.

$$\mathbf{ARP}_{COA} = \begin{bmatrix} \mathbf{ARP_X}_{COA} \\ \mathbf{ARP_Y}_{COA} \\ \mathbf{ARP_Z}_{COA} \end{bmatrix} = \mathbf{ARP}(t_{COA}) = \mathbf{ARP_Poly}(t_{COA})$$

$$\mathbf{VARP}_{COA} = \begin{bmatrix} \mathbf{VARP_X}_{COA} \\ \mathbf{VARP_Y}_{COA} \\ \mathbf{VARP_Z}_{COA} \end{bmatrix} = \mathbf{VARP}(t_{COA}) = \frac{d}{dt} \mathbf{ARP_Poly}(t_{COA})$$

2.5.3 Bistatic Image: COA APC Positions & Velocities

For a bistatic image, the COA APC positions and velocities are computed by first computing the GRP position at the COA time. The GRP is then used to compute a transmit time, t_{XCOA} , and a receive time, t_{RCOA} . The method for computing the COA APC positions and velocities may be implemented as a simple function as shown. Refer to Figure 2-7.



- (1) Compute the GRP position at time t_{COA} .

$$\mathbf{GRP}_{COA} = \begin{bmatrix} \mathbf{GRP_X}_{COA} \\ \mathbf{GRP_Y}_{COA} \\ \mathbf{GRP_Z}_{COA} \end{bmatrix} = \mathbf{GRP}(t_{COA}) = \mathbf{GRP_Poly}(t_{COA})$$

- (2) Compute the transmit time t_{XCOA} . For a radar signal incident at the GRP at time t_{COA} , time t_{XCOA} is an estimate of the time it was transmitted.

$$\mathbf{X0} = \mathbf{Xmt}(t_{COA}) = \mathbf{Xmt_Poly}(t_{COA}) \quad R_X0 = | \mathbf{X0} - \mathbf{GRP}_{COA} |$$

$$t_{XCOA} = t_{COA} - (1/c) \cdot R_X0$$

- (3) Compute the transmit APC position and velocity at time t_{x_COA} .

$$\mathbf{X}_{mt_{COA}} = \mathbf{X}_{mt}(t_{x_{COA}})$$

$$\mathbf{VX}_{mt_{COA}} = \mathbf{VX}_{mt}(t_{x_{COA}}) = \frac{d}{dt}\mathbf{X}_{mt_Poly}(t_{x_{COA}})$$

- (4) Compute the receive time $t_{r_{COA}}$. For a radar signal incident at the GRP at time t_{COA} , time $t_{r_{COA}}$ is an estimate of the time it would be received at the receive APC.

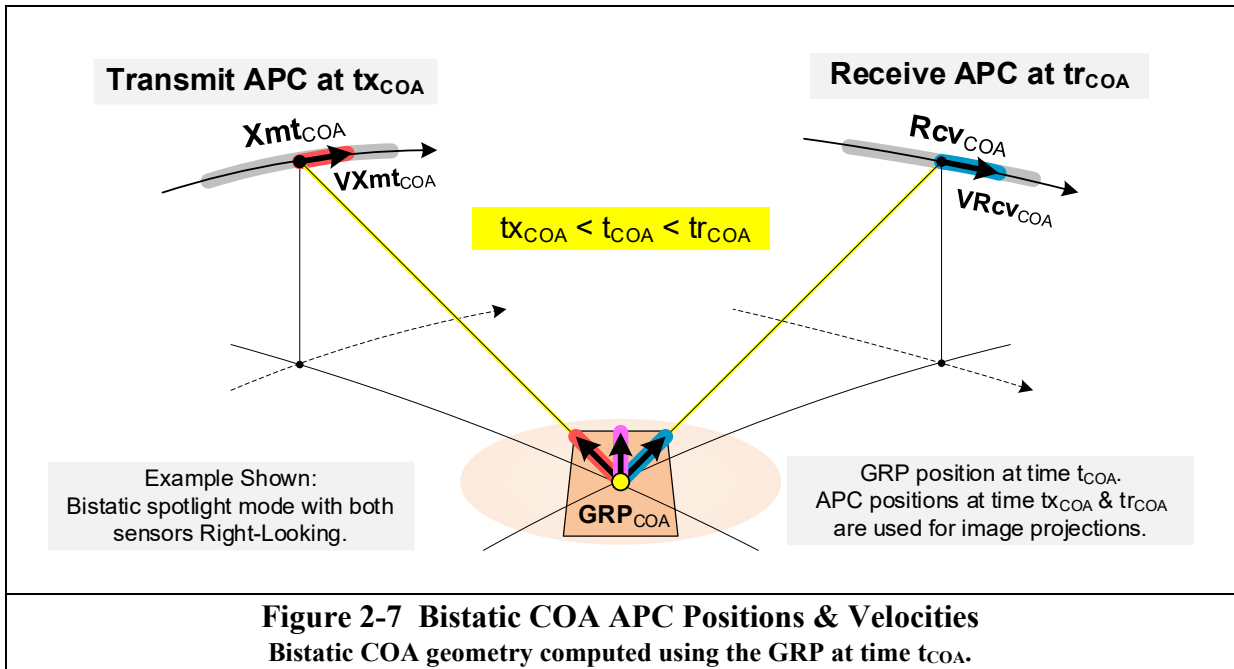
$$\mathbf{R0} = \mathbf{Rcv}(t_{COA}) = \mathbf{Rcv_Poly}(t_{COA}) \quad R_R0 = |\mathbf{R0} - \mathbf{GRP}_{COA}|$$

$$t_{r_{COA}} = t_{COA} + (1/c) \cdot R_R0$$

- (5) Compute the receive APC position and velocity at time $t_{r_{COA}}$.

$$\mathbf{Rcv}_{COA} = \mathbf{Rcv}(t_{r_{COA}})$$

$$\mathbf{VRcv}_{COA} = \mathbf{VRcv}(t_{r_{COA}}) = \frac{d}{dt}\mathbf{Rcv_Poly}(t_{r_{COA}})$$



3 SCP Pixel Projection

The Scene Center Point (SCP) position and image pixel grid location are specified in every image product. The SCP position and pixel location are the base connection point between the image pixel grid and the georeferenced three-dimension scene that was imaged. The SCP position, image pixel location, and Center of Aperture (COA) position and velocity are provided with all products. As described in Section 2.5, the COA position and velocity are dependent on the Collect Type.

3.1 SCP R/Rdot Projection Contour

The SCP R/Rdot projection contour is dependent upon the Collect Type.

3.1.1 Collect Type = Monostatic

For a monostatic image, the SCP Center of Aperture ARP position and velocity are provided in the metadata included with the product. Refer to Table 1-1.

t_{COA}^{SCP}	SCP pixel Center of Aperture time in seconds relative to CST.
$\mathbf{ARP}_{COA}^{SCP}$	ARP position at the time t_{COA}^{SCP} in ECEF coordinates.
$\mathbf{VARP}_{COA}^{SCP}$	ARP velocity at the time t_{COA}^{SCP} in ECEF coordinates.

The SCP COA R/Rdot projection contour of constant range and constant range rate is relative to the COA ARP position and velocity. The SCP COA range and range rate can be computed as shown.

R_{COA}^{SCP}	Range of the ARP to the SCP for the COA geometry.
$Rdot_{COA}^{SCP}$	Range rate for the ARP relative to the SCP for the COA geometry.

Where:

$$R_{COA}^{SCP} = | \mathbf{ARP}_{COA}^{SCP} - \mathbf{SCP} | \quad \mathbf{uPT}_{COA}^{SCP} = \frac{1}{R_{COA}^{SCP}} \cdot (\mathbf{ARP}_{COA}^{SCP} - \mathbf{SCP})$$

$$Rdot_{COA}^{SCP} = \mathbf{VARP}_{COA}^{SCP} \bullet \mathbf{uPT}_{COA}^{SCP}$$

$$\mathbf{uPTDot}_{COA}^{SCP} = \frac{1}{R_{COA}^{SCP}} \cdot \{ \mathbf{VARP}_{COA}^{SCP} - Rdot_{COA}^{SCP} \bullet \mathbf{uPT}_{COA}^{SCP} \}$$

3.1.2 Collect Type = Bistatic

For a bistatic image, the SCP Center of Aperture transmit and receive APC positions and velocities are provided in the metadata included with the product. Also provided are the transmit COA time and receive COA time. Refer to Section 2.5.3.

t_{COA}^{SCP}	SCP pixel Center of Aperture time
tx_{COA}^{SCP}	Transmit time associated with SCP COA geometry.
tr_{COA}^{SCP}	Receive time associated with the SCP COA geometry.
$\mathbf{Xmt}_{COA}^{SCP}$	Transmit APC position at the time tx_{COA}^{SCP} in ECEF coordinates.
$\mathbf{VXmt}_{COA}^{SCP}$	Transmit APC velocity at the time tx_{COA}^{SCP} in ECEF coordinates.
$\mathbf{Rcv}_{COA}^{SCP}$	Receive APC position at the time tr_{COA}^{SCP} in ECEF coordinates.
$\mathbf{VRcv}_{COA}^{SCP}$	Receive APC velocity at the time tr_{COA}^{SCP} in ECEF coordinates.

Where:

$$R_Xmt_{COA}^{SCP} = | \mathbf{Xmt}_{COA}^{SCP} - \mathbf{SCP} | \quad \mathbf{uXmt}_{COA}^{SCP} = \frac{1}{R_Xmt_{COA}^{SCP}} \bullet (\mathbf{Xmt}_{COA}^{SCP} - \mathbf{SCP})$$

$$Rdot_Xmt_{COA}^{SCP} = \mathbf{VXmt}_{COA}^{SCP} \bullet \mathbf{uXmt}_{COA}^{SCP}$$

$$\mathbf{uXmtDot}_{COA}^{SCP} = \frac{1}{R_Xmt_{COA}^{SCP}} \bullet \{ \mathbf{VXmt}_{COA}^{SCP} - Rdot_Xmt_{COA}^{SCP} \bullet \mathbf{uXmt}_{COA}^{SCP} \}$$

$$R_Rcv_{COA}^{SCP} = | \mathbf{Rcv}_{COA}^{SCP} - \mathbf{SCP} | \quad \mathbf{uRcv}_{COA}^{SCP} = \frac{1}{R_Rcv_{COA}^{SCP}} \bullet (\mathbf{Rcv}_{COA}^{SCP} - \mathbf{SCP})$$

$$Rdot_Rcv_{COA}^{SCP} = \mathbf{VRcv}_{COA}^{SCP} \bullet \mathbf{uRcv}_{COA}^{SCP}$$

$$\mathbf{uRcvDot}_{COA}^{SCP} = \frac{1}{R_Rcv_{COA}^{SCP}} \bullet \{ \mathbf{VRcv}_{COA}^{SCP} - Rdot_Rcv_{COA}^{SCP} \bullet \mathbf{uRcv}_{COA}^{SCP} \}$$

The average range and average range rate are computed from the APC ranges and range rates for COA geometry.

$$R_Avg_{COA}^{SCP} = \frac{1}{2} \bullet (R_Xmt_{COA}^{SCP} + R_Rcv_{COA}^{SCP})$$

$$Rdot_Avg_{COA}^{SCP} = \frac{1}{2} \bullet (Rdot_Xmt_{COA}^{SCP} + Rdot_Rcv_{COA}^{SCP})$$

The bistatic pointing vector, $\mathbf{bP}_{COA}^{SCP}$, and it derivative w.r.t. time, $\mathbf{bPDot}_{COA}^{SCP}$, are computed from the unit vectors and their derivatives.

$$\mathbf{bP}_{COA}^{SCP} = \frac{1}{2} \bullet (\mathbf{uXmt}_{COA}^{SCP} + \mathbf{uRcv}_{COA}^{SCP})$$

$$\mathbf{bPDot}_{COA}^{SCP} = \frac{1}{2} \bullet (\mathbf{uXmtDot}_{COA}^{SCP} + \mathbf{uRcvDot}_{COA}^{SCP})$$

3.2 Side of Track

The Side of Track parameter is provided for convenience in all products. Define parameter LOOK that is used in later computations to select ground points on the correct side of the ground track.

SideOfTrack Side of Track parameter. SideOfTrack = L $\Leftrightarrow$ Left Looking
 SideOfTrack = R $\Leftrightarrow$ Right Looking

$$\text{LOOK} = \begin{cases} +1, & \text{if SideOfTrack} = \text{L} \\ -1, & \text{if SideOfTrack} = \text{R} \end{cases}$$

For a monostatic collection, the side of track parameter is clearly defined. For a bistatic collection, the side of track parameter provides the effective side-of-track for the combined motion of the two sensors. The method for setting the value of the SideOfTrack parameter is specified in SICD Volume 1, Section 4.6.3, and Section 4.9.1. For most bistatic collections, the value of the SideOfTrack parameter will indicate the actual side of track for both sensors during the collection.

3.3 SCP COA Slant Plane Normal

The concept of the instantaneous slant plane is applicable for both monostatic and bistatic imaging collections. The method for computing the SCP COA slant plane unit normal is dependent upon the collect type.

3.3.1 Collect Type = Monostatic

For a monostatic collection, the slant plane unit normal, $\mathbf{uSPN}_{\text{COA}}^{\text{SCP}}$, is computed as follows. The LOOK parameter is included to ensure the vector points “upward” for all products.

$$\mathbf{SPN}_{\text{COA}}^{\text{SCP}} = \text{LOOK} \cdot (\mathbf{ARP}_{\text{COA}}^{\text{SCP}} - \mathbf{SCP}) \times \mathbf{VARP}_{\text{COA}}^{\text{SCP}}$$

$$\mathbf{uSPN}_{\text{COA}}^{\text{SCP}} = \frac{1}{|\mathbf{SPN}_{\text{COA}}^{\text{SCP}}|} \cdot \mathbf{SPN}_{\text{COA}}^{\text{SCP}}$$

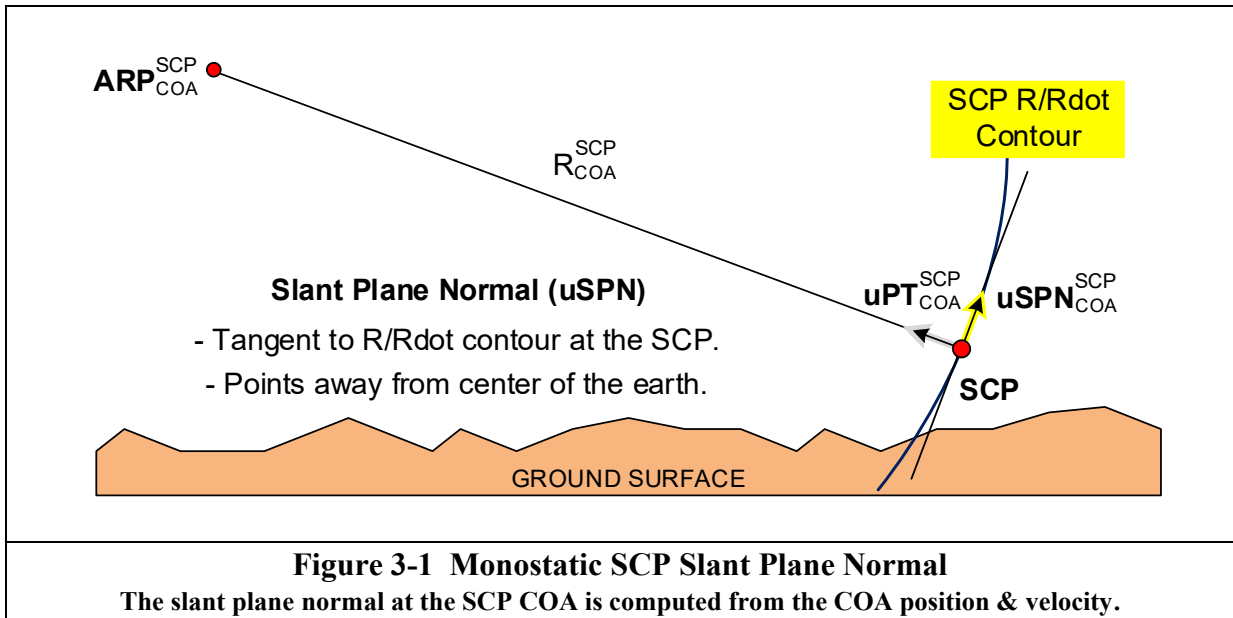
The slant plane unit normal is tangent to R/Rdot contour at the SCP. See Figure 3-1.

3.3.2 Collect Type = Bistatic

For a bistatic collection, the slant plane unit normal, $\mathbf{uSPN}_{\text{COA}}^{\text{SCP}}$, is computed as follows. The LOOK parameter is included to ensure the vector points “upward” for all products.

$$\mathbf{SPN}_{\text{COA}}^{\text{SCP}} = \text{LOOK} \cdot \mathbf{bP}_{\text{COA}}^{\text{SCP}} \times \mathbf{bPDot}_{\text{COA}}^{\text{SCP}}$$

$$\mathbf{uSPN}_{\text{COA}}^{\text{SCP}} = \frac{1}{|\mathbf{SPN}_{\text{COA}}^{\text{SCP}}|} \cdot \mathbf{SPN}_{\text{COA}}^{\text{SCP}}$$



3.4 RIC Coordinates at SCP COA

Applications that estimate platform position and velocity errors and error statistics often work in a platform-centered coordinate frame. A commonly used coordinate frame is a Radial, In-Track, Cross-Track (RIC) frame. The RIC frame is defined at a given time based on the instantaneous position and velocity of the platform. The origin of the RIC frame is the instantaneous position. The radial direction is from the earth center (i.e., the ECEF origin) to the platform pointing away from the earth center. The cross-track direction is normal to the radial direction and the instantaneous velocity. The in-track direction completes a right-handed coordinate frame. For airborne platforms, the ECEF velocity vector is commonly used to define the in-track and cross-track directions. For orbiting platforms, the velocity relative to the Earth Centered Inertial (ECI) coordinate frame is commonly used to define the in-track and cross-track directions.

For all products, all position and velocity metadata parameters are specified in ECEF coordinates. For a product that includes position and velocity error statistics, the error statistics may be specified in any one of three allowed coordinate frames:

- | | |
|------|--|
| ECEF | Earth Centered Earth Fixed: Coordinates are (X, Y, Z). Specified in an XML instance by coordinate frame parameter equal to “ECF”. |
| RICF | Radial, In-Track, Cross-Track, Fixed: Coordinates are (R, I, C). The RICF frame is defined by the sensor position and the ECEF velocity vector at a specified time. Specified in an XML instance by coordinate frame parameter equal to “RIC_ECF”. |
| RICI | Radial, In-Track, Cross-Track, Inertial: Coordinates are (R, I, C). The RICF frame is defined by the sensor position and the ECI velocity vector at a specified time. Specified in an XML instance by coordinate frame parameter equal to “RIC_ECI”. |

For all products, position and velocity error statistics are specified at time $t = t_{COA}^{SCP}$. For statistics specified in an RIC coordinate frame, the sensor position and velocity at $t = t_{COA}^{SCP}$ is used to compute the origin and orientation of the RIC frame relative to the ECEF frame. For a monostatic image, the position and velocity error statistics are specified for the ARP. For a bistatic image, the position and velocity error statistics are specified jointly for the transmit APC and the receive APC.

3.4.1 RIC Basis Vectors

For statistics specified in an RIC frame, the orientation of the RIC coordinate frame relative to the ECEF coordinate frame may be computed using the following method. The input position $\mathbf{P}_{RIC}$ is the sensor position that is the origin of the RIC frame. Input velocity vector $\mathbf{V}_{RIC}$ is used to compute the in-track and cross-track directions. Output matrix $\mathbf{T}_{RIC}^{ECEF}$ is formed from the set of unit basis vectors.

$$\{ \mathbf{P}_{RIC} \ \mathbf{V}_{RIC} \} \rightarrow \boxed{\text{Compute RIC Basis Vectors}} \rightarrow \{ \mathbf{T}_{RIC}^{ECEF} = [\mathbf{uR} \ \mathbf{uI} \ \mathbf{uC}] \}$$

(1) Compute the radial unit vector $\mathbf{uR}$ using the position $\mathbf{P}_{RIC}$. Compute the cross-track unit vector $\mathbf{uC}$ using $\mathbf{uR}$ and velocity vector $\mathbf{V}_{RIC}$. Compute the in-track unit vector $\mathbf{uI}$ from unit vectors $\mathbf{uC}$ and $\mathbf{uR}$.

$$\mathbf{uR} = \frac{1}{|\mathbf{P}_{RIC}|} \cdot \mathbf{P}_{RIC}$$

$$\mathbf{C} = \mathbf{uR} \times \mathbf{V}_{RIC} \quad \mathbf{uC} = \frac{1}{|\mathbf{C}|} \cdot \mathbf{C} \quad \mathbf{uI} = \mathbf{uC} \times \mathbf{uR}$$

(2) Form the 3x3 matrix $\mathbf{T}_{RIC}^{ECEF}$ from the set of vectors:

$$\mathbf{T}_{RIC}^{ECEF} = [\mathbf{uR} \ \mathbf{uI} \ \mathbf{uC}]$$

3.4.2 RIC Basis Vectors for a Monostatic Image

For a monostatic image, the ARP position and velocity at $t = t_{COA}^{SCP}$ are used to compute the RIC basis vectors. Velocity $\mathbf{VARP}_{COA}^{SCP}$ is the ARP velocity relative to the ECEF frame expressed in ECEF coordinates. Define velocity $\mathbf{VARP_ECI}_{COA}^{SCP}$ that is the ARP velocity relative to the ECI frame expressed in ECEF coordinates. Velocity $\mathbf{VARP_ECI}_{COA}^{SCP}$ is computed as shown where parameter ω_3 is the WGS 84 nominal mean angular velocity of the earth.

$$\mathbf{VARP_ECI}_{COA}^{SCP} = \mathbf{VARP}_{COA}^{SCP} + \begin{bmatrix} 0 \\ 0 \\ \omega_3 \end{bmatrix} \times \mathbf{ARP}_{COA}^{SCP} \quad \omega_3 = 7.292,115 \cdot 10^{-5} \text{ rad/sec}$$

Parameter AIF specifies the coordinate frame used to specify the ARP position and velocity error statistics. Refer to Table 1-2. For error statistics specified in RICF coordinates, compute matrix $\mathbf{T}_{RIC_A}^{ECEF}$ using velocity vector $\mathbf{VARP}_{COA}^{SCP}$ as described in Section 3.4.1. Refer to Figure 3-

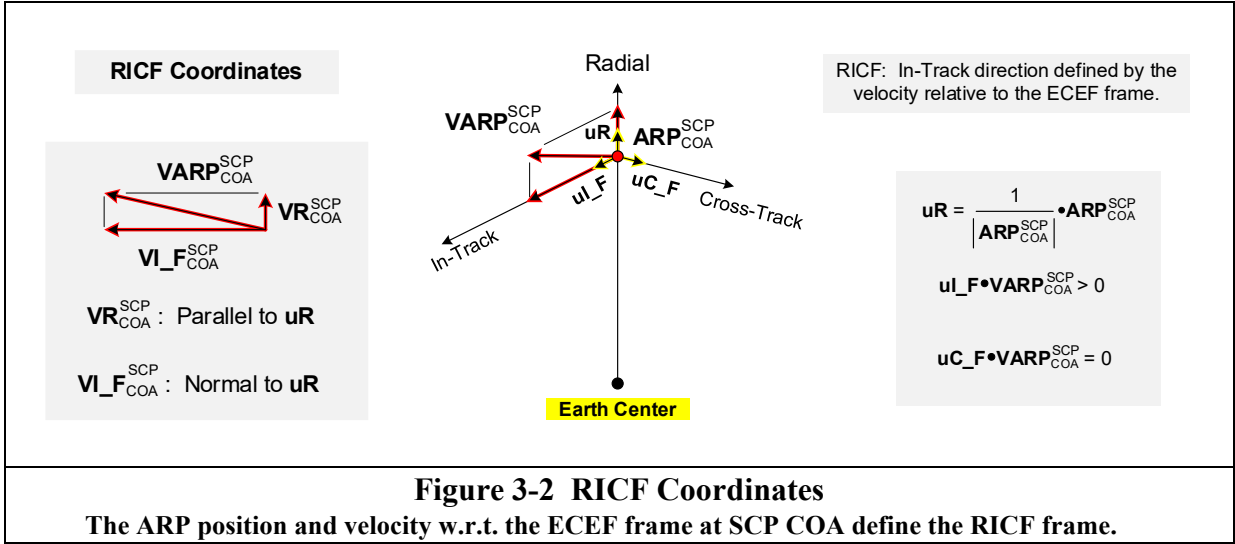
2. For error statistics in RIC coordinates, compute matrix $\mathbf{T}_{RIC_A}^{ECEF}$ using velocity vector $\mathbf{VARP_ECI}_{COA}^{SCP}$.

For AIF = RIC_ECF:

$$\{ \mathbf{ARP}_{COA}^{SCP} \ \mathbf{VARP}_{COA}^{SCP} \} \rightarrow \boxed{\text{Compute RIC Basis Vectors}} \rightarrow \{ \mathbf{T}_{RIC_A}^{ECEF} = [\mathbf{uR} \ \mathbf{uI} \ \mathbf{uC}] \}$$

For AIF = RIC_ECI:

$$\{ \mathbf{ARP}_{COA}^{SCP} \ \mathbf{VARP_ECI}_{COA}^{SCP} \} \rightarrow \boxed{\text{Compute RIC Basis Vectors}} \rightarrow \{ \mathbf{T}_{RIC_A}^{ECEF} = [\mathbf{uR} \ \mathbf{uI} \ \mathbf{uC}] \}$$



3.4.3 RIC Basis Vectors for a Bistatic Image

For a bistatic image, position and velocity error statistics may be expressed in a unique coordinate frame for each sensor. For each sensor with errors specified in an RIC coordinate frame, a unique transformation matrix from RICF or RICI is computed.

For the transmit sensor position and velocity error statistics expressed in an RIC frame, the transmit APC position and velocity at time $t_{RIC} = t_{COA}^{SCP}$ are computed using the transmit APC position polynomial. The position and velocity are then used to compute the basis vectors for the RICF or RICI coordinate frame as described in Section 3.4.1.

Transmit Sensor: $t_{RIC} = t_{COA}^{SCP}$

$$\mathbf{Xmt}_{RIC} = \mathbf{Xmt_Poly}(t_{RIC})$$

$$\mathbf{VXmt}_{RIC} = \frac{d}{dt} \mathbf{Xmt_Poly}(t_{RIC})$$

For XIF = RIC_ECF:

$$\{ \mathbf{Xmt}_{RIC} \quad \mathbf{VXmt}_{RIC} \} \rightarrow \boxed{\text{Compute RIC Basis Vectors}} \rightarrow \{ \mathbf{T}_{RIC_X}^{ECEF} = [\mathbf{uR} \quad \mathbf{uI} \quad \mathbf{uC}] \}$$

For XIF = RIC_ECI:

$$\mathbf{VXmt_ECI}_{RIC} = \mathbf{VXmt}_{RIC} + \begin{bmatrix} 0 \\ 0 \\ \omega_3 \end{bmatrix} \times \mathbf{Xmt}_{RIC}$$

$$\{ \mathbf{Xmt}_{RIC} \quad \mathbf{VXmt_ECI}_{RIC} \} \rightarrow \boxed{\text{Compute RIC Basis Vectors}} \rightarrow \{ \mathbf{T}_{RIC_X}^{ECEF} = [\mathbf{uR} \quad \mathbf{uI} \quad \mathbf{uC}] \}$$

For the receive sensor position and velocity error statistics expressed in an RIC frame, the receive APC position and velocity at time $t_{RIC} = t_{COA}^{SCP}$ are computed using the receive APC position polynomial. The basis vectors are computed as described in Section 3.4.1.

Receive Sensor: $t_{RIC} = t_{COA}^{SCP}$

$$\mathbf{Rcv}_{RIC} = \mathbf{Rcv_Poly}(t_{RIC})$$

$$\mathbf{VRcv}_{RIC} = \frac{d}{dt} \mathbf{Rcv_Poly}(t_{RIC})$$

For RIF = RIC_ECF:

$$\{ \mathbf{Rcv}_{RIC} \quad \mathbf{VRcv}_{RIC} \} \rightarrow \boxed{\text{Compute RIC Basis Vectors}} \rightarrow \{ \mathbf{T}_{RIC_R}^{ECEF} = [\mathbf{uR} \quad \mathbf{uI} \quad \mathbf{uC}] \}$$

For RIF = RIC_ECI:

$$\mathbf{VRcv_ECI}_{RIC} = \mathbf{VRcv}_{RIC} + \begin{bmatrix} 0 \\ 0 \\ \omega_3 \end{bmatrix} \times \mathbf{Rcv}_{RIC}$$

$$\{ \mathbf{Rcv}_{RIC} \quad \mathbf{VRcv_ECI}_{RIC} \} \rightarrow \boxed{\text{Compute RIC Basis Vectors}} \rightarrow \{ \mathbf{T}_{RIC_R}^{ECEF} = [\mathbf{uR} \quad \mathbf{uI} \quad \mathbf{uC}] \}$$

3.4.4 Transforming Errors from RIC to ECEF

The relationship between the ECI coordinate frame and the ECEF coordinate frame is approximated by a single rotation about the z-axis. The rotation angle about the z-axis is denoted $\alpha(t)$ with constant rotation rate ω_3 . The rotation matrix $R_3(t)$ relates the two frames. The matrix Ω_3 may be used to compute the derivative of the $R_3(t)$ matrix, $\frac{d}{dt}(R_3(t))$.

$$\alpha(t) = \alpha_0 + \omega_3 \bullet (t - t_0)$$

$$R_3(t) = \begin{bmatrix} \cos(\alpha(t)) & \sin(\alpha(t)) & 0 \\ -\sin(\alpha(t)) & \cos(\alpha(t)) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \Omega_3 = \begin{bmatrix} 0 & \omega_3 & 0 \\ -\omega_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{d}{dt}(R_3(t)) = \omega_3 \bullet \begin{bmatrix} -\sin(\alpha(t)) & \cos(\alpha(t)) & 0 \\ -\cos(\alpha(t)) & -\sin(\alpha(t)) & 0 \\ 0 & 0 & 0 \end{bmatrix} = \Omega_3 \bullet R_3(t)$$

For a position in ECI coordinates, $\mathbf{P}_{ECI}(t)$, the position in ECEF coordinates, $\mathbf{P}_{ECEF}(t)$, is computed using the rotation matrix $R_3(t)$. For a velocity relative to the ECI frame expressed in ECI coordinates, $\mathbf{V_ECI}_{ECI}(t)$, velocity $\mathbf{V_ECI}_{ECEF}(t)$ is also computed using matrix $R_3(t)$. The velocity relative to the ECEF frame expressed in ECEF coordinates, $\mathbf{V_ECEF}_{ECEF}(t)$, may be computed using matrices $R_3(t)$ and Ω_3 by replacing $\frac{d}{dt}(R_3(t))$ with $\Omega_3 \bullet R_3(t)$.

$$\mathbf{P}_{ECEF}(t) = R_3(t) \bullet \mathbf{P}_{ECI}(t) \quad \mathbf{V_ECI}_{ECEF}(t) = R_3(t) \bullet \mathbf{V_ECI}_{ECI}(t)$$

$$\mathbf{V_ECEF}_{ECEF}(t) = \frac{d}{dt}(R_3(t) \bullet \mathbf{P}_{ECI}(t)) = R_3(t) \bullet \frac{d}{dt}(\mathbf{P}_{ECI}(t)) + \frac{d}{dt}(R_3(t)) \bullet \mathbf{P}_{ECI}(t)$$

$$\mathbf{V_ECEF}_{ECEF}(t) = R_3(t) \bullet \mathbf{V_ECI}_{ECI}(t) + \Omega_3 \bullet R_3(t) \bullet \mathbf{P}_{ECI}(t)$$

$$\mathbf{V_ECEF}_{ECEF}(t) = \mathbf{V_ECI}_{ECEF}(t) + \Omega_3 \bullet \mathbf{P}_{ECEF}(t)$$

For position and velocity errors expressed in RICF coordinates, the matrix $\mathbf{T}_{RICF}^{ECEF}$ is used to transform them into ECEF coordinates as shown below.

$$\Delta \mathbf{P}_{ECEF}(t) = \mathbf{T}_{RICF}^{ECEF} \bullet \Delta \mathbf{P}_{RICF}(t)$$

$$\Delta \mathbf{V_ECEF}_{ECEF}(t) = \mathbf{T}_{RICF}^{ECEF} \bullet \Delta \mathbf{V_ECEF}_{RICF}(t)$$

For position and velocity errors expressed in RIC coordinates, the matrices $\mathbf{T}_{RICI}^{ECEF}$ and Ω_3 are used to transform them into ECEF coordinates as shown below.

$$\Delta \mathbf{P}_{ECEF}(t) = \mathbf{T}_{RICI}^{ECEF} \bullet \Delta \mathbf{P}_{RICI}(t)$$

$$\Delta \mathbf{V_ECEF}_{ECEF}(t) = \mathbf{T}_{RICI}^{ECEF} \bullet \Delta \mathbf{V_ECI}_{RICI}(t) + \Omega_3 \bullet \mathbf{T}_{RICI}^{ECEF} \bullet \Delta \mathbf{P}_{RICI}(t)$$

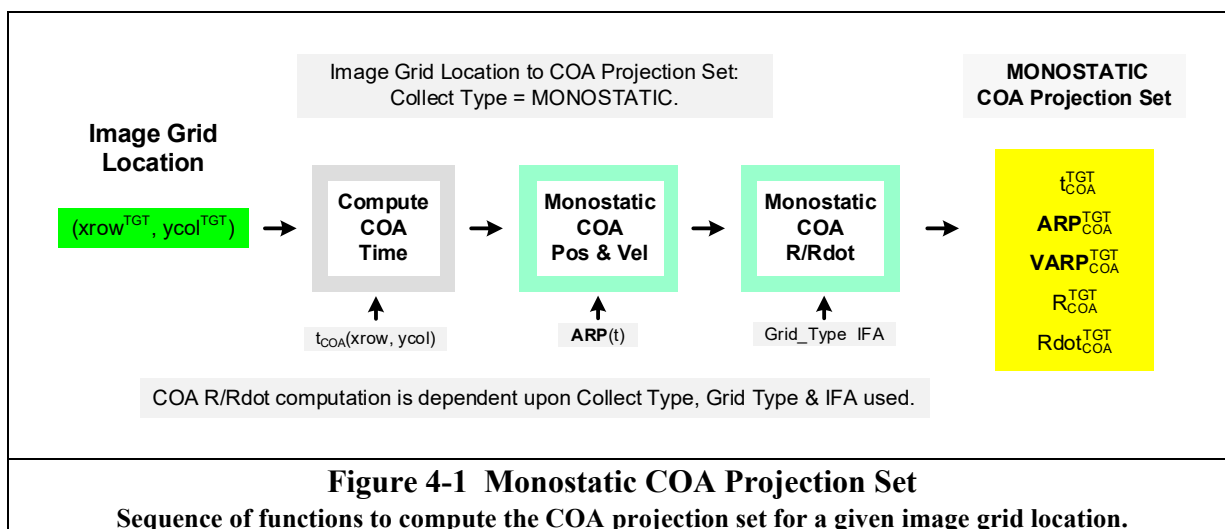
4 Image Grid to R/Rdot Contour

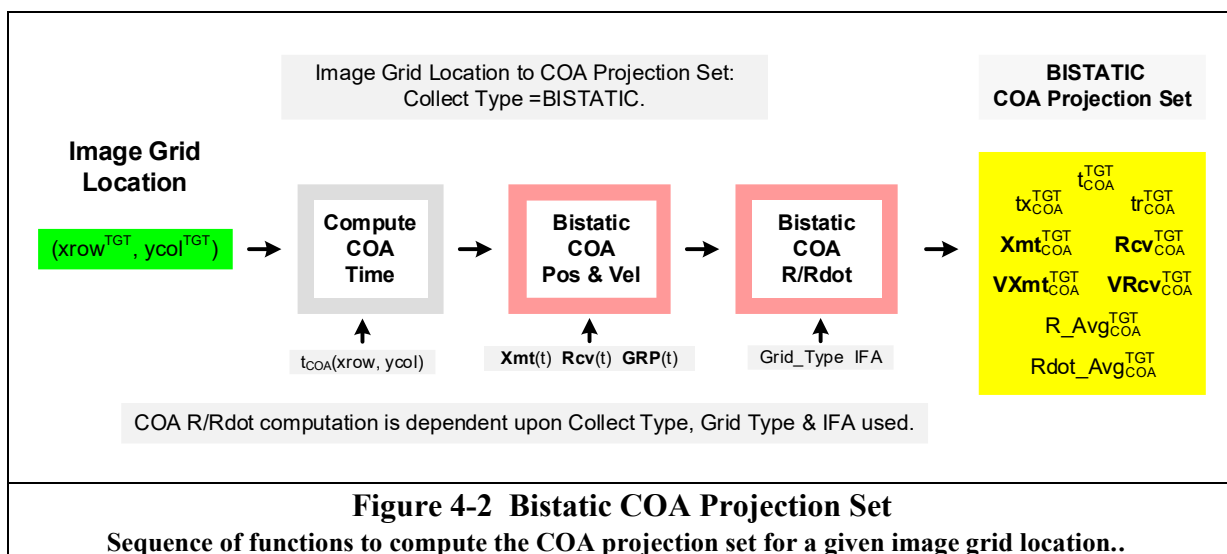
The precise image pixel grid location to R/Rdot projection contour computations are described in following sections. For a selected image grid location, the parameters to be computed include the COA time, the COA positions and velocities, and the COA range and range rate. The set of parameters to be computed are collectively referred to as the Center of Aperture projection set. For a selected image grid location, the COA projection set contains the parameters needed for computing precise image-to-scene projection. The parameters contained in the COA projection set are dependent upon the Collect Type. For a monostatic image, the projection set contains 5 parameters. For a bistatic image, the projection set contains 9 parameters. The parameters contained in the COA projection for each type of collection are as follows. Refer to Figure 1-1 and Figure 1-2.

Monostatic COA Projection Set: $\{ t_{COA}, \mathbf{ARP}_{COA}, \mathbf{VARP}_{COA}, R_{COA}, Rdot_{COA} \}$

Bistatic COA Projection Set: $\{ t_{COA}, tx_{COA}, tr_{COA}, \mathbf{Xmt}_{COA}, \mathbf{VXmt}_{COA}, \mathbf{Rcv}_{COA}, \mathbf{VRcv}_{COA}, R_Avg_{COA}, Rdot_Avg_{COA} \}$

The methods used to compute the COA projection set are dependent upon the Collect Type. For image grid location $(xrow^{TGT}, ycol^{TGT})$, the projection set for a monostatic image is computed as shown in Figure 4-1. For a bistatic image, the COA projection set is computed as shown in Figure 4-2. The method for computing the COA R/Rdot used with a given image is selected based upon two image metadata parameters: Grid_Type and IFA. See Table 1-1. The Grid_Type describes the relative spatial sampling for image pixel grid. The IFA specifies which algorithm-specific metadata is used in the computations. The methods for computing the COA time and the COA positions and velocity are specified in Section 2.5. The methods for computing the COA R/Rdot are specified in Sections 4.1 through 4.6 below.





The allowed values for Grid_Type and IFA are listed below.

Grid_Type	Image grid type represented by the sampling. Allowed values: RGAZIM, RGZERO, XRGYCR, XCTYAT, PLANE.
IFA	Image Formation Algorithm used to produce the grid. Allowed values: PFA, RMA, RGAZCOMP, OTHER

The combined methods used to the compute COA projection sets may be represented by the following two functions. For a monostatic image, the function shown includes the functions shown in Figure 4-1. For a bistatic image, the function shown includes the functions shown in Figure 4-2.

For a monostatic image:

$$\{ (xrow^{TGT}, ycol^{TGT}) \} \rightarrow \boxed{\text{Compute Monostatic Proj Set}} \rightarrow \left\{ \begin{matrix} t_{COA}^{TGT} & \mathbf{ARP}_{COA}^{TGT} & R_{COA}^{TGT} \\ & \mathbf{VARP}_{COA}^{TGT} & Rdot_{COA}^{TGT} \end{matrix} \right\}$$

For a bistatic image:

$$\{ (xrow^{TGT}, ycol^{TGT}) \} \rightarrow \boxed{\text{Compute Bistatic Proj Set}} \rightarrow \left\{ \begin{matrix} tx_{COA}^{TGT} & tr_{COA}^{TGT} & R_Avg_{COA}^{TGT} \\ t_{COA}^{TGT} & \mathbf{Xmt}_{COA}^{TGT} & \mathbf{Rcv}_{COA}^{TGT} \\ & \mathbf{VXmt}_{COA}^{TGT} & \mathbf{VRcv}_{COA}^{TGT} \\ & & Rdot_Avg_{COA}^{TGT} \end{matrix} \right\}$$

4.1 Image Grid To R/Rdot: Grid_Type = RGAZIM & IFA = PFA

The Polar Format Algorithm may be used for monostatic collections and for bistatic image collections with image fully supported when output in SICD format. The method below may be used to compute the precise R/Rdot contour for:

Collect_Type = MONOSTATIC & Collect_Type = BISTATIC

For the RGAZIM grid, the image coordinates are range and azimuth. The row coordinate is the range coordinate, $xrow = rg$. The column coordinate is the azimuth coordinate, $ycol = az$. For image pixel location $(irow^{TGT}, icol^{TGT})$, the image coordinates (rg^{TGT}, az^{TGT}) are computed as follows.

$$rg^{TGT} = Row_SS \cdot irow^{TGT} \quad az^{TGT} = Col_SS \cdot icol^{TGT}$$

For images formed with the Polar Format Algorithm, the R/Rdot computation makes use of the following SICD PFA specific parameters.

PFA Parameters:

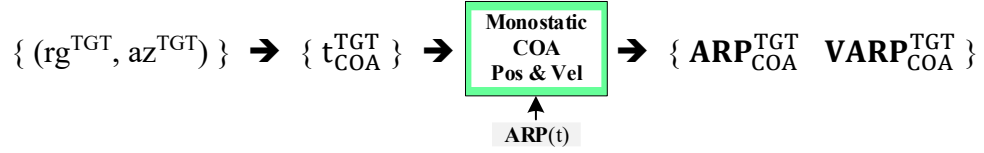
cPA(n)	Polar Angle polynomial coefficients. The polynomial yields polar angle (θ) in radians from collection time. Collection time in seconds ($t = 0$ at Collection Start).
N_PA	Order of the Polar Angle polynomial.
cKSF(n)	Spatial Frequency Scale Factor polynomial coefficients. The polynomial yields the scale factor (KSF) from polar angle (θ). The scale factor is used to map transmit frequency (fx , in Hz) to aperture spatial frequency (Kap , in cycles/m). $Kap(fx, \theta) = fx \cdot \frac{2}{c} \cdot KSF(\theta)$.
N_KSF	Order of the Spatial Frequency Scale Factor polynomial.

The computation of the R/Rdot contour for an arbitrary image grid location is presented with minimal explanation in order that the necessary computations are clearly presented. Refer to Figures 4-3 and 4-4.

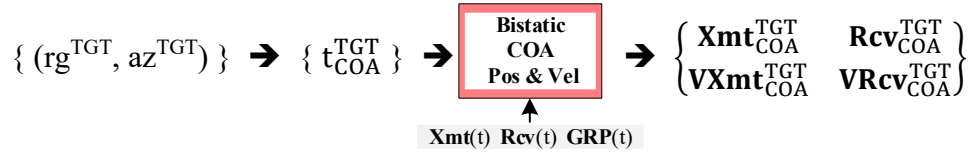
PFA Image Grid Location To R/Rdot Contour

(1) For a target (TGT) with impulse response centered at image grid location (rg^{TGT}, az^{TGT}) , compute the COA time and the COA positions and velocities. Refer to Section 2.5.

For a monostatic image:



For a bistatic image:



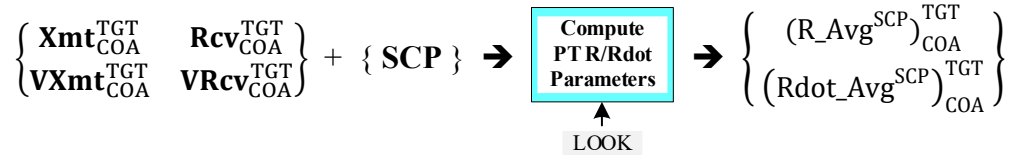
(2) Compute the range and range rate to the SCP relative to the COA positions and velocities. For a monostatic image, compute the range and range rate relative to the ARP as shown. For a bistatic image, compute average range and average range rate to the COA APCs using the Scene Point Bistatic R/Rdot function defined in Section 5.3.1. Only outputs R_Avg and $Rdot_Avg$ are used in the subsequent processing.

For a monostatic image:

$$(R^{SCP})_{COA}^{TGT} = |\mathbf{ARP}_{COA}^{TGT} - \mathbf{SCP}|$$

$$(Rdot^{SCP})_{COA}^{TGT} = \frac{1}{(R^{SCP})_{COA}^{TGT}} \cdot \mathbf{VARP}_{COA}^{TGT} \cdot (\mathbf{ARP}_{COA}^{TGT} - \mathbf{SCP})$$

For a bistatic image:



(3) Compute the polar angle (θ) and its derivative with respect to time ($d\theta/dt$) at the pixel COA time as follows.

$$\theta_{COA}^{TGT} = \sum_{n=0}^{N_{PA}} cPA(n) \cdot (t_{COA}^{TGT})^n \quad \left(\frac{d\theta}{dt} \right)_{COA}^{TGT} = \sum_{n=1}^{N_{PA}} n \cdot cPA(n) \cdot (t_{COA}^{TGT})^{n-1}$$

(4) Compute the polar aperture scale factor (KSF) and its derivative with respect to polar angle ($dKSF/d\theta$) at the pixel COA time as follows.

$$\text{KSF}_{\text{COA}}^{\text{TGT}} = \sum_{n=0}^{N \text{ KSF}} \text{cKSF}(n) \cdot (\theta_{\text{COA}}^{\text{TGT}})^n \left(\frac{d\text{KSF}}{d\theta} \right)_{\text{COA}}^{\text{TGT}} = \sum_{n=1}^{N \text{ KSF}} n \cdot \text{cKSF}(n) \cdot (\theta_{\text{COA}}^{\text{TGT}})^{n-1}$$

(5) Compute the spatial frequency domain phase slopes in the radial (Ka) and cross radial (Kc) directions ($\delta\Phi^{\text{TGT}}/\delta\text{Ka}$ and $\delta\Phi^{\text{TGT}}/\delta\text{Kc}$) for the radial direction at $\theta_{\text{COA}}^{\text{TGT}}$. Note: The sign parameter (SGN) for the phase may be ignored as it is cancelled in a subsequent computation.

$$\left(\frac{\delta\Phi^{\text{TGT}}}{\delta\text{Ka}} \right)_{\text{COA}}^{\text{TGT}} = \text{rg}^{\text{TGT}} \cdot \cos(\theta_{\text{COA}}^{\text{TGT}}) + \text{az}^{\text{TGT}} \cdot \sin(\theta_{\text{COA}}^{\text{TGT}})$$

$$\left(\frac{\delta\Phi^{\text{TGT}}}{\delta\text{Kc}} \right)_{\text{COA}}^{\text{TGT}} = -\text{rg}^{\text{TGT}} \cdot \sin(\theta_{\text{COA}}^{\text{TGT}}) + \text{az}^{\text{TGT}} \cdot \cos(\theta_{\text{COA}}^{\text{TGT}})$$

(6) Compute range relative to the SCP ($\Delta\text{R}^{\text{TGT}}$) at the COA.

$$\Delta\text{R}_{\text{COA}}^{\text{TGT}} = \text{KSF}_{\text{COA}}^{\text{TGT}} \cdot \left(\frac{\delta\Phi^{\text{TGT}}}{\delta\text{Ka}} \right)_{\text{COA}}^{\text{TGT}}$$

(7) Compute the derivative of the range relative to the SCP with respect to polar angle ($d(\Delta\text{R}^{\text{TGT}})/d\theta$) at the COA. Scale by the derivative of the polar angle with respect to time to yield the derivative with respect to time ($d(\Delta\text{R}^{\text{TGT}})/dt = \Delta\text{Rdot}^{\text{TGT}}$).

$$\left(\frac{d(\Delta\text{R}^{\text{TGT}})}{d\theta} \right)_{\text{COA}}^{\text{TGT}} = \left(\frac{d\text{KSF}}{d\theta} \right)_{\text{COA}}^{\text{TGT}} \cdot \left(\frac{\delta\Phi^{\text{TGT}}}{\delta\text{Ka}} \right)_{\text{COA}}^{\text{TGT}} + \text{KSF}_{\text{COA}}^{\text{TGT}} \cdot \left(\frac{\delta\Phi^{\text{TGT}}}{\delta\text{Kc}} \right)_{\text{COA}}^{\text{TGT}}$$

$$\Delta\text{Rdot}_{\text{COA}}^{\text{TGT}} = \left(\frac{d(\Delta\text{R}^{\text{TGT}})}{d\theta} \right)_{\text{COA}}^{\text{TGT}} \cdot \left(\frac{d\theta}{dt} \right)_{\text{COA}}^{\text{TGT}}$$

(8) Compute the range and range rate relative to the COA positions and velocities. For a monostatic image, the range and range rate is relative to the ARP, $\text{R}_{\text{COA}}^{\text{TGT}}$ and $\text{Rdot}_{\text{COA}}^{\text{TGT}}$. For a bistatic image, the range and range rate are the average range and average range rate relative to the COA APCs, $\text{R_Avg}_{\text{COA}}^{\text{TGT}}$ and $\text{Rdot_Avg}_{\text{COA}}^{\text{TGT}}$. The projection to three-dimensional scene point for grid location (rg^{TGT} , az^{TGT}) is along this R/Rdot contour.

For a monostatic image:

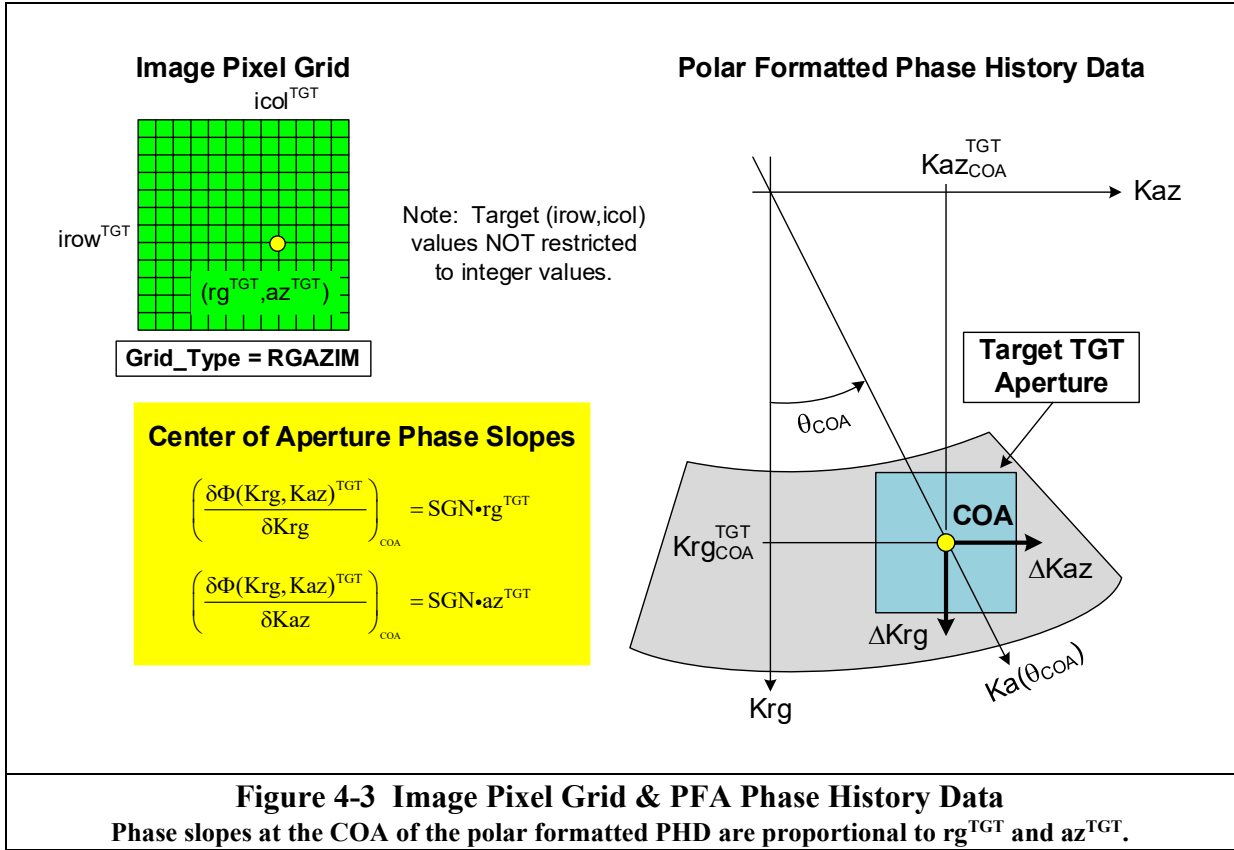
$$\text{R}_{\text{COA}}^{\text{TGT}} = (\text{R}^{\text{SCP}})_{\text{COA}}^{\text{TGT}} + \Delta\text{R}_{\text{COA}}^{\text{TGT}}$$

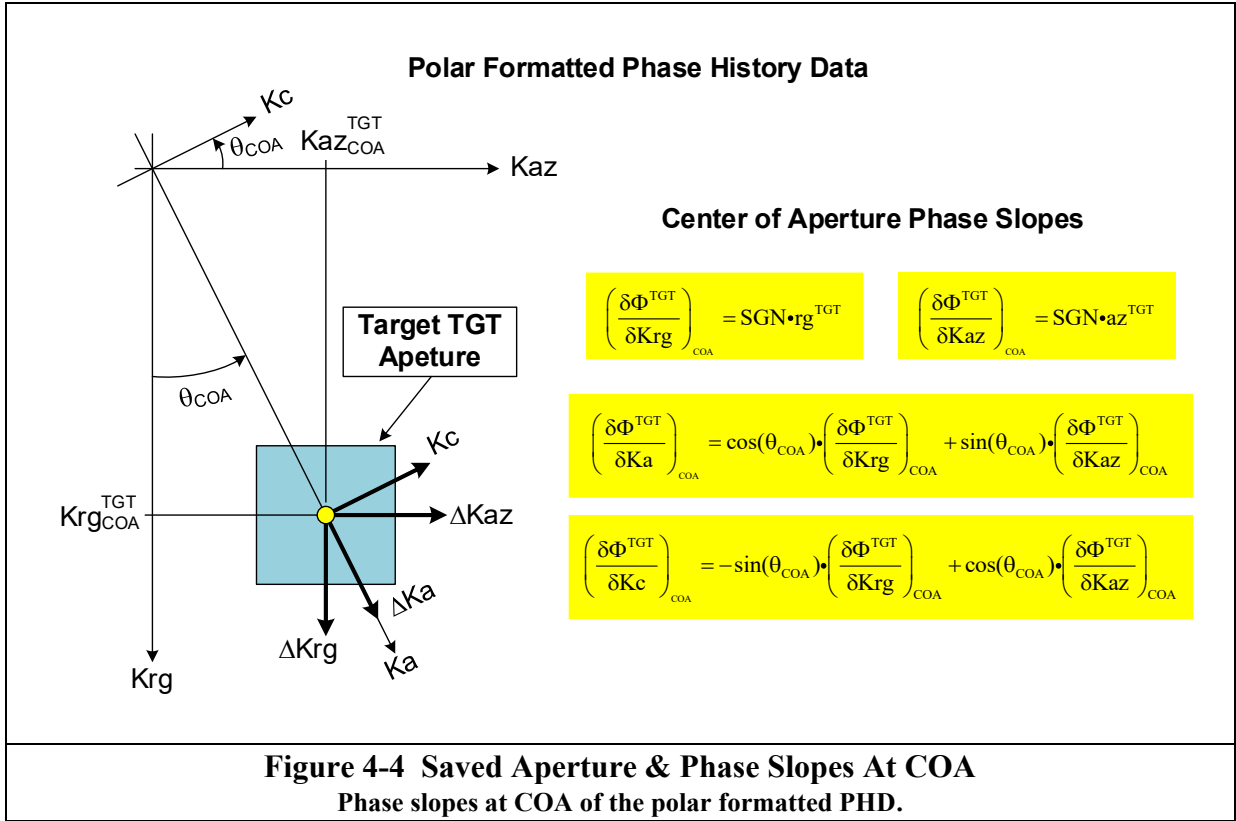
$$\text{Rdot}_{\text{COA}}^{\text{TGT}} = (\text{Rdot}^{\text{SCP}})_{\text{COA}}^{\text{TGT}} + \Delta\text{Rdot}_{\text{COA}}^{\text{TGT}}$$

For a bistatic image:

$$\text{R_Avg}_{\text{COA}}^{\text{TGT}} = (\text{R_Avg}^{\text{SCP}})_{\text{COA}}^{\text{TGT}} + \Delta\text{R}_{\text{COA}}^{\text{TGT}}$$

$$\text{Rdot_Avg}_{\text{COA}}^{\text{TGT}} = (\text{Rdot_Avg}^{\text{SCP}})_{\text{COA}}^{\text{TGT}} + \Delta\text{Rdot}_{\text{COA}}^{\text{TGT}}$$





4.2 Image Grid To R/Rdot: Grid_Type = RGAZIM & IFA = RGAZCOMP

The RGAZCOMP algorithm may only be used for monostatic image formation when the output is provided in SICD format. The method below may only be used to compute the precise R/Rdot for:

Collect_Type = MONOSTATIC

For the RGAZIM grid, the image coordinates are range and azimuth. The row coordinate is the range coordinate, $xrow = rg$. The column coordinate is the azimuth coordinate, $ycol = az$. For image pixel location $(irow^{TGT}, icol^{TGT})$, the image coordinates (rg^{TGT}, az^{TGT}) are computed as follows.

$$rg^{TGT} = Row_SS \cdot irow^{TGT} \quad az^{TGT} = Col_SS \cdot icol^{TGT}$$

For images formed with the Range & Azimuth Compression algorithm, the R/Rdot computation makes use of the following SICD RGAZCOMP specific parameter.

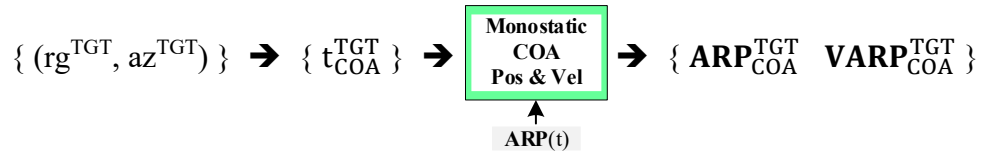
RGAZCOMP Parameter:

AzSF Scale factor that converts azimuth coordinate to an increment in cosine of the DCA at COA.

For an image formed with simple Range & Azimuth Compression, all pixels have a common COA. The resulting image is sometimes referred to as a Range, Doppler image. The rows are samples at constant range at COA and the columns are samples at constant Doppler Cone Angle at COA. Refer to Figure 4-5.

RGAZCOMP Image Grid Location To R/Rdot Contour

(1) For a target (TGT) with impulse response centered at image grid location (rg^{TGT}, az^{TGT}) , compute the COA time and the COA positions and velocities. Refer to Section 2.5.



(2) Compute the range and range rate to the SCP at COA.

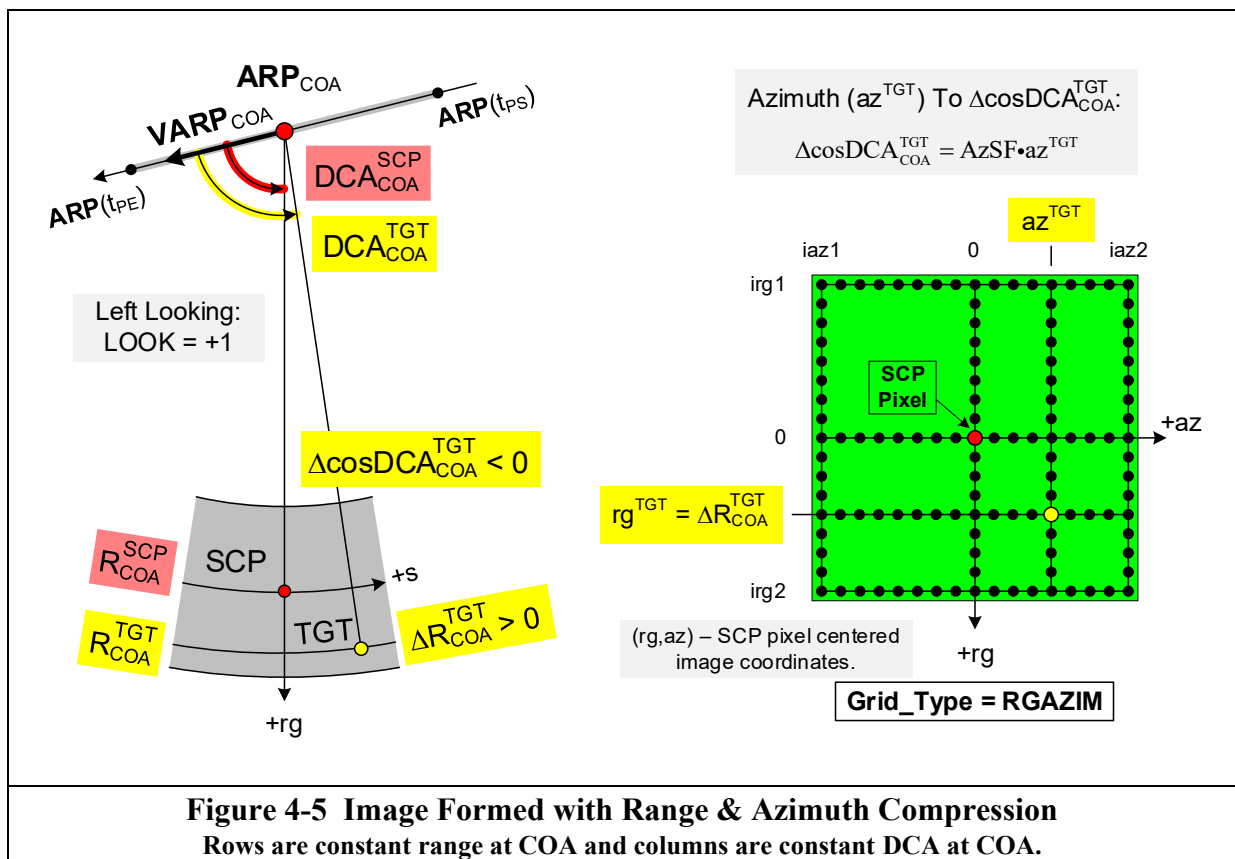
$$R_{COA}^{SCP} = |ARP_{COA}^{TGT} - SCP| \quad Rdot_{COA}^{SCP} = \frac{1}{R_{COA}^{SCP}} \cdot VARP_{COA}^{TGT} \cdot (ARP_{COA}^{TGT} - SCP)$$

- (3) Compute the increment in cosine of the DCA at COA of the target, $\Delta \cos \text{DCA}_{\text{COA}}^{\text{TGT}}$, by scaling the azimuth coordinate by the azimuth to DCA scale factor. Compute the increment in range rate, $\Delta \text{Rdot}_{\text{COA}}^{\text{TGT}}$, by scaling by the magnitude of the velocity vector at COA.

$$\Delta \text{cosDCA}_{\text{COA}}^{\text{TGT}} = \text{AzSF} \cdot \text{az}^{\text{TGT}} \quad \Delta \text{Rdot}_{\text{COA}}^{\text{TGT}} = - \left| \text{VARP}_{\text{COA}}^{\text{TGT}} \right| \cdot \Delta \text{cosDCA}_{\text{COA}}^{\text{TGT}}$$

- (4) Compute the range and range rate to the target at COA as follows.

$$\mathbf{R}_{\text{COA}}^{\text{TGT}} = \mathbf{R}_{\text{COA}}^{\text{SCP}} + \mathbf{rg}^{\text{TGT}} \quad \mathbf{Rdot}_{\text{COA}}^{\text{TGT}} = \mathbf{Rdot}_{\text{COA}}^{\text{SCP}} + \Delta \mathbf{Rdot}_{\text{COA}}^{\text{TGT}}$$



4.3 Image Grid To R/Rdot: Grid_Type = RGZERO

The RGZERO grid may only be used for monostatic images when the output is provided in SICD format. The method below may only be used to compute the precise R/Rdot for:

Collect_Type = MONOSTATIC

For the RGZERO grid, the image coordinates are range and azimuth. The row coordinate is the range coordinate, xrow = rg. The column coordinate is the azimuth coordinate, ycol = az. For image pixel location (irow^{TGT}, icol^{TGT}), the image coordinates (rg^{TGT}, az^{TGT}) are computed as follows.

$$rg^{TGT} = Row_SS \cdot irow^{TGT} \quad az^{TGT} = Col_SS \cdot icol^{TGT}$$

For all images with Grid_Type = RGZERO, the following parameters are also included. The parameters describe the image formation processing with the RMA that formed the image.

RMA INCA Parameters:

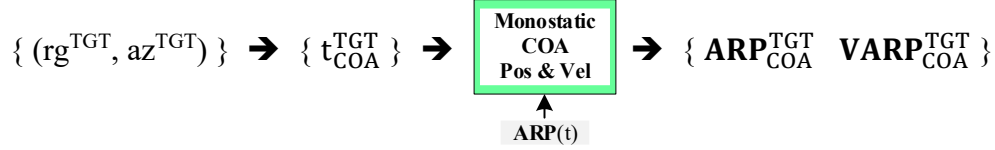
cT_CA(n)	Time of Closest Approach polynomial coefficients. The polynomial yields time of closest approach, t_CA (sec), as a function of image azimuth coordinate (az). Time t = 0 at Collection Start.
N_TCA	Order of the Time of Closest Approach polynomial.
R _{CA} ^{SCP}	Range at Closest Approach for the SCP (m).
cDRSF(m,n)	Doppler Rate Scale Factor polynomial coefficients. The polynomial yields the scale factor (DRSF) from image coordinates (rg,az). The scale factor is used to compute d ² (R)/dt ² at closest approach.
M_DRSF	Order of the DRSF polynomial in the row/range coordinate.
N_DRSF	Order of the DRSF polynomial in the col/azimuth coordinate.

The computation of the R/Rdot contour for an arbitrary image grid location is presented with minimal explanation in order that the necessary computations are clearly presented. Refer to Figure 4-6. For a given RGZERO grid location, the range history that was processed to form the image signal is accurately described by range function RA(t)^{TGT}. The parameters of the RA(t)^{TGT} are computed from the grid location. The range at COA is computed by evaluating the RA(t)^{TGT} function at the t_{COA}^{TGT}. The range rate at COA is computed from the d(RA(t)^{TGT})/dt at t_{COA}^{TGT}.

$$RA(t)^{TGT} = \left[(R_{CA}^{TGT})^2 + DRSF^{TGT} \cdot (VM_{CA}^{TGT})^2 \cdot (t - t_{CA}^{TGT})^2 \right]^{1/2}$$

RGZERO Image Grid Location To R/Rdot Contour

(1) For a target (TGT) with impulse response centered at image grid location (rg^{TGT} , az^{TGT}), compute the COA time and the COA position and velocity. Refer to Section 2.5.



(2) Compute the range at closest approach and the time of closest approach for the image grid location. The range at closest approach, R_{CA}^{TGT} , is computed from the range coordinate. The time of closest approach, t_{CA}^{TGT} , is computed from the azimuth coordinate.

$$R_{CA}^{TGT} = R_{CA}^{SCP} + rg^{TGT} \quad t_{CA}^{TGT} = \sum_{n=0}^{N_{TCA}} cT_CA(n) \cdot (az^{TGT})^n$$

(2) Compute the ARP velocity at t_{CA}^{TGT} , $VARP_{CA}^{TGT}$. Also compute the magnitude of the vector.

$$VARP_{CA}^{TGT} = \begin{bmatrix} VARP_x(t_{CA}^{TGT}) \\ VARP_y(t_{CA}^{TGT}) \\ VARP_z(t_{CA}^{TGT}) \end{bmatrix} \quad VM_{CA}^{TGT} = |VARP_{CA}^{TGT}|$$

(3) Compute the Doppler Rate Scale Factor, $DRSF^{TGT}$, for image grid location (rg^{TGT} , az^{TGT}).

$$DRSF^{TGT} = \sum_{m=0}^{M_{DRSF}} \sum_{n=0}^{N_{DRSF}} cDRSF(m,n) \cdot (rg^{TGT})^m \cdot (az^{TGT})^n$$

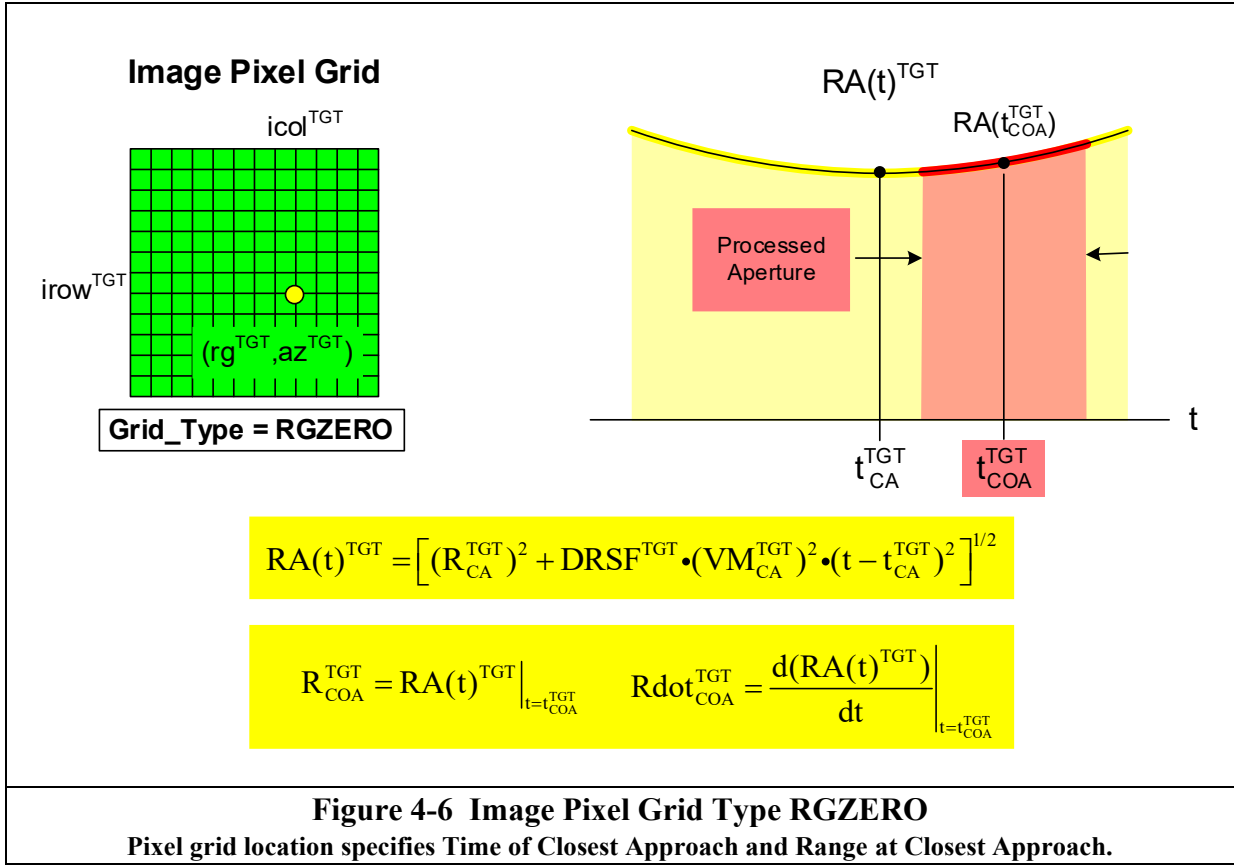
(4) Compute the time difference between the COA time and the CA time, Δt_{COA}^{TGT} .

$$\Delta t_{COA}^{TGT} = t_{COA}^{TGT} - t_{CA}^{TGT}$$

(5) Compute the range and range rate relative to the ARP at COA (R_{COA}^{TGT} and $Rdot_{COA}^{TGT}$). The projection to three-dimensional scene point for grid location (rg^{TGT} , az^{TGT}) is along this R/Rdot contour.

$$R_{COA}^{TGT} = \left[(R_{CA}^{TGT})^2 + DRSF^{TGT} \cdot (VM_{CA}^{TGT})^2 \cdot (\Delta t_{COA}^{TGT})^2 \right]^{1/2}$$

$$Rdot_{COA}^{TGT} = \frac{DRSF^{TGT}}{R_{COA}^{TGT}} \cdot (VM_{CA}^{TGT})^2 \cdot \Delta t_{COA}^{TGT}$$



4.4 Image Grid To R/Rdot: Grid_Type = XRGYCR

The XRGYCR grid may be used for monostatic images or bistatic images. The method below may be used to compute the precise R/Rdot contour for:

Collect_Type = MONOSTATIC & Collect_Type = BISTATIC

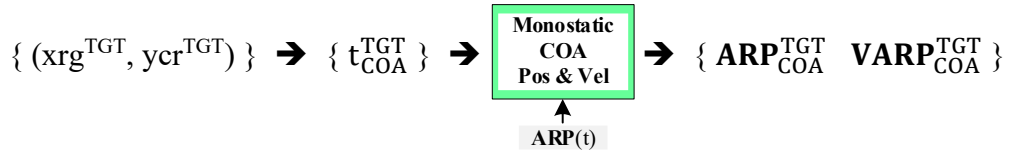
For the XRGYCR grid, the image coordinates are xrg and ycr. The xrg coordinate is the row coordinate, $xrg = xrow$. The ycr coordinate is the column coordinate, $ycr = ycol$. For image pixel location $(irow^{TGT}, icol^{TGT})$, the image coordinates (xrg^{TGT}, ycr^{TGT}) are computed as follows.

$$xrg^{TGT} = Row_SS \cdot irow^{TGT} \quad ycr^{TGT} = Col_SS \cdot icol^{TGT}$$

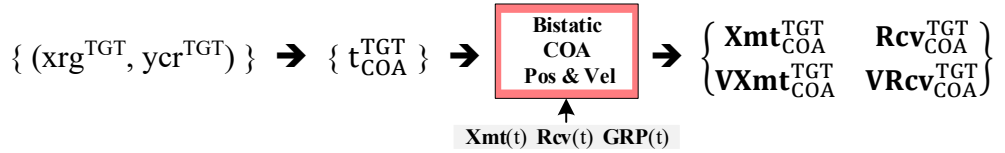
XRGYCR Image Grid Location To R/Rdot Contour

(1) For a target TGT with impulse response centered at image grid location (xrg^{TGT}, ycr^{TGT}) , compute the COA time and the COA positions and velocities. Refer to Section 2.5.

For a monostatic image:



For a bistatic image:



(2) The samples of XRGYCR grid are uniformly spaced locations in the image plane formed by the SCP, and image plane vectors $\mathbf{uRow}$ and $\mathbf{uCol}$. Compute the image plane point for image grid location (xrg^{TGT}, ycr^{TGT}) , $\mathbf{IP_TGT}$.

$$\mathbf{IP_TGT} = \mathbf{SCP} + xrg^{TGT} \cdot \mathbf{uRow} + ycr^{TGT} \cdot \mathbf{uCol}$$

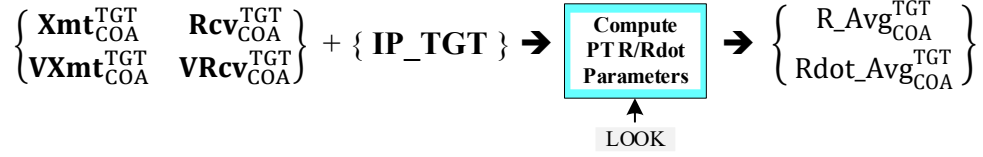
(3) Compute the range and range rate to the image plane point $\mathbf{IP_TGT}$ relative to the COA positions and velocities. For a bistatic image, compute average range and average range rate relative to the COA APCs using the Scene Point Bistatic R/Rdot function defined in Section 5.3.1.

For a monostatic image:

$$R_{COA}^{TGT} = | \mathbf{ARP}_{COA}^{TGT} - \mathbf{IP_TGT} | \quad \mathbf{uPT}_{COA}^{TGT} = \frac{1}{R_{COA}^{TGT}} \cdot (\mathbf{ARP}_{COA}^{TGT} - \mathbf{IP_TGT})$$

$$\mathbf{Rdot}_{COA}^{TGT} = \mathbf{VARP}_{COA}^{TGT} \bullet \mathbf{uPT}_{COA}^{TGT}$$

For a bistatic image:



4.5 Image Grid To R/Rdot: Grid_Type = XCTYAT

The XCTYAT grid may be used for monostatic images or bistatic images. The method below may be used to compute the precise R/Rdot contour for:

$$\text{Collect_Type} = \text{MONOSTATIC} \ \& \ \text{Collect_Type} = \text{BISTATIC}$$

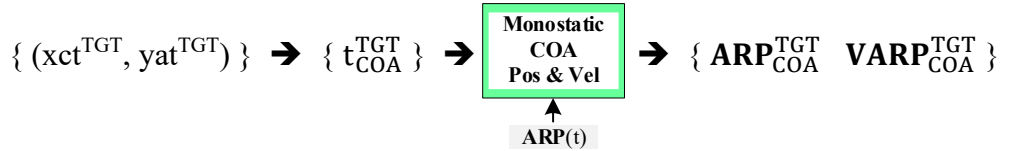
For the XCTYAT grid, the image coordinates are xct and yat.

$$\text{xct}^{TGT} = \text{Row_SS} \bullet \text{irow}^{TGT} \qquad \text{yat}^{TGT} = \text{Col_SS} \bullet \text{icol}^{TGT}$$

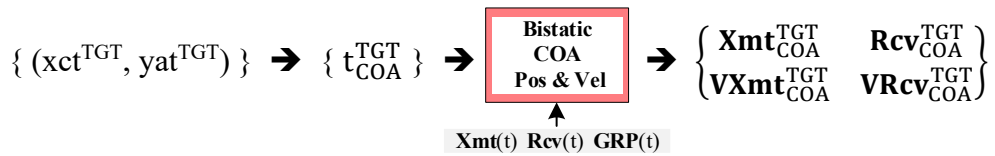
XCTYAT Image Grid Location To R/Rdot Contour

- (1) Compute the COA time and the COA positions and velocities.

For a monostatic image:



For a bistatic image:



- (2) Compute the image plane point for image grid location $(\text{xct}^{TGT}, \text{yat}^{TGT})$, $\mathbf{IP_TGT}$.

$$\mathbf{IP_TGT} = \mathbf{SCP} + \text{xct}^{TGT} \bullet \mathbf{uRow} + \text{yat}^{TGT} \bullet \mathbf{uCol}$$

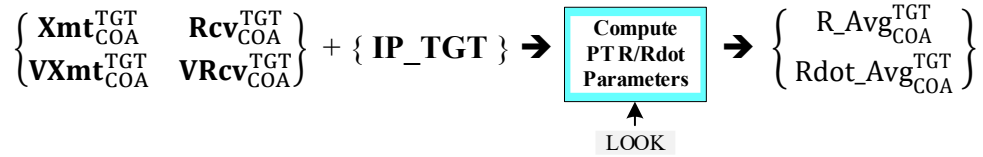
- (3) Compute the range and range rate to the image plane point **IP_TGT** relative to the COA positions and velocities. For a bistatic image, use the Scene Point Bistatic R/Rdot function.

For a monostatic image:

$$R_{COA}^{TGT} = | \mathbf{ARP}_{COA}^{TGT} - \mathbf{IP_TGT} | \quad \mathbf{uPT}_{COA}^{TGT} = \frac{1}{R_{COA}^{TGT}} \cdot (\mathbf{ARP}_{COA}^{TGT} - \mathbf{IP_TGT})$$

$$Rdot_{COA}^{TGT} = \mathbf{VARP}_{COA}^{TGT} \bullet \mathbf{uPT}_{COA}^{TGT}$$

For a bistatic image:



4.6 Image Grid To R/Rdot: Grid_Type = PLANE

The PLANE grid may be used for monostatic images or bistatic images. The method below may be used to compute the precise R/Rdot contour for:

$$\text{Collect_Type} = \text{MONOSTATIC} \ \& \ \text{Collect_Type} = \text{BISTATIC}$$

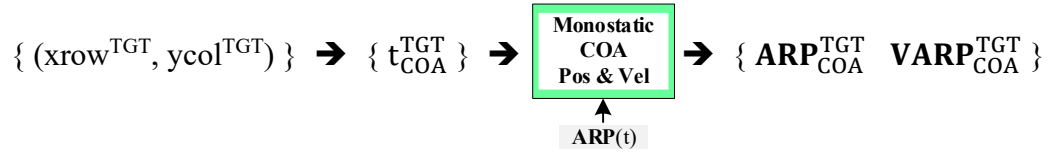
For the PLANE grid, the image coordinates are xrow and ycol.

$$xrow^{TGT} = \text{Row_SS} \cdot irow^{TGT} \quad ycol^{TGT} = \text{Col_SS} \cdot icol^{TGT}$$

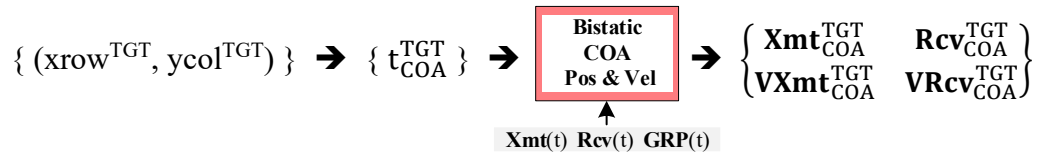
PLANE Image Grid Location To R/Rdot Contour

- (1) Compute the COA time and the COA positions and velocities.

For a monostatic image:



For a bistatic image:



- (2) Compute the image plane point for image grid location ($xrow^{TGT}$, $ycol^{TGT}$), **IP_TGT**.

$$\mathbf{IP_TGT} = \mathbf{SCP} + xrow^{TGT} \bullet \mathbf{uRow} + ycol^{TGT} \bullet \mathbf{uCol}$$

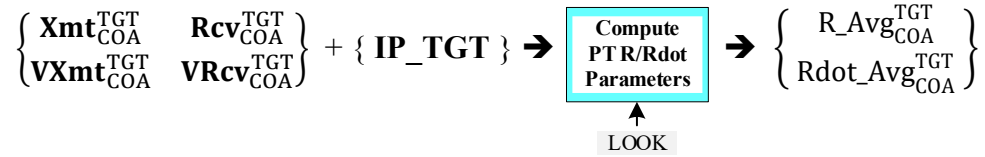
- (3) Compute the range and range rate to the image plane point **IP_TGT** relative to the COA positions and velocities. For a monostatic image, compute the range and range rate relative to the ARP as shown below. For a bistatic image, compute average range and average range rate relative to the COA APCs using the Scene Point Bistatic R/Rdot function.

For a monostatic image:

$$R_{COA}^{TGT} = | \mathbf{ARP}_{COA}^{TGT} - \mathbf{IP_TGT} | \quad \mathbf{uPT}_{COA}^{TGT} = \frac{1}{R_{COA}^{TGT}} \bullet (\mathbf{ARP}_{COA}^{TGT} - \mathbf{IP_TGT})$$

$$Rdot_{COA}^{TGT} = \mathbf{VARP}_{COA}^{TGT} \bullet \mathbf{uPT}_{COA}^{TGT}$$

For a bistatic image:



5 Precise R/Rdot to Ground Plane Projection

The precise Image-to-Scene projection for a selected pixel location to an arbitrary ground plane is specified. For a monostatic image, the projection is along a contour of constant range and range rate relative to the Center of Aperture (COA) ARP position and velocity. The projection method for a monostatic image is described in Section 5.2. For a bistatic image, the projection is along a contour of constant average range and average range rate relative to the COA transmit APC and receive APC positions and velocities. The projection method for a bistatic image is described in Section 5.3.

5.1 Ground Plane & Ground Plane Normal

The ground plane is defined by a reference point in the plane, GREF, and the unit normal vector to the plane, **uGPN**. The reference point and plane orientation may be based upon specific terrain height and slope information for the imaged area. When only a reference point is specified, a ground plane normal may be derived in several ways. Given the position of GREF in ECEF coordinates, **GREF**, two ground planes that contain point GREF are: (1) the Spherical Earth ground plane, and (2) the Earth Tangent Plane (ETP) ground plane.

The Spherical Earth ground plane is tangent to a spherical earth model passing through GREF. The Spherical Earth GP unit normal is computed as follows.

$$\text{Spherical Earth GPN: } \mathbf{uGPN}_{\text{SPH}} = \frac{1}{|\mathbf{GREF}|} \cdot \mathbf{GREF}$$

The ETP ground plane is tangent to the surface of constant geodetic height above the WGS 84 reference ellipsoid passing through GREF. The ETP unit normal is computed as follows by first computing the geodetic latitude and longitude of GREF.

$$\mathbf{GREF} = \begin{bmatrix} \text{GREF.X} \\ \text{GREF.Y} \\ \text{GREF.Z} \end{bmatrix} \rightarrow \begin{array}{|c|} \hline \text{Convert} \\ \text{ECEF to} \\ \text{LLH} \\ \hline \end{array} \rightarrow \mathbf{GREF_LLH(n)} = \begin{bmatrix} \text{GREF.Lat} \\ \text{GREF.Lon} \\ \text{GREF.HAE} \end{bmatrix}$$

$$\text{Earth Tangent Plane GPN: } \mathbf{uGPN}_{\text{ETP}} = \begin{bmatrix} \cos(\text{GREF.Lat}) \cdot \cos(\text{GREF.Lon}) \\ \cos(\text{GREF.Lat}) \cdot \sin(\text{GREF.Lon}) \\ \sin(\text{GREF.Lat}) \end{bmatrix}$$

5.2 Monostatic R/Rdot Contour Ground Plane Intersection

For a monostatic image, the intersection of the R/Rdot contour and a ground plane may be computed in closed form. A closed form method for computing the point of intersection is specified below.

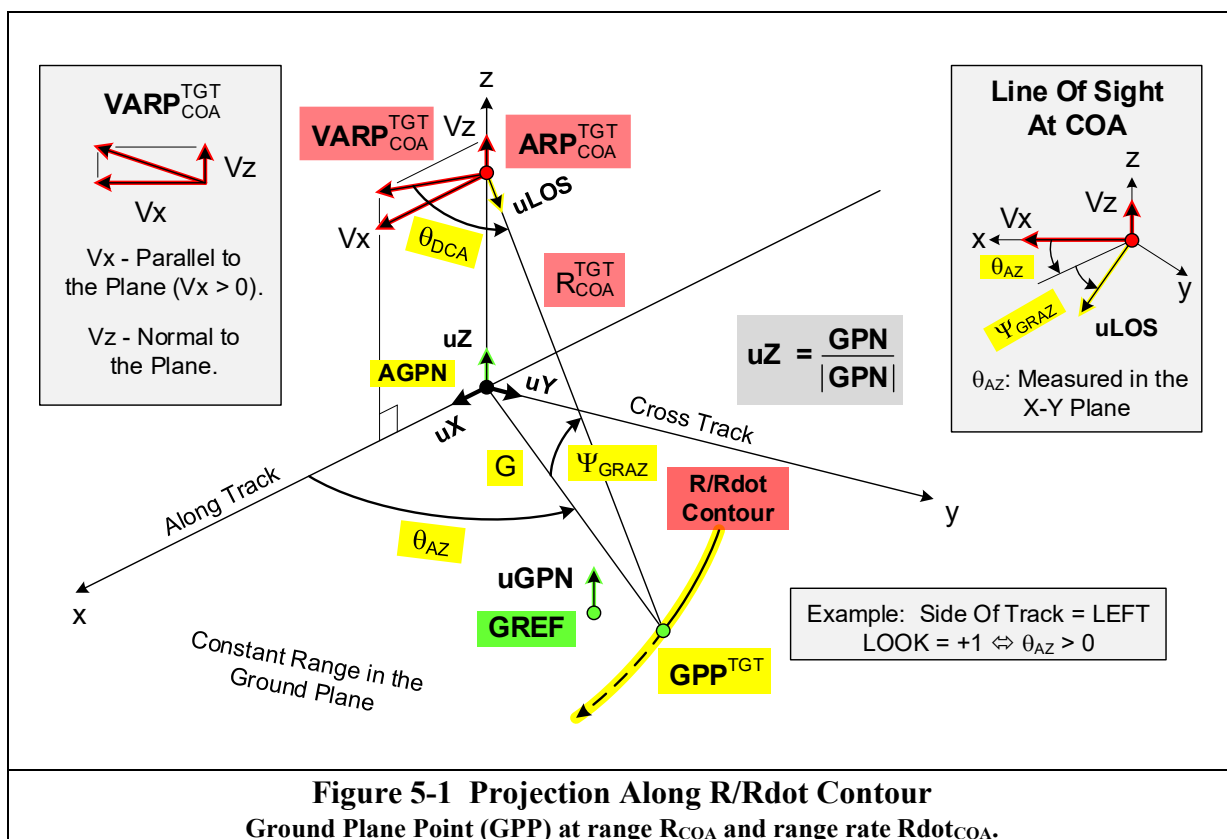
For target TGT with COA range and range rate $R_{\text{COA}}^{\text{TGT}}$ and $\text{Rdot}_{\text{COA}}^{\text{TGT}}$, solve for the intersection of the COA contour and a ground plane. The R/Rdot contour is relative to the COA ARP position and velocity, $\mathbf{ARP}_{\text{COA}}^{\text{TGT}}$ and $\mathbf{VARP}_{\text{COA}}^{\text{TGT}}$. The ground plane is defined by a point in the plane

(GREF) and a unit vector normal to the plane. The intersection of the R/Rdot contour and the ground plane is point GPP^{TGT} . See Figure 5-1.

The computation makes use of a Cartesian coordinate system (XYZ) with origin at the ARP ground plane nadir (AGPN). The X and Y axes are in the ground plane and the Z axis is normal to it. The +X direction is in the Along Track direction. The unit Line of Sight (**uLOS**) vector is defined by the azimuth angle (θ_{AZ}) and the grazing angle (Ψ_{GRAZ}). The range to the ground point determines the grazing angle. The range rate determines the cosine of the azimuth angle. The Side of Track determines the sign of the azimuth angle. The expression for the range rate in terms of the velocity vector and the unit LOS vector in XYZ coordinates is used to solve for the cosine of the azimuth angle.

$$Rdot_{COA}^{TGT} = -\mathbf{VARP}_{COA}^{TGT} \bullet \mathbf{uLOS} = - \begin{bmatrix} V_x \\ 0 \\ V_z \end{bmatrix} \bullet \begin{bmatrix} \cos(\Psi_{GRAZ}) \cdot \cos(\theta_{AZ}) \\ \cos(\Psi_{GRAZ}) \cdot \sin(\theta_{AZ}) \\ -\sin(\Psi_{GRAZ}) \end{bmatrix}$$

$$Rdot_{COA}^{TGT} = -V_x \cdot \cos(\Psi_{GRAZ}) \cdot \cos(\theta_{AZ}) + V_z \cdot \sin(\Psi_{GRAZ})$$



Input COA Projection Parameters – Computed for the image grid location for target TGT

$\mathbf{ARP}_{COA}^{TGT}$	ARP position at the COA time in ECEF coordinates.
$\mathbf{VARP}_{COA}^{TGT}$	ARP velocity at the COA time in ECEF coordinates.
R_{COA}^{TGT}	Range of the R/Rdot contour.
$Rdot_{COA}^{TGT}$	Range rate (dR/dt) of the R/Rdot contour ($Rdot_{COA}^{TGT} < 0 \Leftrightarrow$ Closing geometry).

Image Parameters:

LOOK	Integer parameter indicating the Side of Track for the collection geometry.
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Ground Plane Parameters:

GREF	Ground plane reference point in ECEF coordinates.
uGPN	Ground plane unit normal vector in ECEF coordinates.

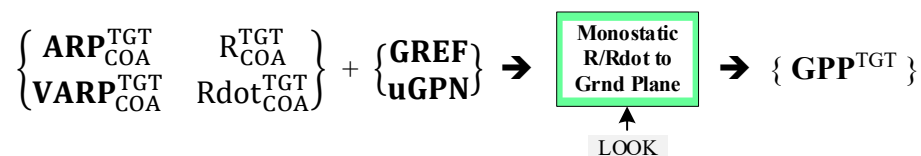
Computed Parameters:

GPP ^{TGT}	Ground Plane Point (GPP ^{TGT}) position in the ground plane and along the R/Rdot contour.
---------------------------	---

All position and velocities are in ECEF coordinates but may be in any common coordinate frame.

Monostatic R/Rdot Contour to Ground Plane Point GPP^{TGT}

The precise projection to the ground plane along the monostatic R/Rdot contour may be implemented as function as shown below. The inputs are the COA ARP position and velocity, the position for a point in the plane, and the ground plane unit normal. The side of track parameter LOOK is also an input.



(1) Assign the unit vector in the +Z direction (normal to the ground plane).

$$\mathbf{uZ} = \mathbf{uGPN}$$

(2) Compute the ARP distance from the plane (ARPz). Also compute the ARP ground plane nadir (AGPN).

$$ARPz = (\mathbf{ARP}_{COA}^{TGT} - \mathbf{GREF}) \bullet \mathbf{uZ} \quad \text{Note: For } |ARPz| > R_{COA}^{TGT} \rightarrow \text{No Solution}$$

$$\mathbf{AGPN} = \mathbf{ARP}_{COA}^{TGT} - ARPz \bullet \mathbf{uZ}$$

(3) Compute the ground plane distance (G) from the ARP nadir to the circle of constant range. Also compute the sine and cosine of the grazing angle (Ψ_{GRAZ}).

$$G = + \left((R_{\text{COA}}^{\text{TGT}})^2 - \text{ARP}_Z^2 \right)^{1/2} \quad \cos_GRAZ = \frac{G}{R_{\text{COA}}^{\text{TGT}}} \quad \sin_GRAZ = \frac{\text{ARP}_Z}{R_{\text{COA}}^{\text{TGT}}}$$

(4) Compute velocity components normal the ground plane (V_Z) and parallel to the ground plane (V_X).

$$V_Z = \mathbf{VARP}_{\text{COA}}^{\text{TGT}} \bullet \mathbf{uZ} \quad V_{\text{Mag}} = \left| \mathbf{VARP}_{\text{COA}}^{\text{TGT}} \right|$$

$$V_X = + (V_{\text{Mag}}^2 - V_Z^2)^{1/2} \quad \text{Note: For } V_X = 0 \rightarrow \text{No Solution}$$

(5) Orient the +X direction in the ground plane such that the $V_X > 0$. Compute unit vectors $\mathbf{uX}$ and $\mathbf{uY}$.

$$\mathbf{uX} = \frac{1}{V_X} \cdot (\mathbf{VARP}_{\text{COA}}^{\text{TGT}} - V_Z \bullet \mathbf{uZ}) \quad \mathbf{uY} = \mathbf{uZ} \times \mathbf{uX}$$

(6) Compute the cosine of the azimuth angle (θ_{AZ}) to the ground plane point.

$$\cos_AZ = \frac{-\text{Rdot}_{\text{COA}}^{\text{TGT}} + V_Z \bullet \sin_GRAZ}{V_X \bullet \cos_GRAZ}$$

Note: For $\cos_AZ < -1.0 \rightarrow$ No Solution. $\text{Rdot}_{\text{COA}}^{\text{TGT}} > \text{Max}\{\text{Rdot}\}$ at ground distance G.

Note: For $\cos_AZ > 1.0 \rightarrow$ No Solution. $\text{Rdot}_{\text{COA}}^{\text{TGT}} < \text{Min}\{\text{Rdot}\}$ at ground distance G.

(7) Compute the sine of the azimuth angle. Use parameter LOOK to establish the correct sign corresponding to the correct Side of Track.

$$\sin_AZ = \text{LOOK} \bullet (1 - \cos_AZ^2)^{1/2}$$

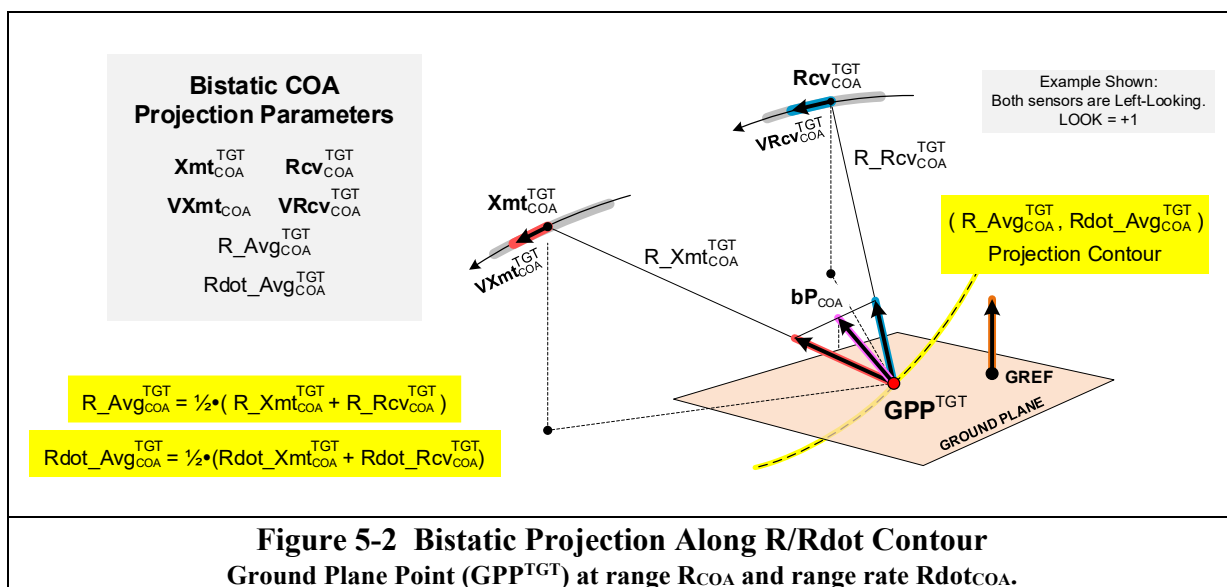
(8) Compute $\mathbf{GPP}^{\text{TGT}}$ at distance G from the AGPN and at the correct azimuth angle.

$$\mathbf{GPP}^{\text{TGT}} = \mathbf{AGPN} + G \bullet \cos_AZ \bullet \mathbf{uX} + G \bullet \sin_AZ \bullet \mathbf{uY}$$

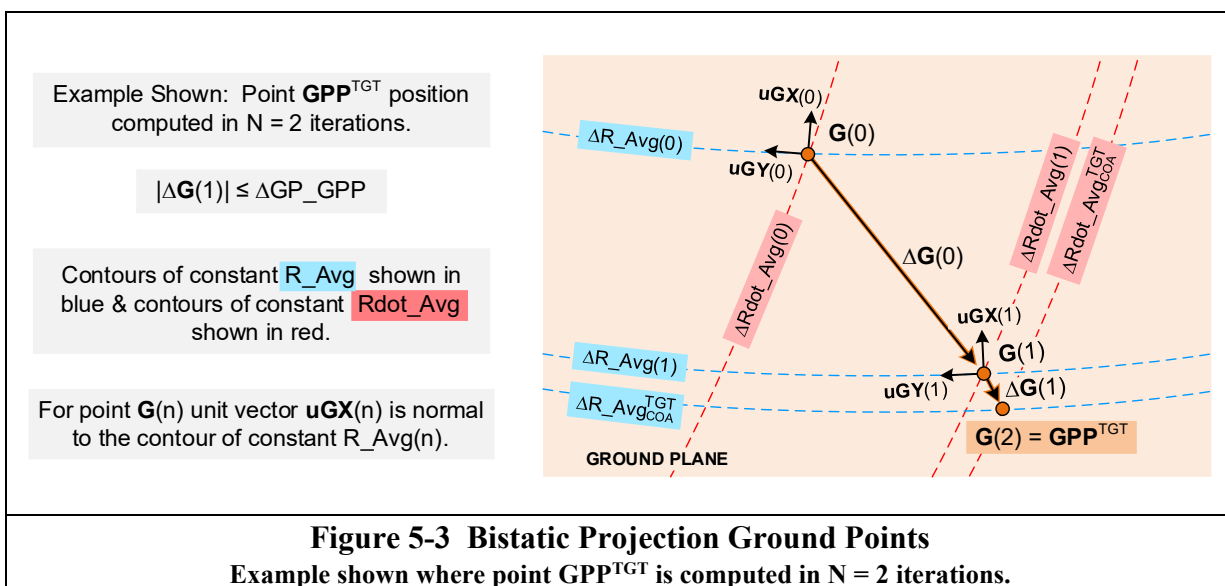
5.3 Bistatic R/Rdot Contour Ground Plane Intersection

For a bistatic image, the intersection of the R/Rdot contour and a ground plane is computed using an iterative approach. An iterative method for computing the point of intersection is specified as follows.

For a bistatic image, the COA projection contour is relative to the COA APC positions and velocities. An example COA geometry is shown in Figure 5-2. For target TGT with COA average range $R_Avg_{COA}^{TGT}$ and average range rate $Rdot_Avg_{COA}^{TGT}$, the R/Rdot contour intersects the selected ground plane at point GPP^{TGT} . For the example shown, both sensors are left-looking.



The method for solving for ground plane point GPP^{TGT} is as follows. An iterative approach computes a sequence of points $\{G(n), n = 0, 1, 2, \dots, N\}$ that lie in the selected ground plane. Refer to Figure 5-3. For point $G(n)$ the position in ECEF coordinates is denoted $G(n)$. An initial ground point $G(0)$ is computed that is nominally located in or near the imaged scene. For each point $G(n)$, the average range and average range rate are computed and compared to the COA range $R_Avg_{COA}^{TGT}$ and range rate $Rdot_Avg_{COA}^{TGT}$. Also computed are the gradients of average range and average range rate for displacements in the ground plane off of point $G(n)$. The gradients are then used to compute an adjustment vector $\Delta G(n)$ that is added to $G(n)$ to yield point $G(n+1)$. When the magnitude of the adjustment vector $\Delta G(n)$ is acceptably small, point $G(n+1)$ is accepted as the precise point of intersection. A method is provided for computing the initial point $G(0)$ and a suggested value is provided for the threshold ΔGP_GPP used for exiting the iteration.



Input COA Projection Parameters – Computed for the image grid location for target TGT

$\mathbf{Xmt}_{\text{COA}}^{\text{TGT}}$	Transmit APC COA velocity for target TGT in ECEF coordinates.
$\mathbf{VXmt}_{\text{COA}}^{\text{TGT}}$	Transmit APC COA velocity for target TGT in ECEF coordinates.
$\mathbf{Rcv}_{\text{COA}}^{\text{TGT}}$	Receive APC COA position for target TGT in ECEF coordinates.
$\mathbf{VRcv}_{\text{COA}}^{\text{TGT}}$	Receive APC COA velocity for target TGT in ECEF coordinates.
$R_Avg_{\text{COA}}^{\text{TGT}}$	Average range for target TGT COA geometry.
$Rdot_Avg_{\text{COA}}^{\text{TGT}}$	Average range rate for target TGT COA geometry.

Image Parameters:

SCP	SCP position in ECEF coordinates.
SCP.Lat	SCP geodetic latitude.
SCP.Lon	SCP geodetic longitude.
LOOK	Integer parameter indicating the Side of Track for the collection geometry.

Ground Plane Parameters:

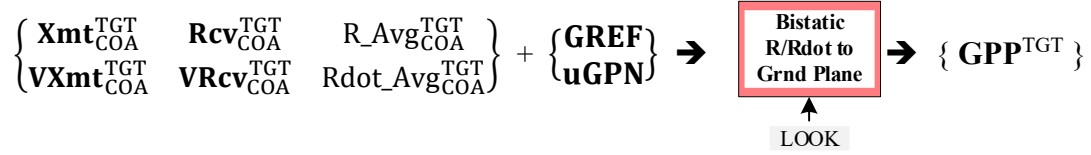
GREF	Ground plane reference point in ECEF coordinates.
uGPN	Ground plane unit normal vector in ECEF coordinates.

Computed Parameters:

$\mathbf{GPP}^{\text{TGT}}$	Ground Plane Point ($\mathbf{GPP}^{\text{TGT}}$) position in the ground plane and along the R/Rdot contour in ECEF coordinates.
---	---

Bistatic R/Rdot Contour to Ground Plane Point GPP^{TGT}

The precise projection to the ground plane along the bistatic R/Rdot contour may be implemented as function as shown below. The inputs are the COA APC positions and velocities, the ground plane reference point, and the ground plane unit normal. The side of track parameter LOOK is also an input.



(1) The position of an initial ground point, $G(0)$, that (1) lies in the selected ground plane, and (2) is located in or near the imaged scene is required. The precise position of the initial point is not critical. Point $G(0)$ lies in the selected ground plane and at the same geodetic latitude and longitude as the SCP. Compute the position $G(0)$ in ECEF coordinates as shown.

$$\mathbf{uUP}^{SCP} = \begin{bmatrix} \cos(SCP.Lat) \cdot \cos(SCP.Lon) \\ \cos(SCP.Lat) \cdot \sin(SCP.Lon) \\ \sin(SCP.Lat) \end{bmatrix}$$

$$Dist_GP = \frac{1}{\mathbf{uUP}^{SCP} \bullet \mathbf{uGPN}} \bullet (\mathbf{GREF} - \mathbf{SCP}) \bullet \mathbf{uGPN}$$

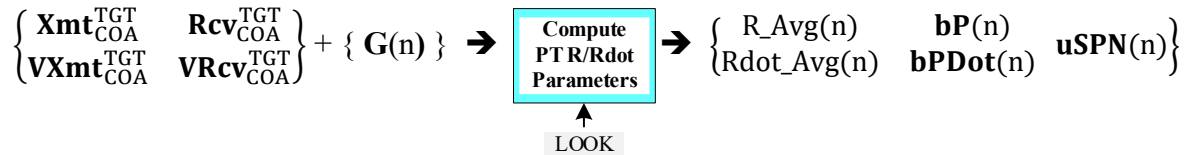
$$\mathbf{G}(0) = \mathbf{SCP} + Dist_GP \bullet \mathbf{uUP}^{SCP}$$

Use the iterative method specified in steps (2) thru (6) to compute a sequence of ground points $\{G(n), n = 0, 1, 2, \dots, N\}$ where point $G(n+1)$ is computed from point $G(n)$. Iterate until the displacement between $G(N-1)$ and $G(N)$ is less than maximum distance ΔGP_GPP .

Suggested threshold value: $\Delta GP_GPP = 0.010$ meters

Initialize $n = 0$

(2) For scene point $G(n)$ compute the range and range rate parameters relative to the COA APC positions and velocities. Use the method specified in Section 5.3.1.



(3) For scene point $G(n)$ compute Ground Plane X,Y parameters using the unit GPN and the pointing vector and its derivative w.r.t. time, $\mathbf{bP}(n)$ and $\mathbf{bPdot}(n)$. Use the method specified in Section 5.3.2.



- (4) Compute the difference in range and range rate for the target COA R/Rdot contour and the R/Rdot contour for ground point G(n).

$$\Delta R_Avg(n) = R_Avg_{COA}^{TGT} - R_Avg(n)$$

$$\Delta Rdot_Avg(n) = Rdot_Avg_{COA}^{TGT} - Rdot_Avg(n)$$

- (5) Compute an estimate of the displacements $\Delta GX(n)$ and $\Delta GY(n)$ from point G(n) to the precise intersection of the target R/Rdot contour. Use the matrix $\mathbf{M}_{RRdot}^{GPXY}(n)$ to convert the difference in average R/Rdot to displacements $\Delta GX(n)$ and $\Delta GY(n)$. Also compute the distance $\Delta GP(n)$.

$$\begin{bmatrix} \Delta GX(n) \\ \Delta GY(n) \end{bmatrix} = \mathbf{M}_{RRdot}^{GPXY}(n) \cdot \begin{bmatrix} \Delta R_Avg(n) \\ \Delta Rdot_Avg(n) \end{bmatrix} \quad \Delta GP(n) = [\Delta GX(n)^2 + \Delta GY(n)^2]^{1/2}$$

- (6) Compute point G(n+1) using displacements $\Delta GX(n)$ and $\Delta GY(n)$. Compare the distance $\Delta GP(n)$ to the threshold ΔGP_GPP . For $\Delta GP(n)$ less than or equal ΔGP_GPP , accept G(n+1) as the precise point of intersection.

$$\mathbf{G}(n+1) = \mathbf{G}(n) + \Delta GX(n) \cdot \mathbf{u}_{GX}(n) + \Delta GY(n) \cdot \mathbf{u}_{GY}(n)$$

If $\Delta GP(n) \leq \Delta GP_GPP$:

$\mathbf{GPP}^{TGT} = \mathbf{G}(n+1)$ Accept point G(n+1) as the precise R/Rdot intersection.

Terminate the iteration.

Else:

$n \leftarrow n + 1$ Increment n

Next iteration

Note 1: Intermediate parameters that are computed for each iteration are indexed by n but are only used to compute G(n+1) from G(n). The index n is used in the description to make clear which parameters are computed for each iteration.

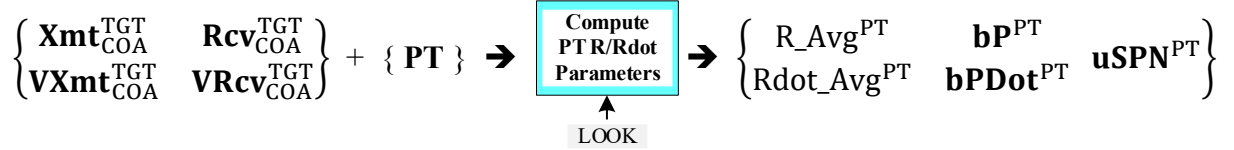
Note 2: The bistatic projection function defined may also be used for a monostatic image. The inputs to the function are shown below. The COA ARP position and velocity are input for each of the APC positions and velocities. The R/Rdot relative to the ARP are input for the average R/Rdot.



5.3.1 Scene Point Range & Range Rate Parameters

The function inputs COA positions and velocities for an image grid location ($xrow^{TGT}$, $ycol^{TGT}$) and a scene point position PT . The computed parameters are the average range and average range rate. Also computed are the bistatic pointing vector, its derivative w.r.t. time, and the bistatic Slant Plane unit normal. The function also inputs the collection side of track parameter LOOK. Refer to Figure 5-4.

For a Bistatic Image:



(1) Compute the range and range rate for the transmit APC relative to the scene point, R_Xmt^{PT} and $Rdot_Xmt^{PT}$. Also compute the unit vector and its derivative w.r.t. time from the scene point to the transmit APC, $\mathbf{uXmt}^{PT}$ and $\mathbf{uXmtDot}^{PT}$.

$$R_Xmt^{PT} = | \mathbf{Xmt}_{COA}^{TGT} - \mathbf{PT} | \quad \mathbf{uXmt}^{PT} = \frac{1}{R_Xmt^{PT}} \cdot (\mathbf{Xmt}_{COA}^{TGT} - \mathbf{PT})$$

$$Rdot_Xmt^{PT} = \mathbf{VXmt}_{COA}^{TGT} \bullet \mathbf{uXmt}^{PT}$$

$$\mathbf{uXmtDot}^{PT} = \frac{1}{R_Xmt^{PT}} \cdot \{ \mathbf{VXmt}_{COA}^{TGT} - Rdot_Xmt^{PT} \cdot \mathbf{uXmt}^{PT} \}$$

(2) Compute the range and range rate for the receive APC relative to the scene point, R_Rcv^{PT} and $Rdot_Rcv^{PT}$. Also compute the unit vector and its derivative w.r.t. time from the scene point to the receive APC, $\mathbf{uRcv}^{PT}$ and $\mathbf{uRcvDot}^{PT}$.

$$R_Rcv^{PT} = | \mathbf{Rcv}_{COA}^{TGT} - \mathbf{PT} | \quad \mathbf{uRcv}^{PT} = \frac{1}{R_Rcv^{PT}} \cdot (\mathbf{Rcv}_{COA}^{TGT} - \mathbf{PT})$$

$$Rdot_Rcv^{PT} = \mathbf{VRcv}_{COA}^{TGT} \bullet \mathbf{uRcv}^{PT}$$

$$\mathbf{uRcvDot}^{PT} = \frac{1}{R_Rcv^{PT}} \cdot \{ \mathbf{VRcv}_{COA}^{TGT} - Rdot_Rcv^{PT} \cdot \mathbf{uRcv}^{PT} \}$$

(3) Compute the average range and average range rate for the scene point, R_Avg^{PT} and $Rdot_Avg^{PT}$.

$$R_Avg^{PT} = \frac{1}{2} \cdot (R_Xmt^{PT} + R_Rcv^{PT})$$

$$Rdot_Avg^{PT} = \frac{1}{2} \cdot (Rdot_Xmt^{PT} + Rdot_Rcv^{PT})$$

(4) Compute the bistatic pointing vector and its derivative w.r.t. time for the scene point, $\mathbf{bP}^{PT}$ and $\mathbf{bPDot}^{PT}$.

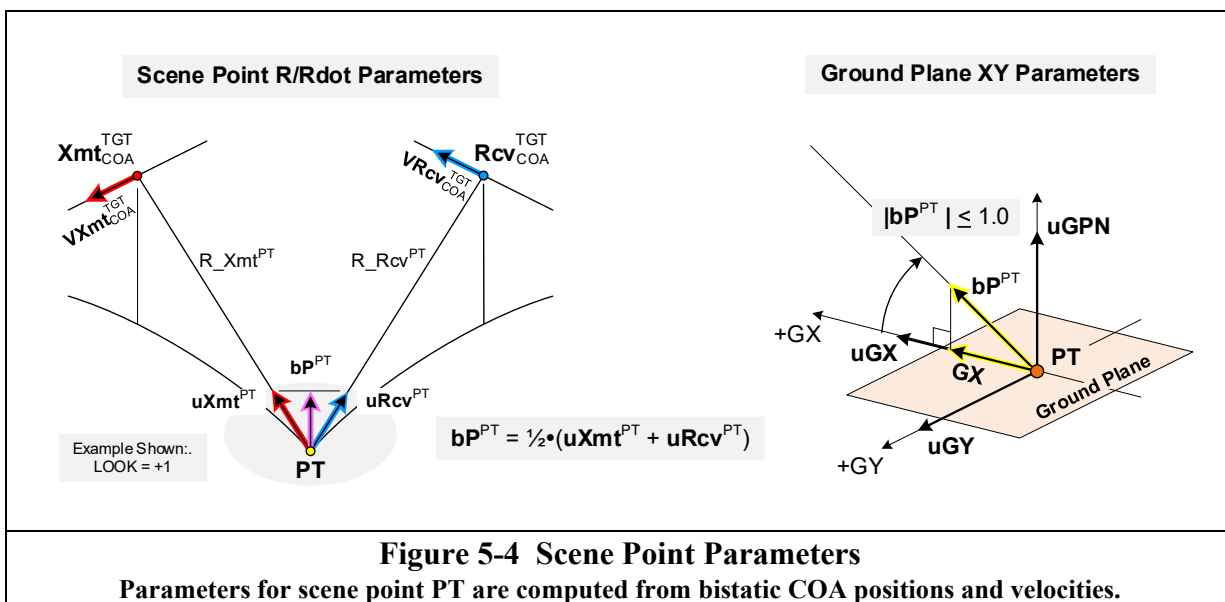
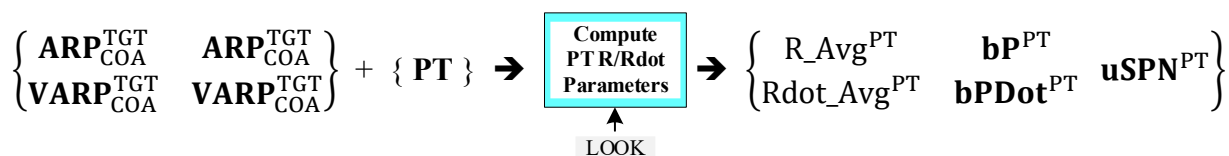
$$\mathbf{bP}^{PT} = \frac{1}{2} \cdot (\mathbf{uXmt}^{PT} + \mathbf{uRcv}^{PT})$$

$$\mathbf{bPDot}^{PT} = \frac{1}{2} \cdot (\mathbf{uXmtDot}^{PT} + \mathbf{uRcvDot}^{PT})$$

(5) Compute the bistatic slant plane unit normal vector, $\mathbf{uSPN}^{PT}$. The side of track parameter LOOK ensures the resulting vector points in the “upward” direction and away from the center of the earth.

$$\mathbf{SPN}^{PT} = \text{LOOK} \cdot \{ \mathbf{bP}^{PT} \times \mathbf{bPDot}^{PT} \} \quad \mathbf{uSPN}^{PT} = \frac{1}{|\mathbf{SPN}^{PT}|} \cdot \mathbf{SPN}^{PT}$$

For a Monostatic Image: Input the COA ARP position and velocity for both COA APC positions and velocities. The resulting range and range rate will be the range and range rate of the ARP relative to the point PT.



5.3.2 Scene Point Ground Plane XY Parameters

The function inputs a scene point position $\mathbf{PT}$ and the bistatic pointing vector and its derivative w.r.t. time. A ground plane (GP) containing the scene point is specified by a GP unit normal. The computed parameters are the basis vectors for a Ground Plane coordinate system. Also computed are the sensitivity matrices that relate changes in GP coordinates (GX, GY) and average range and average range rate. Refer to Figure 5-4.

$$\{ \mathbf{PT} \quad \mathbf{uGPN} \} + \left\{ \begin{array}{c} \mathbf{bP}^{\mathbf{PT}} \\ \mathbf{bPDot}^{\mathbf{PT}} \end{array} \right\} \rightarrow \boxed{\text{Compute GP XY Parameters}} \rightarrow \left\{ \begin{array}{cc} \mathbf{uGX} & \mathbf{M}_{\text{GPXY}}^{\text{RRdot}} \\ \mathbf{uGY} & \mathbf{M}_{\text{GPXY}}^{\text{GPXY}} \end{array} \right\}$$

(1) Compute the unit vector $\mathbf{uGX}$ by projecting the pointing vector $\mathbf{bP}^{\mathbf{PT}}$ into the ground plane along the ground plane normal.

$$\mathbf{GX} = \mathbf{bP}^{\mathbf{PT}} - \{ \mathbf{bP}^{\mathbf{PT}} \bullet \mathbf{uGPN} \} \bullet \mathbf{uGPN} \quad \mathbf{uGX} = \frac{1}{|\mathbf{GX}|} \bullet \mathbf{GX}$$

(2) Compute sign parameter SGN such that SGN times $\mathbf{uGPN}$ points “upward” and away from the center of the earth at point PT.

$$\text{SGN} = \begin{cases} +1, & \text{if } \mathbf{uGPN} \bullet \mathbf{PT} > 0 \\ -1, & \text{otherwise} \end{cases}$$

(3) Compute unit vector $\mathbf{uGY}$ that lies in the ground plane is normal to $\mathbf{uGX}$.

$$\mathbf{GY} = \text{SGN} \bullet \{ \mathbf{uGPN} \times \mathbf{uGX} \} \quad \mathbf{uGY} = \frac{1}{|\mathbf{GY}|} \bullet \mathbf{GY}$$

(4) Compute the elements of the 2 x 2 matrix $\mathbf{M}_{\text{GPXY}}^{\text{RRdot}}$ that is used to compute changes in average R/Rdot due to changes in GX and GY off of the scene point position.

$$\mathbf{M}_{\text{GPXY}}^{\text{RRdot}} = \begin{bmatrix} \frac{\Delta R_{\text{Avg}}}{\Delta GX} & \frac{\Delta R_{\text{Avg}}}{\Delta GY} \\ \frac{\Delta R_{\text{dot_Avg}}}{\Delta GX} & \frac{\Delta R_{\text{dot_Avg}}}{\Delta GY} \end{bmatrix} = (-1) \bullet \begin{bmatrix} \mathbf{bP}^{\mathbf{PT}} \bullet \mathbf{uGX} & 0 \\ \mathbf{bPDot}^{\mathbf{PT}} \bullet \mathbf{uGX} & \mathbf{bPDot}^{\mathbf{PT}} \bullet \mathbf{uGY} \end{bmatrix}$$

(5) Compute the elements of the 2 x 2 matrix $\mathbf{M}_{\text{RRdot}}^{\text{GPXY}}$ that is used to compute changes in GX and GY from change in average R/Rdot from the R/Rdot of the scene point.

$$\mathbf{M}_{\text{RRdot}}^{\text{GPXY}} = \begin{bmatrix} \frac{\Delta GX}{\Delta R_{\text{Avg}}} & \frac{\Delta GX}{\Delta R_{\text{dot_Avg}}} \\ \frac{\Delta GY}{\Delta R_{\text{Avg}}} & \frac{\Delta GY}{\Delta R_{\text{dot_Avg}}} \end{bmatrix} = \{ \mathbf{M}_{\text{GPXY}}^{\text{RRdot}} \}^{-1}$$

Note 1: For $|\mathbf{GX}| = 0$, the collection geometry does not yield resolution in two dimensions for the selected ground plane.

Note 2: For $\mathbf{bPDot}^{\mathbf{PT}} \bullet \mathbf{uGY} = 0$, the collection geometry does not yield resolution in two dimensions for the selected ground plane. The matrix $\mathbf{M}_{\text{GPXY}}^{\text{RRdot}}$ will be a singular matrix.

Note 3: For a monostatic image and $\mathbf{bP}^{\text{PT}}$ and $\mathbf{bPDot}^{\text{PT}}$ computed as specified in Section 5.3.1, the matrices $\mathbf{M}_{\text{GPXY}}^{\text{RRdot}}$ and $\mathbf{M}_{\text{RRdot}}^{\text{GPXY}}$ specify changes in range and range rate relative to the COA ARP position and velocity for changes in GX and GY off of the scene point position.

$$\mathbf{M}_{\text{GPXY}}^{\text{RRdot}} = \begin{bmatrix} \frac{\Delta R_{\text{ARP}}}{\Delta GX} & \frac{\Delta R_{\text{ARP}}}{\Delta GY} \\ \frac{\Delta R_{\text{dot_ARP}}}{\Delta GX} & \frac{\Delta R_{\text{dot_ARP}}}{\Delta GY} \end{bmatrix} \quad \mathbf{M}_{\text{RRdot}}^{\text{GPXY}} = \begin{bmatrix} \frac{\Delta GX}{\Delta R_{\text{ARP}}} & \frac{\Delta GX}{\Delta R_{\text{dot_ARP}}} \\ \frac{\Delta GY}{\Delta R_{\text{ARP}}} & \frac{\Delta GY}{\Delta R_{\text{dot_ARP}}} \end{bmatrix}$$

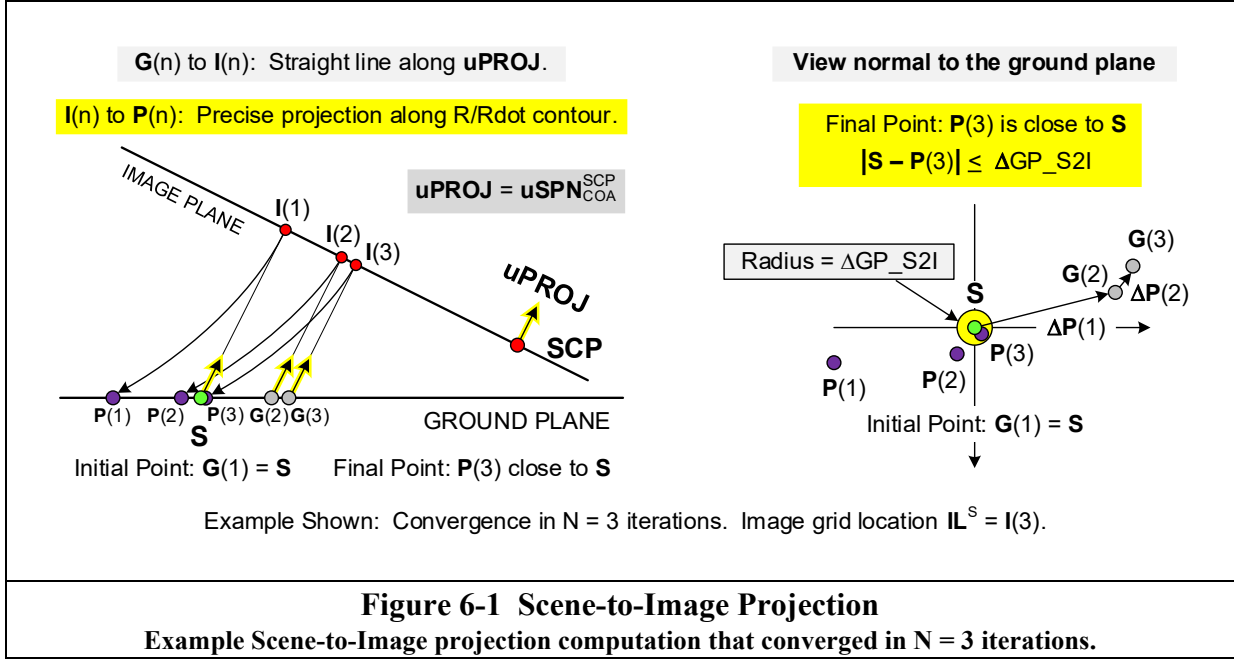
6 Scene-to-Image Grid Projection

The precise Scene Point to Image Grid projection maps a geolocated point in the three-dimensional scene to a single image grid location. The computed image grid location is the center of the two-dimensional image signal response of a fixed point scatterer located at the scene point. The image grid location is computed using an iterative approach. The method may be used with both monostatic images and bistatic images. The scene-to-image projection makes use of many of the computations specified in Sections 2 through 5. Only scene points on the correct Side of Track and at ranges and angles consistent with the imaging collection will yield meaningful results.

For projecting a single scene point to the image grid, the iterative approach is a simple computational task. The approach yields sub-pixel accuracy in a small number of iterations (typically in 2 iterations for ground points at the same height as the SCP). The projection of a single scene point to the image grid is described in Section 6.1. The precise resampling of an entire image to a geolocated surface requires the image grid location to be computed for many scene points. For precise resampling to a complex surface such as a terrain height model of a mountainous scene, all scene points may require the iterative approach. For precise resampling to a smooth surface such as a ground plane, the iterative approach may be applied to only a fraction of the scene points. An approach for resampling to smooth surfaces is described in Section 6.2.

6.1 Scene-to-Image: Single Scene Point

Let point S be the point in the imaged scene for which the image grid location is to be computed. Define a ground plane that contains point S . The iterative approach computes a sequence of ground plane points $G(1), G(2), \dots, G(N)$. The ground plane points $\{G(n)\}$ are projected along straight lines to the Image Plane to produce a sequence of image grid points $I(1), I(2), \dots, I(N)$. Image point $I(N)$ is the final image grid location and is the precise projection of point S . For each image plane point $I(n)$, the precise R/Rdot projection to the ground plane is computed. The precise ground plane projection of image point $I(n)$ is point $P(n)$. Final point $I(N)$ is established when the displacement between point S and point $P(N)$ is less than a user specified tolerance. When the displacement between point S and point $P(n)$ is greater than the tolerance, the displacement is added to point $G(n)$ to yield point $G(n+1)$ for the next iteration. An example projection sequence is shown in Figure 6-1. In the example shown, convergence is achieved in $N = 3$ iterations. Image grid location $I(3) = (xrow(3), ycol(3))$ is accepted as the precise projection of scene point S .



Scene Point S To Image Grid Location $IL^S = (xrow^S, ycol^S)$

(1) For scene point S , compute the spherical earth ground plane unit normal. The choice of ground plane containing point S is not critical. See Section 5.1. Also, choose the ground plane displacement threshold, ΔGP_S2I , for final ground plane point $P(N)$. A recommended value for threshold is provided.

$$u_{GPN} = \frac{1}{|S|} \cdot S \quad \Delta GP_S2I = 0.001 \text{ meters}$$

(2) Ground plane points $\{G(n)\}$ are projected along straight lines to the image plane to establish points $\{I(n)\}$. The GP to IP projection direction is along the SCP COA slant plane normal. Also, compute the image plane unit normal, u_{IPN} . Compute projection scale factor SF as shown.

$$u_{PROJ} = u_{SPN}_{SCP_COA} \quad IPN = u_{Row} \times u_{Col} \quad u_{IPN} = \frac{1}{|IPN|} \cdot IPN$$

$$SF = u_{PROJ} \cdot u_{IPN}$$

(3) Set initial ground plane position $G(1)$ to the scene point position S .

$$G(1) = S \quad \text{Initialize the index: } n = 1.$$

Iterate using steps (4) through (8).

For ground point $G(n)$, compute image plane point $I(n)$ and ground plane point $P(n)$.

- (4) Project ground plane point $G(n)$ to image plane point $I(n)$. The projection distance is $DIST(n)$.

$$DIST(n) = \frac{1}{SF} \bullet (SCP - G(n)) \bullet uIPN \quad I(n) = G(n) + DIST(n) \bullet uPROJ$$

- (5) For image plane point $I(n)$, compute the associated image grid coordinates ($xrow(n)$, $ycol(n)$). See Section 2.4.2.

$$\{ I(n) \} + \{ SCP \quad uRow \quad uCol \} \rightarrow \boxed{\text{Convert IP_PT to (xrow, ycol)}} \rightarrow \{ xrow(n) \quad ycol(n) \}$$

- (6) Compute the COA projection set for image grid location ($xrow(n)$, $ycol(n)$). The method used is dependent upon the Collect Type, Grid_Type, and the IFA used to form the image. See Section 4.

For a monostatic image:

$$\{ xrow(n) \quad ycol(n) \} \rightarrow \boxed{\text{Compute Monostatic Proj Set}} \rightarrow \left\{ t_{COA}(n) \quad \begin{matrix} ARP_{COA}(n) \\ VARP_{COA}(n) \end{matrix} \quad \begin{matrix} R_{COA}(n) \\ Rdot_{COA}(n) \end{matrix} \right\}$$

For a bistatic image:

$$\{ xrow(n) \quad ycol(n) \} \rightarrow \boxed{\text{Compute Bistatic Proj Set}} \rightarrow \left\{ t_{COA}(n) \quad \begin{matrix} tx_{COA}(n) \\ Xmt_{COA}(n) \\ VXmt_{COA}(n) \end{matrix} \quad \begin{matrix} tr_{COA}(n) \\ Rcv_{COA}(n) \\ VRcv_{COA}(n) \end{matrix} \quad \begin{matrix} R_{Avg_{COA}}(n) \\ Rdot_{Avg_{COA}}(n) \end{matrix} \right\}$$

- (7) Compute the precise projection to the ground plane for grid location ($xrow(n)$, $ycol(n)$). The projection uses a subset of the parameters contained in the COA projection set. The result is ground point $P(n)$. The method for projection is dependent upon the Collect Type. See Section 5.

For a monostatic image:

$$\left\{ \begin{matrix} ARP_{COA}(n) & R_{COA}(n) \\ VARP_{COA}(n) & Rdot_{COA}(n) \end{matrix} \right\} + \{ GREF = S \quad uGPN \} \rightarrow \boxed{\text{Monostatic R/Rdot to Grnd Plane}} \rightarrow \{ P(n) \}$$

↑
LOOK

For a bistatic image:

$$\left\{ \begin{matrix} Xmt_{COA}(n) & Rcv_{COA}(n) & R_{Avg_{COA}}(n) \\ VXmt_{COA}(n) & VRcv_{COA}(n) & Rdot_{Avg_{COA}}(n) \end{matrix} \right\} + \{ GREF = S \quad uGPN \} \rightarrow \boxed{\text{Bistatic R/Rdot to Grnd Plane}} \rightarrow \{ P(n) \}$$

↑
LOOK

(8) Compute the displacement between ground plane point $P(n)$ and the scene point S . If the displacement is greater than the threshold (ΔGP_S2I), compute point $G(n+1)$ and repeat the projections in steps (4) through (7) above. If the displacement is less than the threshold, accept image grid location ($xrow(n)$, $ycol(n)$) as the precise image grid location for scene point S .

$$\Delta \mathbf{P}(n) = \mathbf{S} - \mathbf{P}(n) \quad \Delta GP(n) = | \Delta \mathbf{P}(n) |$$

If: $\Delta GP(n) > \Delta GP_S2I$:

$$\mathbf{G}(n+1) = \mathbf{G}(n) + \Delta \mathbf{P}(n)$$

$n \leftarrow n+1$ Increment n , next iteration.

Else: $\Delta GP(n) \leq \Delta GP_S2I$:

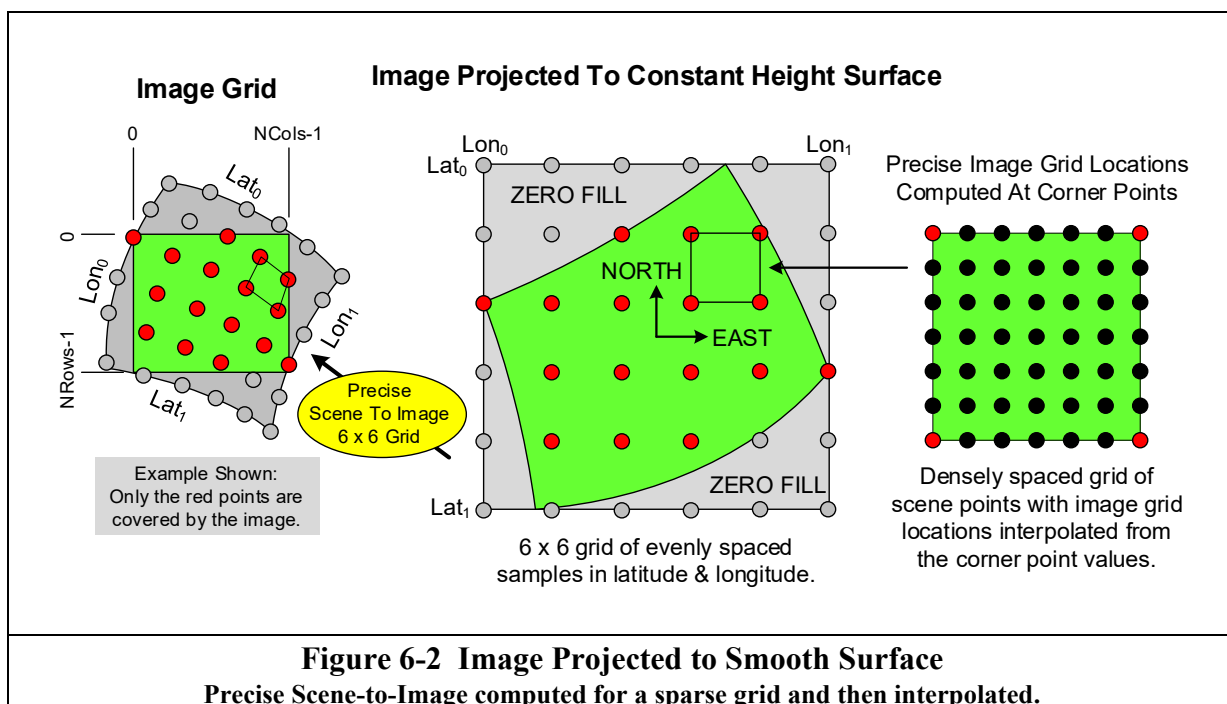
$$(xrow^S, ycol^S) = (xrow(n), ycol(n))$$

Note: The precise projection may yield an image grid location that is off the image pixel grid. Such a solution is valid and indicates that the scene point is not included in the image product.

6.2 Scene-to-Image: Smooth Surface

The precise resampling to a uniformly spaced grid on any geolocated surface may be accomplished by applying the iterative approach described in Section 6.1 to all scene points. However, for smooth surfaces the precise iterative approach may be applied to a subset of the scene points. The approach makes use of the fact that for smooth surfaces, the two-dimensional image grid location varies smoothly across the two-dimensional surface grid.

An example image grid projected to a smooth geolocated surface is shown in Figure 6-2. The smooth surface is at constant height above to the WGS 84 reference ellipsoid. The surface grid points are evenly spaced in latitude and longitude. The extent of the region covered by the image is bounded by corner points (Lat_0, Lon_0) and (Lat_1, Lon_1). These corner points are established by the precise image-to-scene projection of the image grid corner points to the constant height surface. Section 9 describes the projection. In the example shown, a sparse grid of 6 x 6 surface points is selected. The precise image grid location for each of the 36 points is computed using the scene-to-image projection above. The resulting image coordinates are saved for all grid points (including grid points that are not covered by the image). A dense set of image grid coordinates may be rapidly computed by bilinear interpolation of the corner point values. The accuracy of the interpolated image grid coordinates can be determined by the precise scene-to-image projection of the 5 x 5 set of scene points centered in cells formed by the 6 x 6 grid.

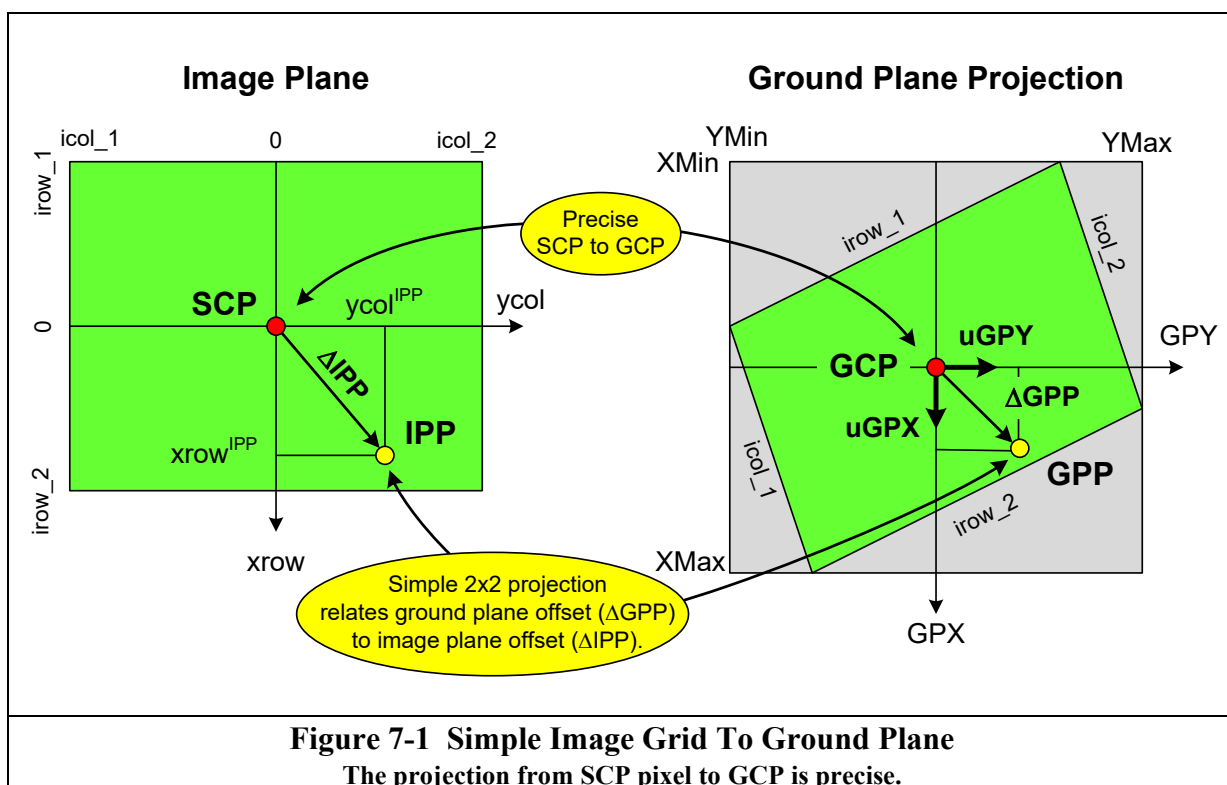


7 Simple Ground Plane Projection

Precise ground plane resampling for large images can be a computational burden. For some applications, a fast but slightly less accurate algorithm is acceptable. A simple method for fast ground plane resampling is described. The fast projection algorithm uses the precise projection of the SCP pixel to the ground plane. For image grid locations displaced from the SCP pixel, a simple computation is used to estimate a ground plane offset from the projected SCP location. The simple projection is highly accurate for pixels near the SCP pixel. Accuracy degrades slowly with displacement from the SCP pixel. The fast algorithm may be applied to any image product independent of the image formation algorithm used to form the image. The simple projection can be used for projection to an arbitrary ground plane.

The simple projection is shown in Figure 7-1. For an arbitrary ground plane, the precise projection of the SCP pixel to the ground plane is computed. The resulting ground plane point is the Ground Center Point (GCP). A pair of orthogonal ground plane axes (GPX, GPY) are defined. For ground plane point GPP at location (GPX^{GPP}, GPY^{GPP}) , image coordinates $(xrow^{GPP}, ycol^{GPP})$ are computed from a 2x2 projection matrix. Image indices $(irow^{GPP}, icol^{GPP})$ are computed from the image coordinates $(xrow^{GPP}, ycol^{GPP})$ by scaling by the sample spacings.

$$\begin{bmatrix} xrow^{GPP} \\ ycol^{GPP} \end{bmatrix} = \mathbf{M}_{GP}^{IC} \cdot \begin{bmatrix} GPX^{GPP} \\ GPY^{GPP} \end{bmatrix} = \begin{bmatrix} GPXtoRow & GPYtoRow \\ GPXtoCol & GPYtoCol \end{bmatrix} \cdot \begin{bmatrix} GPX^{GPP} \\ GPY^{GPP} \end{bmatrix}$$



7.1 Ground Plane Parameters

The ground plane is defined by a reference point in the plane (GREF) and the vector normal to the plane (GPN). Any arbitrary ground plane may be selected.

GREF = Ground plane reference point the ECF coordinates.

uGPN = Ground plane unit normal vector.

For many applications, a logical choice for the ground plane is the plane tangent to the surface of constant height above the ellipsoid at the SCP. See Section 5.1.

$$\mathbf{GREF} = \mathbf{SCP} \quad \mathbf{uGPN} = \mathbf{uGPN}_{\text{GEO}}^{\text{SCP}} = \begin{bmatrix} \cos(\text{SCP.Lat}) \cdot \cos(\text{SCP.Lon}) \\ \cos(\text{SCP.Lat}) \cdot \sin(\text{SCP.Lon}) \\ \sin(\text{SCP.Lat}) \end{bmatrix}$$

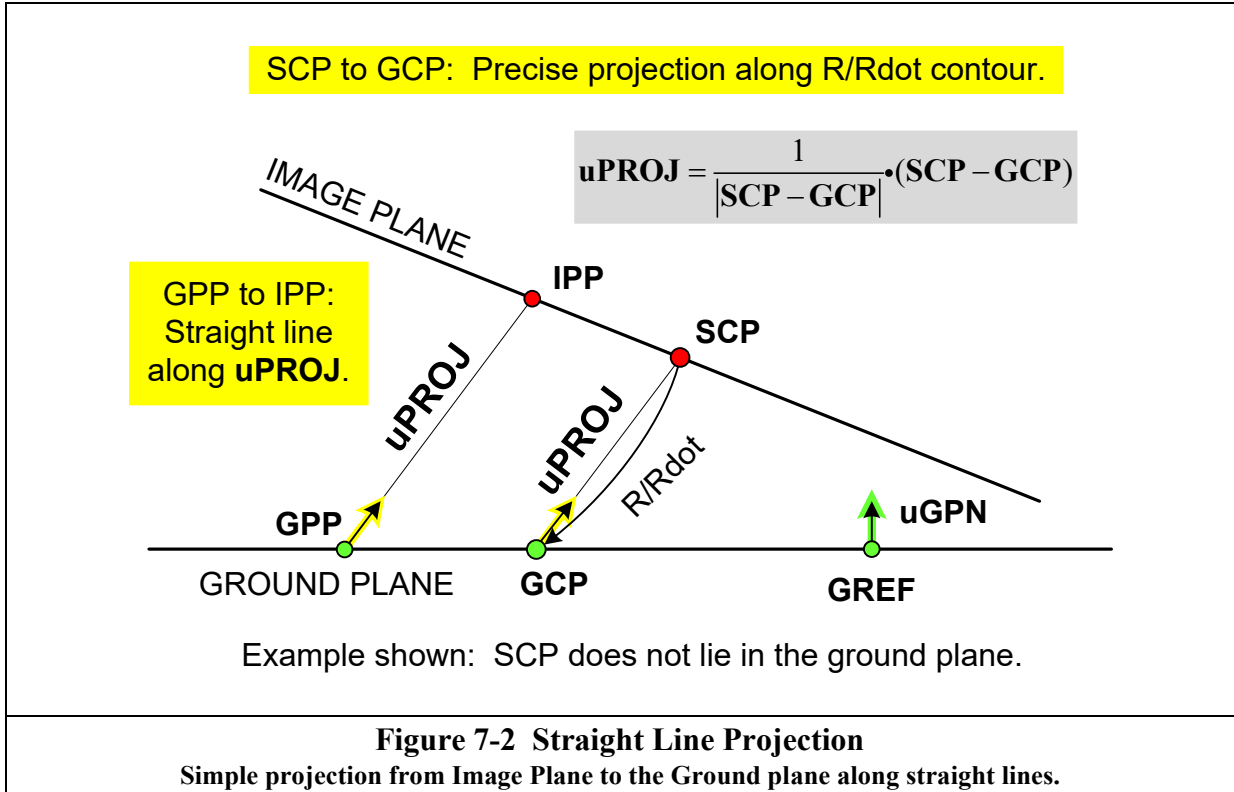
The precise projection of the SCP pixel to the ground plane is computed. The SCP pixel COA parameters and R/Rdot projection contour are described in Section 3. The projection along an R/Rdot contour to a ground plane is described in Section 5. The result of the projection is the Ground Center Point (GCP). If the selected ground plane contains the SCP, the GCP position is set equal to the SCP position. See Figure 7-2. Ground points are projected along straight lines from the ground plane to the image plane. The ground plane to image plane projection direction is determined as follows. For the SCP in the ground plane, the projection direction is along the SCP COA slant plane normal. See Section 3. For the SCP not in the ground plane, the projection direction is from the GCP to the SCP.

For the SCP in the ground plane: $(\mathbf{SCP} - \mathbf{GREF}) \cdot \mathbf{uGPN} = 0$

$$\mathbf{GCP} = \mathbf{SCP} \quad \mathbf{uPROJ} = \mathbf{uSPN}_{\text{COA}}^{\text{SCP}}$$

For the SCP not in ground plane: $(\mathbf{SCP} - \mathbf{GREF}) \cdot \mathbf{uGPN} \neq 0$

$$\begin{array}{|c|} \hline \text{Compute} \\ \text{SCP COA} \\ \text{Parameters} \\ \hline \end{array} \rightarrow \begin{array}{|c|} \hline \text{Project} \\ \text{Along R/Rdot} \\ \text{To GP} \\ \hline \end{array} \rightarrow \mathbf{GCP} \quad \mathbf{uPROJ} = \frac{1}{|\mathbf{SCP} - \mathbf{GCP}|} \cdot (\mathbf{SCP} - \mathbf{GCP})$$



The orthogonal ground plane axes GPX and GPY may have any orientation and are defined by unit vectors $\mathbf{uGPX}$ and $\mathbf{uGPY}$. The only restriction is that the vectors lie in the ground plane.

$\mathbf{uGPX}$ = Unit vector in the GPX direction in ECF coordinates.

$\mathbf{uGPY}$ = Unit vector in the GPY direction in ECF coordinates.

$$\mathbf{uGPX} \cdot \mathbf{uGPN} = 0 \quad \mathbf{uGPY} \cdot \mathbf{uGPN} = 0 \quad \mathbf{uGPX} \cdot \mathbf{uGPY} = 0$$

A logical choice of the GPX direction is in the shadow direction at the GCP. The shadow direction is from the ARP ground plane nadir ($\mathbf{AGPN}$) to the **GCP**. For the ground plane image, GPX is in the increasing row direction and GPY is in the increasing column direction. For a shadows down orientation, compute unit vectors $\mathbf{uGPX}$ and $\mathbf{uGPY}$ as follows.

$$\mathbf{AGPN} = \mathbf{ARP}_{COA}^{SCP} - [(\mathbf{ARP}_{COA}^{SCP} - \mathbf{GCP}) \cdot \mathbf{uGPN}] \cdot \mathbf{uGPN}$$

$$\mathbf{uGPX} = \frac{1}{|\mathbf{GCP} - \mathbf{AGPN}|} \cdot (\mathbf{GCP} - \mathbf{AGPN})$$

$$\mathbf{uGPY} = \mathbf{uGPN} \times \mathbf{uGPX}$$

7.2 Ground Coordinates To Image Coordinates

The projection of a ground plane point to the image grid coordinates is computed as described below. For ground plane point GPP at location (GPX^{GPP}, GPY^{GPP}) , the image coordinates are computed by projecting from the ground plane to the image plane in **uPROJ** direction. The resulting image plane point is IPP. The image plane parameters are described in Section 2.4.

uRow = Unit vector in the increasing row direction in ECF coordinates.

uCol = Unit vector in the increasing column direction in ECF coordinates.

θ_{COL} = Angle between unit vectors **uRow** and **uCol**.

uIPN = Image plane unit normal in ECF coordinates.

The projection of ground plane point GPP at location (GPX^{GPP}, GPY^{GPP}) to image plane position IPP located at image coordinates $(xrow^{GPP}, ycol^{GPP})$ is based on the following relationships.

$$\text{GPP Position:} \quad \mathbf{GPP} = \mathbf{GCP} + \Delta\mathbf{GPP} = \mathbf{GCP} + GPX \cdot \mathbf{uGPX} + GPY \cdot \mathbf{uGPY}$$

$$\text{Projection Distance:} \quad \text{DIST} = \frac{(\mathbf{SCP} - \mathbf{GPP}) \cdot \mathbf{uIPN}}{\mathbf{uPROJ} \cdot \mathbf{uIPN}}$$

$$\text{IPP Position:} \quad \mathbf{IPP} = \mathbf{GPP} + \text{DIST} \cdot \mathbf{uPROJ}$$

The ground coordinate to image coordinate projection matrix, $\mathbf{M}_{GP}^{IC}$, is computed as the product of two matrices. Matrix $\mathbf{M}_{GP}^{IV}$ is the projection of GPP to IPP and then orthogonal projection onto the two image plane vectors. Matrix $\mathbf{M}_{IV}^{IC}$ accounts for possible non-orthogonal image plane vectors.

$$\mathbf{M}_{GP}^{IC} = \begin{bmatrix} GPXtoRow & GPYtoRow \\ GPXtoCol & GPYtoCol \end{bmatrix} = \mathbf{M}_{IV}^{IC} \cdot \mathbf{M}_{GP}^{IV}$$

$$\text{Compute the 2x2 matrix } \mathbf{M}_{GP}^{IV} = \begin{bmatrix} GPXtoRV & GPYtoRV \\ GPXtoCV & GPYtoCV \end{bmatrix}:$$

$$GPXtoRV = \mathbf{uGPX} \cdot \mathbf{uRow} - \frac{(\mathbf{uGPX} \cdot \mathbf{uIPN}) \cdot (\mathbf{uPROJ} \cdot \mathbf{uRow})}{\mathbf{uPROJ} \cdot \mathbf{uIPN}}$$

$$\text{GPYtoRV} = \mathbf{uGPY} \bullet \mathbf{uRow} - \frac{(\mathbf{uGPY} \bullet \mathbf{uIPN}) \cdot (\mathbf{uPROJ} \bullet \mathbf{uRow})}{\mathbf{uPROJ} \bullet \mathbf{uIPN}}$$

$$\text{GPXtoCV} = \mathbf{uGPX} \bullet \mathbf{uCol} - \frac{(\mathbf{uGPX} \bullet \mathbf{uIPN}) \cdot (\mathbf{uPROJ} \bullet \mathbf{uCol})}{\mathbf{uPROJ} \bullet \mathbf{uIPN}}$$

$$\text{GPYtoCV} = \mathbf{uGPY} \bullet \mathbf{uCol} - \frac{(\mathbf{uGPY} \bullet \mathbf{uIPN}) \cdot (\mathbf{uPROJ} \bullet \mathbf{uCol})}{\mathbf{uPROJ} \bullet \mathbf{uIPN}}$$

Compute the 2x2 matrix $\mathbf{M}_{IV}^{IC}$:

$$\mathbf{M}_{IV}^{IC} = \frac{1}{\sin^2(\theta_{COL})} \cdot \begin{bmatrix} 1 & -\cos(\theta_{COL}) \\ -\cos(\theta_{COL}) & 1 \end{bmatrix}$$

Compute the 2x2 matrix $\mathbf{M}_{GP}^{IC} = \mathbf{M}_{IV}^{IC} \cdot \mathbf{M}_{GP}^{IV}$:

$$\mathbf{M}_{GP}^{IC} = \mathbf{M}_{IV}^{IC} \cdot \mathbf{M}_{GP}^{IV} = \frac{1}{\sin^2(\theta_{COL})} \cdot \begin{bmatrix} 1 & -\cos(\theta_{COL}) \\ -\cos(\theta_{COL}) & 1 \end{bmatrix} \cdot \begin{bmatrix} \text{GPXtoRV} & \text{GPYtoRV} \\ \text{GPXtoCV} & \text{GPYtoCV} \end{bmatrix}$$

8 Adjustable Parameters

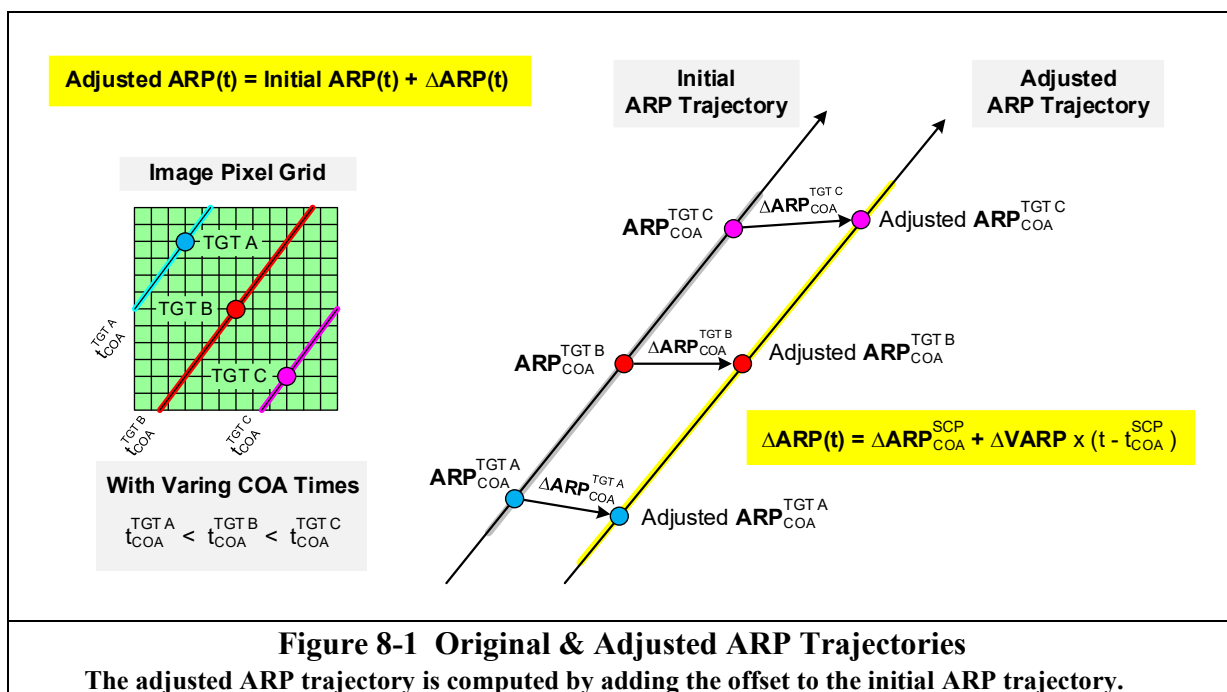
The use of Adjustable Parameter Offsets (APOs) is intended to improve the accuracy for all image projection computations for an image product that contains them. The methods for using APOs are described in the following section. The input set of APOs, if available, is dependent upon the Collect Type of the image. However, the way APOs are incorporated in a projection processing sequence is the same for all images and is independent of the Collect Type of the image.

All projection functions include the computation of one or more COA projection sets in the processing sequence of the function. For an image-to-scene projection, one COA projection set is computed for the selected image grid location. For a scene-to-image projection, a COA projection set may be computed for multiple image grid locations in an iterative computation of the precise image grid location. The input set of APOs, if available, is used to compute a set of offsets that are applied to each COA projection set. For each projection set, the computed offsets are functions of the COA time(s) in the projection set.

The COA projection set is computed based on a selected image grid location. The method for computing the projection set uses only metadata parameters that are required to be included in the product. Refer to Section 4. The projection set computed using the required metadata is referred to as the “initial” COA projection set. The input set of APOs is then used to compute a set of offsets that are added the values of the parameters in the initial COA projection set. The resulting projection set is referred to as the “adjusted” COA projection set.

The convention for computing the input set of APOs and for applying them to the initial COA projection set is as following. For a given parameter, the offset is the value of the error in the parameter in the initial COA projection set. The true value of the parameter is computed by adding the offset to the initial value of the parameter. This convention is demonstrated in the following example. Refer to Figure 8-1. Shown in the left-hand portion of the figure is a monostatic image pixel array that was formed with COA times that vary with image pixel location. Shown are the pixel grid locations for three targets, denoted TGT A, TGT B, and TGT C. The contours of constant COA time are also shown. For the given image, the initial ARP position vs. time used in the image formation processing is determined to be in error. An estimate of error versus time may be provided in the input set of APOs for a monostatic image. For an image grid location, the adjusted ARP position and velocity may be computed using the COA time of grid location. Shown in the right-hand portion of the figure is the initial ARP trajectory and the adjusted ARP trajectory. The adjusted ARP trajectory to taken to be the true trajectory for the ARP.

$$\begin{aligned}
 \underbrace{\text{Adjusted ARP}_{\text{COA}}}_{\text{Ideally Equal To The True Position At Time } t_{\text{COA}}} &= \underbrace{\text{ARP}_{\text{COA}}}_{\text{Computed from the Image Metadata}} + \underbrace{\Delta \text{ARP}_{\text{COA}}^{\text{SCP}} + \Delta \text{VARP} \cdot (t_{\text{COA}} - t_{\text{COA}}^{\text{SCP}})}_{\substack{\Delta \text{ARP}_{\text{COA}} = \text{ARP Offset} \\ \text{Added To Correct The Error}}} \\
 \underbrace{\text{Adjusted VARP}_{\text{COA}}}_{\text{Ideally Equal To The True Velocity At Time } t_{\text{COA}}} &= \underbrace{\text{VARP}_{\text{COA}}}_{\text{Computed from the Image Metadata}} + \underbrace{\Delta \text{VARP}}_{\substack{\text{Constant Offset} \\ \text{For All Pixel Locations}}}
 \end{aligned}$$



8.1 COA Projection Set Offsets

The parameters contained in the COA projection set are dependent upon the Collect Type of the image. The input set of APOs and the method for computing the offsets to be applied are also dependent upon the Collect Type. Refer to Table 1-3. The method for computing the adjusted COA projection set is specified for each Collect Type as follows.

8.1.1 Collect Type = Monostatic

For a monostatic image, the COA projection set is a set of 5 parameters. For a given image grid location, the initial COA projection set contains the following parameters.

Initial COA Projection Set: $\{ t_{COA}, \mathbf{ARP}_{COA}, \mathbf{VARP}_{COA}, R_{COA}, Rdot_{COA} \}$

The input set of APOs for a monostatic image is a set of 4 parameters. The parameters are as follows.

$\Delta \mathbf{ARP}_{COA}^{SCP}$	ARP position offset in ECEF coordinates at time $t = t_{COA}^{SCP}$.
$\Delta \mathbf{VARP}$	ARP velocity offset in ECEF coordinates.
Δt_{COA}^{SCP}	Transmit time offset. A positive value indicates the true transmit times were later than the values used in processing the PHD.
Δt_{COA}^{SCP}	Receive time offset. A positive value indicates the true receive times were later than the values used in processing the PHD.

The input APOs are used to compute the following offsets to be added to the initial COA projection set to form the adjusted COA projection set. Refer to Figure 8-2.

$$\Delta \mathbf{ARP}_{\text{COA}} = \Delta \mathbf{ARP}_{\text{COA}}^{\text{SCP}} + \Delta \mathbf{VARP} \cdot (t_{\text{COA}} - t_{\text{COA}}^{\text{SCP}})$$

$$\Delta R_{\text{ARP}} = (c/2) \cdot (\Delta t_{\text{r}}^{\text{SCP}} - \Delta t_{\text{x}}^{\text{SCP}})$$

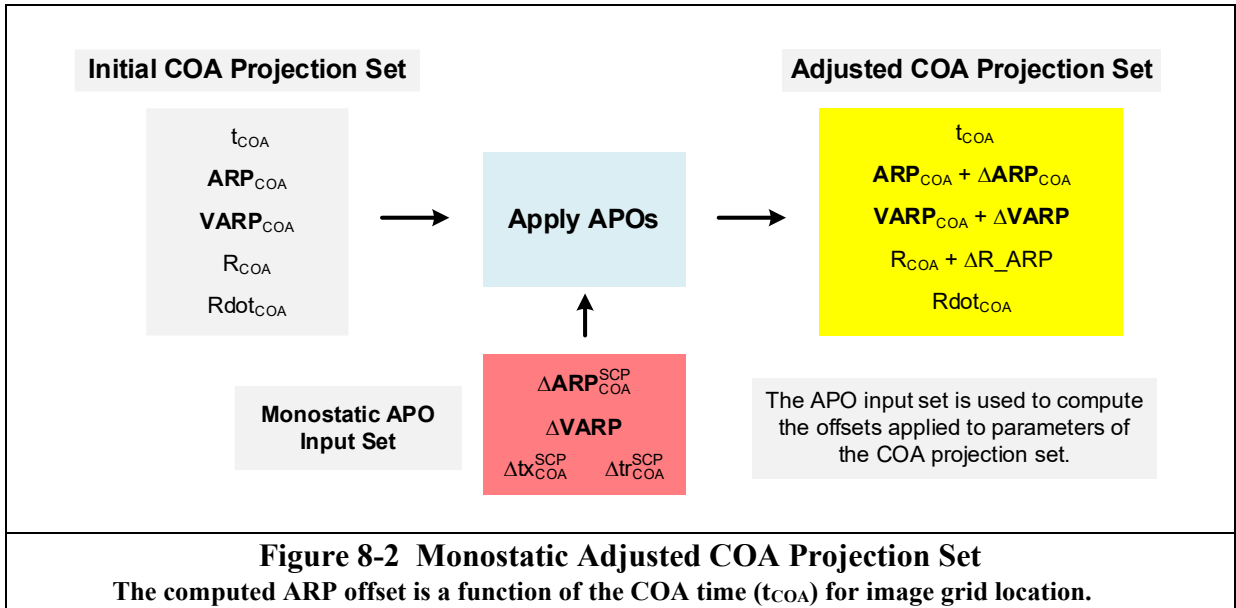
The adjusted COA projection set is formed by adding offsets to the following parameters.

$$\text{Adjusted } \mathbf{ARP}_{\text{COA}} = \mathbf{ARP}_{\text{COA}} + \Delta \mathbf{ARP}_{\text{COA}}$$

$$\text{Adjusted } \mathbf{VARP}_{\text{COA}} = \mathbf{VARP}_{\text{COA}} + \Delta \mathbf{VARP}$$

$$\text{Adjusted } R_{\text{COA}} = R_{\text{COA}} + \Delta R_{\text{ARP}}$$

Note: The values for t_{COA} and R_{dotCOA} are not modified when forming the adjusted projection set.



8.1.2 Collect Type = Bistatic

For a bistatic image, the COA projection set is a set of 9 parameters. For a given image grid location, the initial COA projection set contains the following parameters.

Bistatic COA Projection Set: $\{ t_{COA}, tx_{COA}, tr_{COA}, \mathbf{Xmt}_{COA}, \mathbf{VXmt}_{COA}, \mathbf{Rcv}_{COA}, \mathbf{VRcv}_{COA}, R_Avg_{COA}, Rdot_Avg_{COA} \}$

The input set of APOs for a bistatic image is a set of 8 parameters. The parameters are as follows.

$\Delta \mathbf{Xmt}_{COA}^{SCP}$	Transmit APC position offset in ECEF coordinates at time $t = t_{COA}^{SCP}$.
$\Delta \mathbf{VXmt}$	Transmit APC velocity offset in ECEF coordinates.
Δtx_{COA}^{SCP}	Transmit time offset at time $t = t_{COA}^{SCP}$. For a transmit time $tx = t_{COA}^{SCP}$ computed in processing the PHD, the true value was $tx = t_{COA}^{SCP} + \Delta tx_{COA}^{SCP}$.
$f_Clk_X_SF$	Transmit payload control clock frequency offset expressed as scale factor. A positive value indicates the true clock frequency was greater than the value used in processing the PHD.
$\Delta \mathbf{Rcv}_{COA}^{SCP}$	Receive APC position offset in ECEF coordinates at time $t = t_{COA}^{SCP}$.
$\Delta \mathbf{VRcv}$	Receive APC velocity offset in ECEF coordinates.
Δtr_{COA}^{SCP}	Receive time offset at time $t = t_{COA}^{SCP}$. For a receive time $tr = t_{COA}^{SCP}$ computed in processing the PHD, the true value was $tr = t_{COA}^{SCP} + \Delta tr_{COA}^{SCP}$.
$f_Clk_R_SF$	Receive payload control clock frequency offset expressed as scale factor. A positive value indicates the true clock frequency was greater than the value used in processing the PHD.

For the transmit sensor, the transmit time offset and the clock frequency scale factor are used to compute a transmit time offset, Δtx_{COA} , that is a function of tx_{COA} in the initial projection set. The clock period scale factor, $T_Clk_X_SF$, is computed from the clock frequency scale factor. The transmit time offset is then computed using the initial tx_{COA} and t_{COA}^{SCP} . A positive value of Δtx_{COA} indicates the true transmit time was later than the value in the initial projection set.

$$T_Clk_X_SF = -f_Clk_X_SF \cdot \left(\frac{1}{1+f_Clk_X_SF} \right)$$

$$\Delta tx_{COA} = \Delta tx_{COA}^{SCP} + T_Clk_X_SF \cdot (tx_{COA} - t_{COA}^{SCP})$$

For the receive sensor, the receive time offset and the clock frequency scale factor are used in a similar way.

$$T_Clk_R_SF = -f_Clk_R_SF \cdot \left(\frac{1}{1+f_Clk_R_SF} \right)$$

$$\Delta tr_{COA} = \Delta tr_{COA}^{SCP} + T_Clk_R_SF \cdot (tr_{COA} - t_{COA}^{SCP})$$

The input APOs are used to compute the following offsets to be added to the initial COA projection set to form the adjusted COA projection set. Refer to Figure 8-3.

$$\Delta Xmt_{COA} = VXmt_{COA} \cdot \Delta tx_{COA} + \Delta Xmt_{COA}^{SCP} + \Delta VXmt \cdot (tx_{COA} + \Delta tx_{COA} - t_{COA}^{SCP})$$

$$\Delta Rcv_{COA} = VRcv_{COA} \cdot \Delta tr_{COA} + \Delta Rcv_{COA}^{SCP} + \Delta VRcv \cdot (tr_{COA} + \Delta tr_{COA} - t_{COA}^{SCP})$$

$$\Delta R_Avg_{COA} = (c/2) \cdot (\Delta tr_{COA} - \Delta tx_{COA})$$

$$\Delta Rdot_Avg_{COA} = (c/2) \cdot (T_Clk_R_SF - T_Clk_X_SF)$$

The adjusted COA projection set is formed by adding offsets to the following parameters.

$$\text{Adjusted } tx_{COA} = tx_{COA} + \Delta tx_{COA}$$

$$\text{Adjusted } tr_{COA} = tr_{COA} + \Delta tr_{COA}$$

$$\text{Adjusted } Xmt_{COA} = Xmt_{COA} + \Delta Xmt_{COA}$$

$$\text{Adjusted } Rcv_{COA} = Rcv_{COA} + \Delta Rcv_{COA}$$

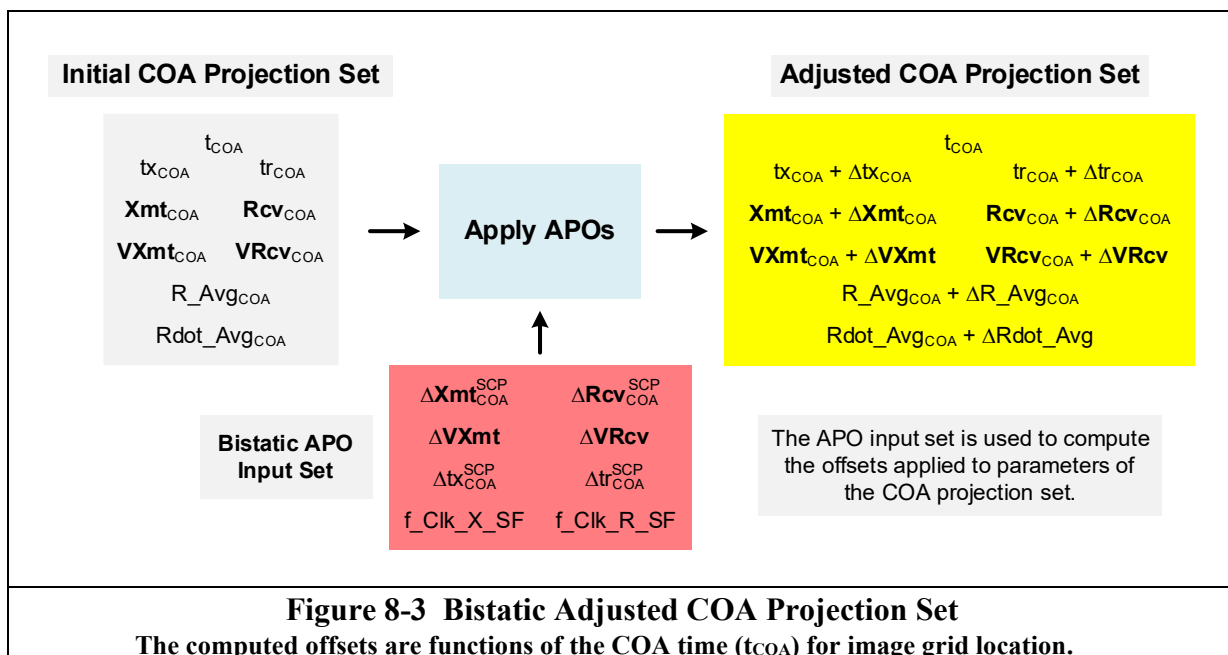
$$\text{Adjusted } VXmt_{COA} = VXmt_{COA} + \Delta VXmt$$

$$\text{Adjusted } VRcv_{COA} = VRcv_{COA} + \Delta VRcv$$

$$\text{Adjusted } R_Avg_{COA} = R_Avg_{COA} + \Delta R_Avg_{COA}$$

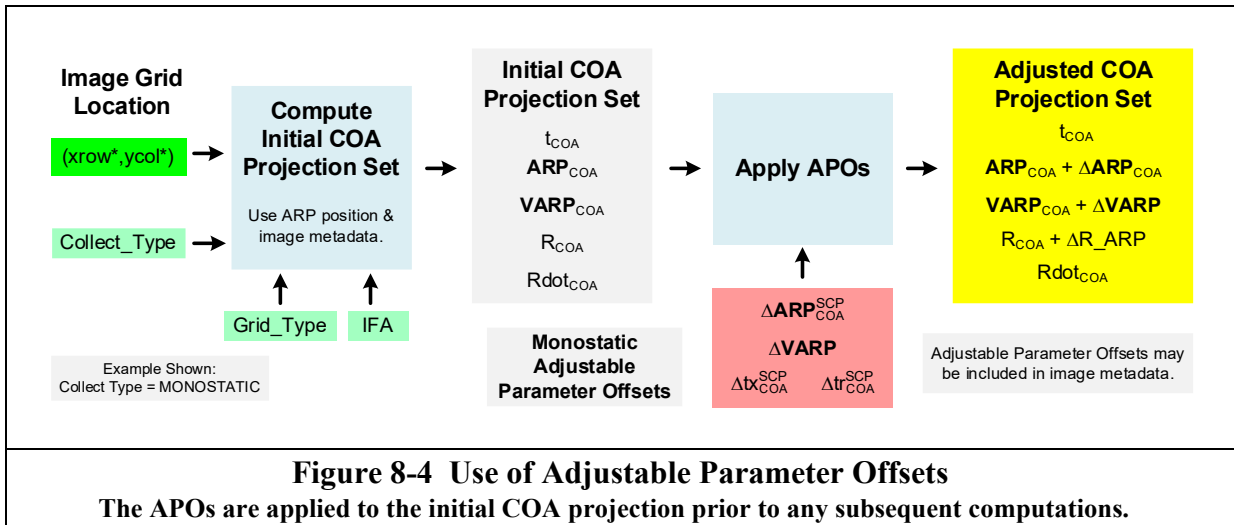
$$\text{Adjusted } Rdot_Avg_{COA} = Rdot_Avg_{COA} + \Delta Rdot_Avg_{COA}$$

Note: The value for t_{COA} is not modified when forming the adjusted projection set.



8.2 Adjusted Image-to-Scene Projection

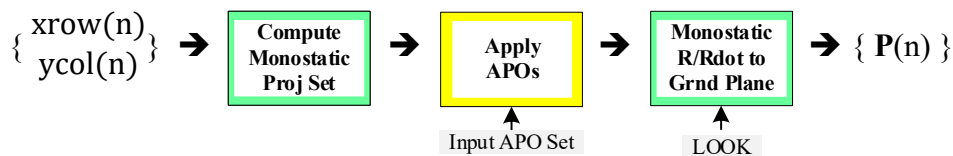
For the Image-to-Scene projection, the initial COA projection set is computed for the selected image grid location. The initial COA projection set is computed as specified in Section 4 based upon the Collect Type, the Grid Type, and the IFA used to form the image. The input APO set is then used to compute the Adjusted COA projection set. The processing sequence for a monostatic image is shown in Figure 8-4.



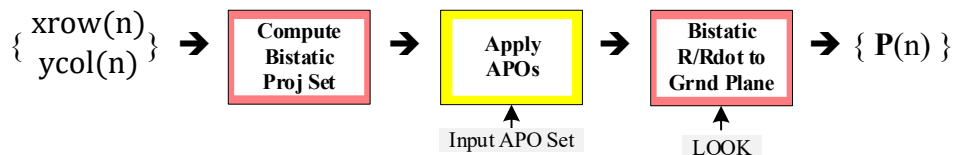
8.3 Adjusted Scene-to-Image Projection

The Scene-to-Image projection computation is an iterative computation. For a selected scene point, a set of N image grid locations are computed $\{I(n), n = 1 \text{ to } N\}$. For each image grid location, the COA projection set is computed. See Section 6.1, Step 6. If input APOs are available, the Adjusted COA projection set is then computed. The Adjusted COA projection set is then used in the R/Rdot projection to the ground plane in Step 7. A functional representation of the modified Step 6 and Step 7 are shown below.

For a monostatic image and image grid location $I(n) = (xrow(n), ycol(n))$:



For a bistatic image and image grid location $I(n) = (xrow(n), ycol(n))$:



9.2 R/Rdot Contour to Surface Intersection

The projection for image pixel grid location ($xrow^{TGT}$, $ycol^{TGT}$) to a constant HAE surface is computed using an iterative algorithm. The algorithm computes the R/Rdot projection to one or more ground planes that are tangent to the constant height surface. Each ground plane projection point computed is slightly above the constant HAE surface. The final surface position SPP^{TGT} is computed by projecting from the final ground plane projection point down to the HAE surface. The inputs to the projection are specified below. The COA Projection parameters are dependent upon the type of collection.

Input COA Projection Parameters – Monostatic Image

ARP_{COA}^{TGT}	ARP position at the COA time in ECEF coordinates.
$VARP_{COA}^{TGT}$	ARP velocity at the COA time in ECEF coordinates.
R_{COA}^{TGT}	Range of the R/Rdot contour.
$Rdot_{COA}^{TGT}$	Range rate (dR/dt) of the R/Rdot contour ($Rdot_{COA}^{TGT} < 0 \Leftrightarrow$ Closing geometry).

Input COA Projection Parameters – Bistatic Image

Xmt_{COA}^{TGT}	Transmit APC COA position for target TGT in ECEF coordinates.
$VXmt_{COA}^{TGT}$	Transmit APC COA velocity for target TGT in ECEF coordinates.
Rcv_{COA}^{TGT}	Receive APC COA position for target TGT in ECEF coordinates.
$VRcv_{COA}^{TGT}$	Receive APC COA velocity for target TGT in ECEF coordinates.
$R_Avg_{COA}^{TGT}$	Average range for target TGT COA geometry.
$Rdot_Avg_{COA}^{TGT}$	Average range rate for target TGT COA geometry.

Image Parameters:

SCP	SCP position in ECEF coordinates.
SCP.Lat	SCP geodetic latitude.
SCP.Lon	SCP geodetic longitude.
SCP.HAE	SCP HAE (m).
LOOK	Integer based on Side of Track parameter. $SideOfTrack = L \Leftrightarrow LOOK = +1$ $SideOfTrack = R \Leftrightarrow LOOK = -1$

Input WGS-84 Height:

HAE ₀	The surface height (m) above the WGS-84 reference ellipsoid for projection point SPP^{TGT} .
------------------	--

R/Rdot Contour to Surface Point $\mathbf{SPP}^{\text{TGT}}$

- (1) Compute the Earth Tangent Plane (ETP) normal at the SCP. Compute the parameters for Ground Plane 1. The reference point position is $\mathbf{GREF}(1)$ and the unit normal is $\mathbf{uGPN}(1)$.

$$\mathbf{uGPN}(1) = \mathbf{uGPN}_{\text{ETP}}^{\text{SCP}} = \begin{bmatrix} \cos(\text{SCP.Lat}) \cdot \cos(\text{SCP.Lon}) \\ \cos(\text{SCP.Lat}) \cdot \sin(\text{SCP.Lon}) \\ \sin(\text{SCP.Lat}) \end{bmatrix}$$

$$\mathbf{GREF}(1) = \mathbf{SCP} + (\text{HAE}_0 - \text{SCP.HAE}) \cdot \mathbf{uGPN}_{\text{ETP}}^{\text{SCP}}$$

Initialize the iteration count to $n = 1$. Iterate over the steps 2 through 5 below.

- (2) Compute the precise projection along the R/Rdot contour to Ground Plane n . The result is ground plane point position $\mathbf{GPP}(n)$. For a monostatic image, the projection to the ground plane may be computed in closed form as specified in Section 5.2. For a bistatic image, the projection to the ground plane is computed iteratively as specified in Section 5.3.

For a monostatic image:

$$\begin{Bmatrix} \mathbf{ARP}_{\text{COA}}^{\text{TGT}} & \mathbf{R}_{\text{COA}}^{\text{TGT}} \\ \mathbf{VARP}_{\text{COA}}^{\text{TGT}} & \mathbf{Rdot}_{\text{COA}}^{\text{TGT}} \end{Bmatrix} + \begin{Bmatrix} \mathbf{GREF}(n) \\ \mathbf{uGPN}(n) \end{Bmatrix} \rightarrow \begin{array}{c} \text{Monostatic} \\ \text{R/Rdot to} \\ \text{Grnd Plane} \end{array} \rightarrow \{ \mathbf{GPP}(n) \}$$

↑
LOOK

For a bistatic image:

$$\begin{Bmatrix} \mathbf{Xmt}_{\text{COA}}^{\text{TGT}} & \mathbf{Rcv}_{\text{COA}}^{\text{TGT}} & \mathbf{R_Avg}_{\text{COA}}^{\text{TGT}} \\ \mathbf{VXmt}_{\text{COA}}^{\text{TGT}} & \mathbf{VRcv}_{\text{COA}}^{\text{TGT}} & \mathbf{Rdot_Avg}_{\text{COA}}^{\text{TGT}} \end{Bmatrix} + \begin{Bmatrix} \mathbf{GREF}(n) \\ \mathbf{uGPN}(n) \end{Bmatrix} \rightarrow \begin{array}{c} \text{Bistatic} \\ \text{R/Rdot to} \\ \text{Grnd Plane} \end{array} \rightarrow \{ \mathbf{GPP}(n) \}$$

↑
LOOK

- (3) For ground point position $\mathbf{GPP}(n)$, convert from ECEF coordinates to WGS 84 geodetic coordinates.

$$\mathbf{GPP}(n) = \begin{bmatrix} \text{GPP.X}(n) \\ \text{GPP.Y}(n) \\ \text{GPP.Z}(n) \end{bmatrix} \rightarrow \begin{array}{c} \text{Convert} \\ \text{ECEF to} \\ \text{LLH} \end{array} \rightarrow \mathbf{GPP_LLH}(n) = \begin{bmatrix} \text{GPP.Lat}(n) \\ \text{GPP.Lon}(n) \\ \text{GPP.HAE}(n) \end{bmatrix}$$

- (4) Compute the unit vector in the increasing height direction at point $\mathbf{GPP}(n)$, $\mathbf{uUP}(n)$. Also compute the height difference at point $\mathbf{GPP}(n)$ relative to the surface height HAE_0 , $\Delta\text{HAE}(n)$.

$$\mathbf{uUP}(n) = \begin{bmatrix} \cos(\text{GPP.Lat}(n)) \cdot \cos(\text{GPP.Lon}(n)) \\ \cos(\text{GPP.Lat}(n)) \cdot \sin(\text{GPP.Lon}(n)) \\ \sin(\text{GPP.Lat}(n)) \end{bmatrix} \quad \Delta\text{HAE}(n) = \text{GPP.HAE}(n) - \text{HAE}_0$$

(5) Test for the point **GPP**(n) sufficiently close the HAE₀ surface ($\Delta\text{HAE}(n) \leq \Delta\text{HAE}_{\text{MAX}}$) and for the maximum number of iterations to be reached ($n = \text{NLIM}$). For either condition, set final index value $N = n$, save parameters **GPP**(n), **uUP**(n) and $\Delta\text{HAE}(n)$ and continue with Steps 6 through 10. Otherwise, compute **GREF**(n+1) and **uGPN**(n+1), increment index n and repeat steps 2 through 5.

IF { $\Delta\text{HAE}(n) \leq \Delta\text{HAE}_{\text{MAX}}$ OR $n = \text{NLIM}$ }:

GPP(N) = **GPP**(n) Final iteration index value $N = n$.
uUP(N) = **uUP**(n) Save values for **GPP**(n), **uUP**(n), and $\Delta\text{HAE}(n)$
 $\Delta\text{HAE}(N) = \Delta\text{HAE}(n)$ Continue with Steps 6 through 10.

ELSE { $\Delta\text{HAE}(n) > \Delta\text{HAE}_{\text{MAX}}$ AND $n < \text{NLIM}$ }:

GREF(n+1) = **GPP**(n) – $\Delta\text{HAE}(n) \bullet \mathbf{uUP}(n)$ Compute **GREF**(n+1)
uGPN(n+1) = **uUP**(n)
 Increment index n: $n \leftarrow n + 1$ Repeat Steps 2 through 5.

(6) Compute the unit slant plane normal vector, **uSPN**, that is tangent to the R/Rdot contour at point **GPP**(N). The method for computing the slant plane normal is dependent upon the Collect Type for the image. Unit vector **uSPN** points away from the center of the earth and in a direction of increasing HAE at **GPP**(N).

For a Monostatic Image: Compute **uSPN** from the COA ARP position & velocity:

$$\mathbf{SPN} = \text{LOOK} \bullet \mathbf{VARP}_{\text{COA}}^{\text{TGT}} \times (\mathbf{GPP}(N) - \mathbf{ARP}_{\text{COA}}^{\text{TGT}}) \quad \mathbf{uSPN} = \frac{1}{|\mathbf{SPN}|} \bullet \mathbf{SPN}$$

For a Bistatic Image: Use the R/Rdot Parameters function defined in Section 5.3.1.

$$\left\{ \begin{matrix} \mathbf{Xmt}_{\text{COA}}^{\text{TGT}} & \mathbf{Rcv}_{\text{COA}}^{\text{TGT}} \\ \mathbf{VXmt}_{\text{COA}}^{\text{TGT}} & \mathbf{VRcv}_{\text{COA}}^{\text{TGT}} \end{matrix} \right\} + \{ \mathbf{GPP}(N) \} \rightarrow \boxed{\text{Compute PTR/Rdot Parameters}} \rightarrow \left\{ \begin{matrix} \text{R_Avg} & \mathbf{bP} \\ \text{Rdot_Avg} & \mathbf{bPDot} \end{matrix} \right\} \mathbf{uSPN}$$

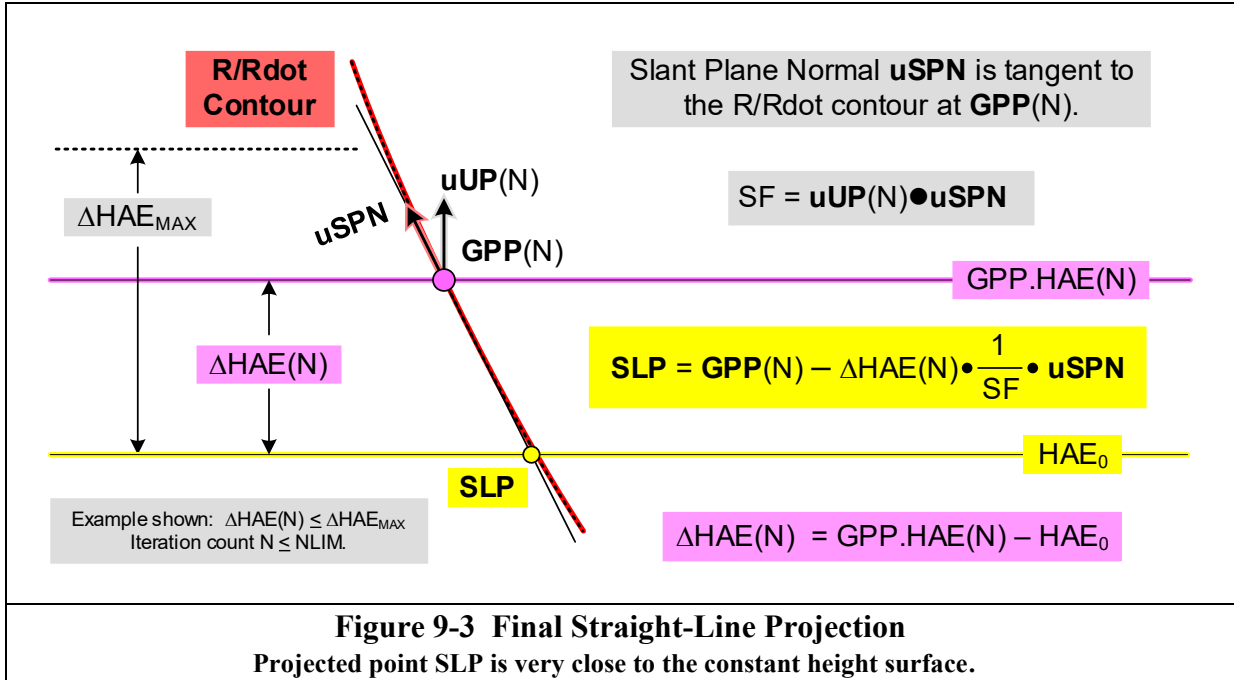
↑
LOOK

(7) Compute the straight-line projection from point **GPP**(N) along the slant plane normal to point SLP. See Figure 9.3. Point SLP is located is very close to the precise R/Rdot contour intersection with the constant height surface.

$$\mathbf{SF} = \mathbf{uUP}(N) \bullet \mathbf{uSPN} \quad \mathbf{SLP} = \mathbf{GPP}(N) - \frac{1}{\mathbf{SF}} \bullet \Delta\text{HAE}(N) \bullet \mathbf{uSPN}$$

(8) Convert the position of point SLP from ECEF coordinates to geodetic coordinates.

$$\mathbf{SLP} = \begin{bmatrix} \text{SLP.X} \\ \text{SLP.Y} \\ \text{SLP.Z} \end{bmatrix} \rightarrow \boxed{\text{Convert ECEF to LLH}} \rightarrow \mathbf{SLP_LLH} = \begin{bmatrix} \text{SLP.Lat} \\ \text{SLP.Lon} \\ \text{SLP.HAE} \end{bmatrix}$$



(9) Assign surface point $\mathbf{SPP}^{TGT}$ position as follows by adjusting the HAE to be on the $\mathbf{HAE}_0$ surface.

$$\mathbf{SPP.Lat}^{TGT} = \mathbf{SLP.Lat} \quad \mathbf{SPP.Lon}^{TGT} = \mathbf{SLP.Lon} \quad \mathbf{SPP.HAE}^{TGT} = \mathbf{HAE}_0$$

(10) Convert position for point $\mathbf{SPP}^{TGT}$ from geodetic coordinates to ECEF coordinates.

$$\mathbf{SPP_LLH}^{TGT} = \begin{bmatrix} \mathbf{SPP.Lat}^{TGT} \\ \mathbf{SPP.Lon}^{TGT} \\ \mathbf{SPP.HAE}^{TGT} \end{bmatrix} \rightarrow \begin{bmatrix} \text{Convert} \\ \text{LLH to} \\ \text{ECEF} \end{bmatrix} \Rightarrow \mathbf{SPP}^{TGT} = \begin{bmatrix} \mathbf{SPP.X}^{TGT} \\ \mathbf{SPP.Y}^{TGT} \\ \mathbf{SPP.Z}^{TGT} \end{bmatrix}$$

9.3 Projection Algorithm Comments

The projection algorithm has been developed with consideration for both accuracy and computational efficiency. The following comments are provided for clarification.

(1) For most projection computations, the number of ground plane projections computed will be either 1 or 2. For the suggested value of $\Delta HAE_{MAX} = 1.0$ m, the iteration stopping criteria is met when the projection point $\mathbf{GPP}(n)$ is displaced less than 3.5 km from the reference point $\mathbf{GREF}(n)$. Even for large area images, the maximum iteration count expected is $N = 2$.

(2) The suggested height threshold of $\Delta HAE_{MAX} = 1.0$ m will yield highly accurate projection results for all SAR imaging geometries. The straight-line projection specified in Step 7 and shown in Figure 9.3 is highly accurate even for very short range and steep grazing angle geometries. Reducing ΔHAE_{MAX} will have negligible improvement in projection accuracy.

- (3) Iteration count limit NLIM is included for safety or for the rare user that chooses a value for ΔHAE_{MAX} less than 1.0 m.
- (4) For the unexpected condition that the R/Rdot contour does not intersect the HAE_0 surface, the condition can be detected by the failure of the initial R/Rdot projection to Ground Plane 1. See Section 5 for this condition. Including the check after each ground plane projection may be included. Such a condition may be due to an error in computing the projection parameters or an error in choosing surface height HAE_0 .

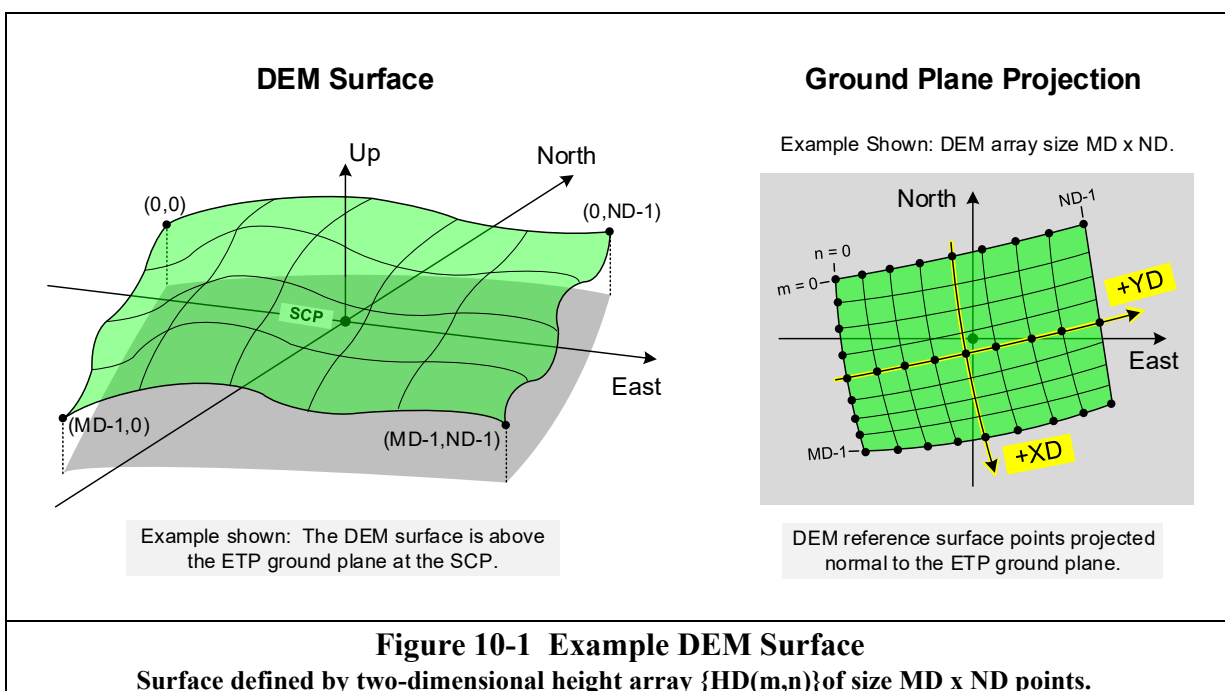
10 Precise R/Rdot to DEM Surface Projection

A method to compute the precise Image-to-Scene projection to a Digital Elevation Model (DEM) of the earth surface is specified. For a monostatic image, the projection is along a contour of constant range and range rate relative to the COA ARP position and velocity. For a bistatic image, the projection is along a contour of constant average range and average range rate relative to the COA transmit APC and receive APC positions and velocities. The projection computation may yield one or more surface points that lie along the R/Rdot contour.

10.1 Surface Model Description

A typical DEM surface is defined by a two-dimensional array of height values, denoted $\{HD(m,n)\}$. The height values are the surface heights relative to a reference surface. The DEM reference surface is spanned by two horizontal coordinates, denoted XD and YD. Surface heights $\{HD(m,n)\}$ are the heights at surface points with horizontal coordinates $\{XD(m)\}$ and $\{YD(n)\}$. The reference surface, also the surface of height $HD = 0$, is approximately parallel to a surface of constant WGS-84 HAE for the region of the earth spanned by the DEM surface. For any position with horizontal coordinates (XD, YD), a unique surface height $HD(XD,YD)$ may be computed by an appropriate interpolation of the height array values.

An example DEM surface is shown in Figure 10-1. The surface is defined by an array of surface points of size MD x ND. The left-hand portion of the figure shows the surface relative to the earth tangent plane defined at the SCP. Shown is an East, North, Up (ENU) coordinate frame with origin at the SCP where the Up direction is normal to the ground plane.



The right-hand portion of the figure shows the region of the DEM reference surface spanned by the two-dimensional array of locations $\{(XD(m), YD(n), 0), m = 0 \text{ to } MD-1, n = 0 \text{ to } ND-1\}$. The locations on the DEM reference surface projected onto the ground plane are an array of approximately equally spaced points.

The coordinate systems used in commonly available DEM data sets include the following examples. Both systems meet the general DEM coordinate system described above.

- (1) The horizontal coordinates (XD, YD) are geodetic latitude and longitude. The height values $\{HD(m,n)\}$ are Height Above Mean Sea Level (HMSL). The height samples are evenly spaced along lines of constant latitude and constant longitude.
- (2) The horizontal coordinates (XD, YD) are UTM coordinates Easting and Northing. The height values $\{HD(m,n)\}$ are either WGS-84 HAE or HMSL. The height samples are evenly spaced along lines of constant Easting and constant Northing for a given UTM zone.

The projection method specified assumes the surface model meets the following set of modest constraints. Let point S be any point on the DEM surface. Point S is located in DEM coordinates at $[S.XD, S.YD, S.HD]^T$ and in geodetic coordinates at $[S.Lat, S.Lon, S.HAE]^T$.

- (1) For the contour of points with constant horizontal coordinates (S.XD, S.YD), there is one and only one surface point (at height S.HD).
- (2) For the contour of points with constant horizontal coordinates (S.XD, S.YD), moving in the increasing HD direction is always in the increasing HAE direction.
- (3) For the line of points with constant latitude and longitude (S.Lat, S.Lon), the line intersects the surface at one and only one point (at height S.HAE).
- (4) For the line of points with constant latitude and longitude (S.Lat, S.Lon), moving in the increasing HAE direction is always in the increasing HD direction.

The projection method specified below assumes the existence of a function mapping a scene point in ECEF coordinates to the height of that point above the DEM surface in meters, where the height above the DEM surface increases in the increasing HAE direction.

$$\mathbf{P}_{ECEF} = \begin{bmatrix} P.X \\ P.Y \\ P.Z \end{bmatrix} \rightarrow \begin{array}{c} \text{Get Height} \\ \text{Above} \\ \text{DEM Surface} \end{array} \rightarrow \Delta HD$$

For $\Delta HD > 0$, point P is above the surface. For $\Delta HD = 0$, point P is on the surface. For $\Delta HD < 0$, point P is below the surface.

10.2 Projection Method Inputs & Outputs

The projection for image pixel grid location ($xrow^{TGT}$, $ycol^{TGT}$) to a DEM surface is computed using the COA projection set and the DEM surface model height array and supporting metadata. The inputs to the projection algorithm are specified below. The COA projection parameters are dependent upon the type of collection.

Input COA Projection Parameters – Monostatic Image

ARP_{COA}^{TGT}	ARP position at the COA time in ECEF coordinates.
$VARP_{COA}^{TGT}$	ARP velocity at the COA time in ECEF coordinates.
R_{COA}^{TGT}	Range of the R/Rdot contour.
$Rdot_{COA}^{TGT}$	Range rate (dR/dt) of the R/Rdot contour ($Rdot_{COA}^{TGT} < 0 \Leftrightarrow$ Closing geometry).

Input COA Projection Parameters – Bistatic Image

Xmt_{COA}^{TGT}	Transmit APC COA position for target TGT in ECEF coordinates.
$VXmt_{COA}^{TGT}$	Transmit APC COA velocity for target TGT in ECEF coordinates.
Rcv_{COA}^{TGT}	Receive APC COA position for target TGT in ECEF coordinates.
$VRcv_{COA}^{TGT}$	Receive APC COA velocity for target TGT in ECEF coordinates.
$R_Avg_{COA}^{TGT}$	Average range for target TGT COA geometry.
$Rdot_Avg_{COA}^{TGT}$	Average range rate for target TGT COA geometry.

Image Parameters:

SCP	SCP position in ECEF coordinates.
LOOK	Integer based on Side of Track parameter. $SideOfTrack = L \Leftrightarrow LOOK = +1$ $SideOfTrack = R \Leftrightarrow LOOK = -1$

DEM Surface Parameters:

HAE_{MIN}	WGS-84 HAE value (m) for which all DEM surface points are on or above the surface of constant $HAE = HAE_{MIN}$.
HAE_{MAX}	WGS-84 HAE value (m) for which all DEM surface points are on or below the surface of constant $HAE = HAE_{MAX}$.
$\Delta DIST_{DEM}$	Maximum horizontal distance between surface points for which the surface is well approximated by a straight line between the points (m). Recommended value: $\Delta DIST_{DEM} = 0.5 \times$ minimum DEM horizontal sample spacing (m). The horizontal spacing is typically referred to as the “post spacing”.

Computed Parameters:

MPTS	Number of projected point(s) on the DEM surface. $MPTS \geq 1$.
$\{S(m)\}$	Set of point(s) where the R/Rdot contour intersects the DEM surface. Points are indexed $m = 1$ to MPTS. The point $S(m)$ is located in ECEF coordinates at $[S.X(m), S.Y(m), S.Z(m)]^T$ and in geodetic coordinates at $[S.Lat(m), S.Lon(m), S.HAE(m)]^T$.

User Selectable Algorithm Parameters:

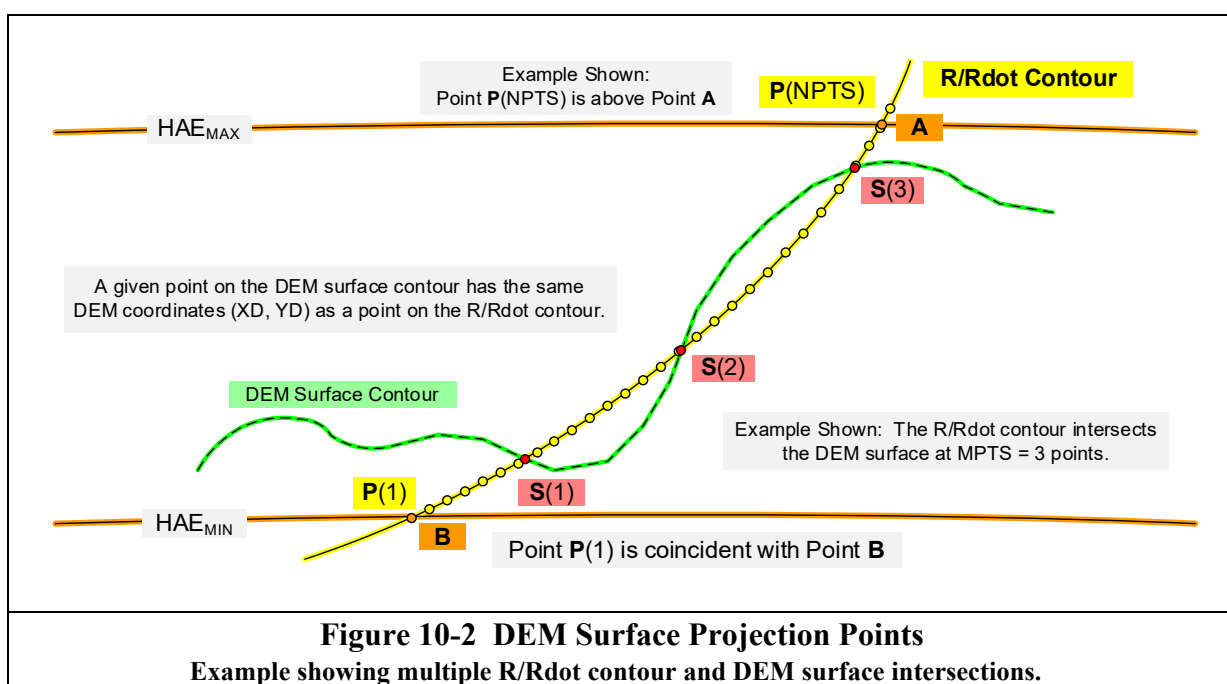
$\Delta DIST_{RRC}$	Maximum distance between adjacent points along the R/Rdot contour. Require $\Delta DIST_{RRC} > 0$. Recommended value: $\Delta DIST_{RRC} = 10.0$ m.
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10.3 Projection to a DEM Surface

The method for computing the projection to a DEM surface consists of two fundamental steps:

- (1) A set of points are computed that span a portion of the R/Rdot contour. The method for computing the contour point positions is dependent on the type of collection (i.e., monostatic or bistatic).
- (2) The set of R/Rdot contour point positions are used to compute a set of one or more points where the contour intersects the DEM surface. The method for computing the surface points is independent of the type of collection.

The set of points on the R/Rdot contour span the portion from HAE_{MIN} to HAE_{MAX} . Contour points A and B are computed via image-to-scene projections to the surfaces of constant $HAE = HAE_{MAX}$ and $HAE = HAE_{MIN}$, respectively. The computed contour point positions are denoted $\{P(n), n = 1 \text{ to } NPTS\}$. Point $P(1)$ is coincident with point B. Point $P(NPTS)$ is coincident with or above point A. An example is depicted in Figure 10-2.



The method for computing the points on the R/Rdot contour is specified in Section 10.4. A method for computing the projection to a surface of constant HAE is specified in Section 9. The precise values of HAE_{MIN} and HAE_{MAX} are not critical as long as they bracket the DEM surface and intersect the R/Rdot contour.

The computation of the points of intersection of the R/Rdot contour and the DEM surface is specified in Section 10.5.

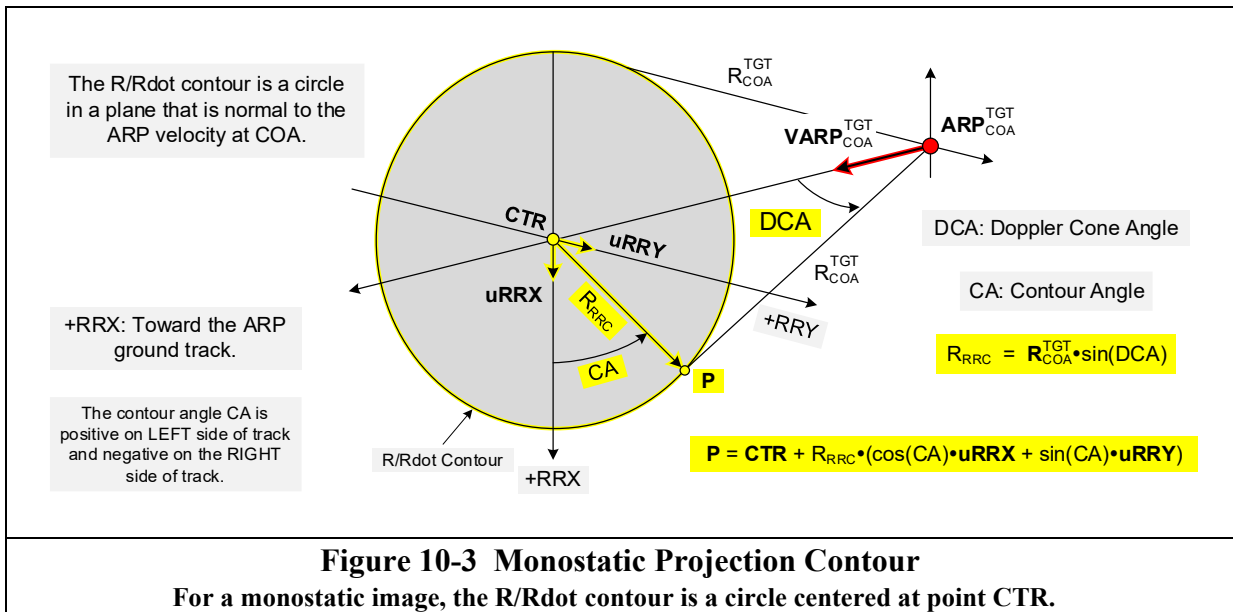
10.4 R/Rdot Contour Points

10.4.1 Monostatic R/Rdot Contour Points

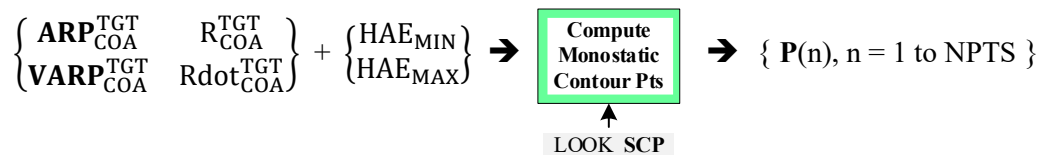
For a monostatic image, the set of contour point positions may be computed in closed form. The R/Rdot contour is a circle that lies in a plane that is normal to the ARP COA velocity vector. Refer to Figure 10-3. The center point of the circle, denoted CTR, is located along the line defined by the ARP position and the ARP velocity vector. The radius of the circle, denoted R_{RRC} , is computed from the range, R_{COA}^{TGT} , and the Doppler Cone Angle (DCA) to the contour. A pair of orthogonal unit vectors that lie in the plane, denoted $\mathbf{u}_{RRX}$ and $\mathbf{u}_{RRY}$, are computed. Unit vector $\mathbf{u}_{RRX}$ points toward the ground track. Unit vector $\mathbf{u}_{RRY}$ points to the left side of the ground track. Points along the contour may be specified by the Contour Angle, denoted CA. The position of a point P on the contour at contour angle CA is computed as shown below. For points on the left side of the ground track, the contour angle CA is positive. For points on the right side of the ground track, the contour angle is negative. For point P, the position in ECEF coordinates is denoted $\mathbf{P}$.

$$R_{RRC} = R_{COA}^{TGT} \cdot \sin(DCA)$$

$$\mathbf{P} = \mathbf{CTR} + R_{RRC} \cdot (\cos(CA) \cdot \mathbf{u}_{RRX} + \sin(CA) \cdot \mathbf{u}_{RRY})$$



The method for computing the set of positions $\{P(n), n = 1 \text{ to } NPTS\}$ located on the R/Rdot contour may be implemented as the function shown below. The SCP position and the LOOK parameter are input to select the portion of the R/Rdot contour on the same side of track as the imaged scene.



(1) Compute the center point, **CTR**, and the radius of the R/Rdot projection contour, R_{RRC} . The center point position is in ECEF coordinates with components in meters. The radius is also in meters.

$$\begin{aligned} VMAG &= | \mathbf{VARP}_{COA}^{TGT} | & \mathbf{uVEL} &= \frac{1}{VMAG} \bullet \mathbf{VARP}_{COA}^{TGT} \\ \cos_DCA &= -\frac{1}{VMAG} \bullet \mathbf{Rdot}_{COA}^{TGT} & \sin_DCA &= +(1 - \cos_DCA^2)^{1/2} \\ \mathbf{CTR} &= \mathbf{ARP}_{COA}^{TGT} + R_{COA}^{TGT} \bullet \cos_DCA \bullet \mathbf{uVEL} \\ R_{RRC} &= R_{COA}^{TGT} \bullet \sin_DCA \end{aligned}$$

(2) Compute the unit vectors **uRRX** and **uRRY** to be used to compute points located on the R/Rdot contour.

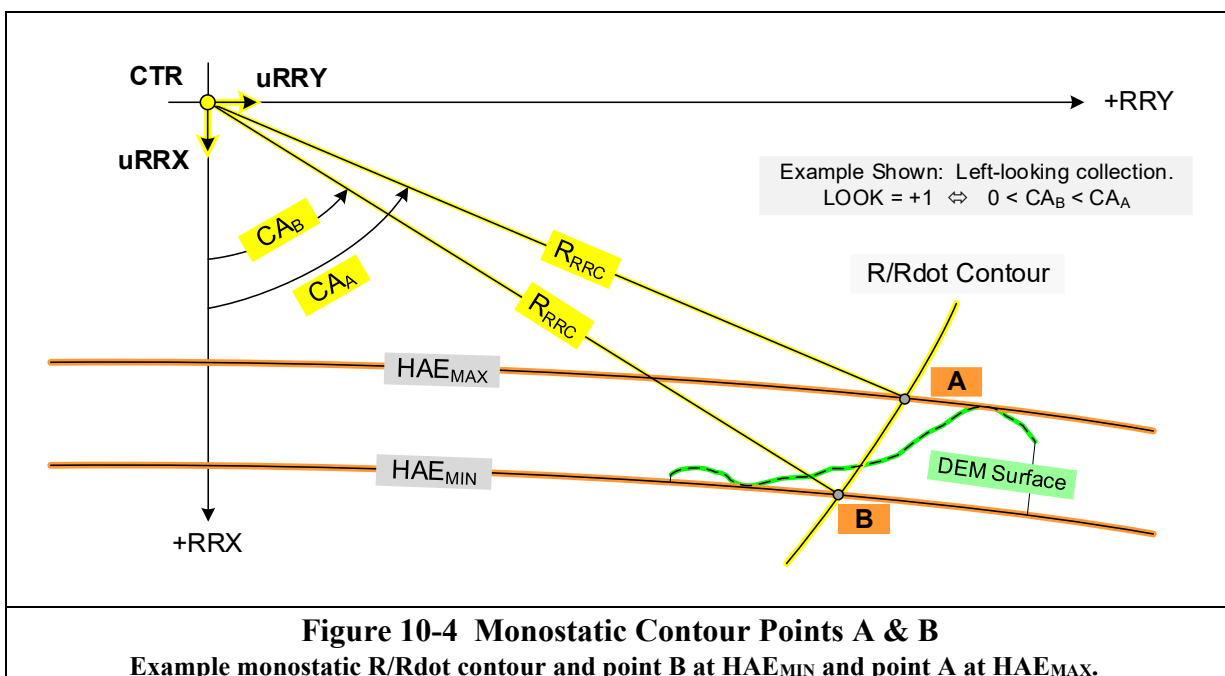
$$\begin{aligned} DEC_ARP &= | \mathbf{ARP}_{COA}^{TGT} | & \mathbf{uUP} &= \frac{1}{DEC_ARP} \bullet \mathbf{ARP}_{COA}^{TGT} \\ \mathbf{RRY} &= \mathbf{uUP} \times \mathbf{uVEL} & \mathbf{uRRY} &= \frac{1}{|\mathbf{RRY}|} \bullet \mathbf{RRY} \\ \mathbf{uRRX} &= \mathbf{uRRY} \times \mathbf{uVEL} \end{aligned}$$

(3) Compute the projection along the R/Rdot contour to the surface of constant HAE at height HAE_{MAX} . The projection point at height HAE_{MAX} is point **A**. Also compute the cosine of the contour angle to point **A**, $\cos_CA_A$. Refer to Figure 10-4.

$$\begin{aligned} \left\{ \begin{array}{l} \mathbf{ARP}_{COA}^{TGT} \\ \mathbf{VARP}_{COA}^{TGT} \end{array} \right. & \left\{ \begin{array}{l} R_{COA}^{TGT} \\ \mathbf{Rdot}_{COA}^{TGT} \end{array} \right\} + \{HAE_{MAX}\} \rightarrow \begin{array}{c} \boxed{\text{Monostatic} \\ \text{R/Rdot to} \\ \text{HAE Surface}} \\ \uparrow \\ \text{LOOK SCP} \end{array} \rightarrow \{ \mathbf{A} = \begin{bmatrix} A.X \\ A.Y \\ A.Z \end{bmatrix} \} \\ \cos_CA_A &= \frac{1}{R_{RRC}} \bullet (\mathbf{A} - \mathbf{CTR}) \bullet \mathbf{uRRX} \end{aligned}$$

(4) Compute the projection along the R/Rdot contour to the surface of constant HAE at height HAE_{MIN} . The projection point at height HAE_{MIN} is point **B**. Also compute the cosine and sine of the contour angle to point **B**, $\cos_CA_B$ and $\sin_CA_B$.

$$\begin{aligned} \left\{ \begin{array}{l} \mathbf{ARP}_{COA}^{TGT} \\ \mathbf{VARP}_{COA}^{TGT} \end{array} \right. & \left\{ \begin{array}{l} R_{COA}^{TGT} \\ \mathbf{Rdot}_{COA}^{TGT} \end{array} \right\} + \{HAE_{MIN}\} \rightarrow \begin{array}{c} \boxed{\text{Monostatic} \\ \text{R/Rdot to} \\ \text{HAE Surface}} \\ \uparrow \\ \text{LOOK SCP} \end{array} \rightarrow \{ \mathbf{B} = \begin{bmatrix} B.X \\ B.Y \\ B.Z \end{bmatrix} \} \\ \cos_CA_B &= \frac{1}{R_{RRC}} \bullet (\mathbf{B} - \mathbf{CTR}) \bullet \mathbf{uRRX} & \sin_CA_B &= \text{LOOK} \bullet (1 - (\cos_CA_B)^2)^{1/2} \end{aligned}$$



(5) The set of NPTS points along the R/Rdot contour to be computed are spaced in equal increments of the cosine of the contour angle. Compute the step size, $\Delta\cos_CA$, as follows.

- (a) Step size $\Delta\cos_RRC$ is computed such that the distance between adjacent points on the R/Rdot contour is approximately equal to ΔDIST_{RRC} .

$$\Delta\cos_RRC = \Delta\text{DIST}_{RRC} \cdot \frac{1}{R_{RRC}} \cdot |\sin_CA_B|$$

- (b) Step size $\Delta\cos_DEM$ is computed such that the horizontal distance between adjacent points on the R/Rdot contour is approximately equal to ΔDIST_{DEM} .

$$\Delta\cos_DEM = \Delta\text{DIST}_{DEM} \cdot \frac{1}{R_{RRC}} \cdot \left(\frac{|\sin CA_B|}{\text{MAX}(\cos CA_B, 0.010)} \right)$$

- (3) Compute $\Delta\cos_CA$ as follows. Note the value of $\Delta\cos_CA < 0$.

$$\Delta\cos_CA = (-1) \cdot \text{MIN}\{ \Delta\cos_RRC, \Delta\cos_DEM \}$$

- (6) Determine the number of points along the R/Rdot contour to be computed, NPTS.

$$\text{NPTS} = \text{FLOOR}\left(\frac{\cos_CA_A - \cos_CA_B}{\Delta\cos_CA}\right) + 2 \quad \text{Note: } \cos_CA_A - \cos_CA_B \leq 0$$

- (7) Compute the set of NPTS along the R/Rdot contour, $\{P(n), \text{ for } n=1, 2, \dots, \text{NPTS}\}$. Initial point $P(1)$ is located on the HAE_{MIN} surface at point B. Final point P_{NPTS} (NPTS) is located above the HAE_{MAX} surface. Point $P(n)$ is computed in ECEF coordinates as follows.

For $n = 1, 2, \dots, \text{NPTS}$:

$$\cos_CA(n) = \cos_CA_B + (n - 1) \cdot \Delta \cos_CA$$

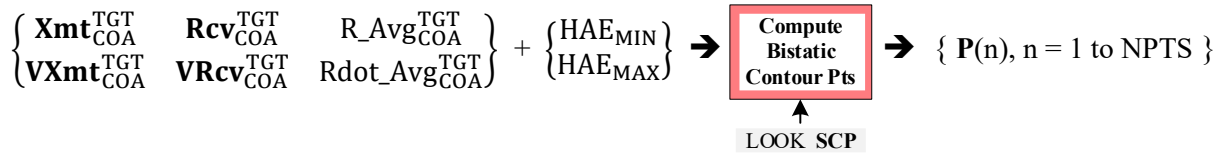
$$\sin_CA(n) = \text{LOOK} \cdot (1 - (\cos_CA(n))^2)^{1/2}$$

$$\mathbf{P}(n) = \mathbf{CTR} + R_{\text{RRC}} \cdot (\cos_CA(n) \cdot \mathbf{uRRX} + \sin_CA(n) \cdot \mathbf{uRRY})$$

10.4.2 Bistatic R/Rdot Contour Points

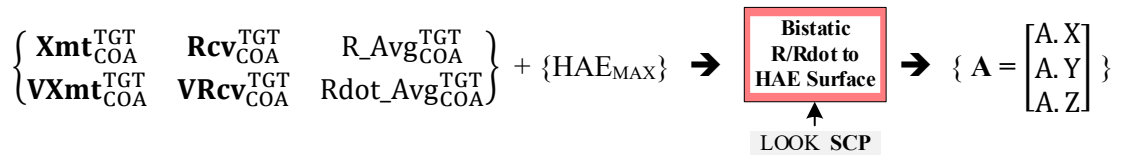
For a bistatic image, the set of contour point positions cannot be computed in closed form. The set of positions are computed using an iterative method. The positions of points A and B are computed at the intersection of the R/Rdot contour and the surfaces of constant HAE at $\text{HAE} = \text{HAE}_{\text{MAX}}$ and at $\text{HAE} = \text{HAE}_{\text{MIN}}$ (as is done for a monostatic image). The set of contour point positions $\{\mathbf{P}(n), n = 1, 2, \dots, \text{NPTS}\}$ are then computed. The position $\mathbf{P}(1)$ is set equal to position **B** located at $\text{HAE} = \text{HAE}_{\text{MIN}}$. The other positions are then computed with position $\mathbf{P}(n)$ computed starting at point $\mathbf{P}(n-1)$. The computation makes use of the bistatic slant plane normal (SPN) at point $\mathbf{P}(n-1)$ followed by a small correction that accounts for the curvature of the R/Rdot contour.

The method for computing the set of positions $\{\mathbf{P}(n), n = 1 \text{ to } \text{NPTS}\}$ located on the R/Rdot contour may be implemented as the function shown below. The SCP position and the LOOK parameter are input to select the portion of the R/Rdot contour on the same side of track as the imaged scene.

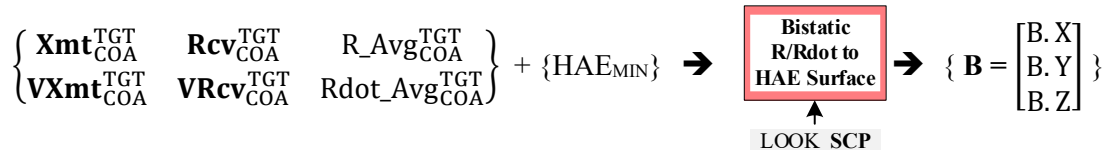


Note: All vectors are in ECEF coordinates. All position components, ranges, HAEs, and distances are in meters. All velocity components are in meters/sec.

(1) Compute the projection along the R/Rdot contour to the surface of constant HAE at height HAE_{MAX} . The projection point at height HAE_{MAX} is point A. See Section 9.



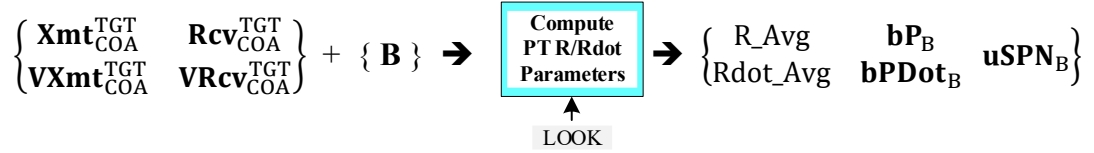
(2) Compute the projection along the R/Rdot contour to the surface of constant HAE at height HAE_{MIN} . The projection point at height HAE_{MIN} is point B.



- (3) At point B, compute the spherical earth ground plane unit normal vector, $\mathbf{uGPN}_B$. See Section 5.1.

$$\mathbf{uGPN}_B = \frac{1}{|B|} \bullet B$$

- (4) At point B, compute the bistatic slant plane normal, $\mathbf{uSPN}_B$ using the method specified in Section 5.3.1.



- (5) The angle between the GPN and the SPN is the slope angle. Compute the cosine and sine of the slope angle at point B, $\cos_SLOPE_B$ and $\sin_SLOPE_B$.

$$\cos_SLOPE_B = \mathbf{uGPN}_B \bullet \mathbf{uSPN}_B \quad \sin_SLOPE_B = +(1 - (\cos_SLOPE_B)^2)^{1/2}$$

- (6) The set of positions $\{P(n)\}$ are nominally equally spaced along the R/Rdot contour. Compute the nominal spacing ΔRRC as follows.

$$\Delta RRC_{DEM} = \Delta DIST_{DEM} \bullet \left(\frac{1}{\text{MAX}(\sin_SLOPE_B, 0.010)} \right)$$

$$\Delta RRC = \text{MIN} \{ \Delta RRC_{DEM}, \Delta DIST_{RRC} \}$$

- (7) Point P(1) is coincident with point B. Set position $P(1)$ equal to position B . Also set the SPN unit normal $\mathbf{uSPN}(1)$ equal to $\mathbf{uSPN}_B$.

$$P(1) = B \quad \mathbf{uSPN}(1) = \mathbf{uSPN}_B$$

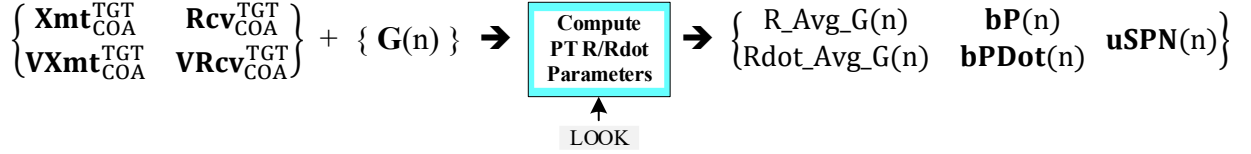
For points $n = 2, 3, \dots, NPTS$, position $P(n)$ is computed from position $P(n-1)$ and $\mathbf{uSPN}(n-1)$. Compute positions $P(2)$ through $P(NPTS)$ as specified in steps (8) through (14). For each computed position, check for point $P(n)$ at or above point A. Point $P(NPTS)$ is the first point that is at or above point A.

Initialize index $n = 2$

- (8) Compute position $G(n)$ by moving a distance ΔRRC from position $P(n-1)$ along the SPN computed for point $P(n-1)$, $\mathbf{uSPN}(n-1)$. See Figure 10-5.

$$G(n) = P(n-1) + \Delta RRC \bullet \mathbf{uSPN}(n-1)$$

- (9) Compute the bistatic range and range rate parameters for point $G(n)$ relative to the COA APC positions and velocities using the method specified in Section 5.3.1.



(10) Compute the difference in average range and average range rate for the R/Rdot contour relative to point $\mathbf{G}(n)$.

$$\Delta \mathbf{R_Avg_G}(n) = \mathbf{R_Avg}_{COA}^{TGT} - \mathbf{R_Avg_G}(n)$$

$$\Delta \mathbf{Rdot_Avg_G}(n) = \mathbf{Rdot_Avg}_{COA}^{TGT} - \mathbf{Rdot_Avg_G}(n)$$

(11) Compute the spherical earth GPN at point $\mathbf{G}(n)$, $\mathbf{uGPN}(n)$. Compute the bistatic ground plane parameters using the method specified in Section 5.3.2.

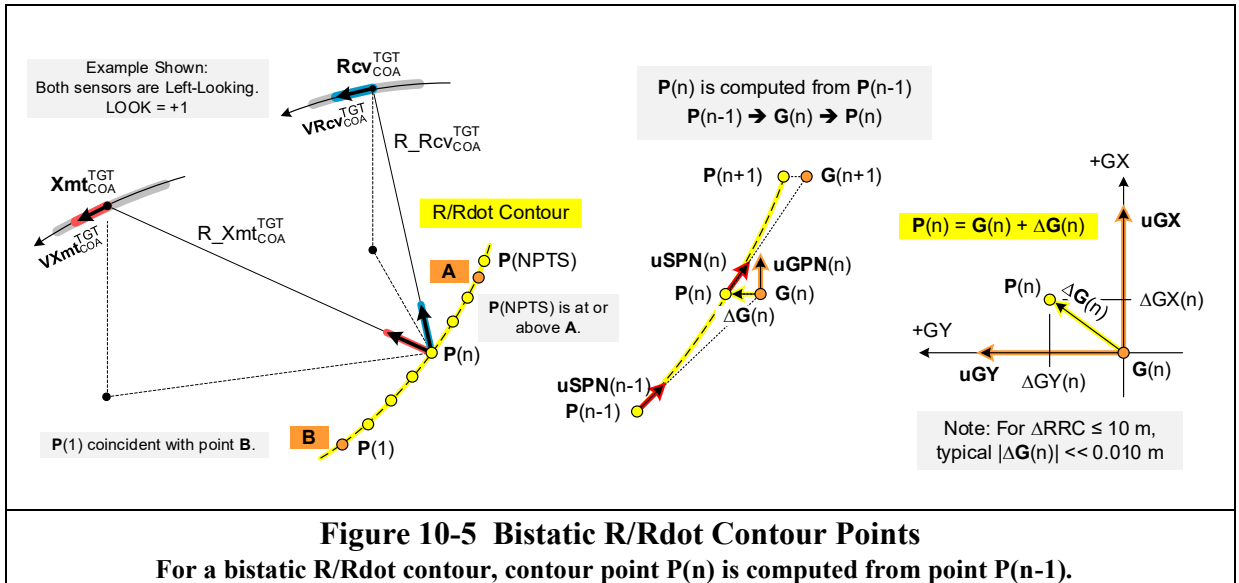
$$\mathbf{uGPN}(n) = \frac{1}{|\mathbf{G}(n)|} \cdot \mathbf{G}(n)$$

$$\left\{ \mathbf{G}(n) \quad \mathbf{uGPN}(n) \right\} + \left\{ \begin{matrix} \mathbf{bP}(n) \\ \mathbf{bPdot}(n) \end{matrix} \right\} \rightarrow \boxed{\text{Compute GPXY Parameters}} \rightarrow \left\{ \begin{matrix} \mathbf{uGX}(n) & \mathbf{M}_{GPXY}^{RRdot}(n) \\ \mathbf{uGY}(n) & \mathbf{M}_{RRdot}^{GPXY}(n) \end{matrix} \right\}$$

(12) Compute displacements $\Delta \mathbf{GX}(n)$ and $\Delta \mathbf{GY}(n)$ that are the shift in the ground plane between contour point $\mathbf{P}(n)$ from the ground point $\mathbf{G}(n)$.

$$\begin{bmatrix} \Delta \mathbf{GX}(n) \\ \Delta \mathbf{GY}(n) \end{bmatrix} = \mathbf{M}_{RRdot}^{GPXY}(n) \cdot \begin{bmatrix} \Delta \mathbf{R_Avg_G}(n) \\ \Delta \mathbf{Rdot_Avg_G}(n) \end{bmatrix}$$

Note: For $\Delta \mathbf{RRC} \leq 10$ m, the displacement $\Delta \mathbf{GP}(n) = [\Delta \mathbf{GX}(n)^2 + \Delta \mathbf{GY}(n)^2]^{1/2} < 0.010$ m.



(13) Compute the vector $\Delta \mathbf{G}(n)$ that is the displacement from $\mathbf{G}(n)$ to the R/Rdot contour in the ground plane. Compute position $\mathbf{P}(n)$ that is in the ground plane and on the R/Rdot contour. Refer to the right-hand portion of Figure 10-5.

$$\Delta \mathbf{G}(n) = \Delta GX(n) \bullet \mathbf{uGX}(n) + \Delta GY(n) \bullet \mathbf{uGY}(n) \quad \mathbf{P}(n) = \mathbf{G}(n) + \Delta \mathbf{G}(n)$$

(14) Compute the distance of point A above the ground plane that contains points $\mathbf{G}(n)$ and $\mathbf{P}(n)$, $A.GZ(n)$. For $A.GZ(n) > 0$, point A is above point $\mathbf{P}(n)$. Increment the index n and compute the next position. For $A.GZ \leq 0$, point $\mathbf{P}(n)$ is at or above point A. Set NPTS equal to n and exit loop.

$$A.GZ(n) = (\mathbf{A} - \mathbf{P}(n)) \bullet \mathbf{uGPN}(n)$$

If $A.GZ(n) > 0$: Point A is above point $\mathbf{P}(n)$.

$n = n + 1$ Increment index n and compute the next position.

Else: $A.GZ(n) \leq 0$ Point A is at or below point $\mathbf{P}(n)$.

$NPTS = n$ Assign NPTS. Exit to compute DEM surface points.

10.5 Compute DEM Surface Points

The method for computing the set of DEM surface points $\{S(m)\}$ from the set of contour points $\{P(n)\}$ is specified below. Refer to Figure 10-2.

- (1) For each of the NPTS points, compute the height of the point above the DEM surface. For point $P(n)$, height above the surface $\Delta HD(n)$ is in meters. Refer to Section 10.1.

For $n = 1, 2, \dots, NPTS$:

$$P(n) = \begin{bmatrix} P.X(n) \\ P.Y(n) \\ P.Z(n) \end{bmatrix} \rightarrow \begin{array}{c} \text{Get Height} \\ \text{Above} \\ \text{DEM Surface} \end{array} \rightarrow \Delta HD(n)$$

- (2) Determine the set of points where the R/Rdot contour intersects the DEM surface, $\{S(m), m = 1, 2, \dots, MPTS\}$. Surface points will be in order of increasing WGS-84 HAE.

Initialize: $m = 0$

Loop over $n = 1, 2, \dots, NPTS - 1$:

If $\Delta HD(n) = 0$: Point $P(n)$ is on the surface.

$m = m + 1$

$S(m) = P(n)$

If $\text{sgn}(\Delta HD(n)) \cdot \text{sgn}(\Delta HD(n+1)) = -1$: Compute position $S(m)$

$m = m + 1$

$$\text{frac} = \frac{\Delta HD(n)}{\Delta HD(n) - \Delta HD(n+1)}$$

$$S(m) = P(n) \cdot (1 - \text{frac}) + P(n+1) \cdot \text{frac}$$

End loop

Note: Function $\text{sgn}(x)$ inputs a real-valued parameter and returns an integer as shown below.

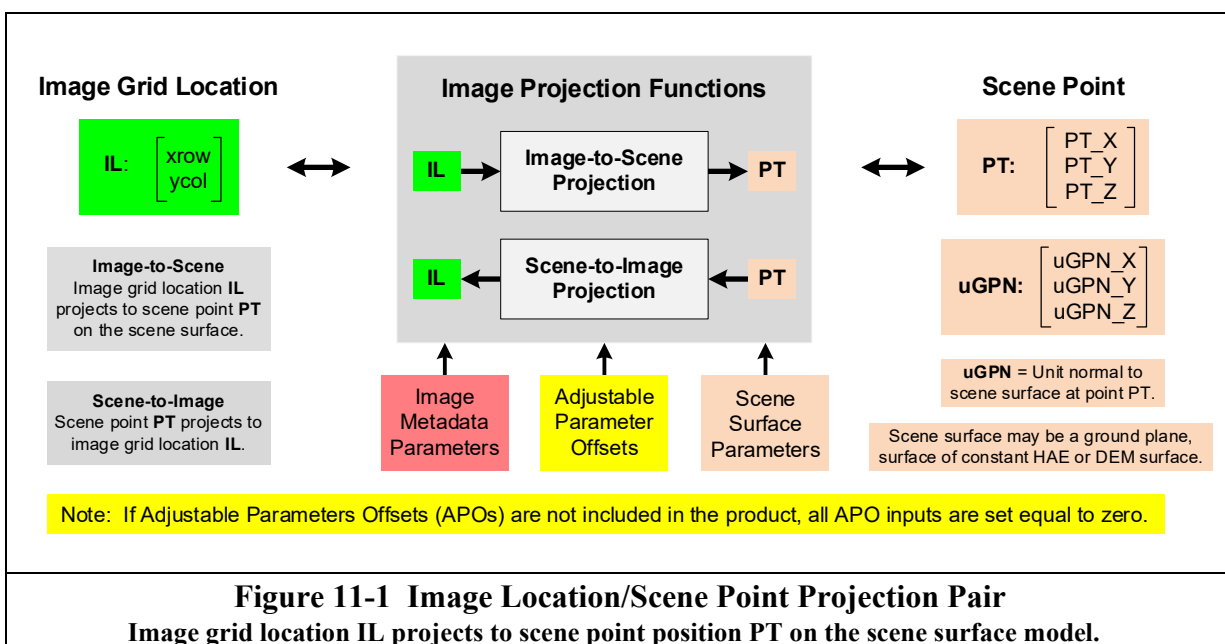
$$\text{sgn}(x) = \begin{cases} +1, & \text{If } x > 0 \\ 0, & \text{If } x = 0 \\ -1, & \text{Otherwise} \end{cases}$$

11 Projection Sensitivity Parameters

Image grid location **IL** is a point in the two-dimensional image grid. Scene point **PT** is a point in the imaged scene that lies along the R/Rdot projection contour for image location **IL**. Given scene point position **PT**, image location **IL** may be computed using the Scene-to-Image projection function. Also, given image location **IL** and a surface model for which scene point **PT** is a point on the surface, scene point **PT** may be computed using the Image-to-Scene projection function. Image location **IL** and scene point **PT** are referred to as an image location/scene point projection pair or, simply, an image/scene projection pair.

The image projection functions that connect the image projection pair **IL** and **PT** are shown in Figure 11-1. The projection functions make use of the required image metadata and a surface model for the imaged scene. The sensor position and velocity and the computed range and range rate may be adjusted via the optional Adjustable Parameter Offsets (APOs). The unit normal to the scene surface model at point **PT** is **uGPN**. For a smoothly varying surface model, the surface near **PT** may be approximated by the ground plane defined by point **PT** and normal vector **uGPN**.

$$\text{Image Location: } \mathbf{IL} = \begin{bmatrix} \text{xrow} \\ \text{ycol} \end{bmatrix} \quad \text{Scene Point (ECEF): } \mathbf{PT} = \begin{bmatrix} \text{PT_X} \\ \text{PT_Y} \\ \text{PT_Z} \end{bmatrix}$$



For a given image/scene projection pair, a set of projection sensitivity parameters may be computed. The sensitivity parameters relate how a change in one parameter will effect changes in another parameter given all other parameters are held constant. For example, let image location **IL1** and scene point **PT1** be an image projection pair. A change in scene point position, $\Delta \mathbf{PT}$, will cause a change in image location, $\Delta \mathbf{IL}$, such that the new image location and

scene point are also an image/scene projection pair. Let the new scene point be **PT2** and the new image location be **IL2**. Scene points and image locations are related as follows.

$$\mathbf{PT1} \rightarrow \mathbf{PT2} = \mathbf{PT1} + \Delta\mathbf{PT} \quad \rightarrow \quad \mathbf{IL1} \rightarrow \mathbf{IL2} = \mathbf{IL1} + \Delta\mathbf{IL}$$

$$\text{where: } \Delta\mathbf{PT} = \begin{bmatrix} \Delta\mathbf{PT_X} \\ \Delta\mathbf{PT_Y} \\ \Delta\mathbf{PT_Z} \end{bmatrix} \quad \Delta\mathbf{IL} = \begin{bmatrix} \Delta\mathbf{xrow} \\ \Delta\mathbf{ycol} \end{bmatrix}$$

The sensitivity parameters are defined in matrix form. For the example above, the change in image location $\Delta\mathbf{IL}$ may be estimated from the change in scene point $\Delta\mathbf{PT}$ using the 2 x 3 sensitivity matrix $\mathbf{M}_{\mathbf{PT}}^{\mathbf{IL}}$. For small changes in scene point position $\Delta\mathbf{PT}$, the change in image location may be accurately computed.

$$\Delta\mathbf{IL} = \mathbf{M}_{\mathbf{PT}}^{\mathbf{IL}} \cdot \Delta\mathbf{PT} \quad \mathbf{M}_{\mathbf{PT}}^{\mathbf{IL}} = \begin{bmatrix} \frac{\Delta\mathbf{xrow}}{\Delta\mathbf{PT_X}} & \frac{\Delta\mathbf{xrow}}{\Delta\mathbf{PT_Y}} & \frac{\Delta\mathbf{xrow}}{\Delta\mathbf{PT_Z}} \\ \frac{\Delta\mathbf{ycol}}{\Delta\mathbf{PT_X}} & \frac{\Delta\mathbf{ycol}}{\Delta\mathbf{PT_Y}} & \frac{\Delta\mathbf{ycol}}{\Delta\mathbf{PT_Z}} \end{bmatrix}$$

For a SAR image formed with either a fixed COA time or smoothly varying COA times, the shape of the R/Rdot projection contours vary smoothly across the imaged scene. In the example above, for image locations near **IL1** that project to scene points near **PT1**, the projection contours near **PT1** may be approximated by straight lines that are normal to the COA slant plane for scene point **PT1**. See Section 11.2. A given projection contour can be specified by its slant plane projection point (SPX, SPY). Hence, the slant plane projection point, denoted **SPXY**, may be used to relate scene points near **PT1** and image grid locations near **IL1**.

$$\Delta\mathbf{PT} = \begin{bmatrix} \Delta\mathbf{PT_X} \\ \Delta\mathbf{PT_Y} \\ \Delta\mathbf{PT_Z} \end{bmatrix} \quad \Leftrightarrow \quad \mathbf{SPXY} = \begin{bmatrix} \mathbf{SPX} \\ \mathbf{SPY} \end{bmatrix} \quad \Leftrightarrow \quad \Delta\mathbf{IL} = \begin{bmatrix} \Delta\mathbf{xrow} \\ \Delta\mathbf{ycol} \end{bmatrix}$$

In the example above, parameter $\Delta\mathbf{PT}$ is referred to as the independent parameter and parameter $\Delta\mathbf{IL}$ is referred to as the dependent parameter. For the sensitivity matrices to be computed, the independent parameters include the image grid location, the scene point position, and the sensor position and velocity. The dependent parameters include the image grid location and the scene point position. The parameters are summarized Table 11-1 below.

Table 11-1 Projection Sensitivity Parameters	
$\Delta\mathbf{IL} = \begin{bmatrix} \Delta\mathbf{xrow} \\ \Delta\mathbf{ycol} \end{bmatrix}$	A change in image grid location specified in SCP-centered image coordinates (in meters).
$\Delta\mathbf{PT} = \begin{bmatrix} \Delta\mathbf{PT_X} \\ \Delta\mathbf{PT_Y} \\ \Delta\mathbf{PT_Z} \end{bmatrix}$	A change in scene point position specified in ECEF coordinates (in meters).

Table 11-1 Projection Sensitivity Parameters	
$\Delta\mathbf{GPXY} = \begin{bmatrix} \Delta\text{GPX} \\ \Delta\text{GPY} \end{bmatrix}$	A change in scene point position that is constrained to the ground plane in ground plane coordinates GPX and GPY (in meters). See Section 11.2.
$\Delta\mathbf{SPXY} = \begin{bmatrix} \Delta\text{SPX} \\ \Delta\text{SPY} \end{bmatrix}$	A change in the slant plane projection point in slant plane coordinates SPX and SPY (in meters). See Section 11.2.
For Collect_Type = MONOSTATIC:	
$\Delta\mathbf{ARP}_{\text{COA}}^{\text{SCP}} = \begin{bmatrix} \Delta\text{ARP_X} \\ \Delta\text{ARP_Y} \\ \Delta\text{ARP_Z} \end{bmatrix}$	A change in ARP position in ECEF coordinates (meters) at $t = t_{\text{COA}}^{\text{SCP}}$.
$\Delta\mathbf{VARP} = \begin{bmatrix} \Delta\text{VARP_X} \\ \Delta\text{VARP_Y} \\ \Delta\text{VARP_Z} \end{bmatrix}$	A change in ARP velocity in ECEF coordinates (meters/sec).
$\Delta\mathbf{RRdot} = \begin{bmatrix} \Delta\text{R_ARP} \\ \Delta\text{Rdot_ARP} \end{bmatrix}$	A change in COA range (meters) and range rate (meters/sec) relative to the COA ARP position and velocity.
For Collect_Type = BISTATIC:	
$\Delta\mathbf{Xmt}_{\text{COA}}^{\text{SCP}} = \begin{bmatrix} \Delta\text{Xmt_X} \\ \Delta\text{Xmt_Y} \\ \Delta\text{Xmt_Z} \end{bmatrix}$	A change in transmit APC position in ECEF coordinates (in meters) at $t = t_{\text{COA}}^{\text{SCP}}$.
$\Delta\mathbf{VXmt} = \begin{bmatrix} \Delta\text{VXmt_X} \\ \Delta\text{VXmt_Y} \\ \Delta\text{VXmt_Z} \end{bmatrix}$	A change in transmit APC velocity in ECEF coordinates (meters/sec).
$\Delta\mathbf{Rcv}_{\text{COA}}^{\text{SCP}} = \begin{bmatrix} \Delta\text{Rcv_X} \\ \Delta\text{Rcv_Y} \\ \Delta\text{Rcv_Z} \end{bmatrix}$	A change in receive APC position in ECEF coordinates (in meters) at $t = t_{\text{COA}}^{\text{SCP}}$.
$\Delta\mathbf{VRcv}_{\text{COA}}^{\text{SCP}} = \begin{bmatrix} \Delta\text{VRcv_X} \\ \Delta\text{VRcv_Y} \\ \Delta\text{VRcv_Z} \end{bmatrix}$	A change in receive APC velocity in ECEF coordinates (meters/sec).
$\Delta\mathbf{RRdot} = \begin{bmatrix} \Delta\text{R_Avg} \\ \Delta\text{Rdot_Avg} \end{bmatrix}$	A change in COA average range (meters) and average range rate (meters/sec) relative to the APC positions and velocities.

For an image grid location **IL** expressed in image coordinates (xrow, ycol), let image grid location **IIDX** be the same grid location expressed in image indices (irow, icol). Grid locations **IL** and **IIDX** are related by matrices the $\mathbf{M}_{\text{IIDX}}^{\text{IL}}$ and $\mathbf{M}_{\text{IL}}^{\text{IIDX}}$. The image row spacing, Row_SS, and image column spacing, Col_SS, are in meters.

$$\mathbf{IL} = \begin{bmatrix} \text{xrow} \\ \text{ycol} \end{bmatrix} \Leftrightarrow \mathbf{IIDX} = \begin{bmatrix} \text{irow} \\ \text{icol} \end{bmatrix} \quad \mathbf{IL} = \mathbf{M}_{\text{IIDX}}^{\text{IL}} \cdot \mathbf{IIDX} \quad \& \quad \mathbf{IIDX} = \mathbf{M}_{\text{IL}}^{\text{IIDX}} \cdot \mathbf{IL}$$

$$\mathbf{M}_{\text{IIDX}}^{\text{IL}} = \begin{bmatrix} \text{Row_SS} & 0 \\ 0 & \text{Col_SS} \end{bmatrix} \quad \mathbf{M}_{\text{IL}}^{\text{IIDX}} = \begin{bmatrix} \frac{1}{\text{Row_SS}} & 0 \\ 0 & \frac{1}{\text{Col_SS}} \end{bmatrix}$$

For applications that work with image locations in pixel coordinates, the sensitivity matrices may be scaled to yield changes in image location in pixel indices. For a sensitivity matrix for which image grid location **IL** is the dependent parameter, the matrix is pre-multiplied by $\mathbf{M}_{\text{IL}}^{\text{IIDX}}$ to yield the image location sensitivity in pixels. For a sensitivity matrix for which image grid location is the independent parameter, the matrix is post-multiplied by $\mathbf{M}_{\text{IIDX}}^{\text{IL}}$ to allow image changes in pixel coordinates. For example, the matrices relating changes in image grid location and ground plane position are modified as follows.

$$\Delta \mathbf{IIDX} \text{ Dependent:} \quad \mathbf{M}_{\text{GPXY}}^{\text{IIDX}} = \mathbf{M}_{\text{IL}}^{\text{IIDX}} \cdot \mathbf{M}_{\text{GPXY}}^{\text{IL}} \quad \rightarrow \quad \Delta \mathbf{IIDX} = \mathbf{M}_{\text{GPXY}}^{\text{IIDX}} \cdot \Delta \mathbf{GPXY}$$

$$\Delta \mathbf{IIDX} \text{ Independent:} \quad \mathbf{M}_{\text{IIDX}}^{\text{GPXY}} = \mathbf{M}_{\text{IL}}^{\text{GPXY}} \cdot \mathbf{M}_{\text{IIDX}}^{\text{IL}} \quad \rightarrow \quad \Delta \mathbf{GPXY} = \mathbf{M}_{\text{IIDX}}^{\text{GPXY}} \cdot \Delta \mathbf{IIDX}$$

In the sections that follow, scene point positions **PT** are specified in ECEF coordinates except where noted. All are expressed in units of meters. Image grid locations **IL** are specified in image coordinates (xrow, ycol). Image coordinates are also in units of meters. Refer to Section 2.3 for the conventions used for all SICD products.

11.1 Image/Scene Pair Projection Parameters

For a given image/scene projection pair, the projection sensitivity matrices are computed for the COA projection geometry. An example projection pair **IL0** and **PT0** are shown in Figure 11-2. The example shown is for a monostatic image. The COA time for image location **IL0** is t_{COA}^{IL0} which, for the example shown, is not equal to the time t_{COA}^{SCP} .

$$\text{Image Location: } \mathbf{IL0} = \begin{bmatrix} xrow_0 \\ ycol_0 \end{bmatrix} \quad \text{Scene Point (ECEF): } \mathbf{PT0} = \begin{bmatrix} PT0_X \\ PT0_Y \\ PT0_Z \end{bmatrix}$$

For a monostatic image, the COA projection set for image location **IL0** is a set of 5 parameters. See Section 4. The projection set may or may not have had APOs applied.

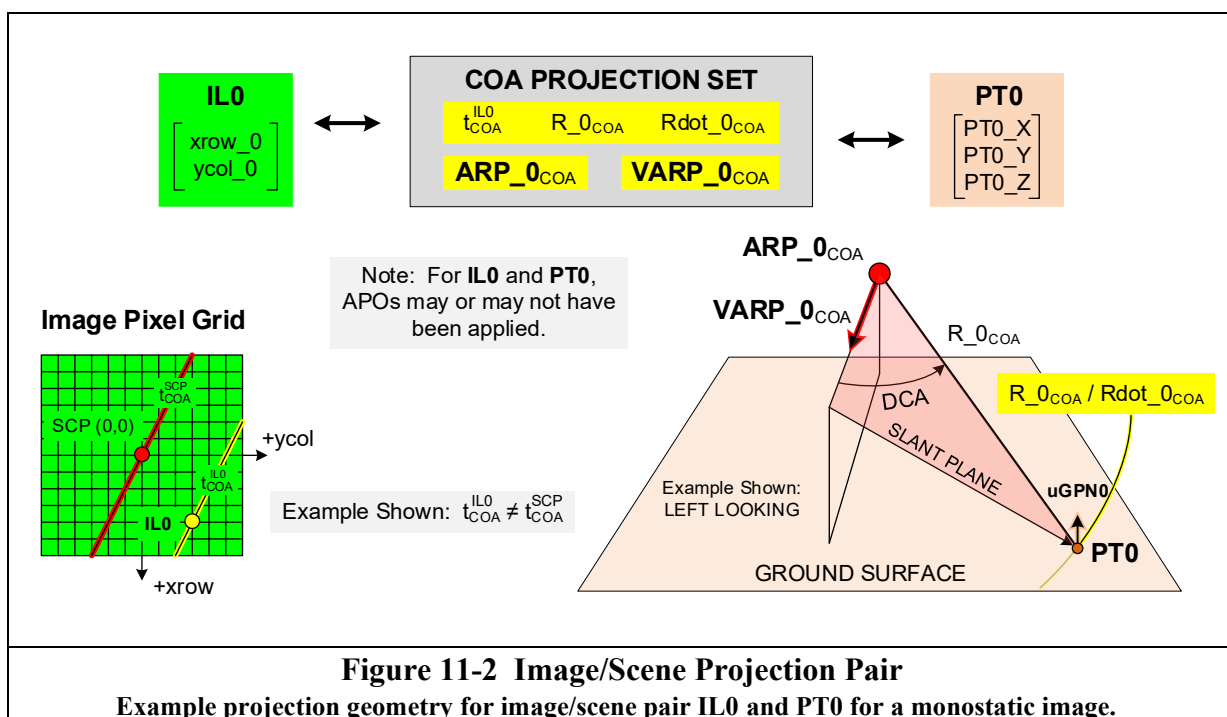
Monostatic Projection Set:

$$\{ t_{COA}^{IL0}, \mathbf{ARP_0}_{COA}, \mathbf{VARP_0}_{COA}, R_{0COA}, Rdot_{0COA} \}$$

For a bistatic image, the COA projection set is a set of 9 parameters. The projection set may or may not have had APOs applied.

Bistatic Projection Set:

$$\{ t_{COA}^{IL0}, tx_{COA}^{IL0}, tr_{COA}^{IL0}, \mathbf{Xmt_0}_{COA}, \mathbf{VXmt_0}_{COA}, \\ \mathbf{Rcv_0}_{COA}, \mathbf{VRcv_0}_{COA}, R_{Avg_0COA}, Rdot_{Avg_0COA} \}$$



At scene point **PT0**, the unit normal to the scene surface is **uGPN0**. The unit vector in the “up” direction is **uUP0** and is normal to surface of constant HAE at point **PT0**. The surface normal may or may not be aligned with the up direction but is always oriented “upward” and away from the center of the earth.

$$\mathbf{PT0} = \begin{bmatrix} \text{PT0_X} \\ \text{PT0_Y} \\ \text{PT0_Z} \end{bmatrix} \rightarrow \begin{bmatrix} \text{PT0_Lat} \\ \text{PT0_Lon} \\ \text{PT0_HAE} \end{bmatrix} \quad \mathbf{uUP0} = \begin{bmatrix} \cos(\text{PT0_Lat}) \cdot \cos(\text{PT0_Lon}) \\ \cos(\text{PT0_Lat}) \cdot \sin(\text{PT0_Lon}) \\ \sin(\text{PT0_Lat}) \end{bmatrix}$$

$$\mathbf{uGPN0} = \begin{bmatrix} \text{uGPN0_X} \\ \text{uGPN0_Y} \\ \text{uGPN0_Z} \end{bmatrix} \quad \mathbf{uGPN0} \text{ oriented “upward”}: \mathbf{uGPN0} \cdot \mathbf{uUP0} > 0$$

11.2 Projection Geometry Parameters

For the image/scene projection pair **IL0/PT0**, the COA projection set is used to compute an additional set of projection geometry parameters. Some parameters to be computed are dependent on the Collect Type for the image.

11.2.1 Collect Type = Monostatic

Use the monostatic COA projection set and the scene point position **PT0** to compute a set of monostatic-only parameters.

$$\{ \mathbf{uPT}, \mathbf{uPTDot} \}$$

(1) Use the ARP position and velocity to compute unit pointing vector **uPT** and its derivative with respect to time, **uPTDot**.

$$\mathbf{uPT} = \frac{1}{R_{0\text{COA}}} \cdot (\mathbf{ARP_0_{COA}} - \mathbf{PT0})$$

$$\mathbf{uPTDot} = \frac{1}{R_{0\text{COA}}} \cdot (\mathbf{VARP_0_{COA}} - R_{\text{dot}0\text{COA}} \cdot \mathbf{uPT})$$

11.2.2 Collect Type = Bistatic

Use the bistatic COA projection set and the scene point position **PT0** to compute a set of bistatic-only parameters.

$$\{ R_Xmt_0_{COA}, Rdot_Xmt_0_{COA}, R_Rcv_0_{COA}, Rdot_Rcv_0_{COA}, \\ \mathbf{uXmt}, \mathbf{uXmtDot}, \mathbf{uRcv}, \mathbf{uRcvDot}, \mathbf{bP}, \mathbf{bPDot}, TRTX_0, TRTXdot_0 \}$$

(1) Use the transmit APC position and velocity to compute unit vector **uXmt** and its derivative with respect. to time, **uXmtDot**. Compute the transmit APC acceleration at time $t = tx_{COA}^{IL0}$, **AXmt_0COA**. Also compute the second derivative of the range w.r.t. time, **Rddot_Xmt_0COA**.

$$R_Xmt_0_{COA} = | \mathbf{Xmt_0_{COA}} - \mathbf{PT0} | \quad \mathbf{uXmt} = \frac{1}{R_Xmt_0_{COA}} \bullet (\mathbf{Xmt_0_{COA}} - \mathbf{PT0})$$

$$Rdot_Xmt_0_{COA} = \mathbf{VXmt_0_{COA}} \bullet \mathbf{uXmt}$$

$$\mathbf{uXmtDot} = \frac{1}{R_Xmt_0_{COA}} \bullet (\mathbf{VXmt_0_{COA}} - Rdot_Xmt_0_{COA} \bullet \mathbf{uXmt})$$

$$\mathbf{AXmt_0_{COA}} = \frac{d^2}{dt^2} \mathbf{Xmt_Poly}(tx_{COA}^{IL0})$$

$$Rddot_Xmt_0_{COA} = \mathbf{AXmt_0_{COA}} \bullet \mathbf{uXmt} \\ + \frac{1}{R_Xmt_0_{COA}} \bullet (|\mathbf{VXmt_0_{COA}}|^2 - (Rdot_Xmt_0_{COA})^2)$$

(2) Use the receive APC position and velocity to compute unit vector **uRcv** and its derivative with respect. to time, **uRcvDot**. Compute the receive APC acceleration at time $t = tr_{COA}^{IL0}$, **ARcv_0COA**. Also compute the second derivative of the range w.r.t. time, **Rddot_Rcv_0COA**.

$$R_Rcv_0_{COA} = | \mathbf{Rcv_0_{COA}} - \mathbf{PT0} | \quad \mathbf{uRcv} = \frac{1}{R_Rcv_0_{COA}} \bullet (\mathbf{Rcv_0_{COA}} - \mathbf{PT0})$$

$$Rdot_Rcv_0_{COA} = \mathbf{VRcv_0_{COA}} \bullet \mathbf{uRcv}$$

$$\mathbf{uRcvDot} = \frac{1}{R_Rcv_0_{COA}} \bullet (\mathbf{VRcv_0_{COA}} - Rdot_Rcv_0_{COA} \bullet \mathbf{uRcv})$$

$$\mathbf{ARcv_0_{COA}} = \frac{d^2}{dt^2} \mathbf{Rcv_Poly}(tr_{COA}^{IL0})$$

$$Rddot_Rcv_0_{COA} = \mathbf{ARcv_0_{COA}} \bullet \mathbf{uRcv} \\ + \frac{1}{R_Rcv_0_{COA}} \bullet (|\mathbf{VRcv_0_{COA}}|^2 - (Rdot_Rcv_0_{COA})^2)$$

- (3) Compute the bistatic pointing vector, **bP**, and its derivative with respect to time, **bPDot**. Also compute the bistatic slant plane unit normal vector, **uSPN**.

$$\mathbf{bP} = \frac{1}{2} \bullet (\mathbf{uXmt} + \mathbf{uRcv}) \quad \mathbf{bPDot} = \frac{1}{2} \bullet (\mathbf{uXmtDot} + \mathbf{uRcvDot})$$

- (4) Compute parameters TRTX_0 and TRTXdot_0 shown. Parameter TRTX_0 is used to scale a change in transmit time at time $t = tx_{COA}^{ILO}$ to yield the change in time of arrival at the receive APC. Parameter TRTXdot_0 is the derivative of TRTX_0 w.r.t. slow time.

$$TRTX_0 = \frac{c + Rdot_Xmt_0_{COA}}{c - Rdot_Rcv_0_{COA}}$$

$$TRTXdot_0 = \frac{1}{c} \bullet (Rddot_Xmt_0_{COA} + Rddot_Rcv_0_{COA})$$

11.2.3 SPC & GPC Parameters

For all images, compute parameters that specify orientation of the Slant Plane Coordinate (SPC) frame and the Ground Plane Coordinate (GPC) frame relative to the ECEF frame. The method of computing the SPC unit basis vectors is dependent upon the type of collection.

- (1) Assign or compute vectors **SPX** and **SPZ** that point in the +SPX and +SPZ directions. The inputs used are dependent upon the collect type.

For Collect_Type = MONOSTATIC:

$$\mathbf{SPX} = \mathbf{uPT} \quad \mathbf{SPZ} = \text{LOOK} \bullet (\mathbf{uPT} \times \mathbf{uPTDot})$$

For Collect_Type = BISTATIC:

$$\mathbf{SPX} = \mathbf{bP} \quad \mathbf{SPZ} = \text{LOOK} \bullet (\mathbf{bP} \times \mathbf{bPDot})$$

- (2) Compute the SPC system unit basis vectors **uSPX**, **uSPY**, and **uSPZ**. Unit vector **uSPZ** is normal to the COA slant plane. Shown in Figure 11-3 is an example monostatic collection geometry. Shown in Figure 11-4 is an example bistatic collection geometry.

$$\mathbf{uSPX} = \frac{1}{|\mathbf{SPX}|} \bullet \mathbf{SPX} \quad \mathbf{uSPZ} = \frac{1}{|\mathbf{SPZ}|} \bullet \mathbf{SPZ} \quad \mathbf{uSPY} = \mathbf{uSPZ} \times \mathbf{uSPX}$$

- (3) Compute the GPC system unit basis vectors **uGPX**, **uGPY**, and **uGPZ**.

$$\mathbf{uGPZ} = \mathbf{uGPN0}$$

$$\mathbf{GPY} = \mathbf{uGPZ} \times \mathbf{uSPX} \quad \mathbf{uGPY} = \frac{1}{|\mathbf{GPY}|} \bullet \mathbf{GPY} \quad \mathbf{uGPX} = \mathbf{uGPY} \times \mathbf{uGPZ}$$

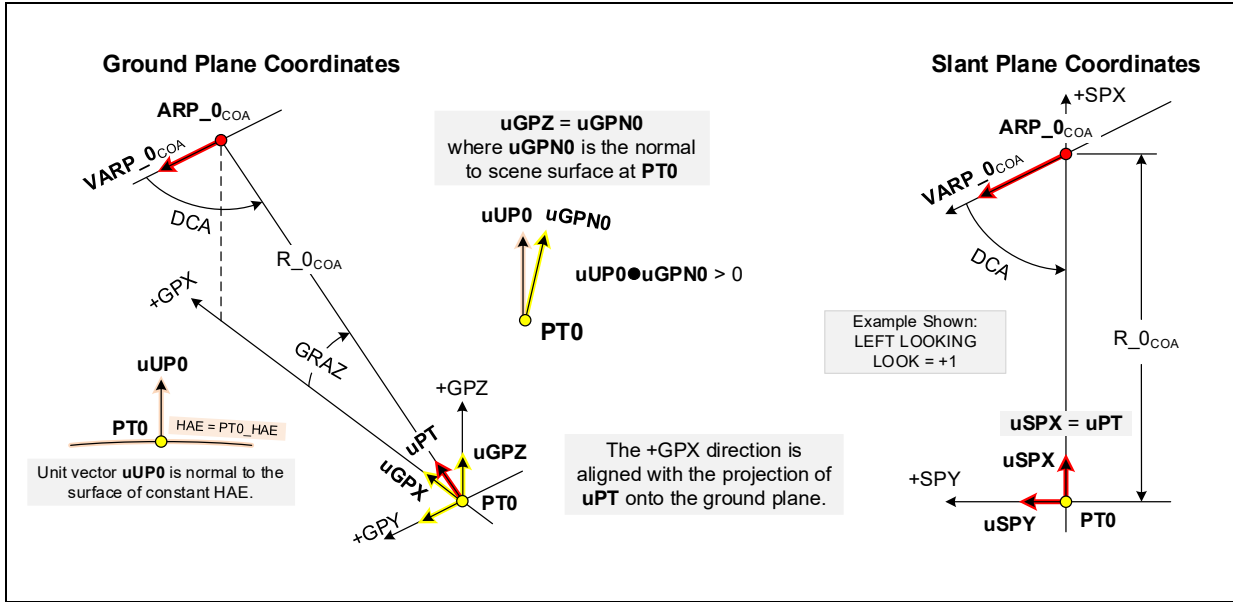


Figure 11-3 Monostatic GPC and SPC Frames
 The GPC frame and SPC frame are computed from vectors uPT and $uPTDot$.

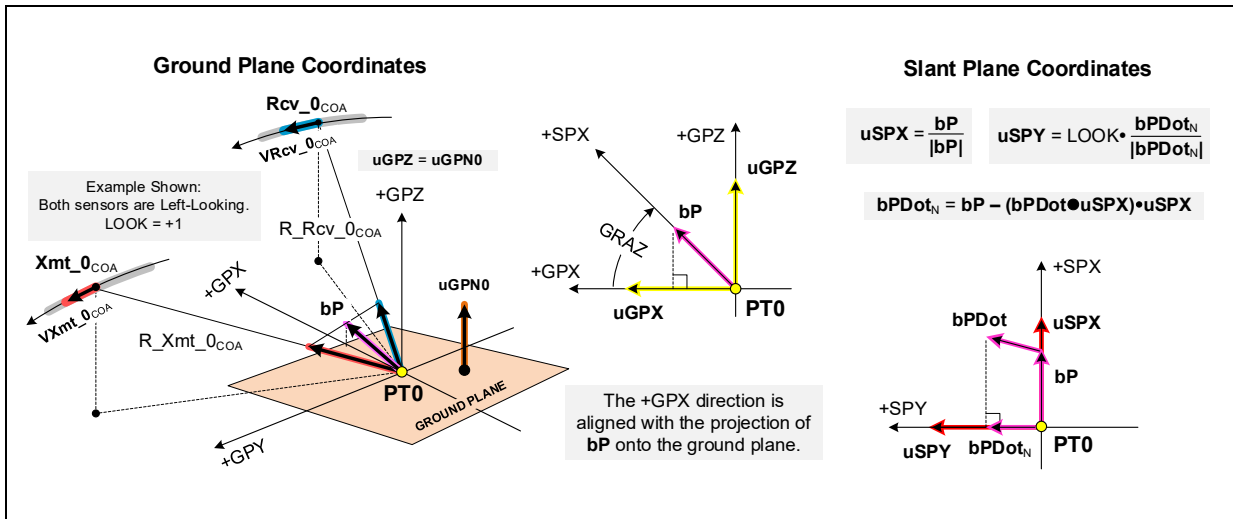


Figure 11-4 Bistatic GPC and SPC Frames
 The GPC frame and SPC frame are computed from vectors bP and $bPDot$.

(4) The grazing angle, GRAZ, and the twist angle, TWST, are used to project points located in the ground plane into the slant plane with the projection along the slant plane normal. Compute cosine, sine, and tangent of the grazing angle. Compute the cosine, sine, and tangent of the twist angle.

$$\cos_GRAZ = uSPX \bullet uGPX$$

$$\sin_GRAZ = uSPX \bullet uGPZ$$

$$\cos_TWST = uSPY \bullet uGPY$$

$$\sin_TWST = -uSPZ \bullet uGPY$$

$$\tan_GRAZ = \frac{\sin_GRAZ}{\cos_GRAZ}$$

$$\tan_TWST = \frac{\sin_TWST}{\cos_TWST}$$

11.3 Slant Plane Sensitivity Matrices

The sensitivity matrices that relate changes in scene point position to slant plane projection point are defined in Table 11-2. Also listed are the matrices that relate changes in COA range and range rate to changes in slant plane projection point. For a monostatic image, the changes in range and range rate are relative to the COA projection set range and range rate relative to ARP. For a bistatic image, the changes in range and range rate are relative to the COA projection set average range and average range rate relative to the transmit and receive APCs.

Table 11-2 Slant Plane Sensitivity Matrices		
Sensitivities computed for COA projection set positions and velocities held constant. If APOs are applied, they are applied to the COA projection set for all image grid locations.		
$\Delta\mathbf{PT} = \begin{bmatrix} \Delta\mathbf{PT}_X \\ \Delta\mathbf{PT}_Y \\ \Delta\mathbf{PT}_Z \end{bmatrix} \quad \Delta\mathbf{GPXY} = \begin{bmatrix} \Delta\mathbf{GPX} \\ \Delta\mathbf{GPY} \end{bmatrix} \quad \Delta\mathbf{SPXY} = \begin{bmatrix} \Delta\mathbf{SPX} \\ \Delta\mathbf{SPY} \end{bmatrix}$		
Matrix	Size	Description
$\mathbf{M}_{\mathbf{PT}}^{\mathbf{SPXY}}$	2 x 3	Sensitivity matrix to compute the change in slant plane projection point ($\Delta\mathbf{SPXY}$) due to a change in scene point position in ECEF coordinates ($\Delta\mathbf{PT}$). Projection along the slant plane normal. $\Delta\mathbf{SPXY} = \mathbf{M}_{\mathbf{PT}}^{\mathbf{SPXY}} \cdot \Delta\mathbf{PT}$
$\mathbf{M}_{\mathbf{GPXY}}^{\mathbf{SPXY}}$	2 x 2	Sensitivity matrix to compute the change in slant plane projection point ($\Delta\mathbf{SPXY}$) due to a change in ground plane point position ($\Delta\mathbf{GPXY}$). Projection along the slant plane normal. $\Delta\mathbf{SPXY} = \mathbf{M}_{\mathbf{GPXY}}^{\mathbf{SPXY}} \cdot \Delta\mathbf{GPXY}$
$\mathbf{M}_{\mathbf{SPXY}}^{\mathbf{GPXY}}$	2 x 2	Sensitivity matrix to compute the change in ground plane point position ($\Delta\mathbf{GPXY}$) due to a change in slant plane projection point ($\Delta\mathbf{SPXY}$). Projection along the slant plane normal. $\Delta\mathbf{GPXY} = \mathbf{M}_{\mathbf{SPXY}}^{\mathbf{GPXY}} \cdot \Delta\mathbf{SPXY}$
$\mathbf{M}_{\mathbf{GPXY}}^{\mathbf{PT}}$	3 x 2	Sensitivity matrix to compute the change in scene point in ECEF coordinates ($\Delta\mathbf{PT}$) due to a change in ground plane point position in GP coordinates ($\Delta\mathbf{GPXY}$). $\Delta\mathbf{PT} = \mathbf{M}_{\mathbf{GPXY}}^{\mathbf{PT}} \cdot \Delta\mathbf{GPXY}$
$\mathbf{MIL}_{\mathbf{HAE}}^{\mathbf{PT}}$	3 x 1	Sensitivity matrix to compute the change in scene point in ECEF coordinates ($\Delta\mathbf{PT}$) due to a change in scene surface height ($\Delta\mathbf{HAE}$). The displacement is along the slant plane normal ($\mathbf{uSPZ}$). $\Delta\mathbf{PT} = \mathbf{MIL}_{\mathbf{HAE}}^{\mathbf{PT}} \cdot \Delta\mathbf{HAE}$
MONOSTATIC: $\Delta\mathbf{RRdot} = \begin{bmatrix} \Delta\mathbf{R_ARP} \\ \Delta\mathbf{Rdot_ARP} \end{bmatrix}$ BISTATIC: $\Delta\mathbf{RRdot} = \begin{bmatrix} \Delta\mathbf{R_Avg} \\ \Delta\mathbf{Rdot_Avg} \end{bmatrix}$		

Table 11-2 Slant Plane Sensitivity Matrices

$\mathbf{M}_{\text{SPXY}}^{\text{RRdot}}$	2×2	Sensitivity matrix to compute the change in COA range and range rate (ΔRRdot) due to a change in slant plane projection point (ΔSPXY). $\Delta\text{RRdot} = \mathbf{M}_{\text{SPXY}}^{\text{RRdot}} \bullet \Delta\text{SPXY}$
$\mathbf{M}_{\text{RRdot}}^{\text{SPXY}}$	2×2	Sensitivity matrix to compute the change in slant plane projection point (ΔSPXY) due to a change in COA range and range rate (ΔRRdot). $\Delta\text{SPXY} = \mathbf{M}_{\text{RRdot}}^{\text{SPXY}} \bullet \Delta\text{RRdot}$

- (1) Form the 2×3 sensitivity matrix used to compute the change in slant plane projection point ΔSPXY due to a change in scene point position in ECEF coordinates ΔPT .

$$\mathbf{M}_{\text{PT}}^{\text{SPXY}} = \begin{bmatrix} \mathbf{uSPX}^T \\ \mathbf{uSPY}^T \end{bmatrix}$$

- (2) Form the 2×2 sensitivity matrix used to compute the change in slant plane projection point ΔSPXY due to a change in ground plane plane position in GPX and GPY coordinates ΔGPXY .

$$\mathbf{M}_{\text{GPXY}}^{\text{SPXY}} = \begin{bmatrix} \cos_GRAZ & 0 \\ -\sin_GRAZ \bullet \sin_TWST & \cos_TWST \end{bmatrix}$$

- (3) Form the 2×2 sensitivity matrix used to compute the change in ground plane point position ΔGPXY due to a change in slant plane projection point ΔSPXY .

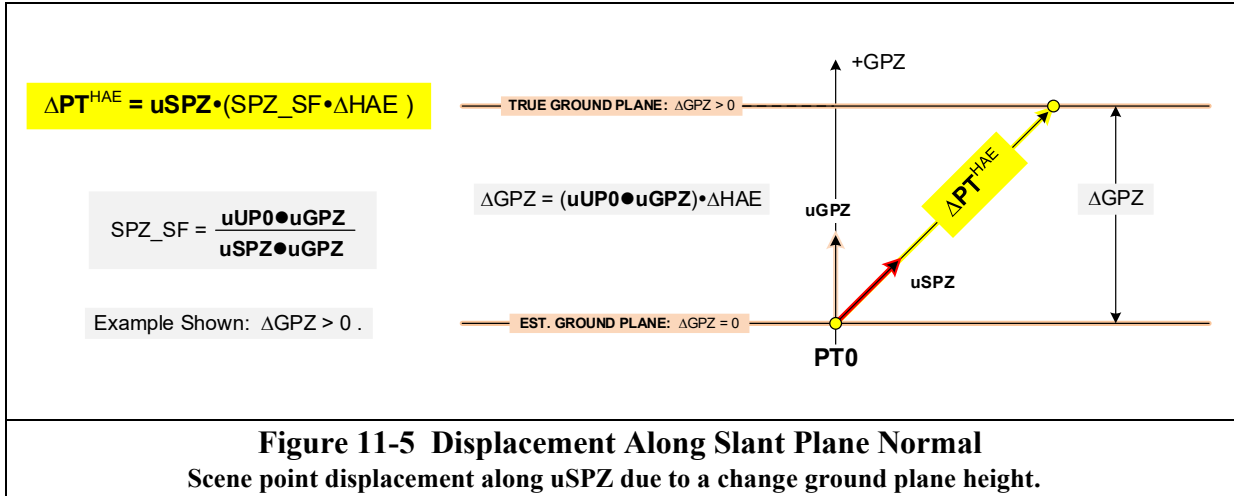
$$\mathbf{M}_{\text{SPXY}}^{\text{GPXY}} = \begin{bmatrix} \frac{1}{\cos_GRAZ} & 0 \\ \tan_GRAZ \bullet \tan_TWST & \frac{1}{\cos_TWST} \end{bmatrix}$$

- (4) Form the 3×2 sensitivity matrix used to compute a change in position in ECEF coordinates ΔPT from a change in ground plane point position expressed in ground plane coordinates GPX and GPY, ΔGPXY .

$$\mathbf{M}_{\text{GPXY}}^{\text{PT}} = [\mathbf{uGPX} \quad \mathbf{uGPY}]$$

- (5) Form the 3×1 sensitivity matrix $\mathbf{MIL}_{\text{HAE}}^{\text{PT}}$ used to compute a change in position in ECEF coordinates ΔPT due to a change in scene surface height ΔHAE . Refer to Figure 11-5. The displacement is along the slant plane normal ($\mathbf{uSPZ}$). Scale factor SF_SPZ accounts for: (1) the difference between the ETP normal and the ground plane normal (if any), and (2) the difference between the ground plane normal and the slant plane normal.

$$\text{SPZ_SF} = \frac{\mathbf{uUP0} \bullet \mathbf{uGPZ}}{\mathbf{uSPZ} \bullet \mathbf{uGPZ}} \quad \mathbf{MIL}_{\text{HAE}}^{\text{PT}} = \text{SPZ_SF} \bullet \mathbf{uSPZ}$$



- (6) Assign vectors **P** and **PDot** based on the type of collection.

For Collect_Type = MONOSTATIC:

$$\mathbf{P} = \mathbf{uPT} \quad \mathbf{PDot} = \mathbf{uPTDot}$$

For Collect_Type = BISTATIC:

$$\mathbf{P} = \mathbf{bP} \quad \mathbf{PDot} = \mathbf{bPDot}$$

- (7) Form the 2 x 2 sensitivity matrix used to compute the change in COA range and range rate $\Delta RRdot$ due to a change in slant plane projection point $\Delta SPXY$.

$$\mathbf{M}_{SPXY}^{RRdot} = \begin{bmatrix} \frac{\Delta R}{\Delta SPX} & \frac{\Delta R}{\Delta SPY} \\ \frac{\Delta Rdot}{\Delta SPX} & \frac{\Delta Rdot}{\Delta SPY} \end{bmatrix} = (-1) \cdot \begin{bmatrix} \mathbf{P} \bullet \mathbf{uSPX} & 0 \\ \mathbf{PDot} \bullet \mathbf{uSPX} & \mathbf{PDot} \bullet \mathbf{uSPY} \end{bmatrix}$$

- (8) Form the 2 x 2 sensitivity matrix used to compute the change in slant plane projection point $\Delta SPXY$ from a change in COA range and range rate $\Delta RRdot$.

$$\mathbf{M}_{RRdot}^{SPXY} = \begin{bmatrix} \frac{\Delta SPX}{\Delta R} & \frac{\Delta SPX}{\Delta Rdot} \\ \frac{\Delta SPY}{\Delta R} & \frac{\Delta SPY}{\Delta Rdot} \end{bmatrix} = \{\mathbf{M}_{SPXY}^{RRdot}\}^{-1}$$

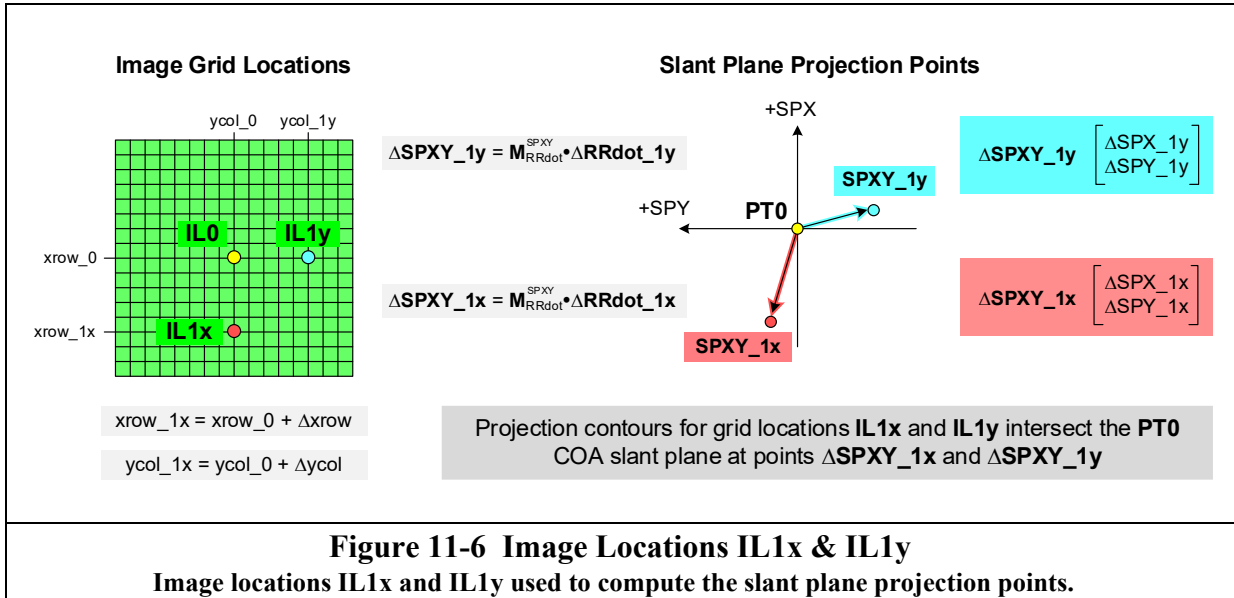
11.4 Image Location Sensitivity Matrices

Sensitivity matrices that relate changes in image grid location to changes in scene point position and slant plane projection point are listed in Table 11-3. Also listed are the matrices that relate changes in image grid location to changes in COA range and range rate.

Table 11-3 Image Location Sensitivity Matrices		
Sensitivities computed for smoothly varying COA time versus image grid location. If APOs are applied, they are applied to the COA projection set for all image grid locations.		
$\Delta \mathbf{IL} = \begin{bmatrix} \Delta x_{row} \\ \Delta y_{col} \end{bmatrix} \quad \Delta \mathbf{SPXY} = \begin{bmatrix} \Delta SPX \\ \Delta SPY \end{bmatrix} \quad \Delta \mathbf{GPXY} = \begin{bmatrix} \Delta GPX \\ \Delta GPY \end{bmatrix}$		
Matrix	Size	Description
$\mathbf{M}_{PT}^{IL}$	2 x 3	Sensitivity matrix to compute the change in image grid location ($\Delta \mathbf{IL}$) due to a change in scene point position ($\Delta \mathbf{PT}$). $\Delta \mathbf{IL} = \mathbf{M}_{PT}^{IL} \cdot \Delta \mathbf{PT}$
$\mathbf{M}_{IL}^{GPXY}$	2 x 2	Sensitivity matrix to compute the change in ground plane point position ($\Delta \mathbf{GPXY}$) due to a change in image grid location ($\Delta \mathbf{IL}$). $\Delta \mathbf{GPXY} = \mathbf{M}_{IL}^{GPXY} \cdot \Delta \mathbf{IL}$
$\mathbf{M}_{IL}^{SPXY}$	2 x 2	Sensitivity matrix to compute the change in slant plane projection point ($\Delta \mathbf{SPXY}$) due to a change in image grid location ($\Delta \mathbf{IL}$). $\Delta \mathbf{SPXY} = \mathbf{M}_{IL}^{SPXY} \cdot \Delta \mathbf{IL}$
$\mathbf{M}_{SPXY}^{IL}$	2 x 2	Sensitivity matrix to compute the change in image grid location ($\Delta \mathbf{IL}$) due to a change in slant plane point position ($\Delta \mathbf{SPXY}$). $\Delta \mathbf{IL} = \mathbf{M}_{SPXY}^{IL} \cdot \Delta \mathbf{SPXY}$
MONOSTATIC: $\Delta \mathbf{RRdot} = \begin{bmatrix} \Delta R_{ARP} \\ \Delta Rdot_{ARP} \end{bmatrix}$ BISTATIC: $\Delta \mathbf{RRdot} = \begin{bmatrix} \Delta R_{Avg} \\ \Delta Rdot_{Avg} \end{bmatrix}$		
$\mathbf{M}_{IL}^{RRdot}$	2 x 2	Sensitivity matrix to compute the change in COA range and range rate ($\Delta \mathbf{RRdot}$) due to a change in image grid location ($\Delta \mathbf{IL}$). $\Delta \mathbf{RRdot} = \mathbf{M}_{IL}^{RRdot} \cdot \Delta \mathbf{IL}$
$\mathbf{M}_{RRdot}^{IL}$	2 x 2	Sensitivity matrix to compute the change in image grid location ($\Delta \mathbf{IL}$) due to a change in COA range and range rate ($\Delta \mathbf{RRdot}$). $\Delta \mathbf{IL} = \mathbf{M}_{RRdot}^{IL} \cdot \Delta \mathbf{RRdot}$

The sensitivity matrix that relates a change in image location $\Delta \mathbf{IL}$ to a change in COA range and range rate $\Delta \mathbf{RRdot}$ is computed as follows. Refer to Figure 11-6. First, the change in COA range and range rate, $\Delta \mathbf{RRdot_1x}$, due to a change in the row coordinate, $\Delta xrow$, is computed with the column coordinate held constant. Second, the change in COA range and range rate, $\Delta \mathbf{RRdot_1y}$, due to a change in the column coordinate, $\Delta ycol$, is computed with the row coordinate held constant. The 2 x 2 sensitivity matrix $\mathbf{M}_{IL}^{RRdot}$ is formed from the vector $\Delta \mathbf{RRdot_1x}$ scaled by $1/\Delta xrow$ and the vector $\Delta \mathbf{RRdot_1y}$ scaled by $1/\Delta ycol$.

$$\mathbf{M}_{IL}^{RRdot} = \begin{bmatrix} \frac{1}{\Delta xrow} \cdot \Delta \mathbf{RRdot_1x} & \frac{1}{\Delta ycol} \cdot \Delta \mathbf{RRdot_1y} \end{bmatrix}$$



11.4.1 Compute IL1x and IL1y

- (1) Select an increment in the row coordinate, $\Delta xrow$. Compute image grid location **IL1x** from location **IL0** by adding increment $\Delta xrow$ to the $xrow$ coordinate. Suggested value: $\Delta xrow = \text{Min}(\text{Row_SS}, 1.0 \text{ m})$.

$$\mathbf{IL1x} = \mathbf{IL0} + \begin{bmatrix} \Delta xrow \\ 0 \end{bmatrix} = \begin{bmatrix} xrow_0 + \Delta xrow \\ ycol_0 \end{bmatrix}$$

- (2) Select an increment in the column coordinate, $\Delta ycol$. Compute image grid location **IL1y** from location **IL0** by adding increment $\Delta ycol$ to the $ycol$ coordinate. Suggested value: $\Delta ycol = \text{Min}(\text{Col_SS}, 1.0 \text{ m})$.

$$\mathbf{IL1y} = \mathbf{IL0} + \begin{bmatrix} 0 \\ \Delta ycol \end{bmatrix} = \begin{bmatrix} xrow_0 \\ ycol_0 + \Delta ycol \end{bmatrix}$$

The method for computing the changes in COA range and range rate, $\Delta\mathbf{RRdot_1x}$ and $\Delta\mathbf{RRdot_1y}$, is dependent on the Collect Type.

11.4.2 Collect Type = Monostatic

For a monostatic image, vectors of $\Delta\mathbf{RRdot_1x}$ and $\Delta\mathbf{RRdot_1y}$ are defined as shown. Compute the elements of each vector using the method specified below.

$$\Delta\mathbf{RRdot_1x} = \begin{bmatrix} \Delta R_1x \\ \Delta Rdot_1x \end{bmatrix} \quad \Delta\mathbf{RRdot_1y} = \begin{bmatrix} \Delta R_1y \\ \Delta Rdot_1y \end{bmatrix}$$

- (1) Compute the monostatic COA projection set for image location **IL1x**. Apply APOs if available and if applied to the COA projection set for **IL0**.

$$\{ t_{COA}^{IL1x}, \mathbf{ARP_1x_{COA}}, \mathbf{VARP_1x_{COA}}, R_1x_{COA}, Rdot_1x_{COA} \}$$

- (2) Compute the change in ARP COA position and velocity for image location **IL1x** relative to **IL0**.

$$\Delta\mathbf{ARP_1x_{COA}} = \mathbf{ARP_1x_{COA}} - \mathbf{ARP_0_{COA}}$$

$$\Delta\mathbf{VARP_1x_{COA}} = \mathbf{VARP_1x_{COA}} - \mathbf{VARP_0_{COA}}$$

- (3) Compute the effective change in range and range rate for the **IL1x** projection contour relative to the **IL0** projection contour.

$$\Delta R_1x = R_1x_{COA} - R_0_{COA} - (\Delta\mathbf{ARP_1x_{COA}} \bullet \mathbf{u_{PT}})$$

$$\begin{aligned} \Delta Rdot_1x = & Rdot_1x_{COA} - Rdot_0_{COA} \\ & - (\Delta\mathbf{ARP_1x_{COA}} \bullet \mathbf{u_{PTDot}} + \Delta\mathbf{VARP_1x_{COA}} \bullet \mathbf{u_{PT}}) \end{aligned}$$

- (4) Compute the monostatic COA projection set for image location **IL1y**. Apply APOs if available and if applied to the COA projection set for **IL0**.

$$\{ t_{COA}^{IL1y}, \mathbf{ARP_1y_{COA}}, \mathbf{VARP_1y_{COA}}, R_1y_{COA}, Rdot_1y_{COA} \}$$

- (5) Compute the change in ARP COA position and velocity for image location **IL1y** relative to **IL0**.

$$\Delta\mathbf{ARP_1y_{COA}} = \mathbf{ARP_1y_{COA}} - \mathbf{ARP_0_{COA}}$$

$$\Delta\mathbf{VARP_1y_{COA}} = \mathbf{VARP_1y_{COA}} - \mathbf{VARP_0_{COA}}$$

- (6) Compute the effective change in range and range rate for the **IL1y** projection contour relative to the **IL0** projection contour.

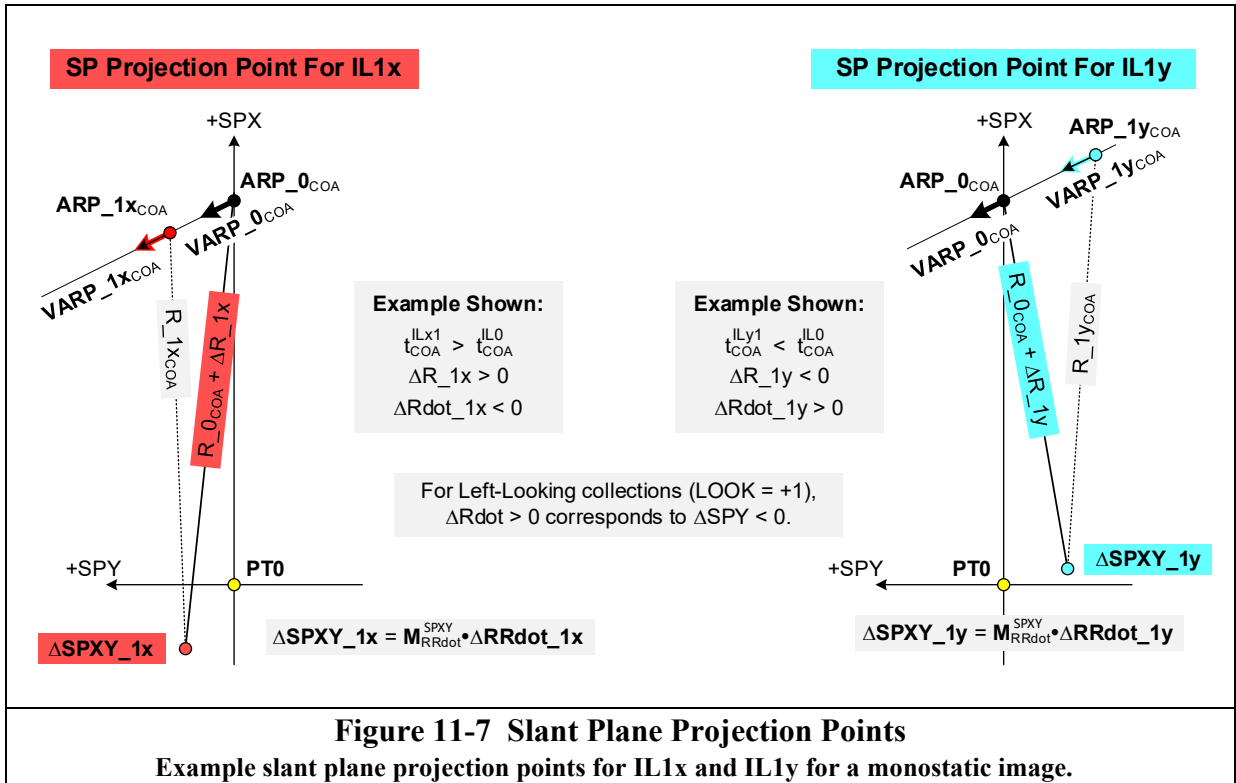
$$\Delta R_{1y} = R_{1y_{COA}} - R_{0_{COA}} - (\Delta ARP_{1y_{COA}} \bullet u_{PT})$$

$$\Delta Rdot_{1y} = Rdot_{1y_{COA}} - Rdot_{0_{COA}} - (\Delta ARP_{1y_{COA}} \bullet u_{PTDot} + \Delta VARP_{1y_{COA}} \bullet u_{PT})$$

- (7) Compute the elements of the matrix M_{IL}^{RRdot} as shown below. Matrix M_{RRdot}^{IL} is computed as by taking the inverse.

$$M_{IL}^{RRdot} = \begin{bmatrix} \frac{\Delta R_{1x}}{\Delta x_{row}} & \frac{\Delta R_{1y}}{\Delta y_{col}} \\ \frac{\Delta Rdot_{1x}}{\Delta x_{row}} & \frac{\Delta Rdot_{1y}}{\Delta y_{col}} \end{bmatrix} \quad M_{RRdot}^{IL} = (M_{IL}^{RRdot})^{-1}$$

Shown in Figure 11-7 are example slant plane projection points for a left-looking monostatic collection. Image grid locations **IL1x** and **IL1y** and slant plane projection points $\Delta SPXY_{1x}$ and $\Delta SPXY_{1y}$ are as shown in Figure 11-5. For the example shown, the COA time varies with image grid location and for which $t_{COA}^{IL1y} < t_{COA}^{IL0} < t_{COA}^{IL1x}$.



11.4.3 Collect Type = Bistatic

For a bistatic image, vectors of $\Delta\mathbf{RRdot_1x}$ and $\Delta\mathbf{RRdot_1y}$ are defined as shown. Compute the elements of each vector using the method specified below.

$$\Delta\mathbf{RRdot_1x} = \begin{bmatrix} \Delta R_Avg_1x \\ \Delta Rdot_Avg_1x \end{bmatrix} \quad \Delta\mathbf{RRdot_1y} = \begin{bmatrix} \Delta R_Avg_1y \\ \Delta Rdot_Avg_1y \end{bmatrix}$$

- (1) Compute the bistatic COA projection set for image location **IL1x**. Apply APOs if available and if applied to the COA projection set for **IL0**.

$$\{ t_{COA}^{IL1x}, tx_{COA}^{IL1x}, tr_{COA}^{IL1x}, \mathbf{Xmt_1x_{COA}}, \mathbf{VXmt_1x_{COA}}, \mathbf{Rcv_1x_{COA}}, \mathbf{VRcv_1x_{COA}}, R_Avg_1x_{COA}, Rdot_Avg_1x_{COA} \}$$

- (2) Compute the change in APC COA positions and velocities for image location **IL1x** relative to **IL0**.

$$\Delta\mathbf{Xmt_1x_{COA}} = \mathbf{Xmt_1x_{COA}} - \mathbf{Xmt_0_{COA}}$$

$$\Delta\mathbf{VXmt_1x_{COA}} = \mathbf{VXmt_1x_{COA}} - \mathbf{VXmt_0_{COA}}$$

$$\Delta\mathbf{Rcv_1x_{COA}} = \mathbf{Rcv_1x_{COA}} - \mathbf{Rcv_0_{COA}}$$

$$\Delta\mathbf{VRcv_1x_{COA}} = \mathbf{VRcv_1x_{COA}} - \mathbf{VRcv_0_{COA}}$$

- (3) Compute the effective change in average range and average range rate for the **IL1x** projection contour relative the **IL0** projection contour.

$$\begin{aligned} \Delta R_Avg_1x &= R_Avg_1x_{COA} - R_Avg_0_{COA} \\ &\quad - \frac{1}{2} \cdot (\Delta\mathbf{Xmt_1x_{COA}} \bullet \mathbf{uXmt} + \Delta\mathbf{Rcv_1x_{COA}} \bullet \mathbf{uRcv}) \end{aligned}$$

$$\begin{aligned} \Delta Rdot_Avg_1x &= Rdot_Avg_1x_{COA} - Rdot_Avg_0_{COA} \\ &\quad - \frac{1}{2} \cdot (\Delta\mathbf{Xmt_1x_{COA}} \bullet \mathbf{uXmtDot} + \Delta\mathbf{VXmt_1x_{COA}} \bullet \mathbf{uXmt}) \\ &\quad - \frac{1}{2} \cdot (\Delta\mathbf{Rcv_1x_{COA}} \bullet \mathbf{uRcvDot} + \Delta\mathbf{VRcv_1x_{COA}} \bullet \mathbf{uRcv}) \end{aligned}$$

- (4) Compute the bistatic COA projection set for image location **IL1y**. Apply APOs if available and if applied to the COA projection set for **IL0**.

$$\{ t_{COA}^{IL1y}, tx_{COA}^{IL1y}, tr_{COA}^{IL1y}, \mathbf{Xmt_1y_{COA}}, \mathbf{VXmt_1y_{COA}}, \mathbf{Rcv_1y_{COA}}, \mathbf{VRcv_1y_{COA}}, R_Avg_1y_{COA}, Rdot_Avg_1y_{COA} \}$$

- (5) Compute the change in APC COA positions and velocities for image location **IL1y** relative to **IL0**.

$$\Delta \mathbf{X}_{mt_1y_{COA}} = \mathbf{X}_{mt_1y_{COA}} - \mathbf{X}_{mt_0_{COA}}$$

$$\Delta \mathbf{V}_{Xmt_1y_{COA}} = \mathbf{V}_{Xmt_1y_{COA}} - \mathbf{V}_{Xmt_0_{COA}}$$

$$\Delta \mathbf{R}_{cv_1y_{COA}} = \mathbf{R}_{cv_1y_{COA}} - \mathbf{R}_{cv_0_{COA}}$$

$$\Delta \mathbf{V}_{Rcv_1y_{COA}} = \mathbf{V}_{Rcv_1y_{COA}} - \mathbf{V}_{Rcv_0_{COA}}$$

- (6) Compute the effective change in average range and average range rate for the **IL1y** projection contour relative the **IL0** projection contour.

$$\Delta R_Avg_1y = R_Avg_1y_{COA} - R_Avg_0_{COA}$$

$$- \frac{1}{2} \bullet (\Delta \mathbf{X}_{mt_1y_{COA}} \bullet \mathbf{u}_{Xmt} + \Delta \mathbf{R}_{cv_1y_{COA}} \bullet \mathbf{u}_{Rcv})$$

$$\Delta Rdot_Avg_1y = Rdot_Avg_1y_{COA} - Rdot_Avg_0_{COA}$$

$$- \frac{1}{2} \bullet (\Delta \mathbf{X}_{mt_1y_{COA}} \bullet \mathbf{u}_{XmtDot} + \Delta \mathbf{V}_{Xmt_1y_{COA}} \bullet \mathbf{u}_{Xmt})$$

$$- \frac{1}{2} \bullet (\Delta \mathbf{R}_{cv_1y_{COA}} \bullet \mathbf{u}_{RcvDot} + \Delta \mathbf{V}_{Rcv_1y_{COA}} \bullet \mathbf{u}_{Rcv})$$

- (7) Compute the elements of the matrix $\mathbf{M}_{IL}^{RRdot}$ as shown below. Matrix $\mathbf{M}_{RRdot}^{IL}$ is computed as by taking the inverse.

$$\mathbf{M}_{IL}^{RRdot} = \begin{bmatrix} \frac{\Delta R_Avg_1x}{\Delta x_{row}} & \frac{\Delta R_Avg_1y}{\Delta y_{col}} \\ \frac{\Delta Rdot_Avg_1x}{\Delta x_{row}} & \frac{\Delta Rdot_Avg_1y}{\Delta y_{col}} \end{bmatrix} \quad \mathbf{M}_{RRdot}^{IL} = (\mathbf{M}_{IL}^{RRdot})^{-1}$$

11.4.4 Image Location & Slant Plane Sensitivity

Compute the matrices that relate changes in image location to changes in slant plane projection points and scene point positions using the matrices $\mathbf{M}_{IL}^{RRdot}$ and $\mathbf{M}_{RRdot}^{IL}$ and matrices $\mathbf{M}_{SPXY}^{RRdot}$ and $\mathbf{M}_{RRdot}^{SPXY}$.

- (1) Compute the 2 x 2 matrices that relate changes in image location $\Delta \mathbf{IL}$ to changes in slant plane projection point $\Delta \mathbf{SPXY}$.

$$\mathbf{M}_{IL}^{SPXY} = \mathbf{M}_{RRdot}^{SPXY} \bullet \mathbf{M}_{IL}^{RRdot} \quad \mathbf{M}_{SPXY}^{IL} = \mathbf{M}_{RRdot}^{IL} \bullet \mathbf{M}_{SPXY}^{RRdot}$$

- (2) Compute the 2 x 2 matrix that relates a change in image location $\Delta \mathbf{IL}$ to a change in ground plane point $\Delta \mathbf{GPXY}$.

$$\mathbf{M}_{\mathbf{IL}}^{\mathbf{GPXY}} = \mathbf{M}_{\mathbf{SPXY}}^{\mathbf{GPXY}} \cdot \mathbf{M}_{\mathbf{IL}}^{\mathbf{SPXY}}$$

- (3) Compute the 2 x 3 matrix that relates a change in scene point position $\Delta \mathbf{PT}$ to a change in image location $\Delta \mathbf{IL}$.

$$\mathbf{M}_{\mathbf{PT}}^{\mathbf{IL}} = \mathbf{M}_{\mathbf{SPXY}}^{\mathbf{IL}} \cdot \mathbf{M}_{\mathbf{PT}}^{\mathbf{SPXY}}$$

11.5 Position/Velocity/Timing Sensitivity Matrices

The matrices that relate changes in sensor position, velocity, and control clock timing to an effective change in COA range and range rate are computed based upon the collect type for the image.

11.5.1 Monostatic P/V/T Matrices

For a monostatic image, the matrices that relate changes in sensor position, velocity, and control clock frequency to an equivalent change in COA range and range rate are listed in Table 11-4. The independent variables include a change in ARP position at time $t = t_{\text{COA}}^{\text{SCP}}$ and a change in ARP velocity that is constant across the slow time period processed to form the image.

Table 11-4 Monostatic Position/Velocity/Timing Sensitivity Matrices		
Sensitivities for a given image location/scene point pair. If APOs are applied, they are applied to the COA projection set for all image grid locations.		
$\Delta \mathbf{ARP}_{\text{COA}}^{\text{SCP}} = \begin{bmatrix} \Delta \text{ARP_X} \\ \Delta \text{ARP_Y} \\ \Delta \text{ARP_Z} \end{bmatrix} \quad \Delta \mathbf{VARP} = \begin{bmatrix} \Delta \text{VARP_X} \\ \Delta \text{VARP_Y} \\ \Delta \text{VARP_Z} \end{bmatrix} \quad \Delta \mathbf{RRdot} = \begin{bmatrix} \Delta \text{R_ARP} \\ \Delta \text{Rdot_ARP} \end{bmatrix}$ For image/scene pair IL0/PT0 with COA time $t_{\text{COA}}^{\text{ILO}}$, the change in ARP position is: $\Delta \mathbf{ARP}(t_{\text{COA}}^{\text{ILO}}) = \Delta \mathbf{ARP}_{\text{COA}}^{\text{SCP}} + \Delta \mathbf{VARP} \cdot (t_{\text{COA}}^{\text{ILO}} - t_{\text{COA}}^{\text{SCP}})$		
Matrix	Size	Description
$\mathbf{M}_{\Delta \text{ARP}}^{\text{RRdot}}$	2 x 3	Sensitivity matrix to compute the change in COA range and range rate ($\Delta \mathbf{RRdot}$) due to a change in ARP position at $t = t_{\text{COA}}^{\text{SCP}}$ ($\Delta \mathbf{ARP}_{\text{COA}}^{\text{SCP}}$). $\Delta \mathbf{RRdot} = \mathbf{M}_{\Delta \text{ARP}}^{\text{RRdot}} \cdot \Delta \mathbf{ARP}_{\text{COA}}^{\text{SCP}}$
$\mathbf{M}_{\Delta \text{VARP}}^{\text{RRdot}}$	2 x 3	Sensitivity matrix to compute the change in COA range and range rate ($\Delta \mathbf{RRdot}$) due to a change in ARP velocity ($\Delta \mathbf{VARP}$). $\Delta \mathbf{RRdot} = \mathbf{M}_{\Delta \text{VARP}}^{\text{RRdot}} \cdot \Delta \mathbf{VARP}$

Table 11-4 Monostatic Position/Velocity/Timing Sensitivity Matrices

$\mathbf{M}_{\text{CLK_SF}}^{\text{RRdot}}$	2×1	Sensitivity matrix to compute the change in COA range and range rate ($\Delta\mathbf{RRdot}$) due to a change in control clock frequency expressed as a scale factor ($\text{CLK_SF} = \Delta f_{\text{CLK}} / f_{\text{Clk}_0}$). $\Delta\mathbf{RRdot} = \mathbf{M}_{\text{CLK_SF}}^{\text{RRdot}} \cdot \text{CLK_SF}$
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- (1) Form the 2×3 sensitivity matrix used to compute a change in COA range and range rate $\Delta\mathbf{RRdot}$ due to a change in ARP position at time $t = t_{\text{COA}}^{\text{SCP}}$, $\Delta\mathbf{ARP}_{\text{COA}}^{\text{SCP}}$.

$$\mathbf{M}_{\Delta\mathbf{ARP}}^{\text{RRdot}} = (-1) \cdot \begin{bmatrix} \mathbf{uPT}^T \\ \mathbf{uPTDot}^T \end{bmatrix}$$

- (2) Form the 2×3 sensitivity matrix used to compute a change in COA range and range rate $\Delta\mathbf{RRdot}$ due to a change in ARP velocity $\Delta\mathbf{VARP}$ that is constant across the slow time period processed to form the image.

$$\mathbf{M}_{\Delta\mathbf{VARP}}^{\text{RRdot}} = (-1) \cdot \begin{bmatrix} \mathbf{uPT}^T \cdot (t_{\text{COA}}^{\text{ILO}} - t_{\text{COA}}^{\text{SCP}}) \\ \mathbf{uPT}^T + \mathbf{uPTDot}^T \cdot (t_{\text{COA}}^{\text{ILO}} - t_{\text{COA}}^{\text{SCP}}) \end{bmatrix}$$

- (3) Form the 2×1 sensitivity matrix used to compute a change in COA range and range rate $\Delta\mathbf{RRdot}$ due to a change in control clock frequency. The change in control clock frequency is expressed as a scale factor, CLK_SF .

$$c\text{RSF} = \frac{\delta(R)}{\delta(\text{CLK_SF})} = (-1) \cdot R_{0\text{COA}} \quad c\text{RdotSF} = \frac{\delta(\text{Rdot})}{\delta(\text{CLK_SF})} = 0$$

$$\mathbf{M}_{\text{CLK_SF}}^{\text{RRdot}} = \begin{bmatrix} c\text{RSF} \\ c\text{RdotSF} \end{bmatrix}$$

11.5.2 Bistatic P/V/T Matrices

For a bistatic image, the matrices that relate changes in sensor position, velocity, and control clock timing to equivalent changes in COA average range and average range rate are listed in Table 11-5. The independent variables include changes in APC positions at $t = t_{COA}^{SCP}$ and changes in APC velocities that are constant across the slow time period processed to form the image.

Table 11-5 Bistatic Position/Velocity/Timing Sensitivity Matrices		
Sensitivities for a given image location/scene point pair.		
If APOs are applied, they are applied to the COA projection set for all image grid locations.		
$\Delta \mathbf{Xmt}_{COA}^{SCP} = \begin{bmatrix} \Delta Xmt_X \\ \Delta Xmt_Y \\ \Delta Xmt_Z \end{bmatrix} \quad \Delta \mathbf{VXmt} = \begin{bmatrix} \Delta VXmt_X \\ \Delta VXmt_Y \\ \Delta VXmt_Z \end{bmatrix} \quad \Delta \mathbf{RRdot} = \begin{bmatrix} \Delta R_Avg \\ \Delta Rdot_Avg \end{bmatrix}$ For image/scene pair IL0/PT0 with COA transmit time tx_{COA}^{ILO}, the change in APC position is: $\Delta \mathbf{Xmt}(tx_{COA}^{ILO}) = \Delta \mathbf{Xmt}_{COA}^{SCP} + \Delta \mathbf{VXmt} \cdot (tx_{COA}^{ILO} - t_{COA}^{SCP})$		
Matrix	Size	Description
$\mathbf{M}_{\Delta Xmt}^{RRdot}$	2 x 3	Sensitivity matrix to compute the change in COA range and range rate ($\Delta \mathbf{RRdot}$) due to a change in Xmt APC position at $t = t_{COA}^{SCP}$ ($\Delta \mathbf{Xmt}_{COA}^{SCP}$). $\Delta \mathbf{RRdot} = \mathbf{M}_{\Delta Xmt}^{RRdot} \cdot \Delta \mathbf{Xmt}_{COA}^{SCP}$
$\mathbf{M}_{\Delta VXmt}^{RRdot}$	2 x 3	Sensitivity matrix to compute the change in COA range and range rate ($\Delta \mathbf{RRdot}$) due to a change in Xmt APC velocity ($\Delta \mathbf{VXmt}$). $\Delta \mathbf{RRdot} = \mathbf{M}_{\Delta VXmt}^{RRdot} \cdot \Delta \mathbf{VXmt}$
$\mathbf{M}_{XTF}^{RRdot}$	2 x 2	Sensitivity matrix to compute the change in COA range and range rate ($\Delta \mathbf{RRdot}$) due to a change in transmit time at $t = t_{COA}^{SCP}$ (Δtx_{COA}^{SCP}) or a change in transmit control clock frequency expressed as a scale factor ($f_Clk_X_SF = \Delta f_CLK_X / f_Clk_X_0$). $\Delta \mathbf{RRdot} = \mathbf{M}_{XTF}^{RRdot} \cdot \Delta \mathbf{XTF}$
$\Delta \mathbf{Rcv}_{COA}^{SCP} = \begin{bmatrix} \Delta Rcv_X \\ \Delta Rcv_Y \\ \Delta Rcv_Z \end{bmatrix} \quad \Delta \mathbf{VRcv} = \begin{bmatrix} \Delta VXmt_X \\ \Delta VXmt_Y \\ \Delta VXmt_Z \end{bmatrix} \quad \Delta \mathbf{RRdot} = \begin{bmatrix} \Delta R_Avg \\ \Delta Rdot_Avg \end{bmatrix}$ For image/scene pair IL0/PT0 with COA receive time tr_{COA}^{ILO}, the change in APC position is: $\Delta \mathbf{Rcv}(tr_{COA}^{ILO}) = \Delta \mathbf{Rcv}_{COA}^{SCP} + \Delta \mathbf{VRcv} \cdot (tr_{COA}^{ILO} - t_{COA}^{SCP})$		

Table 11-5 Bistatic Position/Velocity/Timing Sensitivity Matrices		
$\mathbf{M}_{\Delta Rcv}^{RRdot}$	2 x 3	Sensitivity matrix to compute the change in COA range and range rate ($\Delta RRdot$) due to a change in Rcv APC position at $t = t_{COA}^{SCP}$ (ΔRcv_{COA}^{SCP}). $\Delta RRdot = \mathbf{M}_{\Delta Rcv}^{RRdot} \bullet \Delta Rcv_{COA}^{SCP}$
$\mathbf{M}_{\Delta VRcv}^{RRdot}$	2 x 3	Sensitivity matrix to compute the change in COA range and range rate ($\Delta RRdot$) due to a change in Rcv APC velocity ($\Delta VRcv$). $\Delta RRdot = \mathbf{M}_{\Delta VRcv}^{RRdot} \bullet \Delta VRcv$
$\mathbf{M}_{RTF}^{RRdot}$	2 x 2	Sensitivity matrix to compute the change in COA range and range rate ($\Delta RRdot$) due to a change in receive time at $t = t_{COA}^{SCP}$ (Δtr_{COA}^{SCP}) or a change in receive control clock frequency expressed as a scale factor ($f_Clk_R_SF = \Delta f_CLK_R / f_Clk_R_0$). $\Delta RRdot = \mathbf{M}_{RTF}^{RRdot} \bullet \Delta RTF$

- (1) Form the 2 x 3 sensitivity matrix used to compute a change in COA average range and average range rate $\Delta RRdot$ due to a change in transmit APC position at time $t = t_{COA}^{SCP}$, ΔXmt_{COA}^{SCP} .

$$\mathbf{M}_{\Delta Xmt}^{RRdot} = (-1/2) \bullet \begin{bmatrix} \mathbf{uXmt}^T \\ \mathbf{uXmtDot}^T \end{bmatrix}$$

- (2) Form the 2 x 3 sensitivity matrix used to compute a change in COA average range and average range rate $\Delta RRdot$ due to a change in transmit APC velocity $\Delta VXmt$ that is constant across the slow time period processed to form the image.

$$\mathbf{M}_{\Delta VXmt}^{RRdot} = (-1/2) \bullet \begin{bmatrix} \mathbf{uXmt}^T \bullet (tx_{COA}^{ILO} - t_{COA}^{SCP}) \\ \mathbf{uXmt}^T + \mathbf{uXmtDot}^T \bullet (tx_{COA}^{ILO} - t_{COA}^{SCP}) \end{bmatrix}$$

- (3) Form the 2 x 2 sensitivity matrix $\mathbf{M}_{XTF}^{RRdot}$ that is used to compute a change in COA average range and average range rate $\Delta RRdot$ due to a change in transmit time at time $t = t_{COA}^{SCP}$, Δtx_{COA}^{SCP} , and/or a change in transmit control clock frequency expressed as a scale factor, $f_Clk_X_SF$. The 2 x 1 vector $\mathbf{XTF}$ is defined as shown.

$$\mathbf{XTF} = \begin{bmatrix} \Delta tx_{COA}^{SCP} \\ f_Clk_X_SF \end{bmatrix} \quad \text{where: } f_Clk_X_SF = \Delta f_Clk_X / f_Clk_X_0$$

For image location $IL0$ with transmit time $tx = tx_{COA}^{ILO}$, matrix $\mathbf{M}_{XTF}^{RRdot}$ is formed as shown.

$$\mathbf{M}_{XTF}^{RRdot} = (c/2) \bullet \begin{bmatrix} -TRTX_0 & + (tx_{COA}^{ILO} - t_{COA}^{SCP}) \\ -TRTXdot_0 & + 1 \end{bmatrix}$$

- (4) Form the 2 x 3 sensitivity matrix used to compute a change in COA average range and average range rate $\Delta \mathbf{RRdot}$ due to a change in receive APC position at time $t = t_{COA}^{SCP}$, $\Delta \mathbf{Rcv}_{COA}^{SCP}$.

$$\mathbf{M}_{\Delta Rcv}^{RRdot} = (-1/2) \cdot \begin{bmatrix} \mathbf{uRcv}^T \\ \mathbf{uRcvDot}^T \end{bmatrix}$$

- (5) Form the 2 x 3 sensitivity matrix used to compute a change in COA average range and average range rate $\Delta \mathbf{RRdot}$ due to a change in receive APC velocity $\Delta \mathbf{VRcv}$ that is constant across the slow time period processed to form the image.

$$\mathbf{M}_{\Delta VRcv}^{RRdot} = (-1/2) \cdot \begin{bmatrix} \mathbf{uRcv}^T \cdot (\text{tr}_{COA}^{ILO} - t_{COA}^{SCP}) \\ \mathbf{uRcv}^T + \mathbf{uRcvDot}^T \cdot (\text{tr}_{COA}^{ILO} - t_{COA}^{SCP}) \end{bmatrix}$$

- (6) Form the 2 x 2 sensitivity matrix $\mathbf{M}_{RTF}^{RRdot}$ that is used to compute a change in COA average range and average range rate $\Delta \mathbf{RRdot}$ due to a change in receive time at time $t = t_{COA}^{SCP}$, $\Delta \text{tr}_{COA}^{SCP}$, and/or a change in receive control clock frequency expressed as a scale factor $f_Clk_R_SF$. The 2 x 1 vector $\mathbf{RTF}$ is defined as shown.

$$\mathbf{RTF} = \begin{bmatrix} \Delta \text{tr}_{COA}^{SCP} \\ f_Clk_R_SF \end{bmatrix} \quad \text{where: } f_Clk_R_SF = \Delta f_Clk_R / f_Clk_R_0$$

For image location $\mathbf{ILO}$ with receive time $\text{tr} = \text{tr}_{COA}^{ILO}$, matrix $\mathbf{M}_{RTF}^{RRdot}$ is formed as shown.

$$\mathbf{M}_{RTF}^{RRdot} = (c/2) \cdot \begin{bmatrix} +1 & -(\text{tr}_{COA}^{ILO} - t_{COA}^{SCP}) \\ 0 & -1 \end{bmatrix}$$

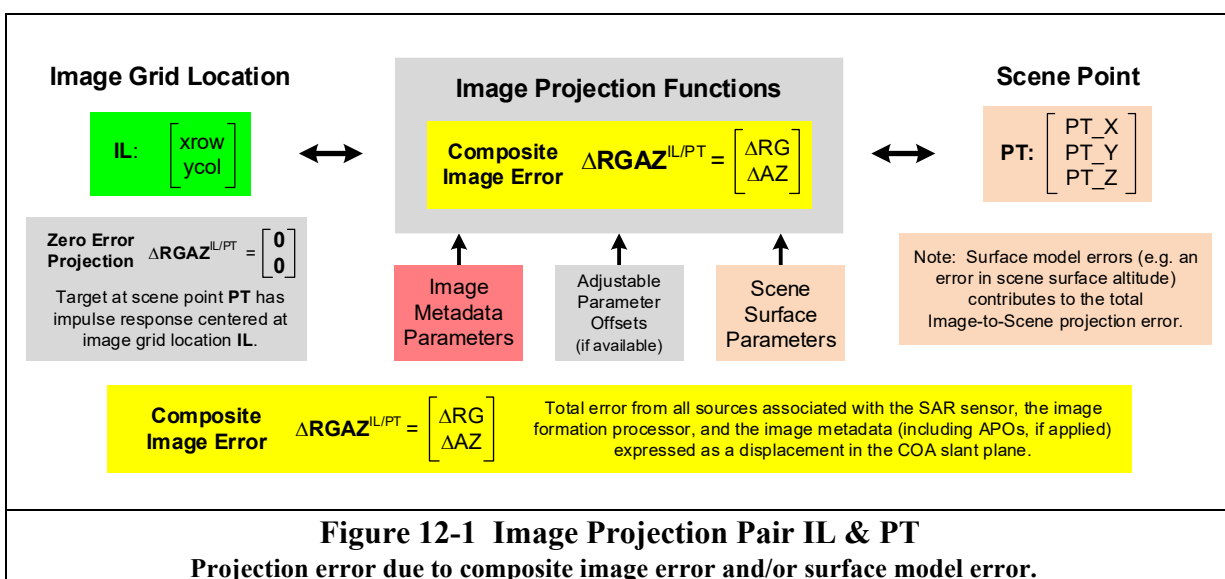
12 Projection Error Propagation

The process of mapping image grid locations to geolocated points in the imaged scene is a fundamental step in the exploitation of SAR complex image products. Projection and resampling to a ground plane pixel grid is needed to enable even the most basic image exploitation. For a given application, the absolute accuracy of the projection may or may not be of concern. For any image product, optional ErrorStatistics may be included and used to compute the expected accuracy of the image-to-scene and scene-to-image projections. The error parameters that may be provided are summarized in Table 1-2 and Table 1-4.

For any image, the fundamental projection computation is the mapping from an image grid location to a scene point along an R/Rdot contour. Errors in the collection and/or processing will result in errors in both the image-to-scene and the scene-to-image projections. Errors in the scene surface model will result in errors in the image-to-scene projection. The following sections specify methods to compute the expected error statistics for the image-to-scene and scene-to-image projections. In the description that follows, errors for all parameters are defined using the following convention. For a given parameter with true value X and an estimated value $E:X$, the error ΔX is defined as the true value minus the estimated value. The expected value of the error $E\{\Delta X\}$ is assumed to be zero.

$$\text{Error} = \text{True Value} - \text{Estimated Value} \quad \Delta X = X - E:X \quad E\{\Delta X\} = 0$$

Consider an image location/scene point projection pair **IL** and **PT**. The image projection functions that connect the projection pair are shown in Figure 12-1. The errors associated with the image collection and processing are referred to as the “image” errors. For a given image /scene pair, the combined effects of all image errors may be expressed as the composite image error, denoted $\Delta \text{RGZ}^{\text{IL/PT}}$. The ErrorStatistics parameters may be used to compute the expected statistics for the composite image error for any image location/scene point pair.

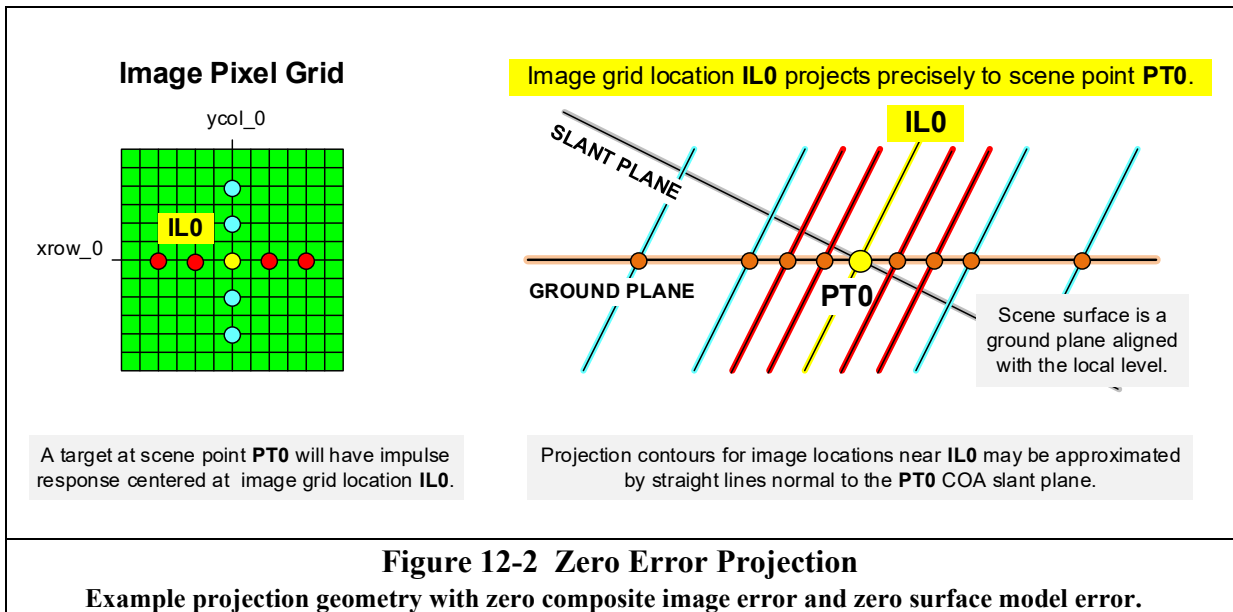


Zero Error Projection

Consider the image location/scene point projection pair **IL0** and **PT0** shown in Figure 12-2. The example shown is referred to as the “zero-error” projection in that all sources of error are zero. For the example shown, the imaged scene is a locally level area. The scene surface model is a ground plane with zero surface model error. Image location **IL0** is specified in image coordinates and scene point position **PT0** is specified in ECEF coordinates.

$$\text{Image Location: } \mathbf{IL0} = \begin{bmatrix} \text{xrow_0} \\ \text{ycol_0} \end{bmatrix} \quad \text{Scene Point (ECEF): } \mathbf{PT0} = \begin{bmatrix} \text{PT0_X} \\ \text{PT0_Y} \\ \text{PT0_Z} \end{bmatrix}$$

The image grid near **IL0** is shown in the left side of the figure. The ground plane and scene point **PT0** are shown in the right side of the figure. Also shown is the COA slant plane for point **PT0**. For zero error projection, an ideal point scatterer located at scene point **PT0** will have its image signal response centered at image grid location **IL0**. Shown in the left side of the figure are image grid locations near **IL0** but displaced in the xrow coordinate (blue) or the ycol coordinate (red). Shown in the right side of the figure are the projection contours for these image grid locations as they pass through the COA slant plane and scene surface/ground plane near point **PT0**. The projection contours for image grid locations near **IL0** are approximated by straight lines as they pass near scene point **PT0**. The straight line approximations are oriented normal to **PT0** COA slant plane.



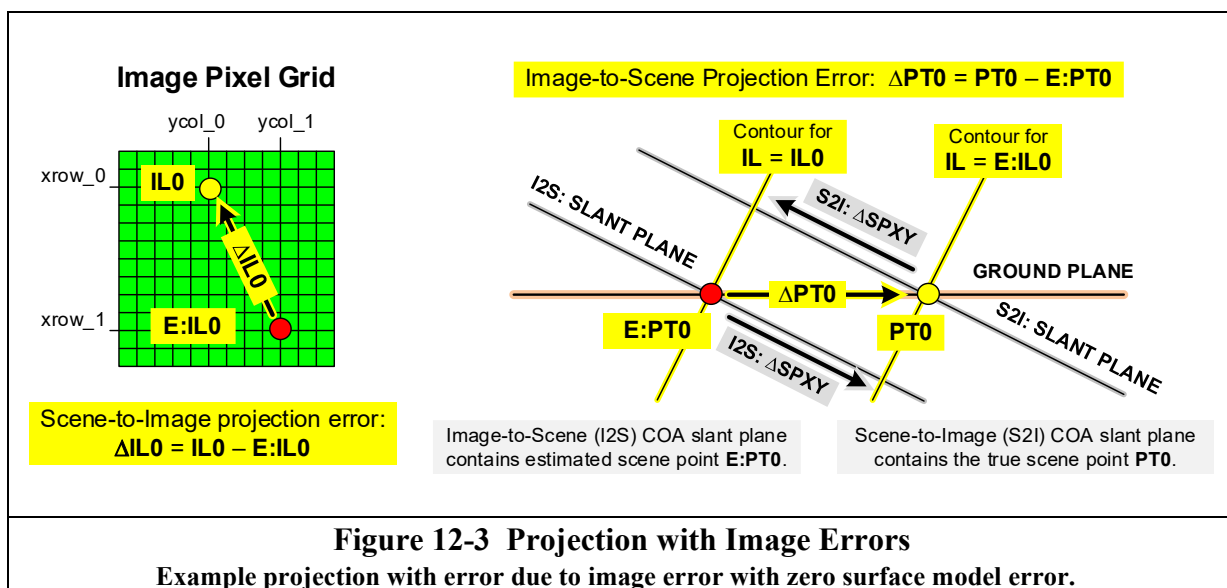
Projection w/ Image Errors

Consider the image location/scene point projections example shown in Figure 12-3. For the example shown, the projections are in error due to errors in the image metadata. Similar to the example in Figure 12-2, the imaged scene is a locally level area and the scene surface model is a

ground plane with zero surface model error. For an ideal point scatterer at scene point **PT0**, the image signal response is centered at image location **IL0**. Given true image location **IL0**, the image-to-scene (I2S) projection yields scene point **E:PT0** equal to an “estimate” of the scene point position. Given the true scene point position **PT0**, the scene-to-image (S2I) projection yields image location **E:IL0** equal to an “estimate” of the image location. For the I2S projection, the error in estimated scene point is ΔPT0 . For the S2I projection, the error in estimated image location is ΔIL0 .

$$\text{Image-to-Scene Projection Error: } \Delta\text{PT0} = \text{PT0} - \text{E:PT0}$$

$$\text{Scene-to-Image Projection Error: } \Delta\text{IL0} = \text{IL0} - \text{E:IL0}$$



For the I2S projection shown in Figure 12-3, true image location **IL0** and estimated scene point **E:PT0** are a projection pair. For the **IL0/E:PT0** pair, we can compute the I2S slant plane coordinates SPX and SPY for which **E:PT0** is located at the origin. See Section 11.2. For true scene point **PT0**, the true projection contour intersects the I2S slant plane at **I2S: ΔSPXY** = I2S: (ΔSPXI , ΔSPYI). For the S2I projection, estimated image location **E:IL0** and true scene point **PT0** are a projection pair. For the **E:IL0/PT0** pair, we can compute the S2I slant plane coordinate system for which **PT0** is located at the origin. For true image location **IL0**, the projection contour intersects the S2I slant plane at **S2I: ΔSPXY** = (ΔSPXS , ΔSPYS). As indicated in Figure 12-3, the slant plane displacements due to the image error are opposite in sign and approximately equal in magnitude.

$$\text{I2S: } \Delta\text{SPXY} \approx - \text{S2I: } \Delta\text{SPXY} \quad \begin{bmatrix} \Delta\text{SPXI} \\ \Delta\text{SPYI} \end{bmatrix} \approx - \begin{bmatrix} \Delta\text{SPXS} \\ \Delta\text{SPYS} \end{bmatrix}$$

For a given image projection pair, the combined effect of all sources of image error is to displace the “correct” projection contour relative to the “computed” projection contour. The

displacement is relative to the COA slant plane origin. For the I2S projection, the correct contour (through true scene point **PT0**) is displaced by **I2S:ΔSPXY** from the I2S estimated scene point **E:PT0**. For the S2I projection, the correct contour (for true image location **IL0**) is displaced by **S2I:ΔSPXY** from the S2I scene point **PT0**.

Composite Image Error

The composite image error is a displacement in the COA slant plane that combines the effects of all image errors. The methods for computing projection error statistics specified in the following sections make use of the expected statistics of the composite image error. For a given image/scene point pair, the composite image error is defined as follows.

(1) Consider an imaging collection with an ideal point scatterer located in the imaged scene at position **PT0**. For the point scatterer, the center of the image signal response is at true image location **IL0**. Consider the projection pair **IL0/E:PT0** for which **E:PT0** is computed using the ETP at the true scene point position **PT0** (i.e., zero error in the knowledge of the scene surface that contains point **PT0**).

(2) For estimated scene point **E:PT0**, the I2S projection error **ΔPT0** is defined as follows.

$$\text{I2S(IL0)} \rightarrow \text{E:PT0} \quad \Delta\text{PT0} = \text{PT0} - \text{E:PT0}$$

(3) For image location **IL0**, the parameters in the COA projection are dependent upon the collect type for the image.

Monostatic Projection Set:

$$\{ t_{\text{COA}}^{\text{IL0}}, \text{ARP_0}_{\text{COA}}, \text{VARP_0}_{\text{COA}}, R_{\text{0COA}}, R_{\text{dot_0COA}} \}$$

Bistatic Projection Set:

$$\{ t_{\text{COA}}^{\text{IL0}}, tx_{\text{COA}}^{\text{IL0}}, tr_{\text{COA}}^{\text{IL0}}, \text{Xmt_0}_{\text{COA}}, \text{VXmt_0}_{\text{COA}}, \\ \text{Rcv_0}_{\text{COA}}, \text{VRcv_0}_{\text{COA}}, R_{\text{Avg_0COA}}, R_{\text{dot_Avg_0COA}} \}$$

(4) Using the positions and velocities in the COA projection set and the true scene point position **PT0**, a “correct” range and range rate may be computed that would have resulted in **E:PT0 = PT0** (i.e., resulted in zero projection error).

Correct Monostatic R/Rdot: (R_{1COA} , $R_{\text{dot_1COA}}$)

$$R_{\text{1COA}} = |\text{ARP_0}_{\text{COA}} - \text{PT0}| \quad \mathbf{uPT} = \frac{1}{R_{\text{1COA}}} \bullet (\text{ARP_0}_{\text{COA}} - \text{PT0})$$

$$R_{\text{dot_1COA}} = \text{VARP_0}_{\text{COA}} \bullet \mathbf{uPT}$$

Correct Bistatic R/Rdot: ($R_{\text{Avg_1COA}}$, $R_{\text{dot_Avg_1COA}}$)

$$R_{\text{Xmt_1COA}} = |\text{Xmt_0}_{\text{COA}} - \text{PT0}| \quad \mathbf{uXmt} = \frac{1}{R_{\text{Xmt_1COA}}} \bullet (\text{Xmt_0}_{\text{COA}} - \text{PT0})$$

$$\text{Rdot_Xmt_1}_{\text{COA}} = \mathbf{VXmt_0}_{\text{COA}} \bullet \mathbf{uXmt}$$

$$\text{R_Rcv_1}_{\text{COA}} = |\mathbf{Rcv_0}_{\text{COA}} - \mathbf{PT0}| \quad \mathbf{uRcv} = \frac{1}{\text{R_Rcv_1}_{\text{COA}}} \bullet (\mathbf{Rcv_0}_{\text{COA}} - \mathbf{PT0})$$

$$\text{Rdot_Rcv_1}_{\text{COA}} = \mathbf{VRcv_0}_{\text{COA}} \bullet \mathbf{uRcv}$$

$$\text{R_Avg_1}_{\text{COA}} = \frac{1}{2} \bullet (\text{R_Xmt_1}_{\text{COA}} + \text{R_Rcv_1}_{\text{COA}})$$

$$\text{Rdot_Avg_1}_{\text{COA}} = \frac{1}{2} \bullet (\text{Rdot_Xmt_1}_{\text{COA}} + \text{Rdot_Rcv_1}_{\text{COA}})$$

(4) For the image/scene pair **IL0/E:PT0**, the composite error may be expressed as a difference in COA range and range rate, denoted $\Delta \mathbf{RRdot}^{\text{IL/PT}}$.

For a monostatic image – composite error:

$$\Delta \mathbf{RRdot}^{\text{IL/PT}} = \begin{bmatrix} \Delta \text{R_ARP} \\ \Delta \text{Rdot_ARP} \end{bmatrix} = \begin{bmatrix} \text{R_1}_{\text{COA}} - \text{R_0}_{\text{COA}} \\ \text{Rdot_1}_{\text{COA}} - \text{Rdot_0}_{\text{COA}} \end{bmatrix}$$

For a bistatic image – composite error:

$$\Delta \mathbf{RRdot}^{\text{IL/PT}} = \begin{bmatrix} \Delta \text{R_Avg} \\ \Delta \text{Rdot_Avg} \end{bmatrix} = \begin{bmatrix} \text{R_Avg_1}_{\text{COA}} - \text{R_0}_{\text{COA}} \\ \text{Rdot_Avg_1}_{\text{COA}} - \text{Rdot_Avg_0}_{\text{COA}} \end{bmatrix}$$

(5) The composite error may also be expressed as a displacement in the COA slant plane, denoted $\Delta \mathbf{RGAZ}^{\text{IL/PT}}$. The composite error $\Delta \mathbf{RGAZ}^{\text{IL/PT}}$ is computed using the sensitivity matrix $\mathbf{M}_{\text{RRdot}}^{\text{SPXY}}$ for image/scene pair **IL0/E:PT0**. See Section 11.3 and Table 11-2.

$$\Delta \mathbf{RGAZ}^{\text{IL/PT}} = \begin{bmatrix} \Delta \text{RG} \\ \Delta \text{AZ} \end{bmatrix} = (-1) \bullet \mathbf{M}_{\text{RRdot}}^{\text{SPXY}} \bullet \Delta \mathbf{RRdot}^{\text{IL/PT}}$$

Composite Error → Slant Plane Displacement

The composite image error expressed as a slant plane displacement, $\Delta \mathbf{RGAZ}^{\text{IL/PT}}$, allows for the direct conversion to slant plane displacement **I2S:ΔSPXY** and **S2I:ΔSPXY**. Shown in Figure 12-4 are the image-to-scene (left) and scene-to-image (right) slant plane displacements for a given composite error.

A commonly used pixel grid is the slant plane (rg,az) grid (i.e., Grid Type RGAZIM). Shown in Figure 12-5 is an example slant plane grid with a non-zero composite image error. The true image grid location **IL0** is the center of the image signal response for a scatterer at scene point **PT0**. The effect of the composite RGAZ error may be observed directly in the range, azimuth image grid: $\Delta \mathbf{IL0} = -\mathbf{RGAZ}^{\text{IL/PT}}$.

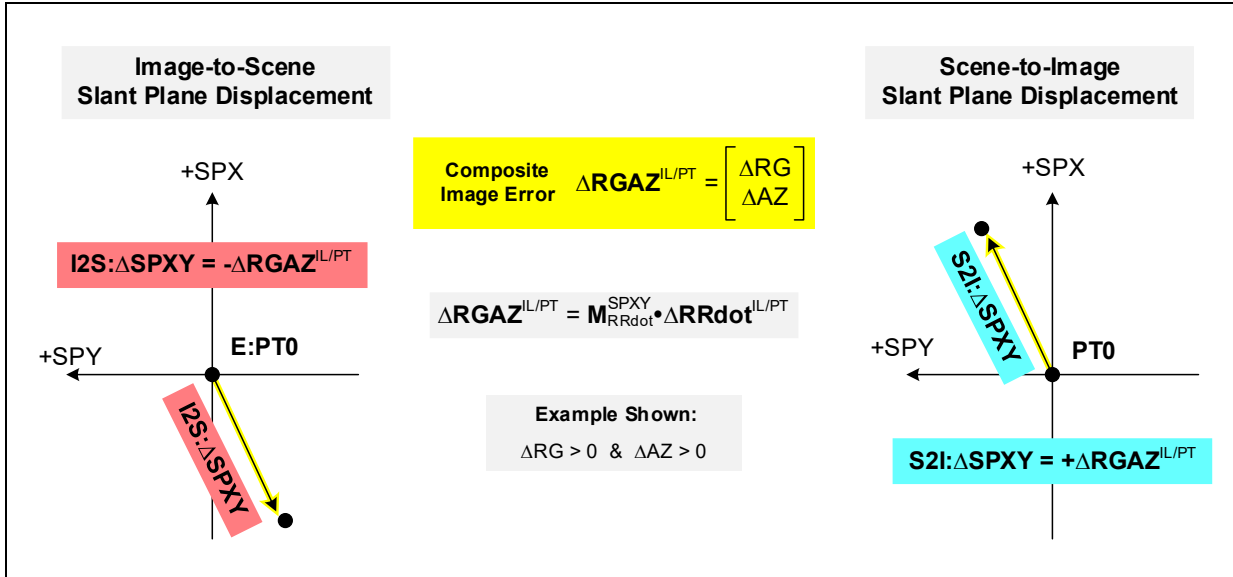


Figure 12-4 Displacement Due to Composite Error
 Example I2S & S2I slant plane displacement due to a composite image error $\Delta\text{RGAZ}^{\text{IL/PT}}$.

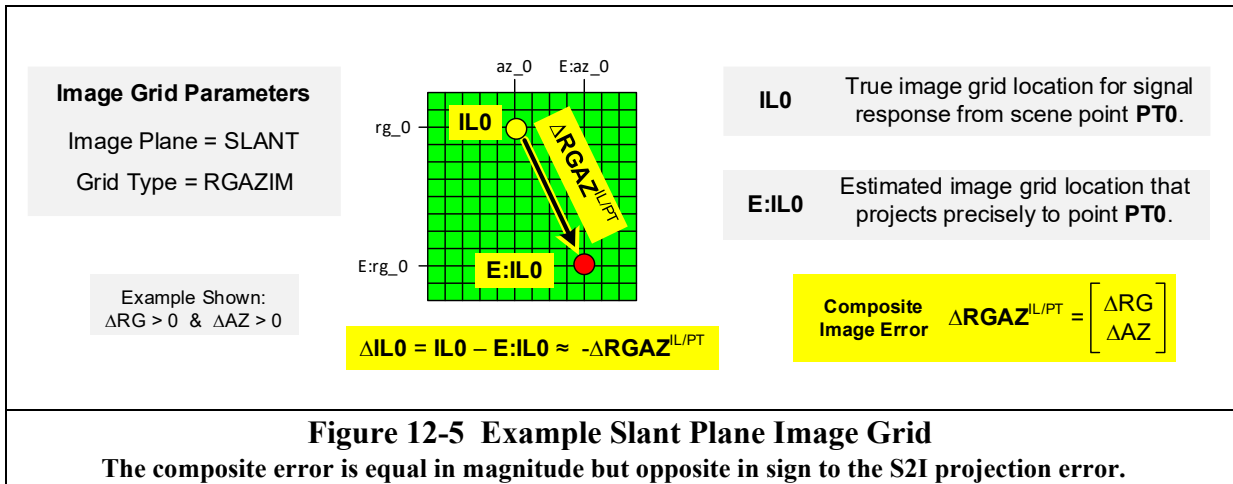


Figure 12-5 Example Slant Plane Image Grid
 The composite error is equal in magnitude but opposite in sign to the S2I projection error.

Sensor Position & Velocity Error

The projection errors that arise due to sensor position and velocity errors are computed for the following idealized error model. The position error versus time in ECEF coordinates is specified by a position error at time $t = t_{\text{COA}}^{\text{SCP}}$ and a velocity error that is constant during the collection time for the phase history data that was used to form the image. For all products, all position and velocity errors are specified with the following convention.

Position Error = True Position – Estimated Position

$$\Delta\mathbf{P}(t) = \mathbf{P}(t) - \mathbf{E}:\mathbf{P}(t)$$

Velocity Error = True Velocity – Estimated Velocity

$$\Delta\mathbf{V}(t) = \mathbf{V}(t) - \mathbf{E}:\mathbf{V}(t)$$

For a monostatic image, the error in the ARP position versus time $\Delta\text{ARP}(t)$ is assumed to be of the following form.

$$\Delta\text{ARP}(t) = \Delta\text{ARP}_{\text{COA}}^{\text{SCP}} + \Delta\text{VARP} \cdot (t - t_{\text{COA}}^{\text{SCP}})$$

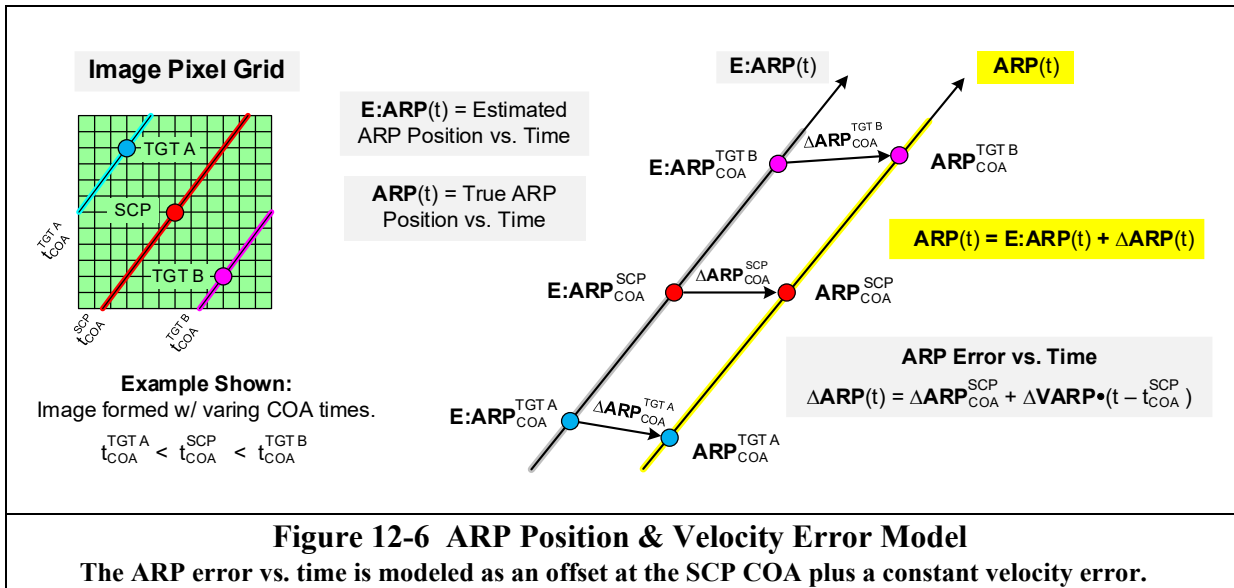
$\Delta\text{ARP}_{\text{COA}}^{\text{SCP}}$

ARP position error at $t = t_{\text{COA}}^{\text{SCP}}$ in ECEF coordinates (meters).

ΔVARP

ARP velocity error in ECEF coordinates (meters/sec). Assumed constant during collection period for the PHD used to form the image.

An example ARP position error versus time is shown in Figure 12-6. For the example shown, the COA time varies across the image grid. For a given image grid location, the ARP COA position error varies with COA time.



For a bistatic image, the error in transmit APC position, $\Delta\text{Xmt}(t)$, and receive APC position, $\Delta\text{Rcv}(t)$, are assumed to be of the following form.

$$\Delta\text{Xmt}(t) = \Delta\text{Xmt}_{\text{COA}}^{\text{SCP}} + \Delta\text{VXmt} \cdot (t - t_{\text{COA}}^{\text{SCP}})$$

$$\Delta\text{Rcv}(t) = \Delta\text{Rcv}_{\text{COA}}^{\text{SCP}} + \Delta\text{VRcv} \cdot (t - t_{\text{COA}}^{\text{SCP}})$$

$\Delta\text{Xmt}_{\text{COA}}^{\text{SCP}}$

Xmt APC position error at $t = t_{\text{COA}}^{\text{SCP}}$ in ECEF coordinates (meters).

ΔVXmt

Xmt APC velocity error in ECEF coordinates (meters/sec). Assumed constant during collection period for the PHD used to form the image.

$\Delta\text{Rcv}_{\text{COA}}^{\text{SCP}}$

Rcv APC position error at $t = t_{\text{COA}}^{\text{SCP}}$ in ECEF coordinates (meters).

ΔVRcv

Rcv APC velocity error in ECEF coordinates (meters/sec). Assumed constant during collection period for the PHD used to form the image.

Note: The position and velocity error model used for computing errors in the image projection functions is matched to the model used for specifying the Adjustable Parameter Offsets. See Table 1-3 and Section 8. For a given pair of position and velocity error values, the computed projection errors are precisely equal to the errors that will be compensated using the same position and velocity values for the adjustable parameter offsets.

Error Propagation Computations

The methods for computing projection error statistics include methods for computing two types of errors. The first type of method computes composite error statistics from input component error statistics or input APO error statistics. The second type of method computes the projection error statistics from composite error statistics. For a given image, the methods used are dependent upon the set of error statistics provided. Refer to Table 1-2 and Table 1-4.

For a product that includes component error statistics and/or APO error statistics, both types of methods are used. For a given image/scene pair, the expected composite error statistics are computed for the specific projection geometry. The I2S or S2I projection error statistics are then computed from the computed composite error statistics. For a product that only includes composite error statistics, the I2S or S2I projection error statistics are computed using the provided composite statistics.

12.1 Image/Scene Pair Parameters

For the image/scene projection pair, compute the projection geometry parameters described in Section 11.1 and Section 11.2. For an image-to-scene projection pair, **IL0/E:PT0**, the COA projection set is computed for image location **IL0** and scene point **E:PT0** is used to compute scene surface unit normal **uGPN0** and unit vector **uUP0**. For a scene-to-image projection, the COA projection set is computed for image location **E:IL0** and scene point **PT0** is used to compute scene surface unit normal **uGPN0** and unit vector **uUP0**. The set of parameters to be computed is dependent upon the Collect Type for the image.

12.1.1 Collect Type = Monostatic

(1) Compute the monostatic COA projection set that links the image/scene pair. The projection set may or may not have had APOs applied.

$$\{ t_{COA}^{IL0}, ARP_0_{COA}, VARP_0_{COA}, R_0_{COA}, Rdot_0_{COA} \}$$

(2) Use the projection set and the scene point to compute the set of monostatic-specific projection geometry parameters specified in Section 11.2.1.

$$\{ uPT, uPTDot \}$$

12.1.2 Collect Type = Bistatic

(1) Compute the bistatic COA projection set that links the image/scene pair. The projection set may or may not have had APOs applied.

$$\{ t_{COA}^{IL0}, tx_{COA}^{IL0}, tr_{COA}^{IL0}, Xmt_0_{COA}, VXmt_0_{COA}, \\ Rcv_0_{COA}, VRcv_0_{COA}, R_Avg_0_{COA}, Rdot_Avg_0_{COA} \}$$

(2) Use the projection set and the scene point to compute the set of bistatic-specific projection geometry parameters specified in Section 11.2.2.

$$\{ R_Xmt_0_{COA}, Rdot_Xmt_0_{COA}, R_Rcv_0_{COA}, Rdot_Rcv_0_{COA}, \\ uXmt, uXmtDot, uRcv, uRcvDot, bP, bPDot, TRTX_0, TRTXdot_0 \}$$

12.1.3 SPC & GPC Parameters

For all image/scene pairs, compute the basis vectors that specify the SPC frame and the GPC frame. Compute the cosine, sine, and tangent for the grazing angle and the twist angle. Refer to Section 11.2.3.

$$\{ uSPX, uSPY, uSPZ, uGPX, uGPY, uGPZ, \\ cos_GRAZ, sin_GRAZ, tan_GRAZ, cos_TWST, sin_TWST, tan_TWST \}$$

12.2 Projection Sensitivity Parameters

The methods for computing I2S and S2I error statistics make use of multiple sensitivity matrices. For the image/scene pair, the sensitivity matrices to be computed are specified as follows. For products that include component error statistics or APO error statistics, the matrices to be computed are dependent upon the collect type for the image.

12.2.1 I2S & S2I Sensitivity Matrices

For image/scene point pair, compute the following matrices used for computing I2S and S2I error statistics.

- (1) Form the set of sensitivity matrices listed in Table 11-2 as specified in Section 11-3.

$$\{ \mathbf{M}_{PT}^{SPXY}, \mathbf{M}_{GPXY}^{SPXY}, \mathbf{M}_{SPXY}^{GPXY}, \mathbf{M}_{GPXY}^{PT}, \mathbf{MIL}_{HAE}^{PT}, \mathbf{M}_{SPXY}^{RRdot}, \mathbf{M}_{RRdot}^{SPXY} \}$$

- (2) Form the set of sensitivity matrices listed in Table 11-3 as specified in Section 11-4.

$$\{ \mathbf{M}_{PT}^{IL}, \mathbf{M}_{IL}^{GPXY}, \mathbf{M}_{IL}^{SPXY}, \mathbf{M}_{SPXY}^{IL}, \mathbf{M}_{IL}^{RRdot}, \mathbf{M}_{RRdot}^{IL} \}$$

- (3) Form the 2 x 2 matrix used to compute the composite error expressed as a slant plane displacement, $\Delta\mathbf{RGAZ}$, from the composite error expressed as a difference in range and range rate, $\Delta\mathbf{RRdot}$.

$$\mathbf{M}_{RRdot}^{RGAZ} = (-1) \cdot \mathbf{M}_{RRdot}^{SPXY}$$

- (4) Form a pair of 2 x 2 matrices used to compute the slant plane displacement from the composite error. For an I2S projection, matrix $\mathbf{MIL}_{RGAZ}^{SPXY}$ is used to compute the slant plane displacement with the image location (IL) held constant. For an S2I projection, matrix $\mathbf{MPT}_{RGAZ}^{SPXY}$ is used to compute the slant plane displacement for the scene point (PT) held constant.

$$\mathbf{MIL}_{RGAZ}^{SPXY} = (-1) \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \mathbf{MPT}_{RGAZ}^{SPXY} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

12.2.2 Component Matrices: Collect Type = Monostatic

For a monostatic image for which component error statistics or APO error statistics are provided, form the following sensitivity matrices. Refer to Table 11-4 and Section 11.5.1.

- (1) Form the pair of 2 x 3 sensitivity matrices used to compute the contribution of ARP position and velocity errors to composite error. Also form the 2 x 1 sensitivity matrix used to compute the contribution of control clock frequency error to composite error.

$$\{ \mathbf{M}_{\Delta\text{ARP}}^{\text{RRdot}}, \mathbf{M}_{\Delta\text{VARP}}^{\text{RRdot}}, \mathbf{M}_{\text{CLK_SF}}^{\text{RRdot}} \}$$

Use matrices $\mathbf{M}_{\Delta\text{ARP}}^{\text{RRdot}}$ and $\mathbf{M}_{\Delta\text{VARP}}^{\text{RRdot}}$ to form a 2 x 6 matrix $\mathbf{M}_{\text{APV}}^{\text{RRdot}}$.

$$\mathbf{M}_{\text{APV}}^{\text{RRdot}} = [\mathbf{M}_{\Delta\text{ARP}}^{\text{RRdot}} \quad \mathbf{M}_{\Delta\text{VARP}}^{\text{RRdot}}]$$

- (2) Form the 6 x 6 matrix $\mathbf{T}_{\text{AIF}}^{\text{ECEF}}$ that is used to compute ARP position and velocity error statistics in the ECEF coordinate frame from the input ARP position and velocity error statistics provided in the product. The coordinate frame for the input error statistics is specified by parameter AIF. Refer to Table 2-1.

For AIF = ECF:

$$\mathbf{T}_{\text{AIF}}^{\text{ECEF}} = \mathbf{I}_{6 \times 6} \quad \mathbf{I}_{6 \times 6} = \text{Identity matrix} - \text{size } 6 \times 6.$$

For AIF = RIC_ECF:

Compute 3 x 3 matrix $\mathbf{T}_{\text{RICF_A}}^{\text{ECEF}}$ as specified in Section 3.4.2.

$$\text{Form matrix } \mathbf{T}_{\text{AIF}}^{\text{ECEF}} = \begin{bmatrix} \mathbf{T}_{\text{RICF_A}}^{\text{ECEF}} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{T}_{\text{RICF_A}}^{\text{ECEF}} \end{bmatrix} \quad \mathbf{0}_{3 \times 3} = \text{Zero matrix} - \text{size } 3 \times 3.$$

For AIF = RIC_ECI:

Compute 3 x 3 matrix $\mathbf{T}_{\text{RICI_A}}^{\text{ECEF}}$ as specified in Section 3.4.2.

Form 3 x 3 matrix $\mathbf{\Omega}_3$ as specified in Section 3.4.4.

$$\text{Form matrix } \mathbf{T}_{\text{AIF}}^{\text{ECEF}} = \begin{bmatrix} \mathbf{T}_{\text{RICI_A}}^{\text{ECEF}} & \mathbf{0}_{3 \times 3} \\ \mathbf{\Omega}_3 \cdot \mathbf{T}_{\text{RICI_A}}^{\text{ECEF}} & \mathbf{T}_{\text{RICI_A}}^{\text{ECEF}} \end{bmatrix}$$

12.2.3 Component Matrices: Collect Type = Bistatic

For a bistatic image for which component error statistics or APO error statistics are provided, form the following sensitivity matrices. Refer to Table 11-5 and Section 11.5.2.

- (1) Form the pair of 2 x 3 sensitivity matrices used to compute the contribution of transmit APC position and velocity errors to composite error. Also form the 2 x 2 sensitivity matrix used to compute the contribution of transmit control clock timing and frequency error to composite error.

$$\{ \mathbf{M}_{\Delta Xmt}^{RRdot}, \mathbf{M}_{\Delta VXmt}^{RRdot}, \mathbf{M}_{XTF}^{RRdot} \}$$

Use matrices $\mathbf{M}_{\Delta Xmt}^{RRdot}$ and $\mathbf{M}_{\Delta VXmt}^{RRdot}$ to form a 2 x 6 matrix $\mathbf{M}_{XPV}^{RRdot}$.

$$\mathbf{M}_{XPV}^{RRdot} = \begin{bmatrix} \mathbf{M}_{\Delta Xmt}^{RRdot} & \mathbf{M}_{\Delta VXmt}^{RRdot} \end{bmatrix}$$

- (2) Form the pair of 2 x 3 sensitivity matrices used to compute the contribution of receive APC position and velocity errors to composite error. Also form the 2 x 2 sensitivity matrix used to compute the contribution of receive control clock timing and frequency error to composite error.

$$\{ \mathbf{M}_{\Delta Rcv}^{RRdot}, \mathbf{M}_{\Delta VRcv}^{RRdot}, \mathbf{M}_{RTF}^{RRdot} \}$$

Use matrices $\mathbf{M}_{\Delta Rcv}^{RRdot}$ and $\mathbf{M}_{\Delta VRcv}^{RRdot}$ to form a 2 x 6 matrix $\mathbf{M}_{RPV}^{RRdot}$.

$$\mathbf{M}_{RPV}^{RRdot} = \begin{bmatrix} \mathbf{M}_{\Delta Rcv}^{RRdot} & \mathbf{M}_{\Delta VRcv}^{RRdot} \end{bmatrix}$$

- (3) Form the 2 x 2 matrix $\mathbf{M}_{ATM}^{RRdot}$ that is used to compute the contribution of atmospheric delay estimate error to composite error.

$$\mathbf{M}_{ATM}^{RRdot} = (c/2) \cdot \begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix}$$

- (4) Form the 6 x 6 matrix $\mathbf{T}_{XIF}^{ECEF}$ that is used to compute transmit APC position and velocity error statistics in the ECEF coordinate frame from the input transmit APC position and velocity error statistics provided in the product. The coordinate frame for the input error statistics is specified by parameter XIF. Refer to Table 2-1.

For XIF = ECF:

$$\mathbf{T}_{XIF}^{ECEF} = \mathbf{I}_{6 \times 6} \quad \mathbf{I}_{6 \times 6} = \text{Identity matrix} - \text{size } 6 \times 6.$$

For XIF = RIC_ECF:

Compute 3 x 3 matrix $\mathbf{T}_{RIC_X}^{ECEF}$ as specified in Section 3.4.3.

$$\text{Form matrix } \mathbf{T}_{XIF}^{ECEF} = \begin{bmatrix} \mathbf{T}_{RIC_X}^{ECEF} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{T}_{RIC_X}^{ECEF} \end{bmatrix} \quad \mathbf{0}_{3 \times 3} = \text{Zero matrix} - \text{size } 3 \times 3.$$

For XIF = RIC_ECI:

Compute 3 x 3 matrix $\mathbf{T}_{RIC_X}^{ECEF}$ as specified in Section 3.4.3.

Form 3 x 3 matrix $\mathbf{\Omega}_3$ as specified in Section 3.4.4.

$$\text{Form matrix } \mathbf{T}_{XIF}^{ECEF} = \begin{bmatrix} \mathbf{T}_{RIC_X}^{ECEF} & \mathbf{0}_{3 \times 3} \\ \mathbf{\Omega}_3 \cdot \mathbf{T}_{RIC_X}^{ECEF} & \mathbf{T}_{RIC_X}^{ECEF} \end{bmatrix}$$

(5) Form the 6 x 6 matrix $\mathbf{T}_{RIF}^{ECEF}$ that is used to compute receive APC position and velocity error statistics in the ECEF coordinate frame from the input receive APC position and velocity error statistics provided in the product. The coordinate frame for the input error statistics is specified by parameter RIF. Refer to Table 2-1.

For RIF = ECF:

$$\mathbf{T}_{RIF}^{ECEF} = \mathbf{I}_{6 \times 6} \quad \mathbf{I}_{6 \times 6} = \text{Identity matrix} - \text{size } 6 \times 6.$$

For RIF = RIC_ECF:

Compute 3 x 3 matrix $\mathbf{T}_{RIC_R}^{ECEF}$ as specified in Section 3.4.3.

$$\text{Form matrix } \mathbf{T}_{RIF}^{ECEF} = \begin{bmatrix} \mathbf{T}_{RIC_R}^{ECEF} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{T}_{RIC_R}^{ECEF} \end{bmatrix} \quad \mathbf{0}_{3 \times 3} = \text{Zero matrix} - \text{size } 3 \times 3.$$

For RIF = RIC_ECI:

Compute 3 x 3 matrix $\mathbf{T}_{RIC_R}^{ECEF}$ as specified in Section 3.4.3.

Form 3 x 3 matrix $\mathbf{\Omega}_3$ as specified in Section 3.4.4.

$$\text{Form matrix } \mathbf{T}_{RIF}^{ECEF} = \begin{bmatrix} \mathbf{T}_{RIC_R}^{ECEF} & \mathbf{0}_{3 \times 3} \\ \mathbf{\Omega}_3 \cdot \mathbf{T}_{RIC_R}^{ECEF} & \mathbf{T}_{RIC_R}^{ECEF} \end{bmatrix}$$

12.3 Composite Error: Collect Type = Monostatic

For a monostatic image, the methods used to compute the expected composite error statistics for a given image/scene pair are specified as follows. The method used is dependent upon (1) if APOs were applied to the COA projection set, and (2) the error statistics provided with the product. The composite error may be expressed in either of the two defined forms.

$\Delta\mathbf{RRdot}^{IL/PT}$ Expressed as error in range and range rate relative to the ARP. The error covariance matrix is $\mathbf{C}_{RRdot}^{IL/PT}$.

$\Delta\mathbf{RGAZ}^{IL/PT}$ Expressed as a displacement in the COA slant plane. The error covariance matrix is $\mathbf{C}_{RGAZ}^{IL/PT}$.

The composite error statistics are computed as specified in the following subsections. For the input metadata to be used refer to Table 1-2 and Table 1-4.

12.3.1 APOs Not Applied w/ SCP Composite Statistics

For an image/scene pair for which APOs have not been applied and only the composite error statistics for the SCP are provided, set the composite error covariance for the image/scene pair equal to the composite error covariance provided.

$$\mathbf{C}_{RGAZ}^{IL/PT} = \mathbf{C}_{RGAZ}^{SCP}$$

12.3.2 APOs Not Applied w/ Component Statistics

For an image/scene pair for which APOs have not been applied and for which component error statistics are provided, compute the composite error covariance from the component error statistics provided. Also include the contribution from unmodeled image error. The components of the composite error are the following.

$\Delta\mathbf{APV}_{ECEF}$ ARP position error and velocity error at $t = t_{COA}^{SCP}$ in ECEF coordinates. Covariance $\mathbf{C}_{APV}^{ECEF}$ computed from input $\mathbf{C}_{APV}^{AIF}$.

ΔR_BIAS Range error that is constant across the image. Variance VAR_{RB} .

CLK_SF Control clock frequency error – expressed as a scale factor. Variance VAR_{CLK_SF} .

ΔR_TROP Range error due to an error in the estimated tropospheric delay. Variance VAR_{TROP} computed from the collection geometry.

ΔR_IONO Range error due to an error in the estimated ionospheric delay (if applicable). Variance VAR_{IONO} computed from the collection geometry.

$\Delta\mathbf{I}_{UI}$ Unmodeled image error for the selected image grid location. Covariance $\mathbf{C}_{UI}$.

The composite error $\Delta\mathbf{RRdot}^{\text{IL/PT}}$ is equal to the sum of the contribution from each component.

$$\Delta\mathbf{RRdot}^{\text{IL/PT}} = \Delta\mathbf{RRdot}^{\text{APV}} + \Delta\mathbf{RRdot}^{\text{RB}} + \Delta\mathbf{RRdot}^{\text{CLK_SF}} + \Delta\mathbf{RRdot}^{\text{TROP}} + \Delta\mathbf{RRdot}^{\text{IONO}} + \Delta\mathbf{RRdot}^{\text{UI}}$$

The contribution to composite error for each component are computed as follows.

- (1) The contribution to composite error due to the ARP position and velocity error is $\Delta\mathbf{RRdot}^{\text{APV}}$ with covariance matrix $\mathbf{C}_{\mathbf{RRdot}}^{\text{APV}}$. Error $\Delta\mathbf{APV}_{\text{AIF}}$ is the error in the AIF frame. Error $\Delta\mathbf{APV}_{\text{ECEF}}$ is the error in the ECEF frame.

$$\Delta\mathbf{APV}_{\text{ECEF}} = \begin{bmatrix} \Delta\mathbf{ARP}_{\text{COA}}^{\text{SCP}} \\ \Delta\mathbf{VARP} \end{bmatrix} = \mathbf{T}_{\text{AIF}}^{\text{ECEF}} \cdot \Delta\mathbf{APV}_{\text{AIF}}$$

$$\Delta\mathbf{RRdot}^{\text{APV}} = \mathbf{M}_{\text{APV}}^{\mathbf{RRdot}} \cdot \Delta\mathbf{APV}_{\text{ECEF}} = \mathbf{M}_{\text{APV}}^{\mathbf{RRdot}} \cdot \mathbf{T}_{\text{AIF}}^{\text{ECEF}} \cdot \Delta\mathbf{APV}_{\text{AIF}}$$

The covariance is computed as shown.

$$\mathbf{C}_{\mathbf{RRdot}}^{\text{APV}} = \mathbf{M}_{\text{APV}}^{\mathbf{RRdot}} \cdot \mathbf{T}_{\text{AIF}}^{\text{ECEF}} \cdot \mathbf{C}_{\text{APV}}^{\text{AIF}} \cdot (\mathbf{M}_{\text{APV}}^{\mathbf{RRdot}} \cdot \mathbf{T}_{\text{AIF}}^{\text{ECEF}})^T$$

- (2) The contribution to composite error due to a range error that is constant across slow time is $\Delta\mathbf{RRdot}^{\text{RB}}$ with covariance matrix $\mathbf{C}_{\mathbf{RRdot}}^{\text{RB}}$.

$$\Delta\mathbf{RRdot}^{\text{RB}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \Delta\mathbf{R_BIAS}$$

$$\mathbf{C}_{\mathbf{RRdot}}^{\text{RB}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \text{VAR}_{\text{RB}}$$

- (3) The contribution to composite error due to a control clock frequency error is $\Delta\mathbf{RRdot}^{\text{CLK_SF}}$ with covariance matrix $\mathbf{C}_{\mathbf{RRdot}}^{\text{CLK_SF}}$.

$$\Delta\mathbf{RRdot}^{\text{CLK_SF}} = \mathbf{M}_{\text{CLK_SF}}^{\mathbf{RRdot}} \cdot \text{CLK_SF}$$

The covariance is computed as shown.

$$\mathbf{C}_{\mathbf{RRdot}}^{\text{CLK_SF}} = \mathbf{M}_{\text{CLK_SF}}^{\mathbf{RRdot}} \cdot (\mathbf{M}_{\text{CLK_SF}}^{\mathbf{RRdot}})^T \cdot \text{VAR}_{\text{CLK_SF}}$$

- (4) The contribution to composite error due to an error in tropospheric delay estimate is $\Delta\mathbf{RRdot}^{\text{TROP}}$ with covariance matrix $\mathbf{C}_{\mathbf{RRdot}}^{\text{TROP}}$.

$$\Delta\mathbf{RRdot}^{\text{TROP}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \Delta\mathbf{R_TROP}$$

$$\mathbf{C}_{\mathbf{RRdot}}^{\text{TROP}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \text{VAR}_{\text{TROP}}$$

- (5) The contribution to composite error due to an error in the ionospheric delay estimate is $\Delta\mathbf{RRdot}^{\text{IONO}}$ with covariance matrix $\mathbf{C}_{\text{RRdot}}^{\text{IONO}}$.

$$\Delta\mathbf{RRdot}^{\text{IONO}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \bullet \Delta R_{\text{IONO}}$$

$$\mathbf{C}_{\text{RRdot}}^{\text{IONO}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \bullet \text{VAR}_{\text{IONO}}$$

- (6) The contribution to composite error due to an unmodeled image error is $\Delta\mathbf{RRdot}^{\text{UI}}$ with covariance matrix $\mathbf{C}_{\text{RRdot}}^{\text{UI}}$.

$$\Delta\mathbf{RRdot}^{\text{UI}} = \mathbf{M}_{\text{IL}}^{\text{RRdot}} \bullet \Delta\mathbf{IL}_{\text{UI}}$$

$$\mathbf{C}_{\text{RRdot}}^{\text{UI}} = \mathbf{M}_{\text{IL}}^{\text{RRdot}} \bullet \mathbf{C}_{\text{UI}} \bullet (\mathbf{M}_{\text{IL}}^{\text{RRdot}})^T$$

- (7) The composite error covariance $\mathbf{C}_{\text{RRdot}}^{\text{IL/PT}}$ is computed by the sum of the covariances for each of the components.

$$\mathbf{C}_{\text{RRdot}}^{\text{IL/PT}} = \mathbf{C}_{\text{RRdot}}^{\text{APV}} + \mathbf{C}_{\text{RRdot}}^{\text{RB}} + \mathbf{C}_{\text{RRdot}}^{\text{CLK_SF}} + \mathbf{C}_{\text{RRdot}}^{\text{TROP}} + \mathbf{C}_{\text{RRdot}}^{\text{IONO}} + \mathbf{C}_{\text{RRdot}}^{\text{UI}}$$

- (8) The composite error expressed as a displacement in the COA slant plane is $\Delta\mathbf{RRdot}^{\text{IL/PT}}$ with covariance $\mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}}$.

$$\Delta\mathbf{RGAZ}^{\text{IL/PT}} = \mathbf{M}_{\text{RRdot}}^{\text{RGAZ}} \bullet \Delta\mathbf{RRdot}^{\text{IL/PT}}$$

$$\mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}} = \mathbf{M}_{\text{RRdot}}^{\text{RGAZ}} \bullet \mathbf{C}_{\text{RRdot}}^{\text{IL/PT}} \bullet (\mathbf{M}_{\text{RRdot}}^{\text{RGAZ}})^T$$

12.3.3 APOs Applied w/ APO Composite Statistics

For an image/scene pair for which APOs have been applied and only the composite error statistics for the SCP are provided, set the composite error covariance for the image/scene pair equal to the composite error covariance provided.

$$\mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}} = \mathbf{C}_{\text{RGAZ}}^{\text{SCPAPO}}$$

12.3.4 APOs Applied w/ APO Component Statistics

For an image/scene pair for which APOs have been applied and for which APO component error statistics are provided, the composite error statistics may be computed as follows. The monostatic APO vector, denoted $\mathbf{APO}_M$, is formed from the APO input parameters. The monostatic APO error vector, denoted $\Delta\mathbf{APO}_M$, is formed from the error in each APO vector component. The covariance is $\mathbf{C}_{APOM} = E\{ \Delta\mathbf{APO}_M \bullet (\Delta\mathbf{APO}_M)^T \}$

$$\mathbf{APO}_M = \begin{bmatrix} \Delta\mathbf{ARP}_{COA}^{SCP} \\ \Delta\mathbf{VARP} \\ \Delta tx_{COA}^{SCP} \\ \Delta tr_{COA}^{SCP} \end{bmatrix} \text{ Size } 8 \times 1 \quad \Delta\mathbf{APO}_M = \begin{bmatrix} \delta\mathbf{ARP}_{COA}^{SCP} \\ \delta\mathbf{VARP} \\ \delta tx_{COA}^{SCP} \\ \delta tr_{COA}^{SCP} \end{bmatrix} \text{ Size } 8 \times 1$$

- (1) Form the 2×2 sensitivity matrix $\mathbf{M}_{XRT}^{RRdot}$ that is used to compute the contribution to composite error from errors in transmit time, δtx_{COA}^{SCP} , and receive time, δtr_{COA}^{SCP} .

$$\mathbf{M}_{XRT}^{RRdot} = (c/2) \bullet \begin{bmatrix} -1 & +1 \\ 0 & 0 \end{bmatrix} \quad \text{where } \Delta\mathbf{RRdot}^{XRT} = \mathbf{M}_{XRT}^{RRdot} \bullet \begin{bmatrix} \delta tx_{COA}^{SCP} \\ \delta tr_{COA}^{SCP} \end{bmatrix}$$

- (2) Form the 2×8 sensitivity matrix $\mathbf{M}_{APOM}^{RRdot}$ that is used to compute the contribution to composite error from monostatic APO error vector.

$$\mathbf{M}_{APOM}^{RRdot} = [\mathbf{M}_{\Delta ARP}^{RRdot} \quad \mathbf{M}_{\Delta VARP}^{RRdot} \quad \mathbf{M}_{XRT}^{RRdot}]$$

- (3) The composite error due to an error in monostatic APO inputs is $\Delta\mathbf{RRdot}^{IL/PT}$ with covariance matrix $\mathbf{C}_{RRdot}^{IL/PT}$. The composite error is as follows.

$$\Delta\mathbf{RRdot}^{IL/PT} = \mathbf{M}_{APOM}^{RRdot} \bullet \Delta\mathbf{APO}_M$$

$$\mathbf{C}_{RRdot}^{IL/PT} = \mathbf{M}_{APOM}^{RRdot} \bullet \mathbf{C}_{APOM} \bullet (\mathbf{M}_{APOM}^{RRdot})^T$$

- (4) The composite error expressed as a displacement in the COA slant plane is $\Delta\mathbf{RRdot}^{IL/PT}$ with covariance $\mathbf{C}_{RGZ}^{IL/PT}$.

$$\Delta\mathbf{RGZ}^{IL/PT} = \mathbf{M}_{RRdot}^{RGZ} \bullet \Delta\mathbf{RRdot}^{IL/PT}$$

$$\mathbf{C}_{RGZ}^{IL/PT} = \mathbf{M}_{RRdot}^{RGZ} \bullet \mathbf{C}_{RRdot}^{IL/PT} \bullet (\mathbf{M}_{RRdot}^{RGZ})^T$$

12.4 Composite Error: Collect Type = Bistatic

For a bistatic image, the methods used to compute the expected composite error statistics for a given image/scene pair are specified as follows. The method used is dependent upon (1) if APOs were applied to the COA projection set, and (2) the error statistics provided with the product. The composite error may be expressed in either of the two defined forms.

$\Delta \mathbf{RRdot}^{IL/PT}$	Expressed as error in average range and average range rate relative to the transmit and receive APCs. The error covariance matrix is $\mathbf{C}_{RRdot}^{IL/PT}$.
$\Delta \mathbf{RGAZ}^{IL/PT}$	Expressed as a displacement in the COA slant plane. The error covariance matrix is $\mathbf{C}_{RGAZ}^{IL/PT}$.

The composite error statistics are computed as specified in the following subsections. For the input metadata to be used refer to Table 1-2 and Table 1-4.

12.4.1 APOs Not Applied w/ SCP Composite Statistics

For an image/scene pair for which APOs have not been applied and only the bistatic composite error statistics for the SCP are provided, set the composite error covariance for the image/scene pair $\mathbf{C}_{RRdot}^{IL/PT}$ equal to the composite error covariance provided. Compute the covariance $\mathbf{C}_{RGAZ}^{IL/PT}$ for the error expressed as a displacement in the COA slant plane.

$$\mathbf{C}_{RRdot}^{IL/PT} = \mathbf{C}_{RRdot}^{SCP} \quad \mathbf{C}_{RGAZ}^{IL/PT} = \mathbf{M}_{RRdot}^{RGAZ} \cdot \mathbf{C}_{RRdot}^{IL/PT} \cdot (\mathbf{M}_{RRdot}^{RGAZ})^T$$

12.4.2 APOs Not Applied w/ Component Statistics

For an image/scene pair for which APOs have not been applied and for which bistatic component error statistics are provided, compute the composite error covariance from the component error statistics provided. Also include the contribution from unmodeled image error. The components of the composite error are the following.

$\Delta \mathbf{XPV}_{ECEF}$	Transmit APC position error and velocity error at $t = t_{COA}^{SCP}$ in ECEF coordinates. Covariance $\mathbf{C}_{XPV}^{ECEF}$ computed from input $\mathbf{C}_{XPV}^{XIF}$.
$\Delta \mathbf{RPV}_{ECEF}$	Receive APC position error and velocity error at $t = t_{COA}^{SCP}$ in ECEF coordinates. Covariance $\mathbf{C}_{RPV}^{ECEF}$ computed from input $\mathbf{C}_{RPV}^{RIF}$.
$\Delta \mathbf{XRTF}$	Transmit and receive time and control clock frequency errors. Covariance $\mathbf{C}_{XRTF}$.
$\Delta \mathbf{TD_ATM}$	Atmospheric delay estimate errors on transmit and on receive. Covariance $\mathbf{C}_{ATM}$.
$\Delta \mathbf{IL}_{UI}$	Unmodeled image error for the selected image grid location. Covariance $\mathbf{C}_{UI}$.

The composite error $\Delta\mathbf{RRdot}^{\text{IL/PT}}$ is equal to the sum of the contribution from each component.

$$\Delta\mathbf{RRdot}^{\text{IL/PT}} = \Delta\mathbf{RRdot}^{\text{XPV_RPV}} + \Delta\mathbf{RRdot}^{\text{XRTF}} + \Delta\mathbf{RRdot}^{\text{ATM}} + \Delta\mathbf{RRdot}^{\text{UI}}$$

The contribution to composite error for each component is computed as follows.

- (1) The transmit APC position and velocity error at $t = t_{\text{COA}}^{\text{SCP}}$ in the XIF frame is $\Delta\mathbf{XPV}_{\text{XIF}}$ with covariance $\mathbf{C}_{\text{XPV}}^{\text{XIF}}$. The error at $t = t_{\text{COA}}^{\text{SCP}}$ in the ECEF frame is $\Delta\mathbf{XPV}_{\text{ECEF}}$ with covariance $\mathbf{C}_{\text{XPV}}^{\text{ECEF}}$. Compute the 6 x 6 covariance $\mathbf{C}_{\text{XPV}}^{\text{ECEF}}$.

$$\Delta\mathbf{XPV}_{\text{ECEF}} = \begin{bmatrix} \Delta\mathbf{Xmt}_{\text{COA}}^{\text{SCP}} \\ \Delta\mathbf{VXmt} \end{bmatrix} = \mathbf{T}_{\text{XIF}}^{\text{ECEF}} \cdot \Delta\mathbf{XPV}_{\text{XIF}}$$

$$\mathbf{C}_{\text{XPV}}^{\text{ECEF}} = \mathbf{T}_{\text{XIF}}^{\text{ECEF}} \cdot \mathbf{C}_{\text{XPV}}^{\text{XIF}} \cdot (\mathbf{T}_{\text{XIF}}^{\text{ECEF}})^T$$

- (2) The receive APC position and velocity error at $t = t_{\text{COA}}^{\text{SCP}}$ in the RIF frame is $\Delta\mathbf{RPV}_{\text{RIF}}$ with covariance $\mathbf{C}_{\text{RPV}}^{\text{RIF}}$. The error at $t = t_{\text{COA}}^{\text{SCP}}$ in the ECEF frame is $\Delta\mathbf{RPV}_{\text{ECEF}}$ with covariance $\mathbf{C}_{\text{RPV}}^{\text{ECEF}}$. Compute the 6 x 6 covariance $\mathbf{C}_{\text{RPV}}^{\text{ECEF}}$.

$$\Delta\mathbf{RPV}_{\text{ECEF}} = \begin{bmatrix} \Delta\mathbf{Rcv}_{\text{COA}}^{\text{SCP}} \\ \Delta\mathbf{VRcv} \end{bmatrix} = \mathbf{T}_{\text{RIF}}^{\text{ECEF}} \cdot \Delta\mathbf{RPV}_{\text{RIF}}$$

$$\mathbf{C}_{\text{RPV}}^{\text{ECEF}} = \mathbf{T}_{\text{RIF}}^{\text{ECEF}} \cdot \mathbf{C}_{\text{RPV}}^{\text{RIF}} \cdot (\mathbf{T}_{\text{RIF}}^{\text{ECEF}})^T$$

- (3) The cross covariance for the transmit APC position and velocity error $\Delta\mathbf{XPV}_{\text{XIF}}$ and the receive position and velocity error $\Delta\mathbf{RPV}_{\text{RIF}}$ is $\mathbf{CC}_{\text{XPV_RPV}}^{\text{XIF_RIF}}$. Compute the 6 x 6 cross covariance matrix for the errors $\Delta\mathbf{XPV}_{\text{ECEF}}$ and $\Delta\mathbf{RPV}_{\text{ECEF}}$.

$$\mathbf{CC}_{\text{XPV_RPV}}^{\text{ECEF}} = \mathbf{T}_{\text{XIF}}^{\text{ECEF}} \cdot \mathbf{CC}_{\text{XPV_RPV}}^{\text{XIF_RIF}} \cdot (\mathbf{T}_{\text{RIF}}^{\text{ECEF}})^T$$

- (4) The joint transmit/receive position and velocity error vector is $\Delta\mathbf{XPV_RPV}_{\text{ECEF}}$ with covariance matrix $\mathbf{C}_{\text{XPV_RPV}}^{\text{ECEF}}$. Form the 12 x 12 covariance matrix $\mathbf{C}_{\text{XPV_RPV}}^{\text{ECEF}}$ for the combined error vector.

$$\Delta\mathbf{XPV_RPV}_{\text{ECEF}} = \begin{bmatrix} \Delta\mathbf{XPV}_{\text{ECEF}} \\ \Delta\mathbf{RPV}_{\text{ECEF}} \end{bmatrix} \quad \mathbf{C}_{\text{XPV_RPV}}^{\text{ECEF}} = \begin{bmatrix} \mathbf{C}_{\text{XPV}}^{\text{ECEF}} & \mathbf{CC}_{\text{XPV_RPV}}^{\text{ECEF}} \\ (\mathbf{CC}_{\text{XPV_RPV}}^{\text{ECEF}})^T & \mathbf{C}_{\text{RPV}}^{\text{ECEF}} \end{bmatrix}$$

- (5) Form the 2 x 12 sensitivity matrix $\mathbf{M}_{\text{XPV_RPV}}^{\text{RRdot}}$.

$$\mathbf{M}_{\text{XPV_RPV}}^{\text{RRdot}} = [\mathbf{M}_{\text{XPV}}^{\text{RRdot}} \quad \mathbf{M}_{\text{RPV}}^{\text{RRdot}}]$$

(6) The contribution to composite error due to the joint error vector $\Delta\mathbf{XPV_RPV}_{\text{ECEF}}$ is $\Delta\mathbf{RRdot}^{\text{XPV_RPV}}$. The covariance matrix $\mathbf{C}_{\text{RRdot}}^{\text{XPV_RPV}}$ is computed from the joint position/velocity error covariance $\mathbf{C}_{\text{XPV_RPV}}^{\text{ECEF}}$.

$$\Delta\mathbf{RRdot}^{\text{XPV_RPV}} = \mathbf{M}_{\text{XPV_RPV}}^{\text{RRdot}} \bullet \Delta\mathbf{XPV_RPV}_{\text{ECEF}}$$

$$\mathbf{C}_{\text{RRdot}}^{\text{XPV_RPV}} = \mathbf{M}_{\text{XPV_RPV}}^{\text{RRdot}} \bullet \mathbf{C}_{\text{XPV_RPV}}^{\text{ECEF}} \bullet (\mathbf{M}_{\text{XPV_RPV}}^{\text{RRdot}})^T$$

(7) The transmit sensor time and control clock frequency error vector is $\Delta\mathbf{XTF}$. The receive sensor time and control clock frequency error vector is $\Delta\mathbf{RTF}$. The joint time and control clock frequency error vector is $\Delta\mathbf{XRTF}$.

$$\Delta\mathbf{XTF} = \begin{bmatrix} \Delta\text{tx}_{\text{COA}}^{\text{SCP}} \\ \text{f_Clk_X_SF} \end{bmatrix} \quad \Delta\mathbf{RTF} = \begin{bmatrix} \Delta\text{tr}_{\text{COA}}^{\text{SCP}} \\ \text{f_Clk_R_SF} \end{bmatrix} \quad \Delta\mathbf{XRTF} = \begin{bmatrix} \Delta\mathbf{XTF} \\ \Delta\mathbf{RTF} \end{bmatrix}$$

(8) The contribution to composite error due to the joint time and control clock frequency errors is $\Delta\mathbf{RRdot}^{\text{XRTF}}$ with covariance matrix $\mathbf{C}_{\text{RRdot}}^{\text{XRTF}}$. Form the 2 x 4 sensitivity matrix $\mathbf{M}_{\text{XRTF}}^{\text{RRdot}}$ that is used to compute the composite error from the joint error vector and composite error covariance matrix.

$$\mathbf{M}_{\text{XRTF}}^{\text{RRdot}} = [\mathbf{M}_{\text{XTF}}^{\text{RRdot}} \quad \mathbf{M}_{\text{RTF}}^{\text{RRdot}}]$$

$$\Delta\mathbf{RRdot}^{\text{XRTF}} = \mathbf{M}_{\text{XRTF}}^{\text{RRdot}} \bullet \Delta\mathbf{XRTF}$$

$$\mathbf{C}_{\text{RRdot}}^{\text{XRTF}} = \mathbf{M}_{\text{XRTF}}^{\text{RRdot}} \bullet \mathbf{C}_{\text{XRTF}} \bullet (\mathbf{M}_{\text{XRTF}}^{\text{RRdot}})^T$$

(9) The contribution to composite error due to errors in the atmospheric delay estimates is $\Delta\mathbf{RRdot}^{\text{ATM}}$ with covariance matrix $\mathbf{C}_{\text{RRdot}}^{\text{ATM}}$.

$$\Delta\mathbf{RRdot}^{\text{ATM}} = \mathbf{M}_{\text{ATM}}^{\text{RRdot}} \bullet \Delta\mathbf{TD_ATM}$$

$$\mathbf{C}_{\text{RRdot}}^{\text{ATM}} = \mathbf{M}_{\text{ATM}}^{\text{RRdot}} \bullet \mathbf{C}_{\text{ATM}} \bullet (\mathbf{M}_{\text{ATM}}^{\text{RRdot}})^T$$

(10) The contribution to composite error due to an unmodeled image error is $\Delta\mathbf{RRdot}^{\text{UI}}$ with covariance matrix $\mathbf{C}_{\text{RRdot}}^{\text{UI}}$.

$$\Delta\mathbf{RRdot}^{\text{UI}} = \mathbf{M}_{\text{IL}}^{\text{RRdot}} \bullet \Delta\mathbf{IL}_{\text{UI}}$$

$$\mathbf{C}_{\text{RRdot}}^{\text{UI}} = \mathbf{M}_{\text{IL}}^{\text{RRdot}} \bullet \mathbf{C}_{\text{UI}} \bullet (\mathbf{M}_{\text{IL}}^{\text{RRdot}})^T$$

(11) The composite error covariance $\mathbf{C}_{\text{RRdot}}^{\text{IL/PT}}$ is computed by the sum of the covariances for each of the components.

$$\mathbf{C}_{\text{RRdot}}^{\text{IL/PT}} = \mathbf{C}_{\text{RRdot}}^{\text{XPV_RPV}} + \mathbf{C}_{\text{RRdot}}^{\text{XRTF}} + \mathbf{C}_{\text{RRdot}}^{\text{ATM}} + \mathbf{C}_{\text{RRdot}}^{\text{UI}}$$

(12) The composite error expressed as a displacement in the COA slant plane is $\Delta \mathbf{RRdot}^{IL/PT}$ with covariance $\mathbf{C}_{\mathbf{RGAZ}}^{IL/PT}$.

$$\Delta \mathbf{RGAZ}^{IL/PT} = \mathbf{M}_{\mathbf{RRdot}}^{\mathbf{RGAZ}} \bullet \Delta \mathbf{RRdot}^{IL/PT}$$

$$\mathbf{C}_{\mathbf{RGAZ}}^{IL/PT} = \mathbf{M}_{\mathbf{RRdot}}^{\mathbf{RGAZ}} \bullet \mathbf{C}_{\mathbf{RRdot}}^{IL/PT} \bullet (\mathbf{M}_{\mathbf{RRdot}}^{\mathbf{RGAZ}})^T$$

12.4.3 APOs Applied w/ APO Composite Statistics

For an image/scene pair for which APOs have been applied and only the composite error statistics for the SCP are provided, set the composite error covariance for the image/scene pair $\mathbf{C}_{\mathbf{RRdot}}^{IL/PT}$ equal to the composite error covariance provided. Compute the covariance $\mathbf{C}_{\mathbf{RGAZ}}^{IL/PT}$ for the error expressed as a displacement in the COA slant plane.

$$\mathbf{C}_{\mathbf{RGAZ}}^{IL/PT} = \mathbf{C}_{\mathbf{RRdot}}^{\mathbf{SCPAPO}} \qquad \mathbf{C}_{\mathbf{RGAZ}}^{IL/PT} = \mathbf{M}_{\mathbf{RRdot}}^{\mathbf{RGAZ}} \bullet \mathbf{C}_{\mathbf{RRdot}}^{IL/PT} \bullet (\mathbf{M}_{\mathbf{RRdot}}^{\mathbf{RGAZ}})^T$$

12.4.4 APOs Applied w/ APO Component Statistics

For an image/scene pair for which APOs have been applied and for which APO component error statistics are provided, the composite error statistics are computed as follows. The transmit sensor APO vector, denoted $\mathbf{APO}_X$, is formed from the transmit sensor APO input parameters. The transmit sensor APO error vector, denoted $\Delta \mathbf{APO}_X$, is formed from the error in each transmit APO vector component. The receive sensor APO vector, denoted $\mathbf{APO}_R$, is formed from the receive sensor APO input parameters. The receive sensor APO error vector, denoted $\Delta \mathbf{APO}_R$, is formed from the error in each receive APO vector component.

$$\mathbf{APO}_X = \begin{bmatrix} \Delta \mathbf{Xmt}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \Delta \mathbf{VXmt} \\ \Delta \mathbf{tx}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \mathbf{f_Clk_SF_X} \end{bmatrix} \text{ Size } 8 \times 1 \qquad \Delta \mathbf{APO}_X = \begin{bmatrix} \delta \mathbf{Xmt}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \delta \mathbf{VXmt} \\ \delta \mathbf{tx}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \delta \mathbf{f_Clk_SF_X} \end{bmatrix} \text{ Size } 8 \times 1$$

$$\mathbf{APO}_R = \begin{bmatrix} \Delta \mathbf{Rcv}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \Delta \mathbf{VRcv} \\ \Delta \mathbf{tr}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \mathbf{f_Clk_SF_R} \end{bmatrix} \text{ Size } 8 \times 1 \qquad \Delta \mathbf{APO}_R = \begin{bmatrix} \delta \mathbf{Rcv}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \delta \mathbf{VRcv} \\ \delta \mathbf{tr}_{\mathbf{COA}}^{\mathbf{SCP}} \\ \delta \mathbf{f_Clk_SF_R} \end{bmatrix} \text{ Size } 8 \times 1$$

The bistatic APO error vector $\Delta \mathbf{APO}_{XR}$ is formed from the two APO vectors and has covariance $\mathbf{C}_{\mathbf{APOXR}}$.

$$\Delta \mathbf{APO}_{XR} = \begin{bmatrix} \Delta \mathbf{APO}_X \\ \Delta \mathbf{APO}_R \end{bmatrix} \text{ Size } 16 \times 1 \qquad \mathbf{C}_{\mathbf{APOXR}} = E \{ \Delta \mathbf{APO}_{XR} \bullet (\Delta \mathbf{APO}_{XR})^T \}$$

- (1) Form the 2 x 16 sensitivity matrix $\mathbf{M}_{\text{APOXR}}^{\text{RRdot}}$ that is used to compute the contribution to composite error from bistatic APO error vector.

$$\mathbf{M}_{\text{APOXR}}^{\text{RRdot}} = [\mathbf{M}_{\text{XPV}}^{\text{RRdot}} \quad \mathbf{M}_{\text{XTF}}^{\text{RRdot}} \quad \mathbf{M}_{\text{RPV}}^{\text{RRdot}} \quad \mathbf{M}_{\text{RTF}}^{\text{RRdot}}]$$

Note: The matrices $\mathbf{M}_{\text{XTF}}^{\text{RRdot}}$ and $\mathbf{M}_{\text{RTF}}^{\text{RRdot}}$ specified in Section 11.5.2 were derived for computing the contribution of sensor timing and frequency errors to the composite error for a given IL/PT pair. They also provide an accurate approximation for computing the contribution of APO offset errors to the composite error.

- (2) The composite error due to an error in bistatic APOs is $\Delta\mathbf{RRdot}^{\text{IL/PT}}$ with covariance matrix $\mathbf{C}_{\text{RRdot}}^{\text{IL/PT}}$. The composite error is as follows.

$$\Delta\mathbf{RRdot}^{\text{IL/PT}} = \mathbf{M}_{\text{APOXR}}^{\text{RRdot}} \cdot \Delta\mathbf{APO}_{\text{XR}}$$

$$\mathbf{C}_{\text{RRdot}}^{\text{IL/PT}} = \mathbf{M}_{\text{APOXR}}^{\text{RRdot}} \cdot \mathbf{C}_{\text{APOXR}} \cdot (\mathbf{M}_{\text{APOXR}}^{\text{RRdot}})^T$$

- (3) The composite error expressed as a displacement in the COA slant plane is $\Delta\mathbf{RRdot}^{\text{IL/PT}}$ with covariance $\mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}}$.

$$\Delta\mathbf{RGAZ}^{\text{IL/PT}} = \mathbf{M}_{\text{RRdot}}^{\text{RGAZ}} \cdot \Delta\mathbf{RRdot}^{\text{IL/PT}}$$

$$\mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}} = \mathbf{M}_{\text{RRdot}}^{\text{RGAZ}} \cdot \mathbf{C}_{\text{RRdot}}^{\text{IL/PT}} \cdot (\mathbf{M}_{\text{RRdot}}^{\text{RGAZ}})^T$$

12.5 Image-to-Scene Error Propagation

The image-to-scene projection error is defined to be the difference between the true scene position, $\mathbf{PT0}$, minus the estimated scene point position, $\mathbf{E:PT0}$. The true and estimated scene point positions are in ECEF coordinates.

$$\text{Image-to-Scene Projection Error: } \Delta\mathbf{PT0} = \mathbf{PT0} - \mathbf{E:PT0}$$

The projection error, $\Delta\mathbf{PT0}$, and its expected statistics are computed for the following sources of error. Each source of error is assumed to be zero mean. All sources are assumed to be statistically independent.

$\Delta\mathbf{RGAZ}^{\text{IL/PT}}$	Composite RGAZ image error for the given image/scene projection pair (components $\Delta\mathbf{RG}$ and $\Delta\mathbf{AZ}$ are in meters). The predicted error covariance is matrix $\mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}}$.
$\Delta\mathbf{IL}_{\text{SEL}}$	Error in estimated image location (components Δx_{row} and Δy_{col} are in meters). Applicable when the initial image location $\mathbf{E:IL0}$ is estimated or measured for a distributed target or feature in the image. True image location is: $\mathbf{IL0} = \mathbf{E:IL0} + \Delta\mathbf{IL}_{\text{SEL}}$. The predicted error covariance is 2 x 2 matrix $\mathbf{C}_{\text{IL_SEL}}$. All covariance elements are in m^2 .
$\Delta\mathbf{HAE}$	Error in estimated scene surface Height Above the Ellipsoid (HAE) for a scene surface in the neighborhood of the estimated scene point. Error is in meters and is the true height minus the estimated height: $\Delta\mathbf{HAE} = \mathbf{HAE} - \mathbf{E:HAE}$. Height error variance is VAR_{HAE} and is in m^2 .

(1) The image-to-scene projection error due to the composite image error is a displacement in the ground plane. See Figure 12-7. The error expressed in ground plane coordinates GPX and GPY is $\Delta\mathbf{GPXY}^{\text{RGAZ}}$. The error in ECEF coordinates is $\Delta\mathbf{PT0}^{\text{RGAZ}}$.

$$\Delta\mathbf{GPXY}^{\text{RGAZ}} = \begin{bmatrix} \Delta\mathbf{GPX} \\ \Delta\mathbf{GPY} \end{bmatrix} = \mathbf{M}_{\text{SPXY}}^{\text{GPXY}} \cdot \mathbf{MIL}_{\text{RGAZ}}^{\text{SPXY}} \cdot \Delta\mathbf{RGAZ}^{\text{IL/PT}}$$

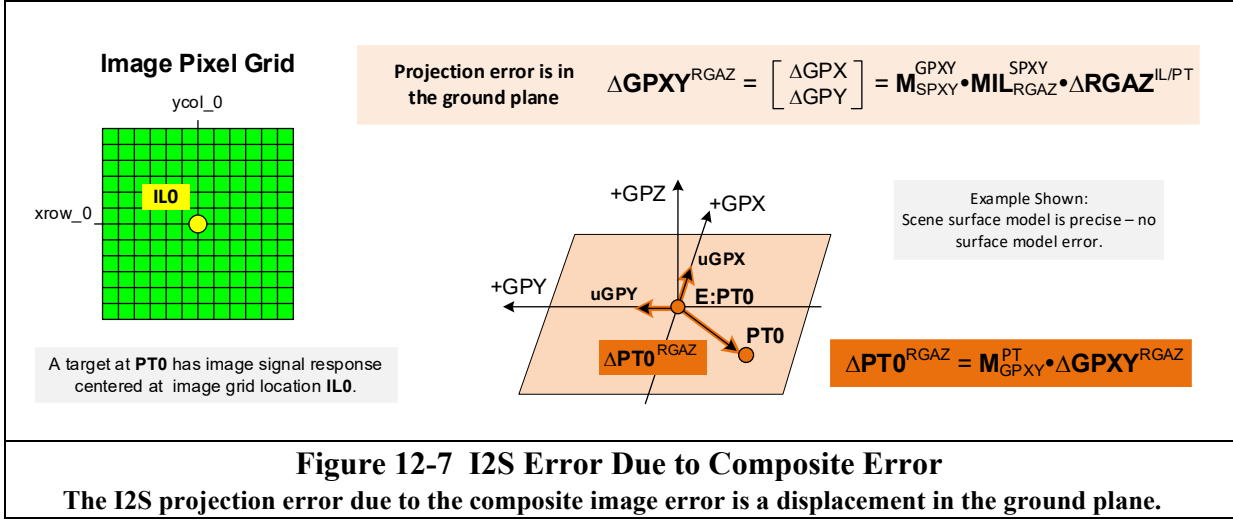
$$\Delta\mathbf{PT0}^{\text{RGAZ}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \Delta\mathbf{GPXY}^{\text{RGAZ}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \mathbf{M}_{\text{SPXY}}^{\text{GPXY}} \cdot \mathbf{MIL}_{\text{RGAZ}}^{\text{SPXY}} \cdot \Delta\mathbf{RGAZ}^{\text{IL/PT}}$$

The error covariance matrix for each form of the error vector is computed as shown.

$$\mathbf{C}_{\text{GPXY}}^{\text{RGAZ}} = \mathbf{M}_{\text{SPXY}}^{\text{GPXY}} \cdot \mathbf{MIL}_{\text{RGAZ}}^{\text{SPXY}} \cdot \mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}} \cdot (\mathbf{M}_{\text{SPXY}}^{\text{GPXY}} \cdot \mathbf{MIL}_{\text{RGAZ}}^{\text{SPXY}})^T$$

$$\mathbf{C}_{\text{PT}}^{\text{RGAZ}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \mathbf{C}_{\text{GPXY}}^{\text{RGAZ}} \cdot (\mathbf{M}_{\text{GPXY}}^{\text{PT}})^T$$

Note: The covariance matrix $\mathbf{C}_{\text{GPXY}}^{\text{RGAZ}}$ may be used to compute ground plane error statistics (e.g., the CE50) for all errors associated with the SAR sensor and the image formation processor.



(2) The image-to-scene projection error due to an error in selecting the initial image grid location is also a displacement in the ground plane. See Figure 12-8. The error expressed in ground plane coordinates GPX and GPY is $\Delta \mathbf{GPXY}^{\text{IL_SEL}}$. The error in ECEF coordinates is $\Delta \mathbf{PT0}^{\text{IL_SEL}}$.

$$\Delta \mathbf{GPXY}^{\text{IL_SEL}} = \mathbf{M}_{\text{IL}}^{\text{GPXY}} \cdot \Delta \mathbf{IL}_{\text{SEL}}$$

$$\Delta \mathbf{PT0}^{\text{IL_SEL}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \Delta \mathbf{GPXY}^{\text{IL_SEL}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \mathbf{M}_{\text{IL}}^{\text{GPXY}} \cdot \Delta \mathbf{IL}_{\text{SEL}}$$

The error covariance matrix for each form of the error vector is computed as shown.

$$\mathbf{C}_{\text{GPXY}}^{\text{IL_SEL}} = \mathbf{M}_{\text{IL}}^{\text{GPXY}} \cdot \mathbf{C}_{\text{IL_SEL}} \cdot (\mathbf{M}_{\text{IL}}^{\text{GPXY}})^T$$

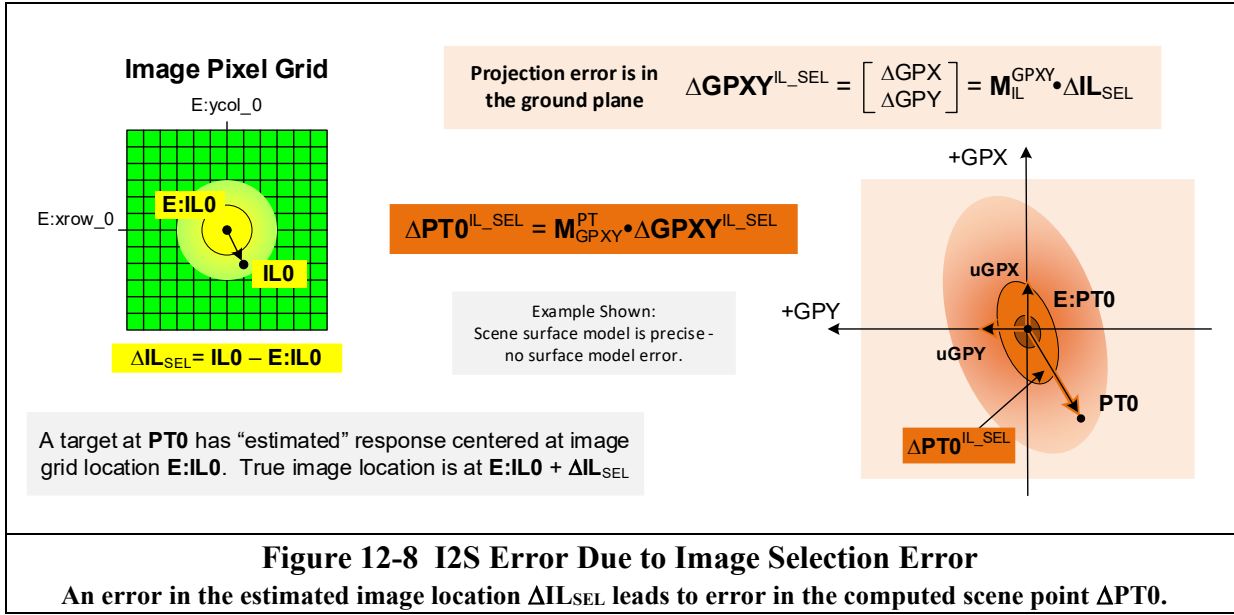
$$\mathbf{C}_{\text{PT}}^{\text{IL_SEL}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \mathbf{C}_{\text{GPXY}}^{\text{IL_SEL}} \cdot (\mathbf{M}_{\text{GPXY}}^{\text{PT}})^T$$

(3) The combined error due to the composite image error and an error in image location selection is the sum of the errors. The combined error is a displacement in the ground plane.

$$\Delta \mathbf{GPXY}^{\text{SUM}} = \Delta \mathbf{GPXY}^{\text{RGZ}} + \Delta \mathbf{GPXY}^{\text{IL_SEL}}$$

The covariance matrix of the combined error is the sum of the covariance matrices.

$$\mathbf{C}_{\text{GPXY}}^{\text{SUM}} = \mathbf{C}_{\text{GPXY}}^{\text{RGZ}} + \mathbf{C}_{\text{GPXY}}^{\text{IL_SEL}}$$

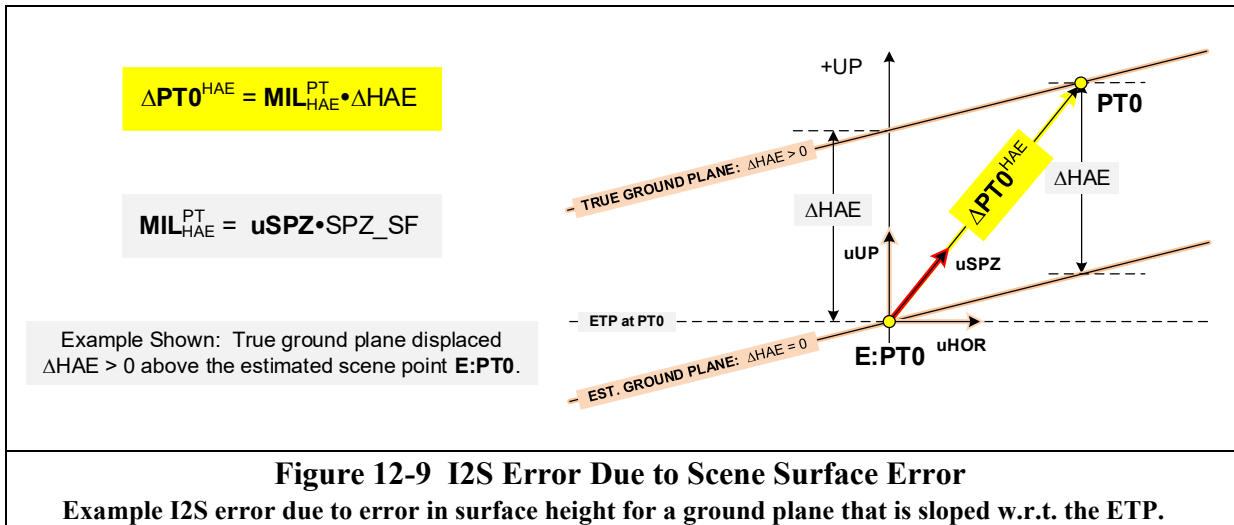


(4) The image-to-scene projection error due to an error in estimated surface height is a displacement along the slant plane normal. For a height error ΔHAE in the estimated scene surface, the contribution to the I2S projection error is $\Delta PT0^{HAE}$. Refer to Figure 12-9.

$$\Delta PT0^{HAE} = MIL_{HAE}^{PT} \cdot \Delta HAE$$

The covariance matrix for error is computed from the variance of the height error.

$$C_{PT}^{HAE} = MIL_{HAE}^{PT} \cdot (MIL_{HAE}^{PT})^T \cdot VAR_{HAE}$$



12.6 Scene-to-Image Error Propagation

The scene-to-image projection error is defined to be the difference between the true image grid location, $\mathbf{IL0}$, minus the estimated image grid location, $\mathbf{E:IL0}$. The true and estimated grid locations are in image coordinates (components xrow and ycol are in meters).

$$\text{Scene-to-Image Projection Error: } \Delta\mathbf{IL0} = \mathbf{IL0} - \mathbf{E:IL0}$$

The projection error, $\Delta\mathbf{IL0}$, and its expected statistics are computed for the following sources of error. Each source of error is assumed to be zero mean. All sources are assumed to be statistically independent.

$\Delta\mathbf{RGAZ}^{\mathbf{IL/PT}}$	Composite RGAZ image error for the given image/scene projection pair (components $\Delta\mathbf{RG}$ and $\Delta\mathbf{AZ}$ are in meters). The predicted error covariance is matrix $\mathbf{C}_{\mathbf{RGAZ}}^{\mathbf{IL/PT}}$.
$\Delta\mathbf{PT}_{\mathbf{SEL}}$	Error in estimated scene point position in ECEF coordinates. Applicable when the initial scene point position is an estimated position, $\mathbf{E:PT0}$. True scene position is: $\mathbf{PT0} = \mathbf{E:PT0} + \Delta\mathbf{PT}_{\mathbf{SEL}}$. The predicted error covariance is 3 x 3 matrix $\mathbf{C}_{\mathbf{PT_SEL}}$. All covariance elements are in m ² .

(1) The scene-to-image projection error due to the composite image error is computed as follows. The error in estimated image location $\Delta\mathbf{IL0}^{\mathbf{RGAZ}}$ is in image coordinates. The covariance matrix is computed as follows.

$$\Delta\mathbf{IL0}^{\mathbf{RGAZ}} = \mathbf{M}_{\mathbf{SPXY}}^{\mathbf{IL}} \cdot \mathbf{MPT}_{\mathbf{RGAZ}}^{\mathbf{SPXY}} \cdot \Delta\mathbf{RGAZ}^{\mathbf{IL/PT}}$$

$$\mathbf{C}_{\mathbf{IL}}^{\mathbf{RGAZ}} = \mathbf{M}_{\mathbf{SPXY}}^{\mathbf{IL}} \cdot \mathbf{MPT}_{\mathbf{RGAZ}}^{\mathbf{SPXY}} \cdot \mathbf{C}_{\mathbf{RGAZ}}^{\mathbf{IL/PT}} \cdot (\mathbf{M}_{\mathbf{SPXY}}^{\mathbf{IL}} \cdot \mathbf{MPT}_{\mathbf{RGAZ}}^{\mathbf{SPXY}})^T$$

(2) The scene-to-image projection error due to an error in the estimated scene point position is computed as follows. The error in estimated image location $\Delta\mathbf{IL0}^{\mathbf{PT_SEL}}$ is in image coordinates. The covariance matrix is computed as follows. Shown in Figure 12-11 is example error in the estimated scene point position that results in an error in estimated image location.

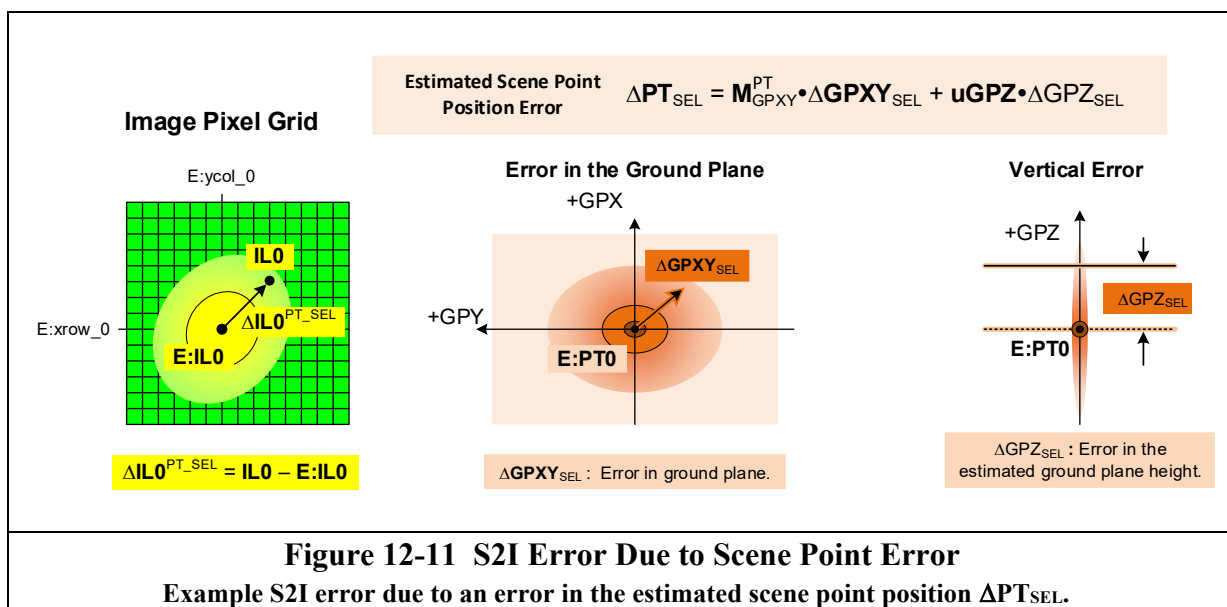
$$\Delta\mathbf{IL0}^{\mathbf{PT_SEL}} = \mathbf{M}_{\mathbf{PT}}^{\mathbf{IL}} \cdot \Delta\mathbf{PT}_{\mathbf{SEL}}$$

$$\mathbf{C}_{\mathbf{IL}}^{\mathbf{PT_SEL}} = \mathbf{M}_{\mathbf{PT}}^{\mathbf{IL}} \cdot \mathbf{C}_{\mathbf{PT_SEL}} \cdot (\mathbf{M}_{\mathbf{PT}}^{\mathbf{IL}})^T$$

(3) The total I2S error is computed by the sum of the contributions from each independent source. The covariance of the total error is the sum of the covariances.

$$\Delta\mathbf{IL0} = \begin{bmatrix} \Delta\mathbf{xrow_0} \\ \Delta\mathbf{ycol_0} \end{bmatrix} = \Delta\mathbf{IL0}^{\mathbf{RGAZ}} + \Delta\mathbf{IL0}^{\mathbf{PT_SEL}}$$

$$\mathbf{C}_{\mathbf{IL}} = \mathbf{E} \{ \Delta\mathbf{IL0} \cdot (\Delta\mathbf{IL0})^T \} = \mathbf{C}_{\mathbf{IL}}^{\mathbf{RGAZ}} + \mathbf{C}_{\mathbf{IL}}^{\mathbf{PT_SEL}}$$



12.7 Error Propagation Summary

For an ideal scatterer at true scene point position **PT0**, the image signal response is centered at image grid location **IL0**. The methods for computing projection errors and the expected error statistics are summarized below. For a given parameter with true value **X** and an estimated or measured value **E:X**, the error ΔX is defined as the true value minus the estimated value. For all parameters, the expected value of the error $E\{\Delta X\}$ is assumed to be zero.

$$\text{Error} = \text{True Value} - \text{Estimated Value} \quad \Delta X = X - E:X \quad E\{\Delta X\} = 0$$

Image-to-Scene Projection Error

The method for computing the error in the image-to-scene projection is summarized below. See Section 12.5. The initial image grid location may be the true image location, **IL0**, or an estimate of the location, **E:IL0**, that may include an error, $\Delta \mathbf{IL}_{\text{SEL}}$.

$$\mathbf{IL0} = \mathbf{E:IL0} + \Delta \mathbf{IL}_{\text{SEL}}$$

The projection error, $\Delta \mathbf{PT0}$, and its expected statistics are computed for the following sources of error. Each source of error is assumed to be zero mean. All sources are assumed to be statistically independent.

$$\text{Image-to-Scene Projection Error:} \quad \Delta \mathbf{PT0} = \mathbf{PT0} - \mathbf{E:PT0}$$

Image-to-Scene Error Sources:

$\Delta \mathbf{RGAZ}^{\text{IL/PT}}$	Composite RGAZ error for the image/scene projection pair (m). The predicted error covariance is matrix $\mathbf{C}_{\text{RGAZ}}^{\text{IL/PT}}$ (m ²). See Section 12.3 and Section 12.4.
$\Delta \mathbf{IL}_{\text{SEL}}$	Error in estimated image grid location when the initial image location is an estimated value E:IL0 . The error covariance is matrix $\mathbf{C}_{\text{IL_SEL}}$ (m ²).
ΔHAE	Error in estimated scene surface Height Above the Ellipsoid (HAE) (m). The height error variance is VAR_{HAE} (m ²).

Image-to-Scene Error, $\Delta \mathbf{PT0}$:

$$\Delta \mathbf{PT0} = \begin{bmatrix} \Delta \mathbf{PT0_X} \\ \Delta \mathbf{PT0_Y} \\ \Delta \mathbf{PT0_Z} \end{bmatrix} = \Delta \mathbf{PT0}^{\text{RGAZ}} + \Delta \mathbf{PT0}^{\text{IL_SEL}} + \Delta \mathbf{PT0}^{\text{HAE}}$$

$$\Delta \mathbf{PT0}^{\text{RGAZ}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \mathbf{M}_{\text{SPXY}}^{\text{GPXY}} \cdot \mathbf{MIL}_{\text{RGAZ}}^{\text{SPXY}} \cdot \Delta \mathbf{RGAZ}^{\text{IL/PT}}$$

$$\Delta \mathbf{PT0}^{\text{IL_SEL}} = \mathbf{M}_{\text{GPXY}}^{\text{PT}} \cdot \mathbf{M}_{\text{IL}}^{\text{GPXY}} \cdot \Delta \mathbf{IL}_{\text{SEL}}$$

$$\Delta \mathbf{PT0}^{\text{HAE}} = \mathbf{MIL}_{\text{HAE}}^{\text{PT}} \cdot \Delta \text{HAE}$$

Image-to-Scene Error Covariance, C_{PT} :

$$C_{PT} = E\{ \Delta PT0 \cdot (\Delta PT0)^T \} = C_{PT}^{RGZ} + C_{PT}^{IL_SEL} + C_{PT}^{HAE}$$

$$C_{PT}^{RGZ} = M_{GPXY}^{PT} \cdot M_{SPXY}^{GPXY} \cdot MIL_{RGZ}^{SPXY} \cdot C_{RGZ}^{IL/PT} \cdot (M_{GPXY}^{PT} \cdot M_{SPXY}^{GPXY} \cdot MIL_{RGZ}^{SPXY})^T$$

$$C_{PT}^{IL_SEL} = M_{GPXY}^{PT} \cdot M_{IL}^{GPXY} \cdot C_{IL_SEL} \cdot (M_{GPXY}^{PT} \cdot M_{IL}^{GPXY})^T$$

$$C_{PT}^{HAE} = MIL_{HAE}^{PT} \cdot (MIL_{HAE}^{PT})^T \cdot VAR_{HAE}$$

Scene-to-Image Projection Error

The method for computing the error in the scene-to-image projection is summarized below. See Section 12.6. The initial scene point position may be the true position, $PT0$, or an estimate of the position, $E:PT0$, that may include an error, ΔPT_{SEL} .

$$PT0 = E:PT0 + \Delta PT_{SEL}$$

The projection error, $\Delta IL0$, and its expected statistics are computed for the following sources of error. Each source of error is assumed to be zero mean. All sources are assumed to be statistically independent.

$$\text{Scene-to-Image Projection Error: } \Delta IL0 = IL0 - E:IL0$$

Scene-to-Image Error Sources:

$\Delta RGZ^{IL/PT}$	Composite RGZ error for the image/scene projection pair (m). The predicted error covariance is matrix $C_{RGZ}^{IL/PT}$ (m^2). See Section 12.3 and Section 12.4.
ΔPT_{SEL}	Error in estimated scene point position when the initial scene point position is an estimated position, $E:PT0$. The error covariance is matrix C_{PT_SEL} .

Scene-to-Image Error, $\Delta IL0$:

$$\Delta IL0 = \begin{bmatrix} \Delta x_{row_0} \\ \Delta y_{col_0} \end{bmatrix} = \Delta IL0^{RGZ} + \Delta IL0^{PT_SEL}$$

$$\Delta IL0^{RGZ} = M_{SPXY}^{IL} \cdot M_{RGZ}^{SPXY} \cdot \Delta RGZ^{IL/PT}$$

$$\Delta IL0^{PT_SEL} = M_{PT}^{IL} \cdot \Delta PT_{SEL}$$

Scene-to-Image Error Covariance, $\mathbf{C}_{IL}$:

$$\mathbf{C}_{IL} = E\{ \Delta \mathbf{IL} \mathbf{0} \bullet (\Delta \mathbf{IL} \mathbf{0})^T \} = \mathbf{C}_{IL}^{RGZ} + \mathbf{C}_{IL}^{PT_SEL}$$

$$\mathbf{C}_{IL}^{RGZ} = \mathbf{M}_{SPXY}^{IL} \bullet \mathbf{MPT}_{RGZ}^{SPXY} \bullet \mathbf{C}_{RGZ}^{IL/PT} \bullet (\mathbf{M}_{SPXY}^{IL} \bullet \mathbf{MPT}_{RGZ}^{SPXY})^T$$

$$\mathbf{C}_{IL}^{PT_SEL} = \mathbf{M}_{PT}^{IL} \bullet \mathbf{C}_{PT_SEL} \bullet (\mathbf{M}_{PT}^{IL})^T$$